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	Item	Description	Page
	MACHINE SERIAL NO.: MODEL: CUSTOMER:	FB14430 BF-32F-60-76F NUSA TAMBANG PRATAMA	
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JOY MINING MACHINERY
STAMLER PRODUCTS

Final Specification Sheet

Date: _____April 12, 2011_____(Revised: April 29, 2011) _____

Send To: _____

SUBJECT: Final Specifications for S/N __14430_____

COMPANY:____NTP (Nusa Tambang Pratama) _____

Mine: _____

Attached are copies of final specification sheets for Stamler Equipment

Model: __BF-32F-60-76F_____.

No changes have been made _____

Please note and check this specification sheet. Notify us if any questions arise as this is the way we intend to build the equipment if everything is correct. Return one copy with your approval, and keep one copy for your records.

SPECIFICATIONS FOR STAMLER SURFACE
FOOT MOUNTED BELT FEEDER-BREAKER MODEL: BF-32F-60-76F

- | | | |
|----|--|--|
| A. | <u>ELECTRICAL</u> | |
| | 1. CONVEYOR PUMP MOTORS | 2-90 KW – 380 VAC |
| | 2. BREAKER MOTOR | 190 KW – 380 VAC |
| | 3. CONTROLLER | BY OTHERS |
| | 4. BREAKER SPEED SWITCH | SOLID STATE |
| | 5. ELECTRICAL EMERGENCY
SHUT OFF SWITCH | (2) ONE ON EACH SIDE OF MACHINE |
| | 6. EMERGENCY START UP HORN | FURNISHED |
| | 7. LOW OIL LEVEL AND HIGH
PRESSURE SWITCHES | SWITCHES WILL “SHUT DOWN” MACHINE
WHEN ACTIVATED |
| B. | <u>GEAR REDUCERS:</u> | |
| | 1. BREAKER | SHAFT MOUNTED (342 KW) |
| C. | <u>COUPLINGS:</u> | |
| | 1. BREAKER | TORQUE LIMITING CLUTCH |
| | 2. HYDRAULIC PUMP DRIVES | STEEL, GEAR TYPE |
| D. | <u>CONVEYOR AND DRIVE:</u> | |
| | 1. WIDTH | 2286 mm |
| | 2. CHAINS (COVERED) (4) | 89 mm PITCH, 99,792 Kgs. ULTIMATE
PER STRAND (399,168 TOTAL) WITH
FLIGHT ATTACHMENT CAPTURE PINS |
| | 3. FLIGHTS | 44 mm X 101 mm SOLID ALLOY STEEL |
| | 4. HEAD SHAFT (2) | 127 mm DIAMETER WITH 113 mm BEARINGS |
| | 5. TAIL SHAFT (2) | 101 mm DIAMETER WITH 101 mm BEARINGS,
REMOVABLE AS A UNIT |
| | 6. TAKE UP | SCREW TYPE |
| | 7. RELIEF | HYDRAULIC RELIEF VALVE |
| | 8. SPEED | VARIABLE |

SPECIFICATIONS FOR STAMLER SURFACE
FOOT MOUNTED BELT FEEDER-BREAKER MODEL: BF-32F-60-76F

D.	<u>CONVEYOR & DRIVE:</u> (CONT.)	
	9. HYDRAULIC MOTOR TO HEAD SHAFT DRIVES (2)	DIRECT - SHAFT MOUNTED
E.	<u>BREAKER:</u>	
	1. DIRECTION OF ROTATION	WITH MATERIAL FLOW
	2. HEIGHT SETTING	ADJUSTABLE
	3. RELIEF	TORQUE LIMITING CLUTCH
	4. REDUCER TO BREAKER SHAFT DRIVE	DIRECT SHAFT MOUNTED
	5. BITS	REPLACEABLE PICK TYPE HEAT TREATED AND CARBIDE INSERTS
	6. BREAKER SHAFT	235 mm DIAMETER SOLID STEEL
	7. BREAKER PICK DIAMETER	1270 mm
	8. SHAFT BEARINGS	203 mm WITH DUO-CONE SEALS
F.	<u>MOUNTING:</u>	FIXED PADS
G.	<u>DIMENSIONS:</u>	SEE DRAWING 37-006-4734
H.	<u>OTHER FEATURES:</u>	
	1. HYDRAULIC PUMPS	(2) PISTON TYPE
	2. HYDRAULIC OPERATED REAR DOOR	FURNISHED
	3. LUBRICATION	MANUAL-CENTRALIZED GREASE BLOCK
	4. HYDRAULIC CONVEYOR DRIVE MOTOR	LOW SPEED, HIGH TORQUE RADIAL PISTON
	5. OPERATING PRESSURE	193 BAR ESTIMATED
	6. OIL FILTRATION	7 MICRON (75 BETA)
	7. MISC. HYDRAULICS	JIC HOSES AND FITTINGS, PRESSURE GAUGES AND TEST FITTINGS
	8. AIR/OIL HEAT EXCHANGER	FURNISHED
	9. SEPARATE POWER UNIT	FURNISHED

SPECIFICATIONS FOR STAMLER SURFACE
FOOT MOUNTED BELT FEEDER-BREAKER MODEL: BF-32F-60-76F

H. OTHER FEATURES: (CONT.)

10. DISCHARGE CHUTE	FURNISHED
11. MATERIAL (COAL) SIZE IN	1000 mm X 1000 MM X 1000 MM MAXIMUM
12. MATERIAL (COAL) SIZE OUT	MINUS 200 mm (ONE DIMENSION ONLY)
13. THROUGHPUT (VARIABLE)	0-2500 MTPH
14. APPROXIMATE WEIGHT	52,000 KGS.



JOY MINING MACHINERY
STAMLER PRODUCTS

P.O. Box 307, 100 Stamler Street
Millersburg, Kentucky 40348

Meeting/Distribution Date: March 29, 2011

2 nd Design Review Required? Yes _____ No _____

Telephone: (859) 484-3431
Facsimile: (859) 484-3336

TO: DISTRIBUTION

SUBJECT: Specification Review S/N 14430 NTP (Nusa Tambang Pratama) BF-32F-60-76F
(Australia)

Electrical:

- 190 KW and (2) 90 KW, 380 VAC, 50 HZ electrics
- Controller by customer (220 VAC control voltage)
- 190 KW, 380 VAC, 50 HZ breaker motor
- (2) 90 KW, 380 VAC, 50 HZ pump motors
- JMM Millersburg to supply standard on-board devices on machine / power unit:
 - NEMA 4X junction box
 - Breaker speed switch (220 VAC input power to motion detector)
 - Start-up horn
 - (4) E-stops (one on each side of machine, two on power unit)
 - Low oil level switch
 - Vacuum switch
 - Charge pressure switch
 - Pump control
 - Amp cards
- Wiring in PVC coated flexible conduit
- JMM Millersburg to supply suggested electrical schematics
- Component locations per general arrangement drawing 98-370064734
- Emphasis to be made on protection of all cables from sharp corners of plates, gussets and covers. This is to be accomplished by beveling edges of plates or use of rubber conduit.

Hydraulics:

- (2) Piston pumps (70 GPM each @ 1500 RPM) (Linde 210 with stroke limiters)
- (2) Gear pumps (16 GPM each @ 1500 RPM)
- (2) Hagglunds CA-140 hydraulic motors (splined) for conveyor drive
- AOVH40 air/oil heat exchanger
- Bulkhead fitting on power unit and machine for hose connection (connections labeled or color coded). Bulkhead fitting located near conveyor motor on machine.
- Gear coupling between hydraulic pump and electric motor
- JIC fittings
- Emphasis to be made on protection of all hoses from sharp corners of plates, gussets and covers. This is to be accomplished by beveling edges of plates or use of rubber conduit.

Power Unit:

- Separate power unit with roof (for conveyor control), foot mounted
- 20 Foot (6 meter) interconnecting hoses supplied by JMM Millersburg
- Power unit mounted to machine for rear door cylinder control
- Emphasis to be made on protection of all hoses and cables from sharp corners of plates, gussets and covers. This is to be accomplished by beveling edges of plates or use of rubber conduit.

Lubrication:

- Manual central lube located on breaker drive side (between reducer and hopper access door).

Conveyor/Conveyor Drive:

- Dual 5.0 inch (127 mm) diameter head shafts with 4.44 inch (113 mm) bearings
- 9 Tooth head shaft sprockets
- Conveyor drive: -76, dual Hagglunds CA-140 splined
- Dual 3.44 inch (87 mm) diameter tail shafts with 3.44 inch (87 mm) bearings
- 3.50 Inch (89 mm) pitch conveyor chain with flights every sixth pitch
- 1.75 Inch x 4.00 inch (44 mm x 102 mm) alloy steel flights with vertical grousers (same as S/N 14244)
- Screw take-up on conveyors
- Throughput: 0-2500 MTPH

Breaker Drive/Breaker:

- Breaker drive: -60, Falk 1210 reducer (41.45:1 ratio)
- UJT328 coupling (electric motor to reducer) set to 4,092 ft-lbf
- 50 Inch (1270 mm) diameter breaker
- 8" (203 mm) Diameter bearings
- Output: -200 mm (one dimension only)
- Breaker cleaner

Main Frame/Hopper:

- AR400 deck
- 12" Channel construction
- Foot mounted
- Vertical hopper sides

Misc.:

- Access door on discharge chute (for maintenance access)
- Access doors on top and both sides between breaker and hopper for clearing
- Hydraulic operated intake end door on hopper with pin to lock in open position
- Discharge flange pattern per general arrangement drawing 98-370064734
- Hopper flange pattern per general arrangement drawing 98-370064734
- Standard Stamler paint specification for overseas machines
- Do not paint hydraulic hoses or heat exchanger core or heat exchanger fan.
- Shipping break down instructions to be supplied to shop
- Fastener kit (all fasteners required for field assembly) to ship with machine
- Emphasis to be made on protection of all cables and hoses from sharp corners of plates, gussets and covers. This is to be accomplished by beveling edges of plates or use of rubber conduit.

Documentation:

- Installation instructions required
- Production schedule with monthly updates and photos

Shipment: July 15, 2011

DISTRIBUTION: Sales, Quality, Engineering, Service, Manufacturing, Materials, Purchasing, Etc.

ATTENDEES: Noel Skilbeck, Mike Chizmar, Mike Boggess, Don Hughes, Tim Edwards, Keith Cole

Nusa Tambang Pratama
Stationary Feeder Breaker
Model: BF-32F-60-76F
Serial No.: FB14430
SAP #: 100477557

July 20, 2011

	Section 1.0 Machine Summary	Material/Document
1.1	SERIAL NUMBER FILE	100477557
	Section 2.0 Mechanical Assemblies	Material/Document
	BELT FEEDER BREAKER	100373701
	FEEDER BREAKER WELDMENT	100374096
2.1	CONVEYOR SECTION	
2.1.1	Headshaft Assembly	98-370760017
2.1.2	Tailshaft Assembly	98-330770003
2.1.3	Conveyor Chain & Flight Assembly	98-330780146
2.1.4	Conveyor Flight Wiper Assembly	98-310110018
2.1.5	Adapter Pump Assembly	98-310510125
2.1.6	Pump/Motor Gear Coupling Assembly	98-330530048
2.2	BREAKER SECTION	
2.2.1	Breaker Assembly	98-370790024
2.2.1.1	L9 Bolt Assembly	98-8583
2.2.2	Breaker Drive Assembly	100373700
2.2.2.1	Torque Limiting Coupling Assembly	98-330530046
2.3	COVERS, GUARDS & GUIDES	
2.3.1	Breaker Cover Assembly	98-370130013
2.3.2	Breaker Side Plate Cover Assembly	98-330130254
2.3.3	Chain Cover Assembly	98-310130111
2.3.4	Wrap Around Headshaft Cover Assembly	98-370130014
2.3.5	Location Takeup Cover Assembly	100374095
2.3.6	Center Frame Discharge End Cover Assembly	98-330130260
2.3.7	Intake End Center Frame Cover Assembly	98-310130424
2.3.8	Removable Chain Cover Assembly	98-310130436
2.3.9	Headshaft Chain Guard Assembly	98-370320001
2.3.10	Shaft Guard Assembly	100373902
2.3.11	Breaker Shaft Guard Assembly	100372455
2.4	MISCELLANEOUS MECHANICAL	
2.4.1	Centralized Lubrication System Assembly	98-370170012
2.4.2	Machine Holddown Assembly	98-310152476
2.4.3	Machine Nameplate Assembly	98-330110174
2.4.4	Power Unit Assembly	100373702
2.4.5	Clevis Assembly	100373510
2.4.6	Power Unit Mounting Bracket Assembly	100373830
2.4.7	Door Cylinder to Power Unit Hydraulic Hose/Fittings	100374689
2.4.8	Breaker Cleaner Assembly	100371554
2.4.9	Discharge Chute Assembly	98-370110129
2.4.10	Bumper Assembly	100371225
2.4.11	Hopper Access Door Assembly	100374099

Nusa Tambang Pratama
Stationary Feeder Breaker
Model: BF-32F-60-76F
Serial No.: FB14430
SAP #: 100477557

July 20, 2011

	Section 3.0 Electrical Assemblies	Material/Document
3.1	Kit - Electrical (190/90/90 kw, 380vac, 50hz)	100498223
3.1.1	Kit - Electrical (Suggested Schematic)	100498379
	Section 4.0 Hydraulic Assemblies	Material/Document
4.1	Kit - Hydraulic	100498224
4.1.1	180cc Series Pump	98-79026F049
4.1.2	Fitting Kit - Hardware	98-390890337
4.1.3	Single Pump	98-33026F004
4.1.4	Fitting Kit - Flushing Valve	98-390890305
4.1.5	Fitting Kit - CA Series	98-390890306
4.1.6	Fitting Kit - Charge Filter	98-390890304
4.1.7	Pall Filter Assembly	98-49052F004
4.1.8	Fitting Kit - Heat Exchanger	100313436
4.1.9	Fitting Kit - Gauge & Switch	98-39089F054
4.1.10	Hydraulic Tank Assembly	98-310180273
4.1.11	Hand Pump	98-730260036



	Description	Page
	Section 1.0 Machine Summary SERIAL NUMBER FILE	1-4-1

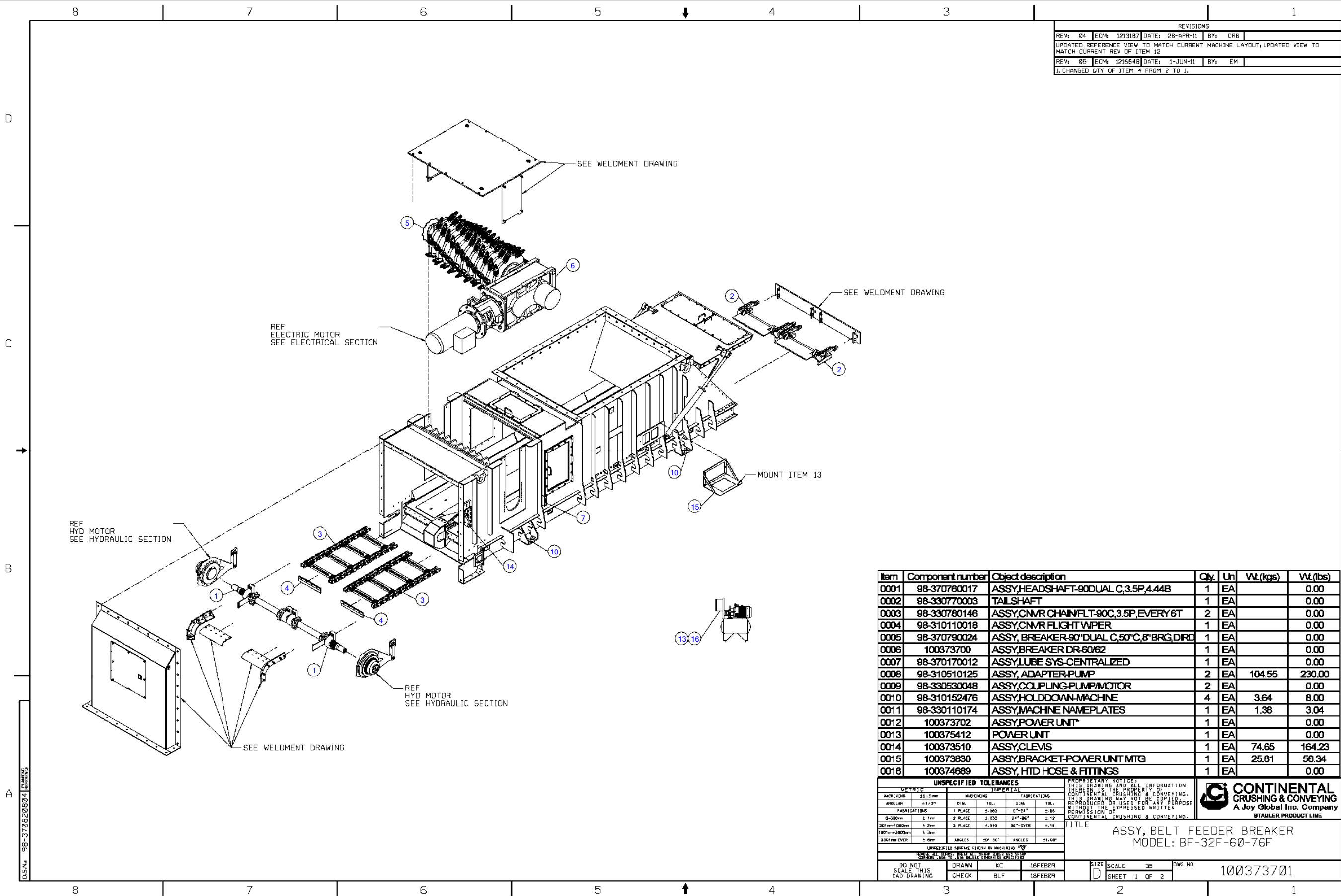


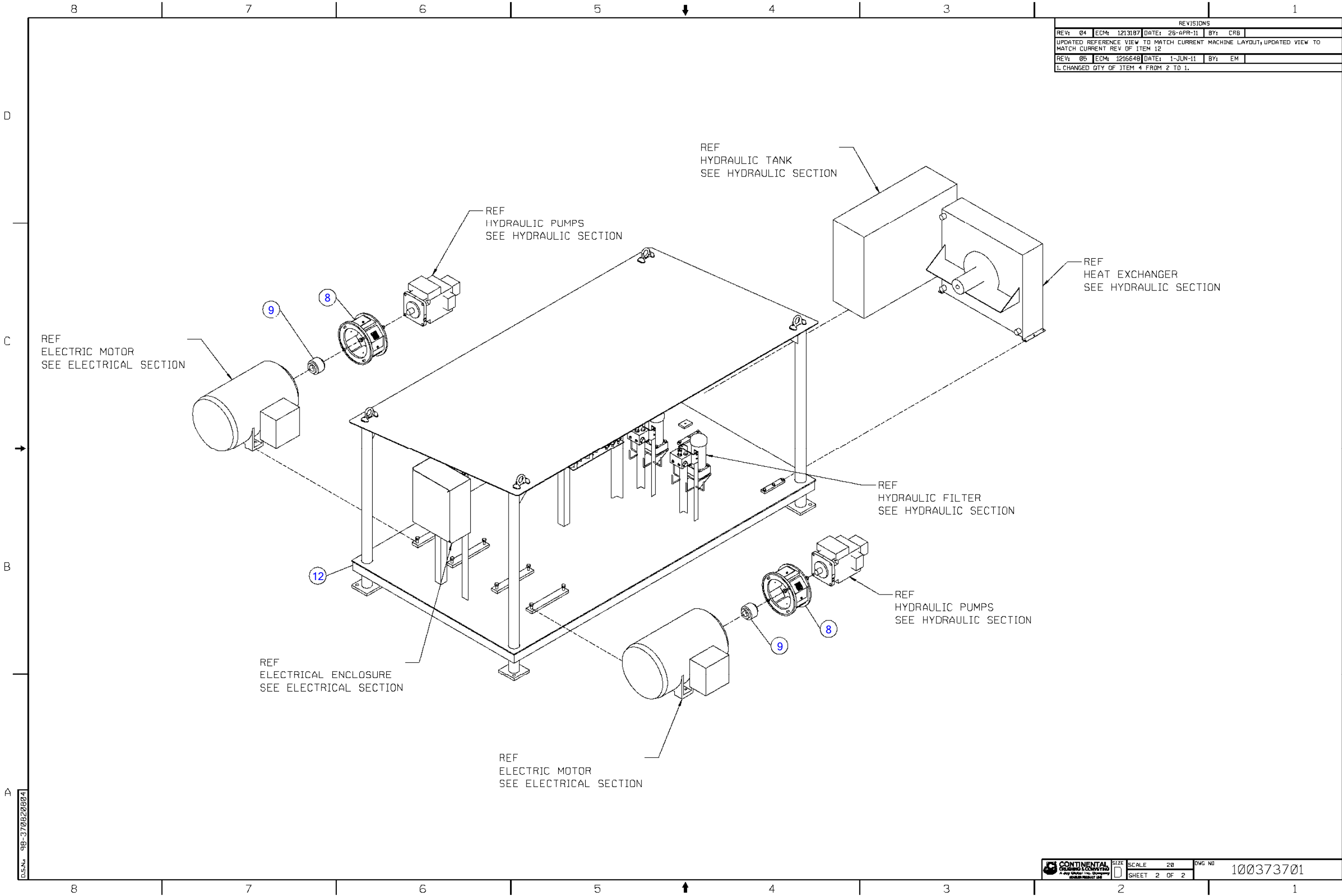
	Item	Part Number	Description	Qty	Units
	ASSY	100477557	FB,14430,BF-32,NTP	1	EA
	ZZ01	100498223	• KIT,ELECTRICAL,190/90/90/KW, 380VAC,50HZ	1	EA
	ZZ02	100373701	• BELT FEEDER BREAKER,BF-32F-60-76F	1	EA
	ZZ03	100498224	• KIT,HYD.	1	EA
	ZZ04	100374096	• WELDMENT,FEEDER BREAKER BF32F	1	EA
	ZZ05	98-370064734	• GEN ARR,FEEDER BREAKER *	1	EA
	ZZ06	98-390110001	• MANUAL	5	EA
	ZZ07	98-390110001CD	• MANUAL,CD	2	EA



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Section 2.0 Mechanical Assemblies	
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• FEEDER BREAKER WELDMENT	1-5-2
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• • Conveyor Flight Wiper Assembly	1-5-6
• • Adapter Pump Assembly	1-5-7
• • Pump/Motor Gear Coupling Assembly	1-5-8
• BREAKER SECTION	
• • Breaker Assembly	1-5-9
• • • L9 Bolt Assembly	1-5-10
• • Breaker Drive Assembly	1-5-11
• • • Torque Limiting Coupling Assembly	1-5-12
• COVERS, GUARDS & GUIDES	
• • Breaker Cover Assembly	1-5-13
• • Breaker Side Plate Cover Assembly	1-5-14
• • Chain Cover Assembly	1-5-15
• • Wrap Around Headshaft Cover Assembly	1-5-16
• • Location Takeup Cover Assembly	1-5-17
• • Center Frame Discharge End Cover Assembly	1-5-18
• • Intake End Center Frame Cover Assembly	1-5-19
• • Removable Chain Cover Assembly	1-5-20
• • Headshaft Chain Guard Assembly	1-5-21
• • Shaft Guard Assembly	1-5-22
• • Breaker Shaft Guard Assembly	1-5-23
• MISCELLANEOUS MECHANICAL	
• • Centralized Lubrication System Assembly	1-5-24
• • Machine Holddown Assembly	1-5-25
• • Machine Nameplate Assembly	1-5-26
• • Power Unit Assembly	1-5-27
• • Clevis Assembly	1-5-28
• • Power Unit Mounting Bracket Assembly	1-5-29
• • Door Cylinder to Power Unit Hydraulic Hose/Fittings	1-5-30
• • Breaker Cleaner Assembly	1-5-31
• • Discharge Chute Assembly	1-5-32
• • Bumper Assembly	1-5-33
• • Hopper Access Door Assembly	1-5-34

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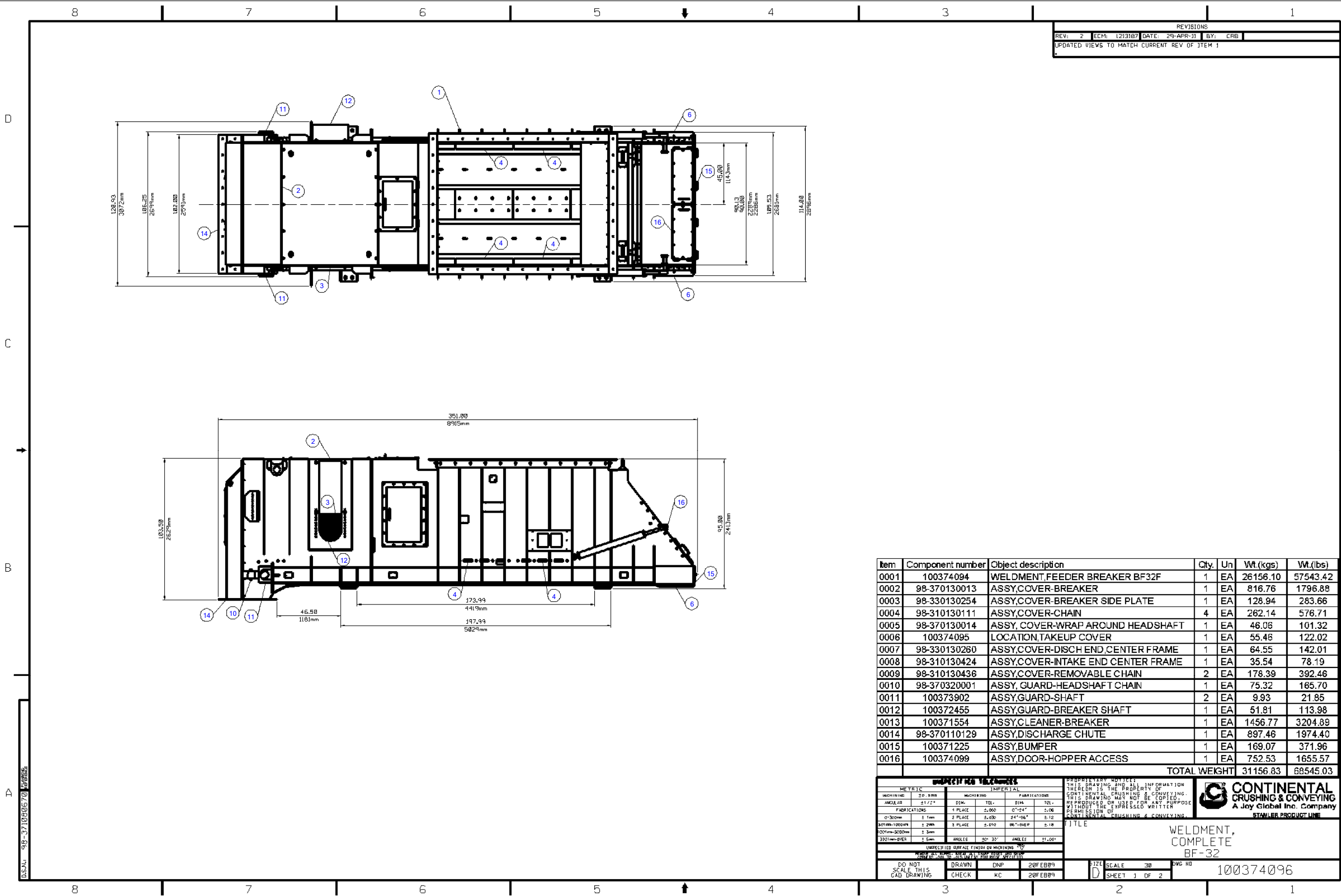


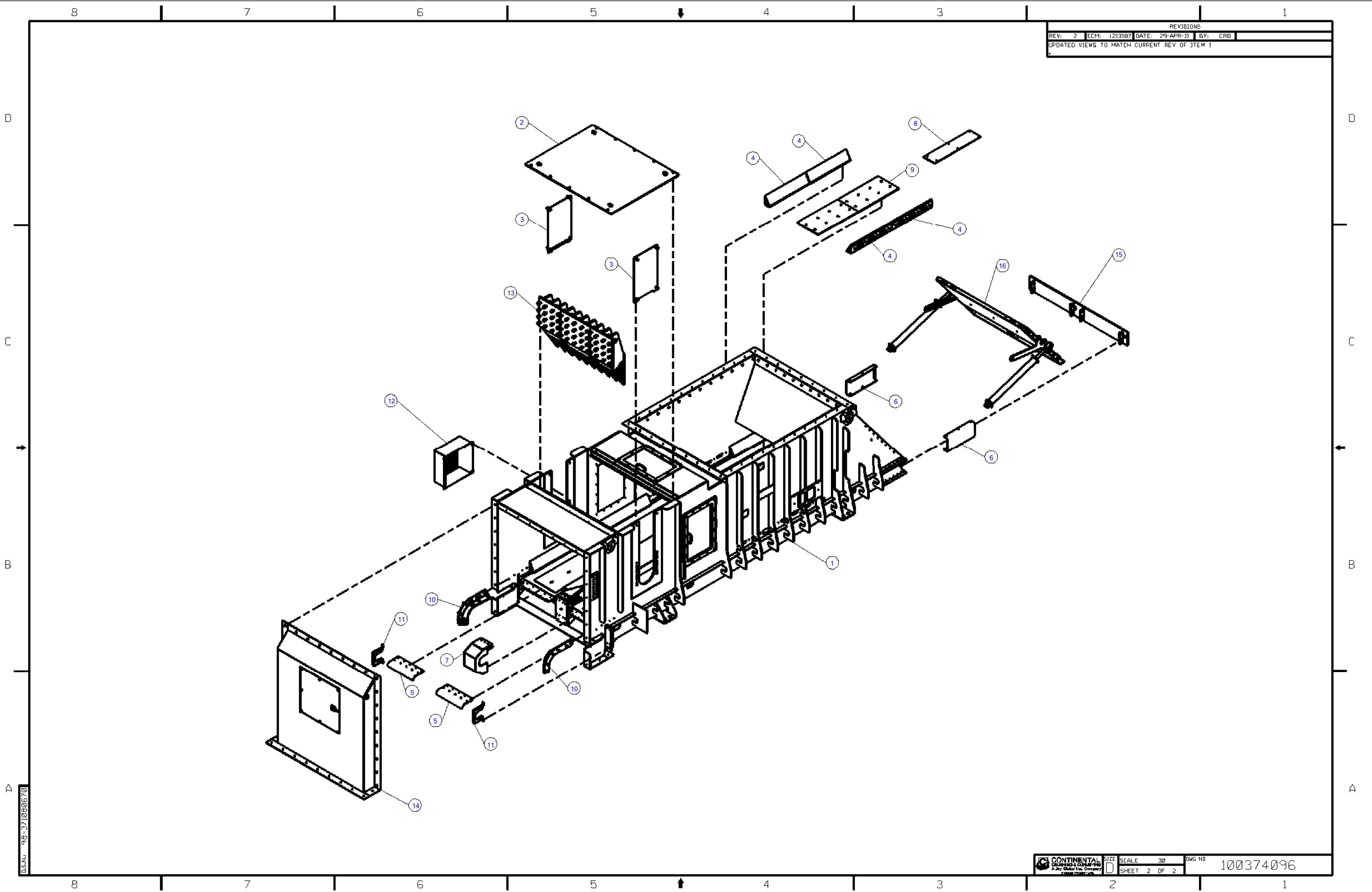




	Item	Part Number	Description	Qty	Units
	ASSY	100373701	BELT FEEDER BREAKER,BF-32F-60-76F	1	EA
	1	98-370760017	• ASSY,HEADSHAFT-90DUAL C,3.5P,4.44B	1	EA
	2	98-330770003	• TAILSHAFT	1	EA
	3	98-330780146	• CHAIN & FLIGHT ASSEMBLY	2	EA
	4	98-310110018	• ASSY,CNVR FLIGHT WIPER	1	EA
	5	98-370790024	• ASSY, BREAKER-90"DUAL C,50"C,8"BRG,DI	1	EA
	6	100373700	• ASSY,BREAKER DR-60/62	1	EA
	7	98-370170012	• ASSY,LUBE SYS-CENTRALIZED	1	EA
	8	98-310510125	• ASSY, ADAPTER-PUMP	2	EA
	9	98-330530048	• ASSY,COUPLING-PUMP/MOTOR	2	EA
	10	98-310152476	• ASSY,HOLDDOWN-MACHINE	4	EA
	11	98-330110174	• ASSY,MACHINE NAMEPLATES	1	EA
	12	100373702	• ASSY,POWER UNIT*	1	EA
	13	100375412	• POWER UNIT, 5HP, 380VAC, 50HZ	1	EA
	14	100373510	• ASSY,CLEVIS	1	EA
	15	100373830	• ASSY,BRACKET-POWER UNIT MTG	1	EA
	16	100374689	• HYDRAULICS,HOSE/FTGS-DOOR CYL TO PWR UN	1	EA

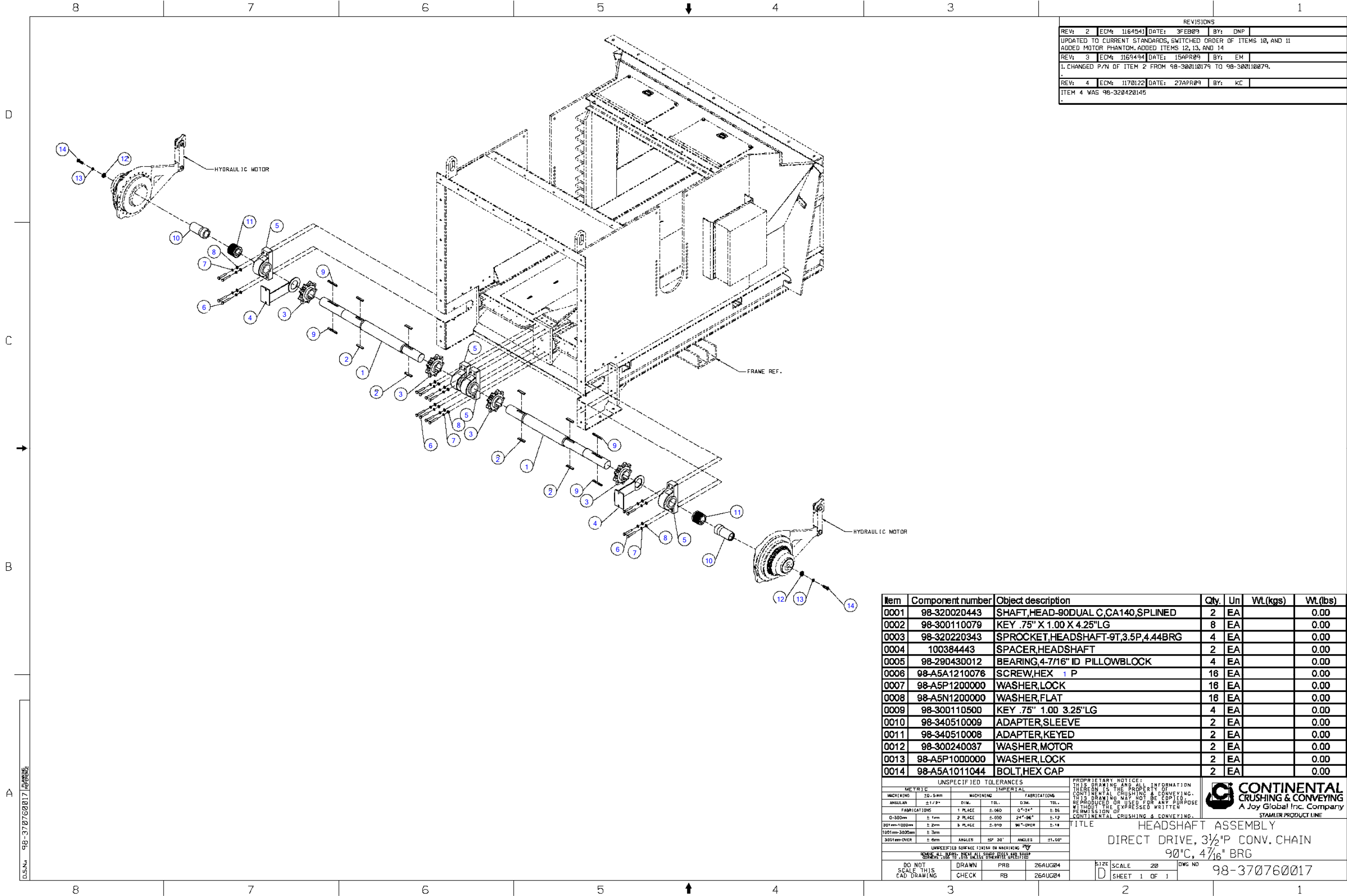
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	Item	Part Number	Description	Qty	Units
	ASSY	100374096	WELDMENT,FEEDER BREAKER BF32F	1	EA
	1	100374094	• WELDMENT,FEEDER BREAKER BF32F	1	EA
	2	98-370130013	• ASSY,COVER-BREAKER	1	EA
	3	98-330130254	• ASSY,COVER-BREAKER SIDE PLATE	1	EA
	4	98-310130111	• ASSY,COVER-CHAIN	4	EA
	5	98-370130014	• ASSY, COVER-WRAP AROUND HEADSHAFT	1	EA
	6	100374095	• LOCATION,TAKEUP COVER	1	EA
	7	98-330130260	• ASSY,COVER-DISCH END,CENTER FRAME	1	EA
	8	98-310130424	• ASSY,COVER-INTAKE END CENTER FRAME	1	EA
	9	98-310130436	• ASSY,COVER-REMOVABLE CHAIN	2	EA
	10	98-370320001	• ASSY, GUARD-HEADSHAFT CHAIN	1	EA
	11	100373902	• ASSY,GUARD-SHAFT	2	EA
	12	100372455	• ASSY,GUARD-BREAKER SHAFT	1	EA
	13	100371554	• ASSY,CLEANER-BREAKER	1	EA
	14	98-370110129	• ASSY,DISCHARGE CHUTE	1	EA
	15	100371225	• ASSY,BUMPER	1	EA
	16	100374099	• ASSY,DOOR-HOPPER ACCESS	1	EA



REVISIONS				
REV:	2	ECM:	1164541	DATE: 3FEB09 BY: DNP
UPDATED TO CURRENT STANDARDS, SWITCHED ORDER OF ITEMS 10, AND 11				
ADDED MOTOR PHANTOM, ADDED ITEMS 12, 13, AND 14				
REV:	3	ECM:	1169494	DATE: 15APR09 BY: EM
1. CHANGED P/N OF ITEM 2 FROM 98-300110179 TO 98-300110079.				
REV:	4	ECM:	1170122	DATE: 27APR09 BY: KC
ITEM 4 WAS 98-320420145				

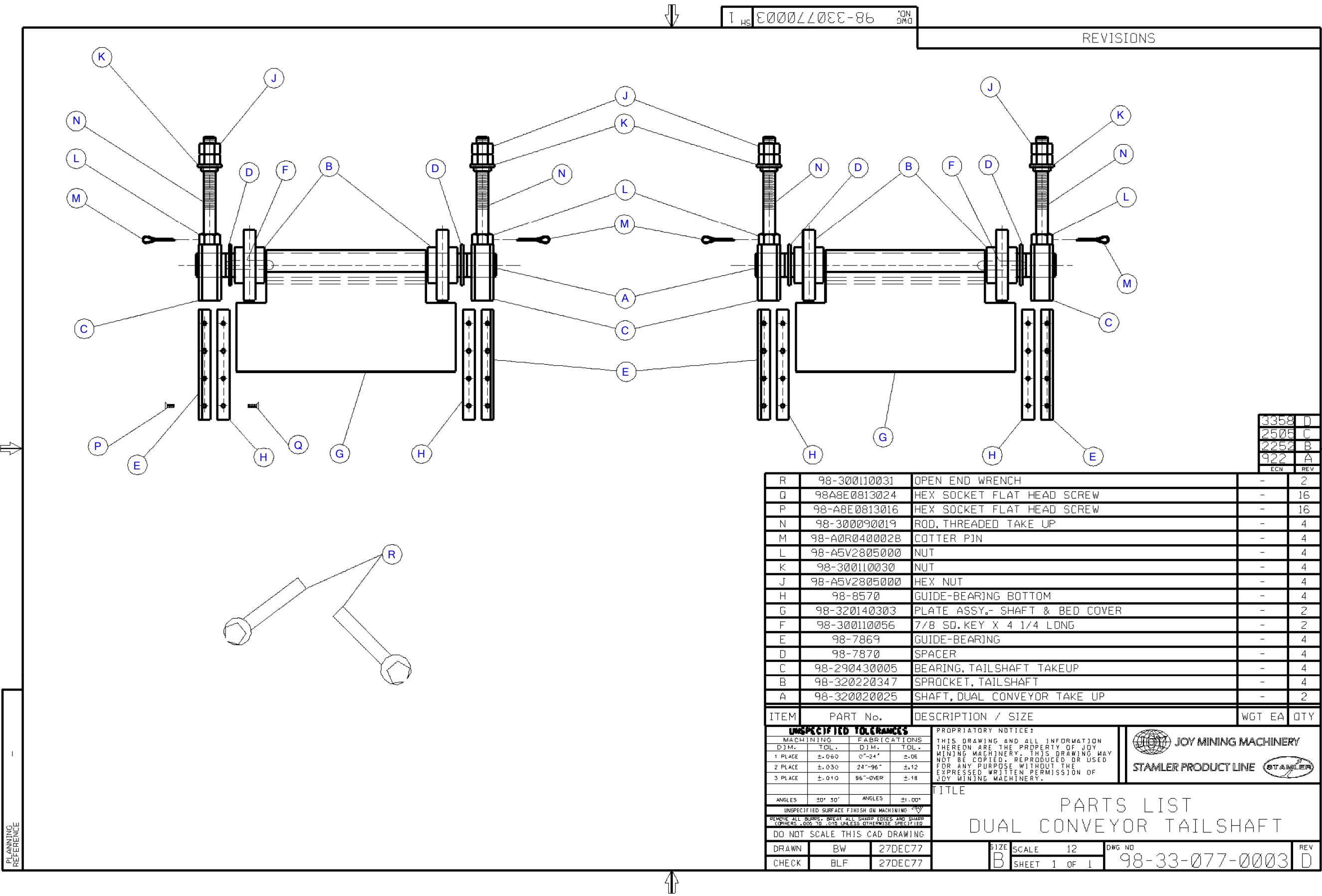
Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	98-320020443	SHAFT,HEAD-90DUAL C,CA140,SPLINED	2	EA		0.00
0002	98-300110079	KEY .75" X 1.00 X 4.25"LG	8	EA		0.00
0003	98-320220343	SPROCKET,HEADSHAFT-9T,3.5P,4.44BRG	4	EA		0.00
0004	100384443	SPACER,HEADSHAFT	2	EA		0.00
0005	98-290430012	BEARING,4-7/16" ID PILLOWBLOCK	4	EA		0.00
0006	98-A5A1210076	SCREW,HEX 1 P	16	EA		0.00
0007	98-A5P1200000	WASHER,LOCK	16	EA		0.00
0008	98-A5N1200000	WASHER,FLAT	16	EA		0.00
0009	98-300110500	KEY .75" 1.00 3.25"LG	4	EA		0.00
0010	98-340510009	ADAPTER,SLEEVE	2	EA		0.00
0011	98-340510008	ADAPTER,KEYED	2	EA		0.00
0012	98-300240037	WASHER,MOTOR	2	EA		0.00
0013	98-A5P1000000	WASHER,LOCK	2	EA		0.00
0014	98-A5A1011044	BOLT,HEX CAP	2	EA		0.00

UNSPECIFIED TOLERANCES			
METRIC		IMPERIAL	
MACHINING	±0.5mm	MACHINING	TOL.
ANGULAR	±1/2°	FABRICATIONS	TOL.
FABRICATIONS		1 PLACE	±.060
0-300mm		2 PLACE	±.030
301mm-1000mm		3 PLACE	±.010
1001mm-3000mm			±.010
3001mm-OVER			±.010
UNSPECIFIED SURFACE FINISH ON MACHINING		ANGLES	±1.0°
DO NOT SCALE THIS CAD DRAWING		DRAWN	PRB
		CHECK	RB
		DATE	26AUG04
		DATE	26AUG04

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CONTINENTAL CRUSHING & CONVEYING A Joy Global Inc. Company STAMLER PRODUCT LINE	
TITLE HEADSHAFT ASSEMBLY DIRECT DRIVE, 3 1/2" P CONV. CHAIN 90°C, 4 7/16" BRG	
SIZE D	SCALE 20
SHEET 1 OF 1	DWG NO 98-370760017

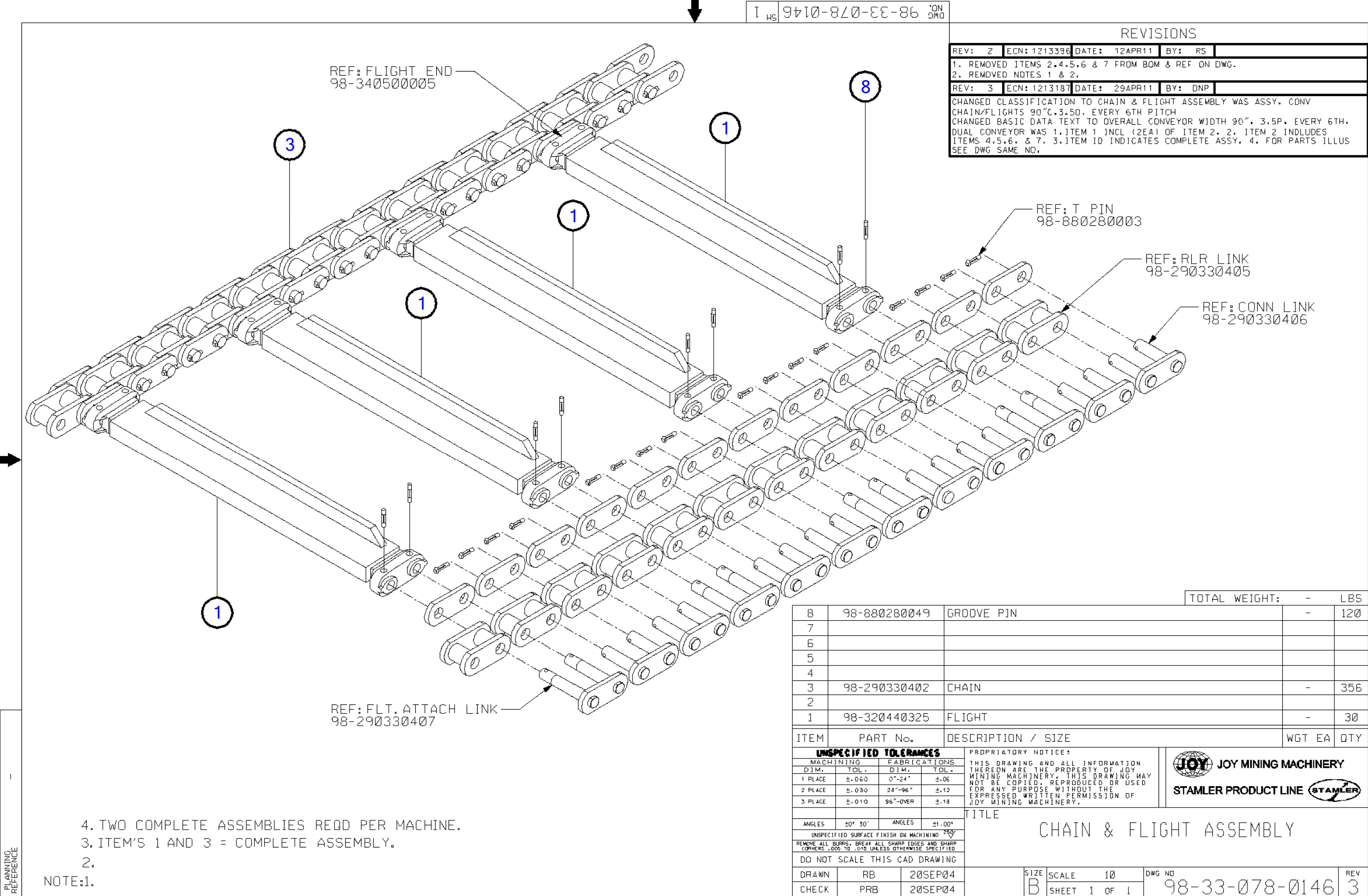


	Item	Part Number	Description	Qty	Units
	ASSY	98-370760017	ASSY,HEADSHAFT-90DUAL C,3.5P,4.44B	1	EA
	1	98-320020443	• SHAFT,HEAD-90DUAL C,CA140,SPLINED	2	EA
	2	98-300110079	• KEY .75" X 1.00 X 4.25"LG	8	EA
	3	98-320220343	• SPROCKET,HEADSHAFT-9T,3.5P,4.44BRG	4	EA
	4	100384443	• SPACER,HEADSHAFT	2	EA
	5	98-290430012	• BEARING,4-7/16" ID PILLOWBLOCK	4	EA
	6	98-A5A1210076	• SCREW,HEX CAP *	16	EA
	7	98-A5P1200000	• WASHER,LOCK *	16	EA
	8	98-A5N1200000	• WASHER,FLAT *	16	EA
	9	98-300110500	• KEY .75" 1.00 3.25"LG	4	EA
	10	98-340510009	• ADAPTER,SLEEVE	2	EA
	11	98-340510008	• ADAPTER,KEYED SPLINED	2	EA
	12	98-300240037	• WASHER,MOTOR *	2	EA
	13	98-A5P1000000	• WASHER,LOCK *	2	EA
	14	98-A5A1011044	• SCREW,HEX CAP *	2	EA





	Item	Part Number	Description	Qty	Units
	ASSY	98-330770003	TAILSHAFT	1	EA
	A	98-320020025	• SHAFT, TAIL-DUAL CNVR	2	EA
	B	98-320220347	• SPROCKET, TAILSHAFT-4T, 8P, 3.5PT	4	EA
	C	98-290430005	• BEARING, 3-7/16"ID	4	EA
	D	98-7870	• SPACER, TAKEUP SHAFT	4	EA
	E	98-7869	• GUIDE, TAKEUP BEARING	4	EA
	F	98-300110056	• KEY, SQUARE ENDS, 0.880", 0.880"	2	EA
	G	98-320140303	• PLATE, TAKEUP-DUAL CNVR	2	EA
	H	98-8570	• GUIDE, TAILSHAFT BOTTOM	4	EA
	J	98-A5V2805000	• NUT, JAM *	4	EA
	K	98-300110030	• NUT, ADJUSTING UP	4	EA
	L	98-A5V2805000	• NUT, JAM *	4	EA
	M	98-A0R0400028	• PIN, COTTER	4	EA
	N	98-300090019	• ROD, THREADED TAKEUP	4	EA
	P	98-A8E0813016	• SCREW, HEX FLAT CAP *	16	EA
	Q	98-A8E0813024	• SCREW, HEX FLAT CAP *	16	EA
	R	98-300110031	• WRENCH, OPEN END	2	EA



98-33-078-0146
1 HS

REVISIONS

REV: 2	ECN: 1213396	DATE: 12APR11	BY: RS
1. REMOVED ITEMS 2,4,5,6 & 7 FROM BOM & REF ON DWG.			
2. REMOVED NOTES 1 & 2.			
REV: 3	ECN: 1213187	DATE: 29APR11	BY: DNP
CHANGED CLASSIFICATION TO CHAIN & FLIGHT ASSEMBLY WAS ASSY. CONV CHAIN/FLIGHTS 90"C,3.50, EVERY 6TH PITCH			
CHANGED BASIC DATA TEXT TO OVERALL CONVEYOR WIDTH 90", 3.5P, EVERY 6TH.			
DUAL CONVEYOR WAS 1. ITEM 1 INCL (2EA) OF ITEM 2. 2. ITEM 2 INCLUDES ITEMS 4,5,6, & 7. 3. ITEM ID INDICATES COMPLETE ASSY. 4. FOR PARTS ILLUS SEE DWG SAME NO.			

TOTAL WEIGHT: - LBS

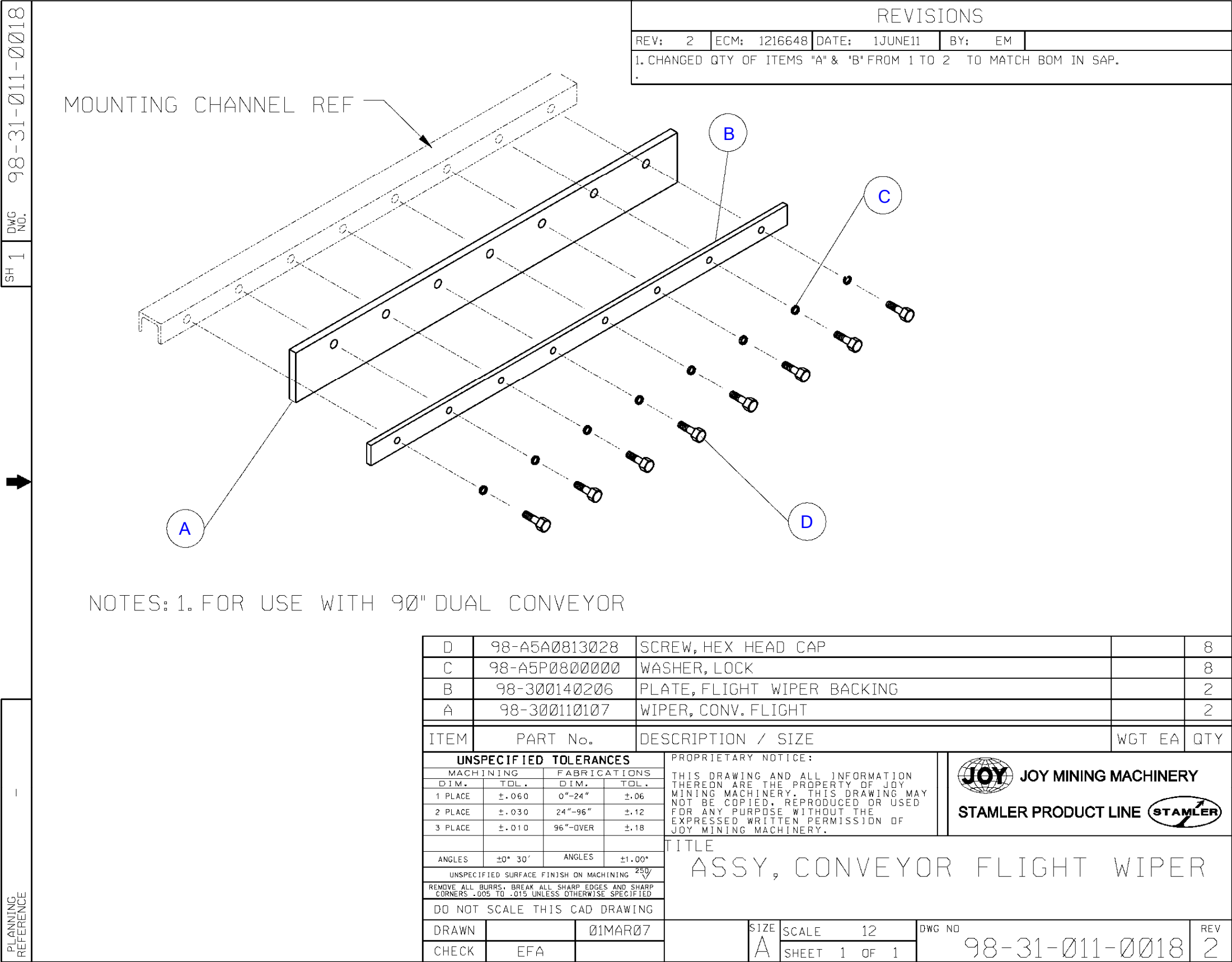
8	98-880280049	GROOVE PIN	-	120
7				
6				
5				
4				
3	98-290330402	CHAIN	-	356
2				
1	98-320440325	FLIGHT	-	30

ITEM	PART No.	DESCRIPTION / SIZE	WGT EA	QTY
UNSPECIFIED TOLERANCES				
MACHINING		FABRICATIONS		
DIM.	TOL.	DIM.	TOL.	
1 PLACE	±.060	0"-24"	±.06	
2 PLACE	±.030	24"-96"	±.12	
3 PLACE	±.010	96"-OVER	±.18	
ANGLES		±0° 30'	±1.00°	
UNSPECIFIED SURFACE FINISH ON MACHINING: 250				
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .015 UNLESS OTHERWISE SPECIFIED				
DO NOT SCALE THIS CAD DRAWING				
DRAWN	RB	20SEP04	SIZE B	SCALE 10
CHECK	PRB	20SEP04	SHEET 1 OF 1	DWG NO 98-33-078-0146
TITLE CHAIN & FLIGHT ASSEMBLY				REV 3

PLANNING
REFERENCE

NOTE:1.
2.
3. ITEM'S 1 AND 3 = COMPLETE ASSEMBLY.
4. TWO COMPLETE ASSEMBLIES REQD PER MACHINE.







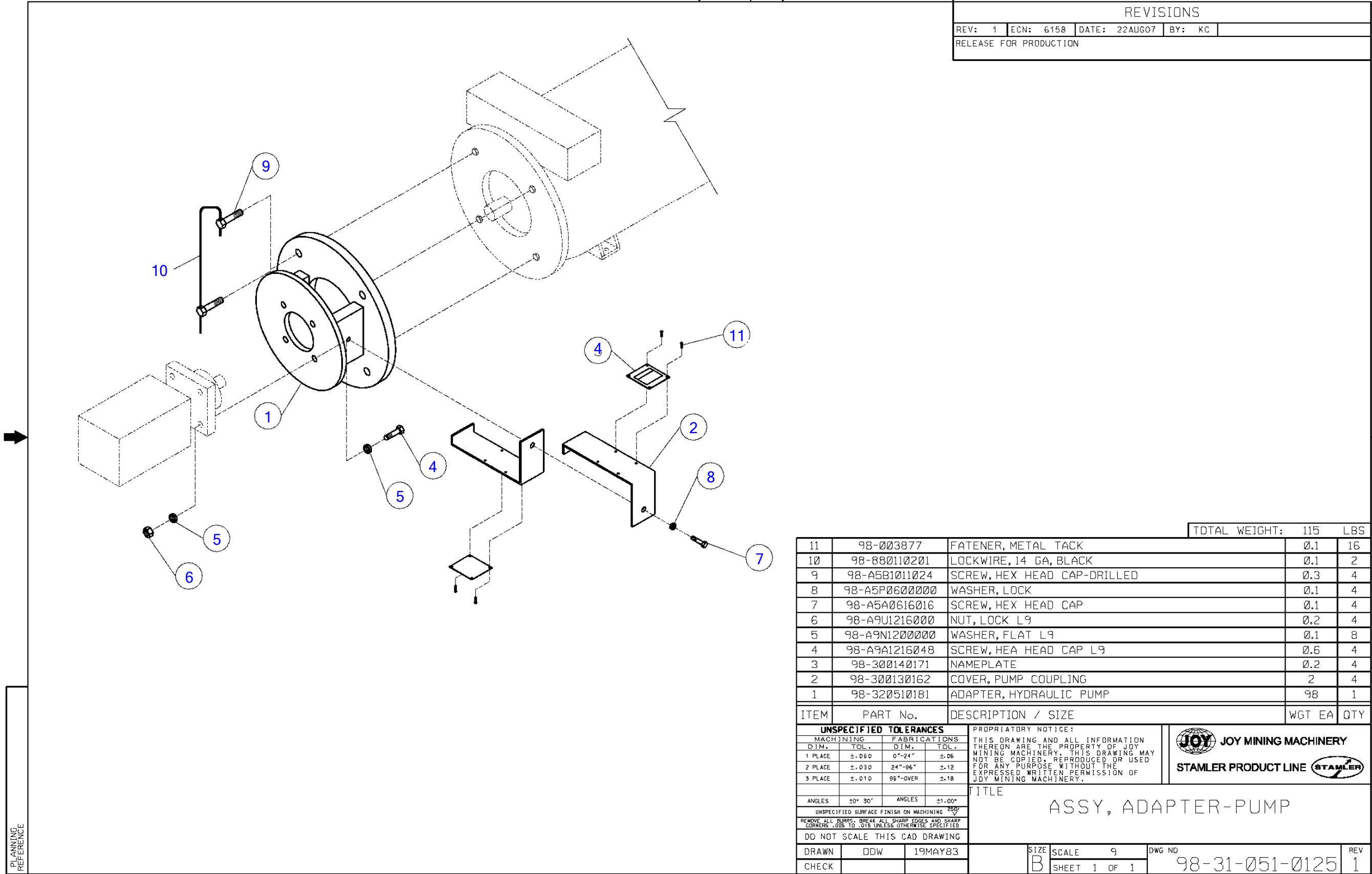
Joy Mining Machinery
FB14430 - BF-32F-60-76F - STATIONARY FEEDER BREAKER - NUSA TAMBANG PRATAMA
ASSY,CNVR FLIGHT WIPER



1 HS 98-310510125 1 ON 0MO

REVISIONS

REV: 1	ECN: 6158	DATE: 22AUG07	BY: KC
RELEASE FOR PRODUCTION			



TOTAL WEIGHT: 115 LBS

11	98-003877	FATENER, METAL TACK	Ø.1	16
10	98-880110201	LOCKWIRE, 14 GA, BLACK	Ø.1	2
9	98-A5B1011024	SCREW, HEX HEAD CAP-DRILLED	Ø.3	4
8	98-A5P0600000	WASHER, LOCK	Ø.1	4
7	98-A5A0616016	SCREW, HEX HEAD CAP	Ø.1	4
6	98-A9U1216000	NUT, LOCK L9	Ø.2	4
5	98-A9N1200000	WASHER, FLAT L9	Ø.1	8
4	98-A9A1216048	SCREW, HEA HEAD CAP L9	Ø.6	4
3	98-300140171	NAMEPLATE	Ø.2	4
2	98-300130162	COVER, PUMP COUPLING	2	4
1	98-320510181	ADAPTER, HYDRAULIC PUMP	98	1

ITEM	PART No.		DESCRIPTION / SIZE		WGT EA	QTY
UNSPECIFIED TOLERANCES				PROPRIATARY NOTICE:		
MACHINING		FABRICATIONS		 JOY MINING MACHINERY STAMLER PRODUCT LINE 		
DIM.	TOL.	DIM.	TOL.			
1 PLACE	±.060	0°-24°	±.06			
2 PLACE	±.030	24°-96°	±.12			
3 PLACE	±.010	96°-OVER	±.18			
ANGLES		±0°-30°	ANGLES		±1.00°	
UNSPECIFIED SURFACE FINISH ON MACHINING				250		
REMOVE ALL BURRS. BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .015 UNLESS OTHERWISE SPECIFIED				TITLE ASSY, ADAPTER-PUMP		
DO NOT SCALE THIS CAD DRAWING						
DRAWN	DDW	19MAY83				
CHECK				SIZE	SCALE	9
				B	SHEET	1 OF 1
				DWG NO	98-31-051-0125	
					REV	1

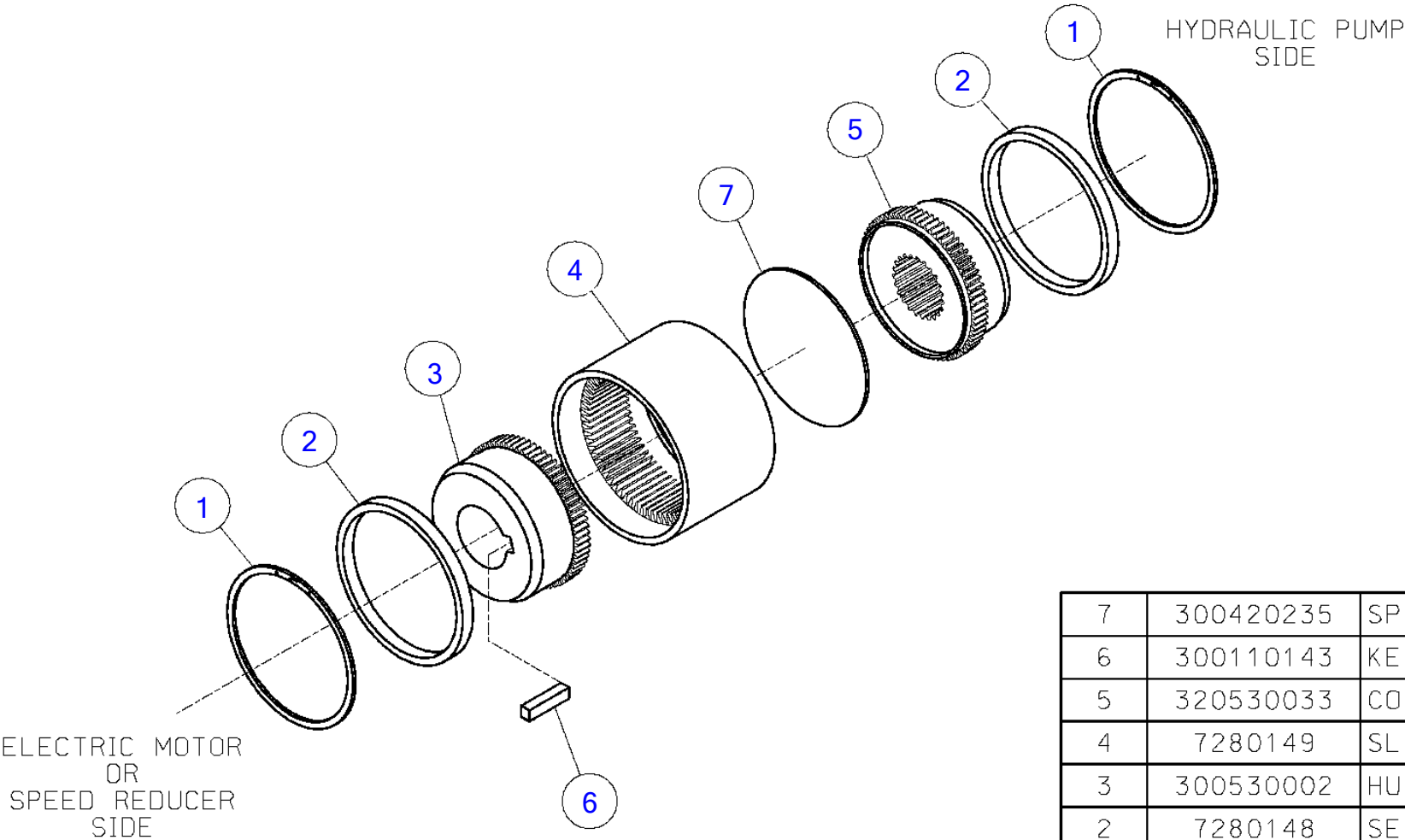




	Item	Part Number	Description	Qty	Units
	ASSY	98-310510125	ASSY, ADAPTER-PUMP	1	EA
	1	98-320510181	• ADAPTER,HYDRAULIC PUMP	1	EA
	2	98-300130162	• COVER,PUMP COUPLING	4	EA
	3	98-300140171	• PLATE,CAUTION *	4	EA
	4	98-A9A1216048	• SCREW,HEX CAP-L9 *	4	EA
	5	98-A9N1200000	• WASHER,FLAT-L9 *	8	EA
	6	98-A9U1216000	• NUT,LOCK-L9 *	4	EA
	7	98-A5A0616016	• SCREW,HEX CAP *	4	EA
	8	98-A5P0600000	• WASHER,LOCK *	4	EA
	9	98-A5B1011024	• SCREW,HEX CAP DRL LKWR	4	EA
	10	98-880110201	• LOCKWIRE,BASIC STEEL	2	EA
	11	98-003877	• FASTENER, METAL TACK	16	EA



REVISIONS				
REV: 2	ECM: 1159140	DATE: 13OCT08	BY: MS	
1.NO CHANGES MADE TO DRAWING JUST CHANGED BORDER FROM OLDENBURG TO CCC				
REV: 3	ECM: 1167795	DATE: 18MAR09	BY: MS	
1. CORRECTED BALLOONS TO MATCH BOM				



7	300420235	SPACER, GEAR COUPLING	1
6	300110143	KEY,.62 SQ X 2.00	1
5	320530033	COUPLING HALF,SPLINED GEAR	1
4	7280149	SLEEVE	1
3	300530002	HUB, CPLG-SIZE 2.50,2.38 BORE	1
2	7280148	SEAL	2
1	7280147	RING, SNAP	2
ITEM	PART NO.	DESCRIPTION	QTY

UNSPECIFIED TOLERANCES					
METRIC		MACHINING		FABRICATIONS	
MACHINING	±0.5 mm	DIM.	TOL.	DIM.	TOL.
ANGULAR	±1/2°				
FABRICATIONS		1 PLACE	±.060	0°-24°	±.06
0-300mm	± 1mm	2 PLACE	±.030	24°-96°	±.12
301mm-1000mm	± 2mm	3 PLACE	±.010	96°-OVER	±.18
1001mm-3000mm	± 3mm				
3001mm-OVER	± 5mm	ANGLES	±0°-30°	ANGLES	±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING: 750/					
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .015 UNLESS OTHERWISE SPECIFIED					
DO NOT SCALE THIS CAD DRAWING	DRAWN	RB	17SEP04		
	CHECK	PRB	17SEP04		

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TITLE
ASSY, COUPLING-PUMP/MOTOR

SIZE	SCALE	6	DWG NO.	98-330530048
B	SHEET	1 OF 1		



	Item	Part Number	Description	Qty	Units
	ASSY	98-330530048	ASSY,COUPLING-PUMP/MOTOR	1	EA
	1	98-7280147	• RING,SNAP	2	EA
	2	98-7280148	• SEAL	2	EA
	3	98-300530002	• COUPLING,HUB	1	EA
	4	98-7280149	• SLEEVE	1	EA
	5	98-320530033	• COUPLING GEAR	1	EA
	6	98-300110143	• KEY,SQUARE ENDS,0.630",0.630"	1	EA
	7	98-300420235	• SPACER,GEAR COUPLING	1	EA



REV: A

ECN: 4305

DATE:

BY:

REVISIONS

TENSIONING SEQUENCE CHART

BOLT NO.	HAND SNUG	¼TURN BOLT	¼TURN BOLT	HALF TORQUE	FULL TORQUE	FULL TORQUE	FULL TORQUE
1							
8							
5							
12							
2							
9							
6							
13							
3							
10							
7							
14							
4							
11							

TIGHTENING SEQUENCE FOR LOCKING SCREWS

STEP 1. CLEAN AND SLIGHTLY OIL SHAFT AND THRUST RING ASSEMBLY.
MAKE SURE TO ALWAYS OIL THE THREADS OF ALL LOCKING SCREWS.
(NEVER USE MOLYBDENUM DISULPHIDE "MOLYKOTE" OR OTHER SIMILAR LUBRICANTS.

2.SLIP THRUST RING ASSEMBLY ONTO THE SHAFT AND LOCATE ALL PARTS IN CORRECT POSITION.

3.DRAW LINES RADIALLY OUT FROM THE BOLTS AND NUMBER AS SHOWN.

4.TIGHTEN LOCKING SCREWS EVENLY IN SEVERAL STAGES FOLLOWING THE TENSIONING SEQUENCE AS SHOWN IN THE CHART ABOVE.
(EXAMPLE:1 , 8 , 5 , 12 ,ETC.)
STAGE 1:TIGTHEN BY HAND UNTIL A SLIGHT POSITIVE CONTACT IS ESTABLISHED.MAKE FINAL ALIGNMENT AND ADJUSTMENT TO CONNECTION.
STAGE 2: BY THE BOLT METHOD TIGHTEN EACH BOLT IN SEQUENCE¼TURN. MAKE TWO COMPLETE PASSES AROUND BEFORE MOVING TO STAGE 3.
STAGE 3:TIGHTEN SCREWS TO APPROXIMATELY HALF THE SPECIFIED TIGHTENING TORQUE USING THE INTERMEDIATE TOOL AND A STANDARD TORQUE WRENCH.(SEE DWG 31-021-0066)
STAGE4:TIGHTEN SCREWS TO FULL TIGHTENING TORQUE USING THE SAME TOOL AND TORQUE WRENCH COMBINATIONS
STAGE 5:CHECK AND MAKE SURE THAT NO SCREW WILL TURN ANY MORE BY APPLYING SPECIFIED TORQUE.

5.AFTER FULL TORQUING ALL BOLTS TWICE, WAIT ONE HOUR, THEN TORQUE ALL BOLTS AGAIN.RECHECK TIGHTENING TORQUE FOLLOWING THE SAME SEQUENCE. ASSEMBLY IS COMPLETED WHEN NO SCREW CAN BE TURNED ANY MORE BY TIGHTENING TORQUE.

2. SEE DWG. 98-310210066 FOR REQD. TORQUE AND WRENCH COMBINATIONS.
NOTE: 1. FOR USE WITH 8.00" DIA BEARINGS.

UNSPECIFIED TOLERANCES

MACHINING	FABRICATIONS		
DIM.	TOL.	DIM.	TOL.
1 PLACE	±.060	0°-24°	±.06
2 PLACE	±.030	24°-96°	±.12
3 PLACE	±.010	96°-OVER	±.18
ANGLES	30°-30°	ANGLES	±1.00°

UNSPECIFIED SURFACE FINISH ON MACHINING 250
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS - DOB TO .010 UNLESS OTHERWISE SPECIFIED

DO NOT SCALE THIS CAD DRAWING

DRAWN DNG

CHECK -

25SEP93

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JOY MINING MACHINERY

STAMLER PRODUCT LINE

STAMLER

TITLE

INSTRUCTION - THRUST RING TORQUE SEQUENCE (14 BOLT)

SIZE A

SCALE 1

SHEET 1

OF 1

DWG NO 98-31-021-0065

REV A

98-310210065

DWG NO.

1

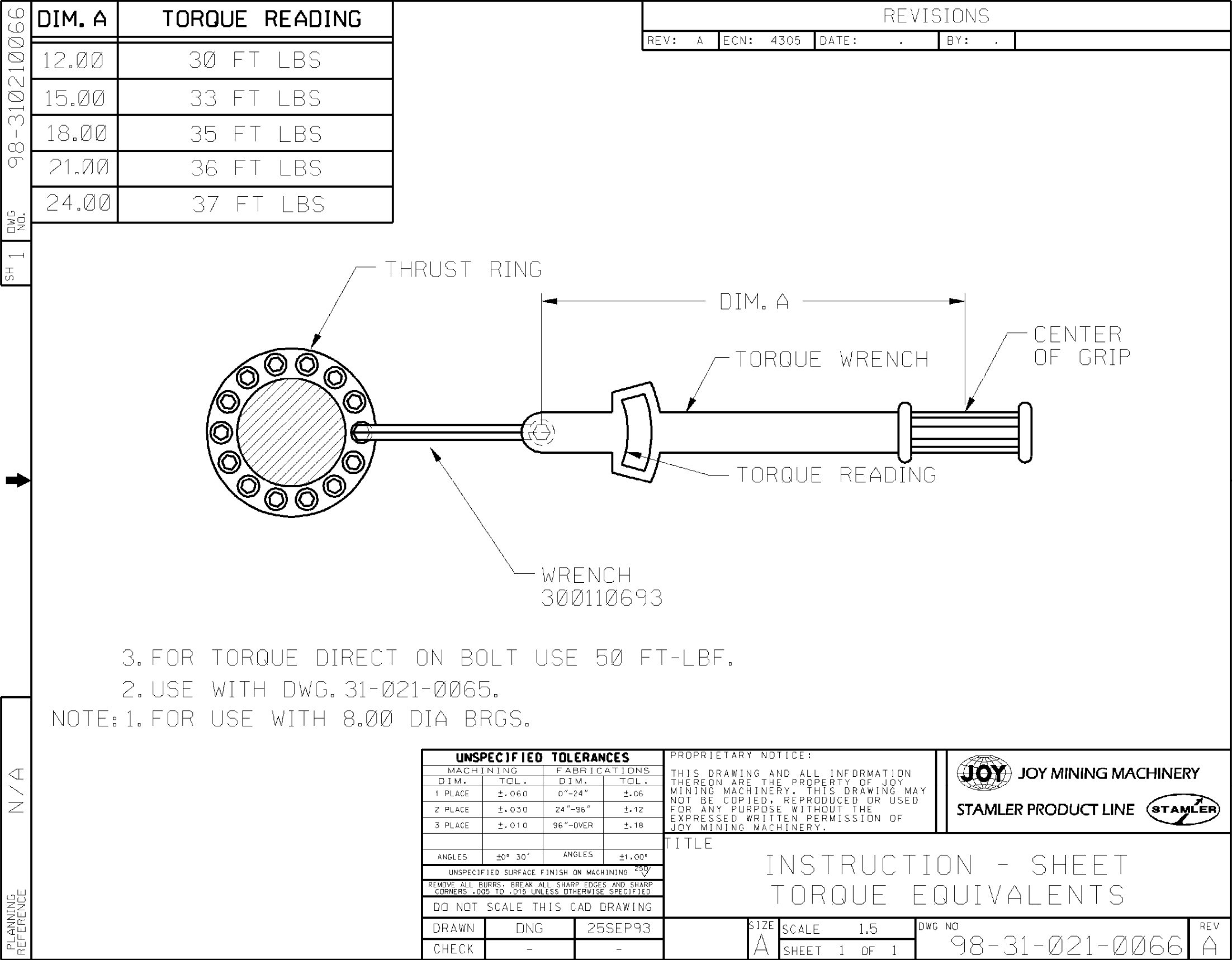
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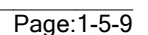
PLANNING REFERENCE

-

Reference:98-370790024

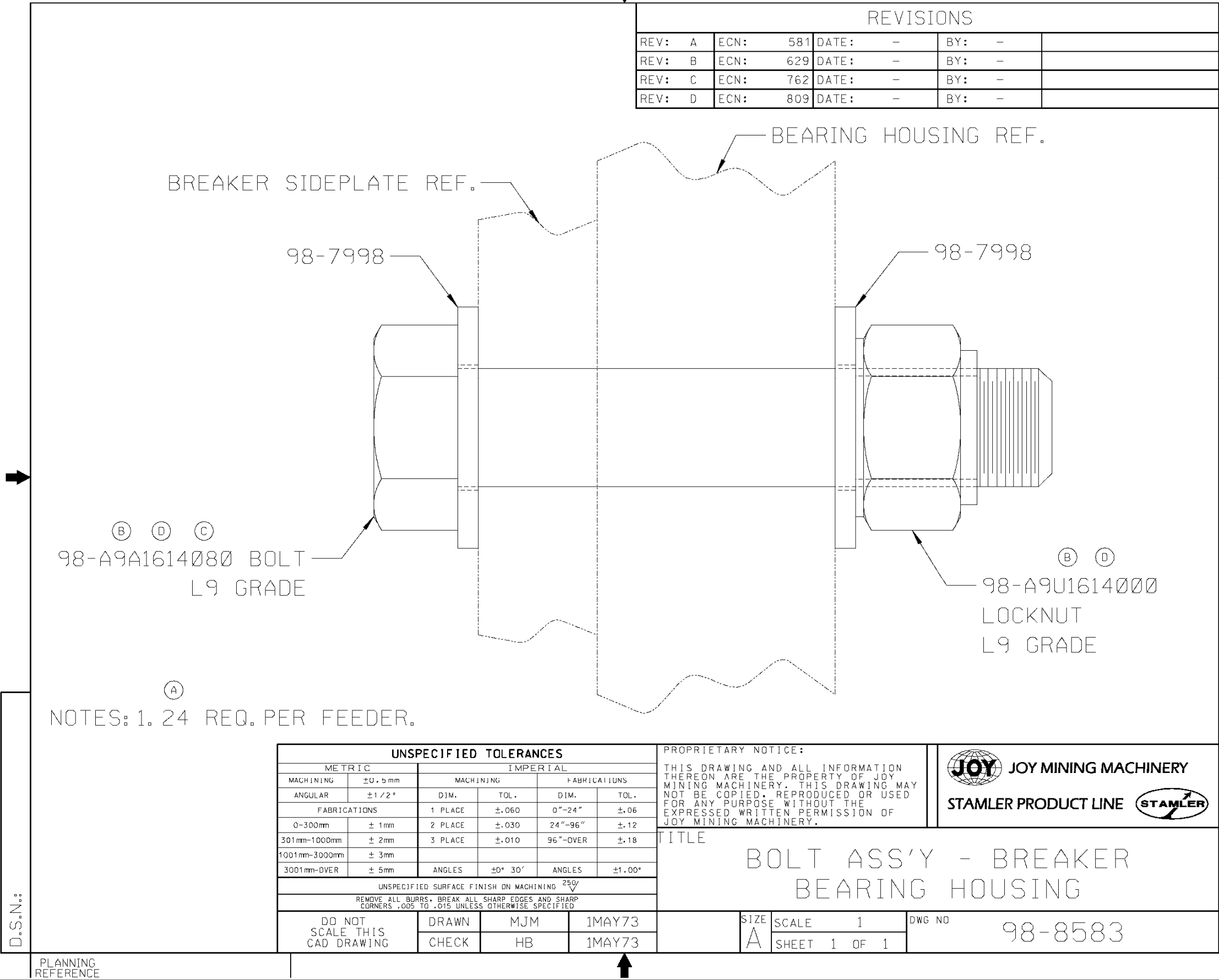
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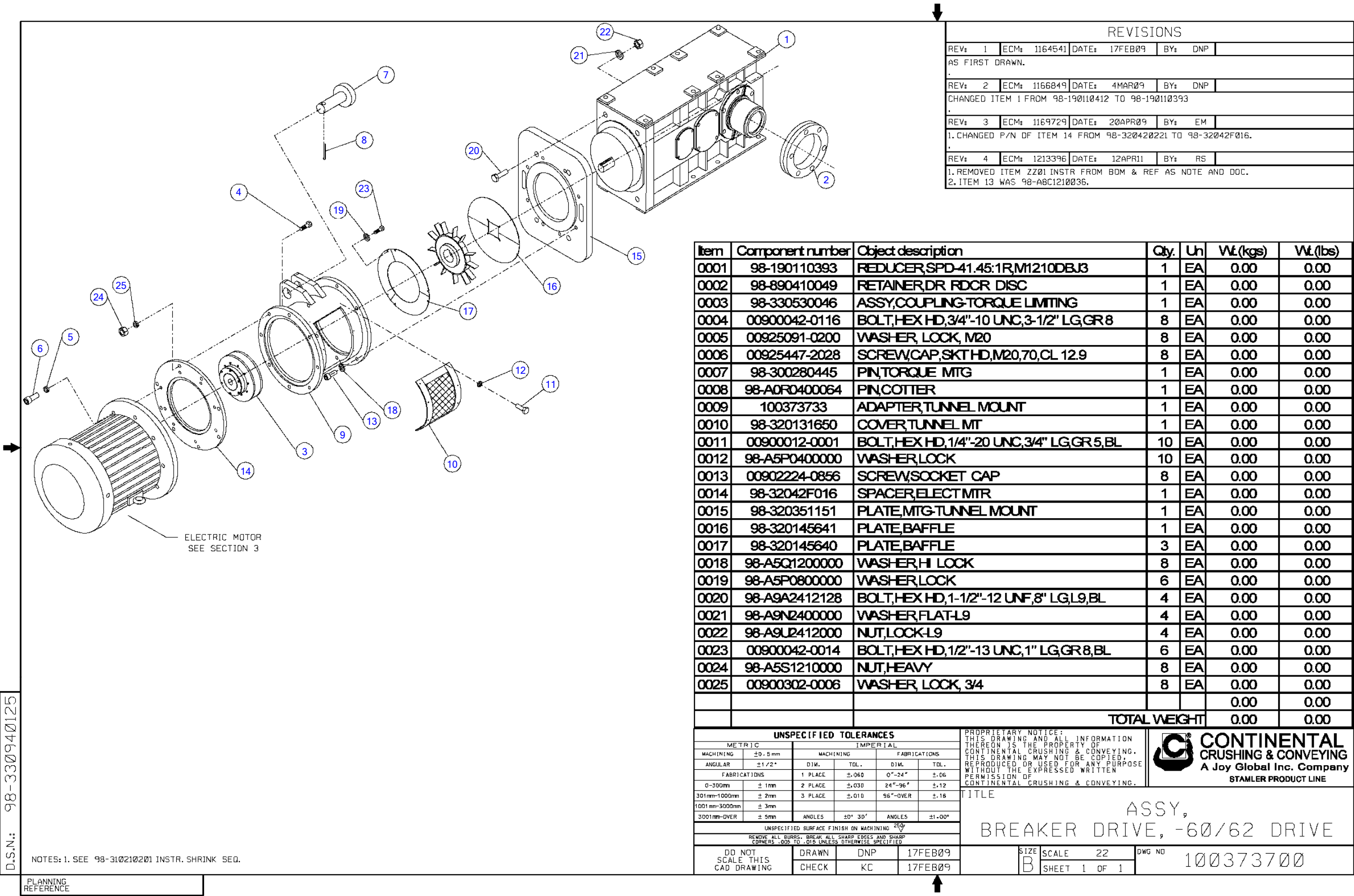
	Item	Part Number	Description	Qty	Units
	ASSY	98-370790024	ASSY, BREAKER-90"DUAL C,50"C,8"BRG,DI	1	EA
	1	98-320030171	• RING,LOCKING	1	EA
	2	98-320030174	• RING,COMPRESSION-6.19"BRG	1	EA
	3	98-320030178	• +ING,LOCKING	1	EA
	4	98-320030309	• RING,WEAR	1	EA
	5	00900042-0033	• BOLT,HEX HD,3/8"-16 UNC,1-1/2" LG,GR 8	16	EA
	6	98-320030179	• RING,WEAR	2	EA
	7	98-300430014	• BEARING,MODIFIED-8.00	2	EA
	8	98-320030175	• RING,THRUST	1	EA
	9	98-320410063	• RETAINER,OUTER SEAL	2	EA
	10	98-360010159	• BREAKER,90C(DUAL),50OD,8BRG	1	EA
	11	98-8583	• ASSY,BOLT L9	24	EA
	12	98-880280049	• PIN,SPIRAL ROLL,SST	88	EA
	14	98-320050140	• PICK,BREAKER	88	EA
	16	98-880110590	• SEAL	2	EA
	17	98-880110442	• SEAL	2	EA
	18	98-300630029	• BOLT,CLAMP RING	28	EA
	19	98-880110201	• LOCKWIRE,BASIC STEEL	2	EA
	20	98-340030125	• RING,BACKING	2	EA
	21	98-320350464	• PLATE,MTG-BRKR BRG HSG	2	EA
	22	98-340270045	• HOUSING,BREAKER 8" *	2	EA
	23	98-340410009	• RETAINER,BREAKER BEARING	2	EA
	24	00902224-0858	• SCREW,CAP,SKT HD,3/4"-10 UNC,2-3/4" LG	12	EA
	25	98-A5Q1200000	• WASHER,LOCK-HIGH COLLAR	12	EA
	26	98-320040118	• HOLDER,PICK	88	EA
	27	98-880480038	• SPRING,DISC,1.25	264	EA
	28	00903113-0012	• NUT,LOCK,HEX COLLAR,1-1/4"-12 UNF	88	EA
	29	98-320030208	• RING,SPEED SENSOR	1	EA
	30	00900042-0024	• BOLT,HEX HD,1/2"-13 UNC,1-1/4" LG,GR 8	2	EA
	31	98-A5P0800000	• WASHER,LOCK	2	EA
	32	98-300030293	• RING,PICK SPACER	88	EA
	33	98-A5P0600000	• WASHER,LOCK *	16	EA
	34	98-300110061	• KEY,SQUARE ENDS,0.880",0.880"	2	EA





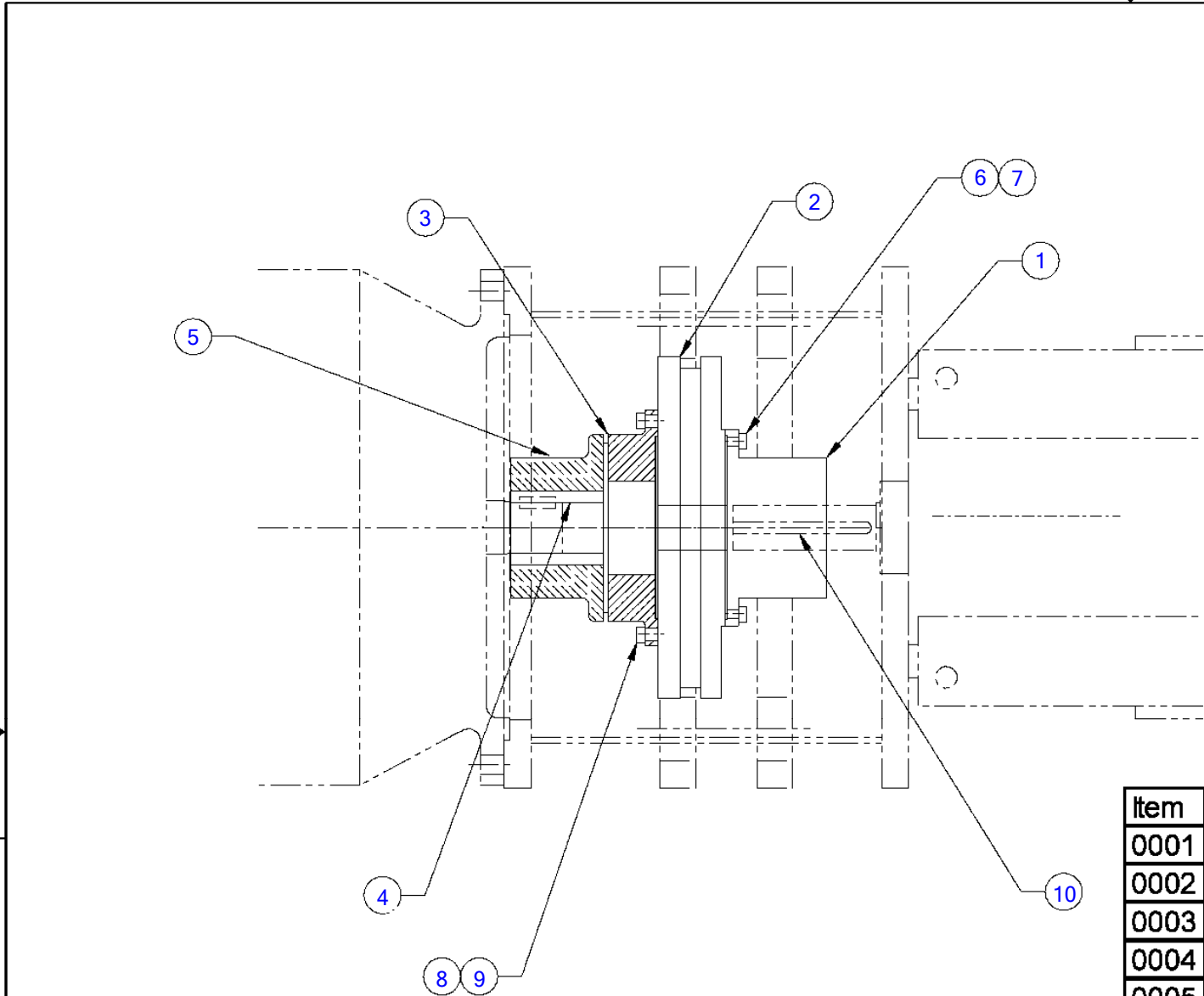
Reference:98-8583

Page:1-5-10





	Item	Part Number	Description	Qty	Units
	ASSY	100373700	ASSY,BREAKER DR-60/62	1	EA
	1	98-190110393	• REDUCER,SPD-41.45:1R,M1210DBJ3	1	EA
	2	98-890410049	• SHRINK DISC DR ONE REDUCER	1	EA
	3	98-330530046	• ASSY,COUPLING-TORQUE LIMITING	1	EA
	4	00900042-0116	• BOLT,HEX HD,3/4"-10 UNC,3-1/2" LG,GR 8	8	EA
	5	00925091-0200	• WASHER, LOCK, M20	8	EA
	6	00925447-2028	• SCREW,CAP,SKT HD,M20,70,CL 12.9	8	EA
	7	98-300280445	• PIN,TORQUE MTG	1	EA
	8	98-A0R0400064	• PIN,COTTER *	1	EA
	9	100373733	• ADAPTER,TUNNEL MOUNT	1	EA
	10	98-320131650	• COVER,TUNNEL MT	1	EA
	11	00900012-0001	• BOLT,HEX HD,1/4"-20 UNC,3/4" LG,GR 5,B	10	EA
	12	98-A5P0400000	• WASHER,LOCK *	10	EA
	13	00902224-0856	• SCREW,CAP,SKT HD,3/4"-10 UNC,2-1/4" LG	8	EA
	14	98-32042F016	• SPACER,ELECTRIC MOTOR	1	EA
	15	98-320351151	• PLATE,MTG-TUNNEL MOUNT	1	EA
	16	98-320145641	• PLATE,BAFFLE	1	EA
	17	98-320145640	• PLATE,BAFFLE	3	EA
	18	98-A5Q1200000	• WASHER,LOCK-HIGH COLLAR	8	EA
	19	98-A5P0800000	• WASHER,LOCK	6	EA
	20	98-A9A2412128	• BOLT,HEX HD,1-1/2"-12 UNF,8" LG,L9,BL	4	EA
	21	98-A9N2400000	• WASHER,FLAT-L9	4	EA
	22	98-A9U2412000	• NUT,LOCK-L9 *	4	EA
	23	00900042-0014	• BOLT,HEX HD,1/2"-13 UNC,1" LG,GR 8,BL	6	EA
	24	98-A5S1210000	• NUT,HEAVY HEX	8	EA
	25	00900302-0006	• WASHER,SPRING,3/4",BLACK	8	EA



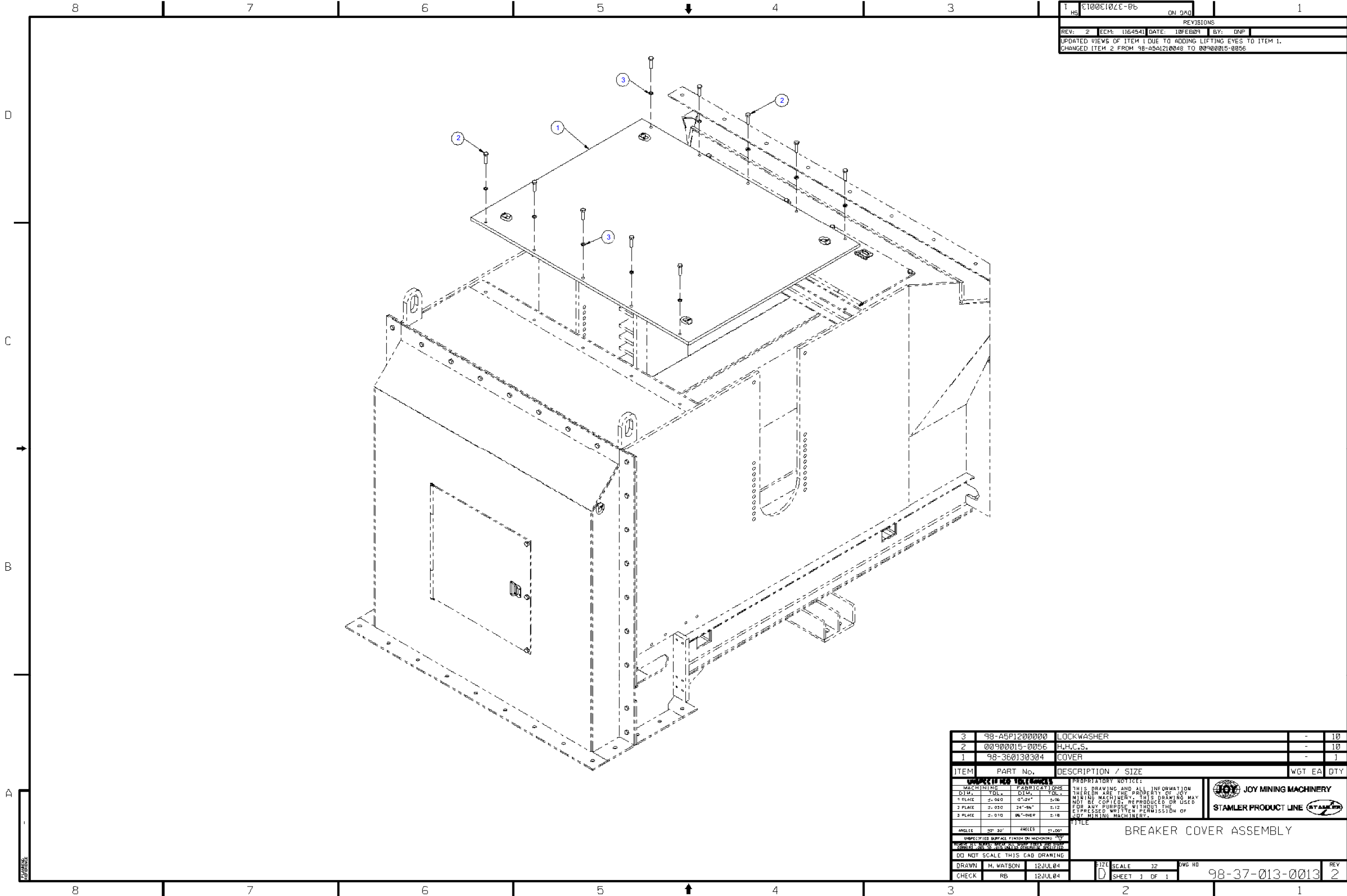
REVISIONS				
REV:	2	ECM:	1149028	DATE: 15AUG08 BY: MS
1. UPDATED BORDER NO CHANGES DONE TO DRAWING				
REV:	3	ECM:	1160886	DATE: 12NOV08 BY: SMB
1. CHANGED ITEM '10' WAS 98-320111302				

Item	Component number	Object description	Qty.	Un
0001	98-320250132	HUB,CLUTCH-INTRF BORED	1	EA
0002	98-210530250	COUPLING,TORQUE *	1	EA
0003	98-320530024	COUPLING,MOD VULCAN DR	1	EA
0004	98-300110074	KEY .63"SQ 4.00"LG	1	EA
0005	98-M4K12BR012	SCREW,HEX HD W/C PT	2	EA
0006	98-A8C0720020	SCREW,SOCKET CAP *	12	EA
0007	98-A5P0700000	WASHER,LOCK *	12	EA
0008	98-M6C12BR030	SCREW,SOCKET CAP *	8	EA
0009	98-M6P1200000	WASHER,LOCK *	8	EA
0010	98-300111163	KEY 16MM 28MM 200MM LG	1	EA

UNSPECIFIED TOLERANCES					PROPRIETARY NOTICE:		 JOY MINING MACHINERY		
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MACHINING	±0.5 mm	MACHINING	FABRICATIONS						
ANGULAR	±1/2°	DIM.	TOL.	DIM.					TOL.
FABRICATIONS		1 PLACE	±.060	0"-24"					±.06
0-300mm		2 PLACE	±.030	24"-96"					±.12
301mm-1000mm		3 PLACE	±.010	96"-OVER					±.18
1001mm-3000mm						TITLE ASSY, COUPLING TORQUE LIMITING			
3001mm-OVER		± 5mm	ANGLES	±0° 30'	ANGLES			±1.00°	
UNSPECIFIED SURFACE FINISH ON MACHINING 250									
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .010 UNLESS OTHERWISE SPECIFIED									
DO NOT SCALE THIS CAD DRAWING		DRAWN	PRB	27AUG04		SIZE B	SCALE	6	DWG NO. 98-33-053-0046
		CHECK	RB	27AUG04			SHEET 1 OF 1		



	Item	Part Number	Description	Qty	Units
	ASSY	98-330530046	ASSY,COUPLING-TORQUE LIMITING	1	EA
	1	98-320250132	• HUB,CLUTCH-INTRF BORED	1	EA
	2	98-210530250	• COUPLING,TORQUE LIMITING	1	EA
	3	98-320530024	• COUPLING,MOD VULKAN DR	1	EA
	4	98-300110074	• KEY,SQUARE ENDS,0.630",0.630"	1	EA
	5	98-M4K12BR012	• SCREW,HEX HD W/C PT	2	EA
	6	98-A8C0720020	• SCREW,CAP,SKT HD,7/16"-20 UNF	12	EA
	7	98-A5P0700000	• WASHER,LOCK	12	EA
	8	98-M6C12BR030	• SCREW,SOCKET CAP *	8	EA
	9	98-M6P1200000	• WASHER,LOCK	8	EA
	10	98-300111163	• KEY 16MM 28MM 200MM LG	1	EA

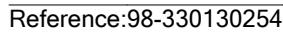


1	HS	98-360130304	ON 040	1
REVISIONS				
REV:	2	ECM:	1164541	DATE: 10FEB09 BY: DNP
UPDATED VIEWS OF ITEM 1 DUE TO ADDING LIFTING EYES TO ITEM 1.				
CHANGED ITEM 2 FROM 98-4541210048 TO 00900015-0056				

3	98-ASP1200000	LOCKWASHER	-	10	
2	00900015-0056	H.H.C.S.	-	10	
1	98-360130304	COVER	-	1	
ITEM	PART No.	DESCRIPTION / SIZE	WGT	EA	QTY
UNPACKING INSTRUCTIONS					
MACHINING		FABRICATIONS		 JOY MINING MACHINERY STAMLER PRODUCT LINE 	
DIM.	TOL.	DIM.	TOL.		
1 PLACE	2.060	0"-24"	±.06		
2 PLACE	2.030	24"-36"	±.12		
3 PLACE	2.010	36"-OVER	±.18	PROPRIETARY NOTICE: THIS DRAWING AND ALL INFORMATION HEREON ARE THE PROPERTY OF JOY MINING MACHINERY. THIS DRAWING MAY NOT BE COPIED, REPRODUCED OR USED FOR ANY PURPOSE, WITHOUT THE EXPRESSED WRITTEN PERMISSION OF JOY MINING MACHINERY.	
ANGLE	20° 30'	FINISH	±1.00'		
UNPACKING INSTRUCTIONS					
DO NOT SCALE THIS CAD DRAWING					
BREAKER COVER ASSEMBLY					
DRAWN	M. WATSON	12JUL04	SIZE	SCALE	32
CHECK	RB	12JUL04	SHEET	1	OF 1
			DWG NO	98-37-013-0013	
			REV	2	

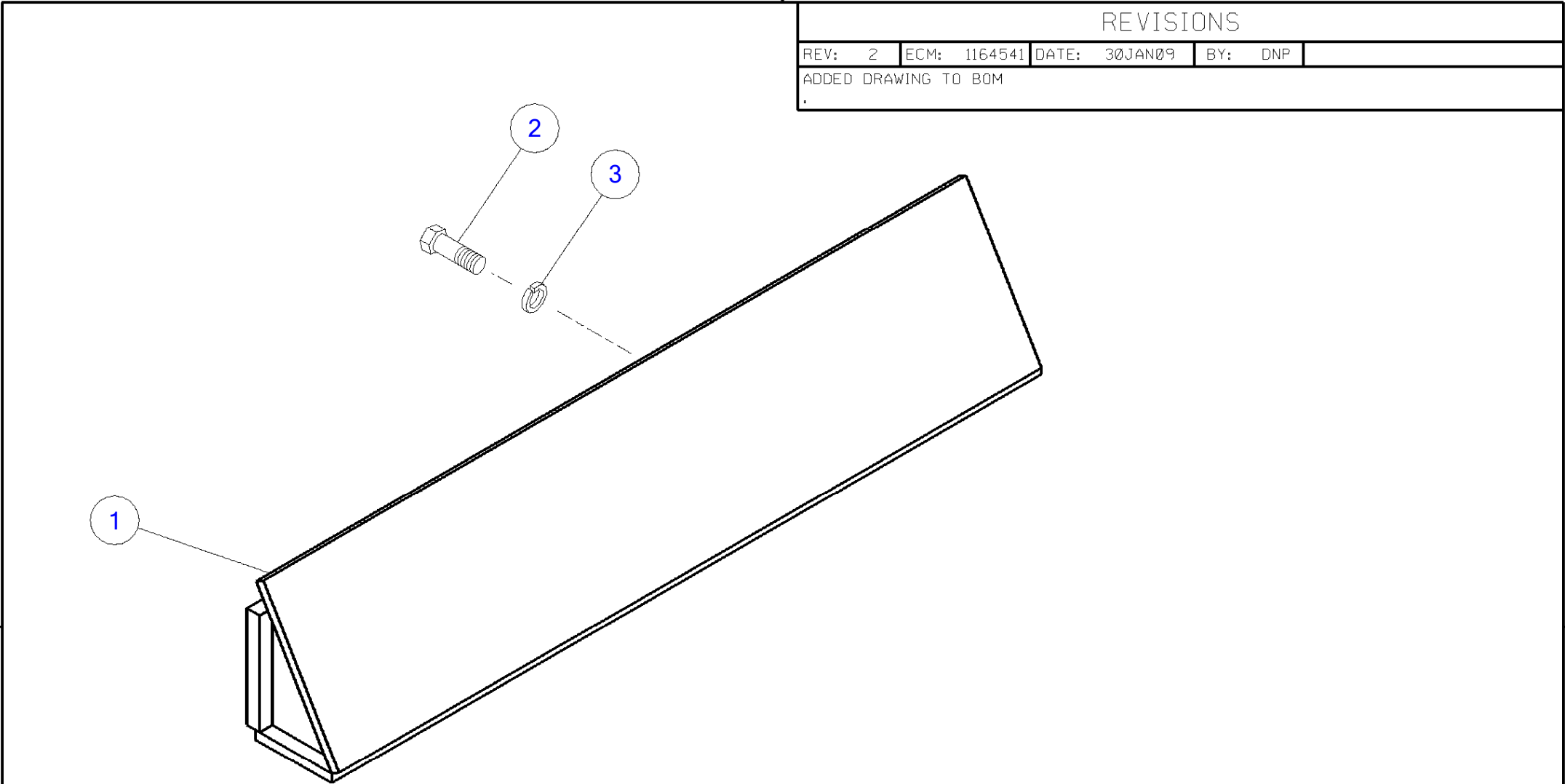


	Item	Part Number	Description	Qty	Units
	ASSY	98-370130013	ASSY,COVER-BREAKER	1	EA
	1	98-360130304	• COVER, BREAKER	1	EA
	2	00900015-0056	• BOLT,HEX HD,3/4"-10 UNC,2" LG,GR 5,BL	10	EA
	3	98-A5P1200000	• WASHER,LOCK *	10	EA





Item	Part Number	Description	Qty	Units
ASSY	98-330130254	ASSY,COVER-BREAKER SIDE PLATE	1	EA
1	98-320132288	• COVER,BREAKER PLATE	2	EA
2	00900042-0098	• BOLT,HEX HD,1"-8 UNC,3" LG,GR 8,BL	8	EA
3	00900302-0008	• WASHER,SPRING,1",BLACK	8	EA
4	98-A8T1608000	• NUT, HEX	8	EA



REVISIONS					
REV:	2	ECM:	1164541	DATE:	30JAN09
		BY:	DNP		
ADDED DRAWING TO BOM					
.					

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	98-340130249	COVER, CHAIN	1	EA		0.00
0002	00900302-0006	WASHER, LOCK, 3/4	4	EA		0.00
0003	00900042-0096	BOLT,HEX HD,3/4"-10 UNC,3" LG,GR 8,BL	4	EA		0.00

UNSPECIFIED TOLERANCES					
METRIC		IMPERIAL			
MACHINING	±0.5mm	MACHINING		FABRICATIONS	
ANGULAR	±1/2°	DIM.	TOL.	DIM.	TOL.
FABRICATIONS		1 PLACE	±.060	0"-24"	±.06
0-300mm	± 1mm	2 PLACE	±.030	24"-96"	±.12
301mm-1000mm	± 2mm	3 PLACE	±.010	96"-OVER	±.18
1001mm-3000mm	± 3mm				
3001mm-OVER	± 5mm	ANGLES	±0° 30'	ANGLES	±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING 250					
REMOVE ALL BURRS. BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .015 UNLESS OTHERWISE SPECIFIED					
DO NOT SCALE THIS CAD DRAWING		DRAWN	DNP	30JAN09	
		CHECK	KC	30JAN09	

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STAMLER PRODUCT LINE

TITLE			ASSY, COVER CHAIN		
SIZE A	SCALE	8	DWG NO	98-310130111	
	SHEET	1 OF 1			

D.S.N.: 98-310130111

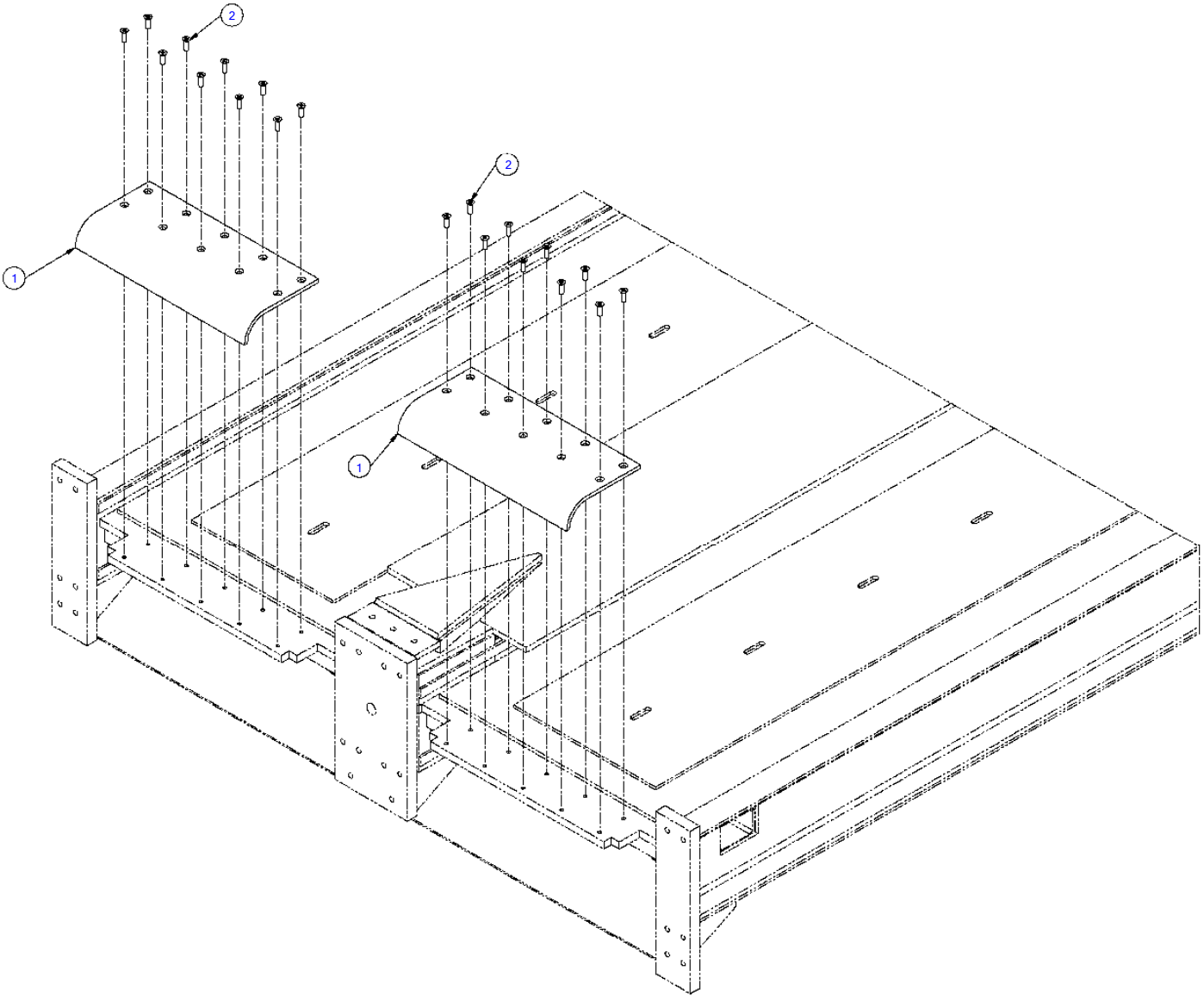
PLANNING
REFERENCE



	Item	Part Number	Description	Qty	Units
	ASSY	98-310130111	ASSY,COVER-CHAIN	1	EA
	1	98-340130249	• COVER, CHAIN	1	EA
	2	00900302-0006	• WASHER,SPRING,3/4",BLACK	4	EA
	3	00900042-0096	• BOLT,HEX HD,3/4"-10 UNC,3" LG,GR 8,BL	4	EA

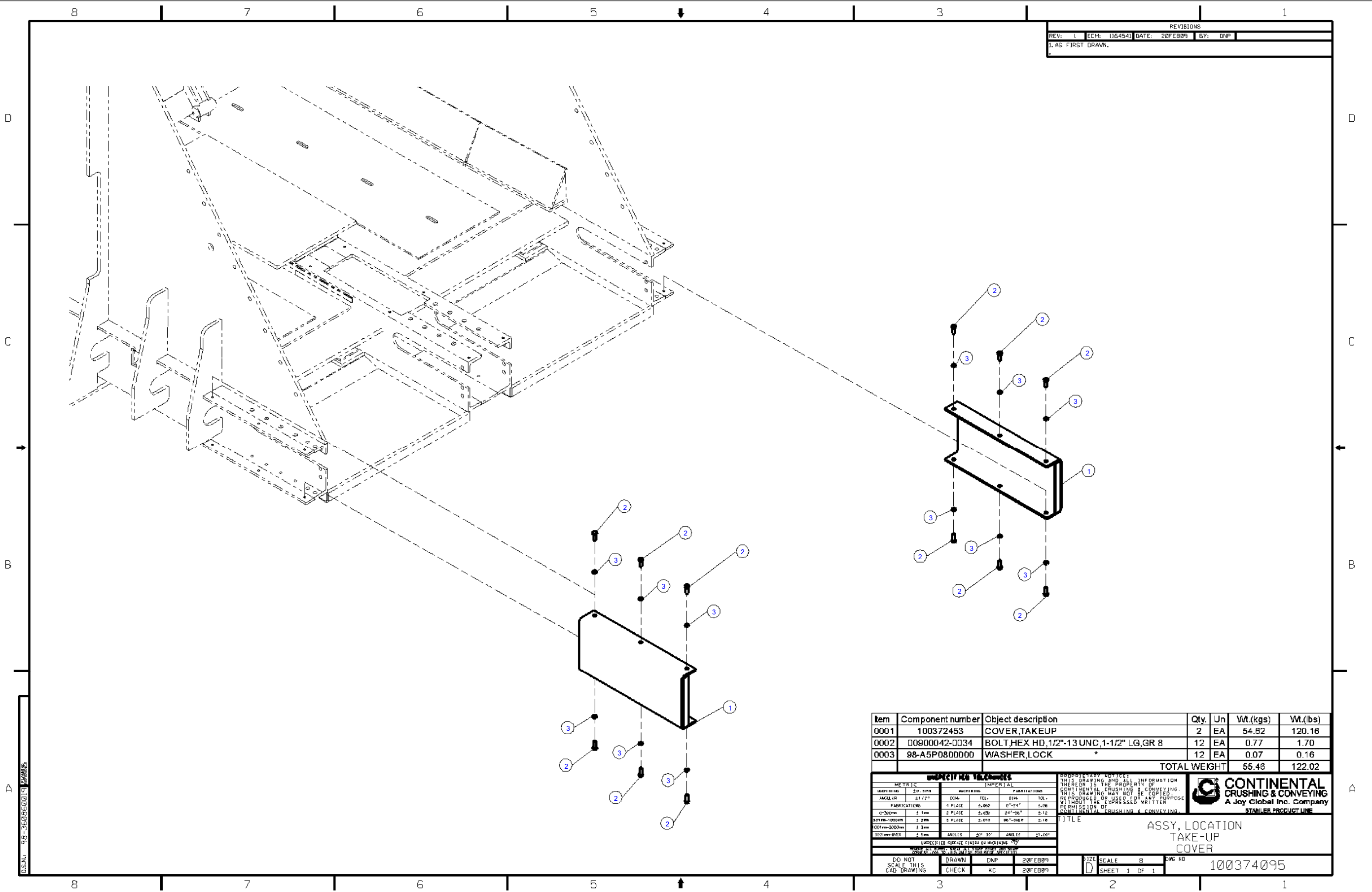


1	HS	98-37-013-0014	ON 040
REVISIONS			



2	98-A5E0813024	F.S.H.C.S.	-	20	
1	98-360130303	COVER	-	2	
ITEM	PART No.	DESCRIPTION / SIZE	WGT	EA	QTY
UNSPECIFIED TOLERANCES			 JOY MINING MACHINERY STAMLER PRODUCT LINE 		
MACHINING FABRICATIONS					
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TITLE					
DIMS. TOL.			WRAP-AROUND HEADSHAFT COVER ASSY		
1 PLACE	±.040	0°-90°	±.06		
2 PLACE	±.030	90°-180°	±.12		
3 PLACE	±.010	180°-360°	±.18		
ANGLES	30° 30'	ANGLES	±1.00°		
UNSPECIFIED SURFACE FINISH OR MACHINING					
REMOVE ALL BURRS FROM ALL SURFACES AND SHARP CORNERS, AND TO THIS UNLESS OTHERWISE SPECIFIED					
DO NOT SCALE THIS CAD DRAWING					
DRAWN	M. WATSON	13JUL04	SIZE	SCALE	B
CHECK	RB	13JUL04	SHEET	1	OF 1
			DWG NO		98-37-013-0014
					REV
					2

[illegible]



REVISIONS			
REV:	1	ECH: 1164541	DATE: 20FEB09 BY: DNP
1. AS FIRST DRAWN.			

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	100372453	COVER,TAKEUP	2	EA	54.82	120.16
0002	00900042-0034	BOLT,HEX HD,1/2"-13 UNC,1-1/2" LG,GR 8	12	EA	0.77	1.70
0003	98-A5P0800000	WASHER,LOCK *	12	EA	0.07	0.16
			TOTAL WEIGHT		55.46	122.02

METRIC				INCHES			
MACHINING	±0.075	MACHINING	±0.005	FABRICATION	±0.005	FABRICATION	±0.005
ANGLE IR	±1/2°	ANGLE IR	±1/2°	ANGLE IR	±1/2°	ANGLE IR	±1/2°
FABRICATIONS				FABRICATIONS			
0-300mm	±1mm	1 PLACE	±0.00	0"-24"	±0.06	1 PLACE	±0.06
300mm-1000mm	±2mm	2 PLACE	±0.00	24"-96"	±0.12	2 PLACE	±0.12
1000mm-3000mm	±3mm	3 PLACE	±0.00	96"-384"	±0.18	3 PLACE	±0.18
3000mm-OVER	±5mm	4 PLACE	±0.00	384"-OVER	±0.24	4 PLACE	±0.24
UNSPECIFIED SURFACE FINISH ON MACHINING				UNSPECIFIED SURFACE FINISH ON MACHINING			
DO NOT SCALE THIS CAD DRAWING				DO NOT SCALE THIS CAD DRAWING			
DRAWN		DNP		20FEB09		SIZE SCALE B	
CHECK		KC		20FEB09		SHEET 1 OF 1	

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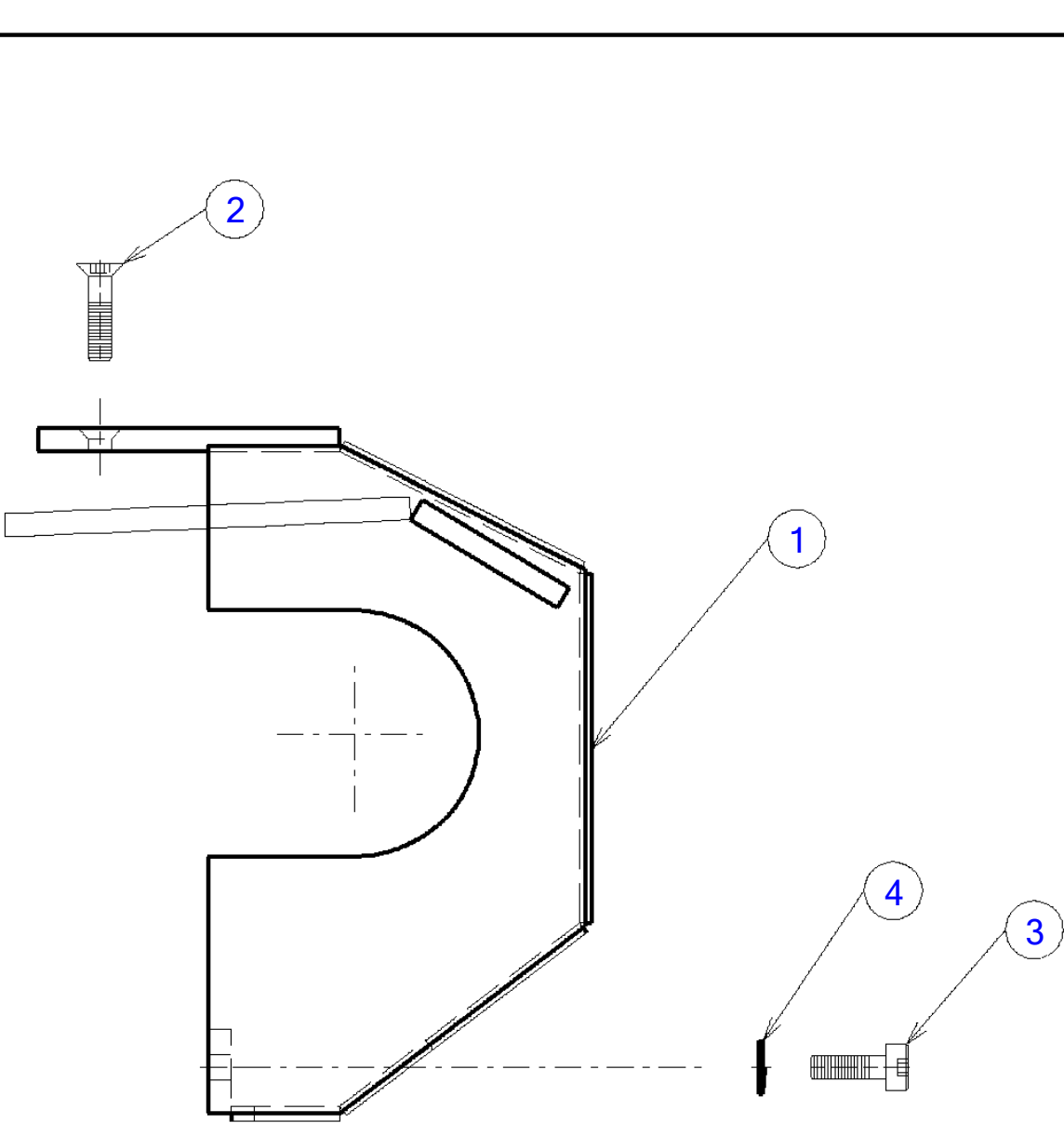
**CONTINENTAL
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STAMLER PRODUCT LINE

TITLE
ASSY, LOCATION
TAKE-UP
COVER

DWG NO
100374095



	Item	Part Number	Description	Qty	Units
	ASSY	100374095	LOCATION,TAKEUP COVER	1	EA
	1	100372453	• COVER,TAKEUP	2	EA
	2	00900042-0034	• BOLT,HEX HD,1/2"-13 UNC,1-1/2" LG,GR 8	12	EA
	3	98-A5P0800000	• WASHER,LOCK	12	EA



REVISIONS				
REV:	2	ECM:	1164541	DATE: 30JAN09 BY: DNP
UPDATED DRAWING TO CURRENT STANDARDS				

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	98-360130337	COVER,CNVR-DUAL	1	EA		0.00
0002	98-A8E1210024	SCREW,HEX FLAT CAP *	3	EA		0.00
0003	98-A8C1210036	SCREW,SOCKET CAP *	2	EA		0.00
0004	98-A5P1200000	WASHER,LOCK *	2	EA		0.00

UNSPECIFIED TOLERANCES					
METRIC			IMPERIAL		
MACHINING	DIM.	TOL.	MACHINING	DIM.	TOL.
ANGULAR	±1/2°		FABRICATIONS		
			1 PLACE	±.060	0°-24° ±.06
			2 PLACE	±.030	24°-86° ±.12
0-300mm	± 1mm		3 PLACE	±.010	86°-OVER ±.18
301mm-1000mm	± 2mm				
1001mm-3000mm	± 3mm				
3001mm-OVER	± 5mm		ANGLES	±0° 30'	ANGLES ±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING: Ra 3.2					
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS LOG TO DIS UNLESS OTHERWISE SPECIFIED					
DD NOT SCALE THIS CAD DRAWING	DRAWN	RB	17SEPT04	SIZE B SCALE 5 SHEET 1 OF 1	
	CHECK	PRB	17SEPT04		

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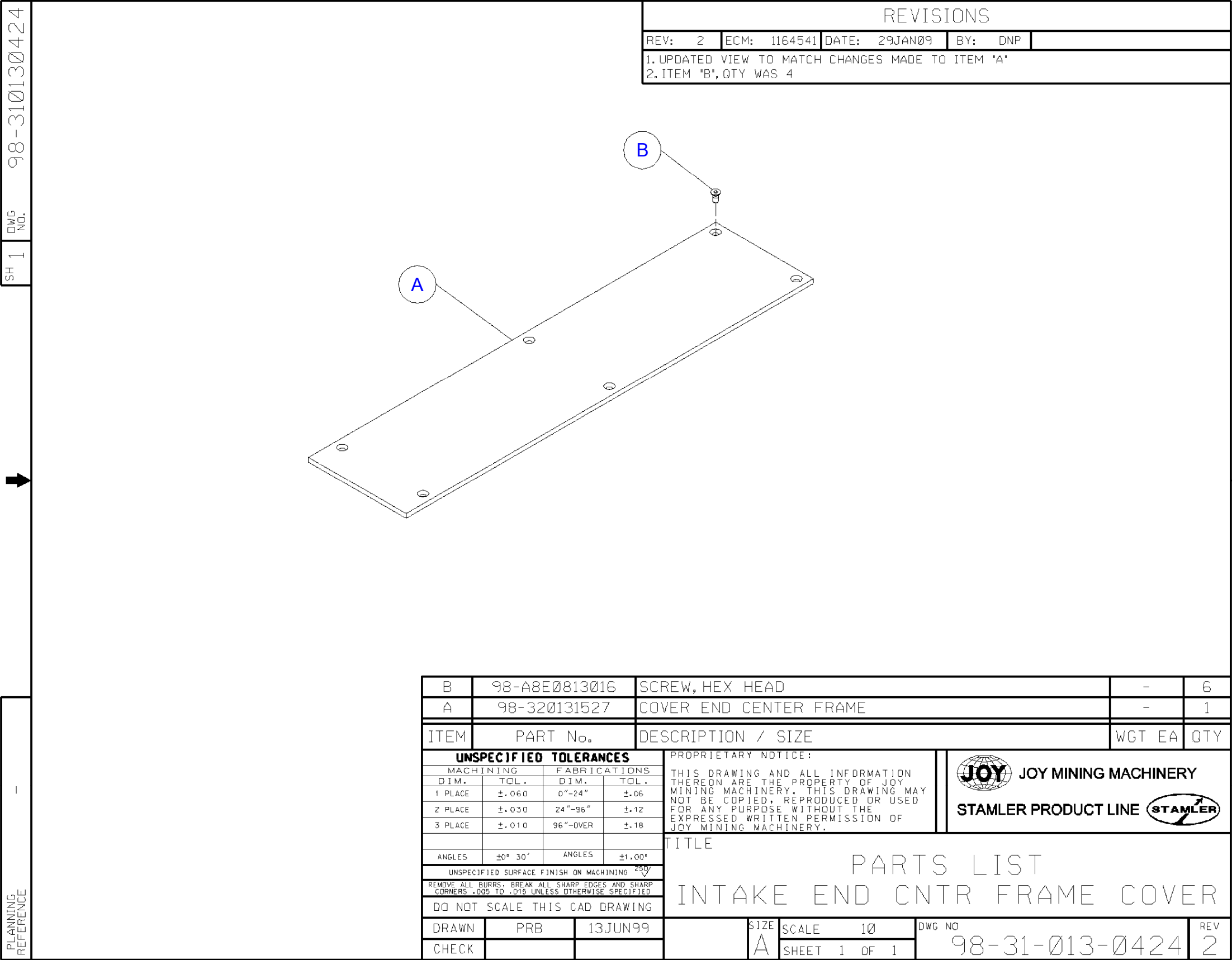
TITLE			DISCHARGE END CENTER FRAME COVER		
DWG NO			98-330130260		

D.S.N.: 98-330130260

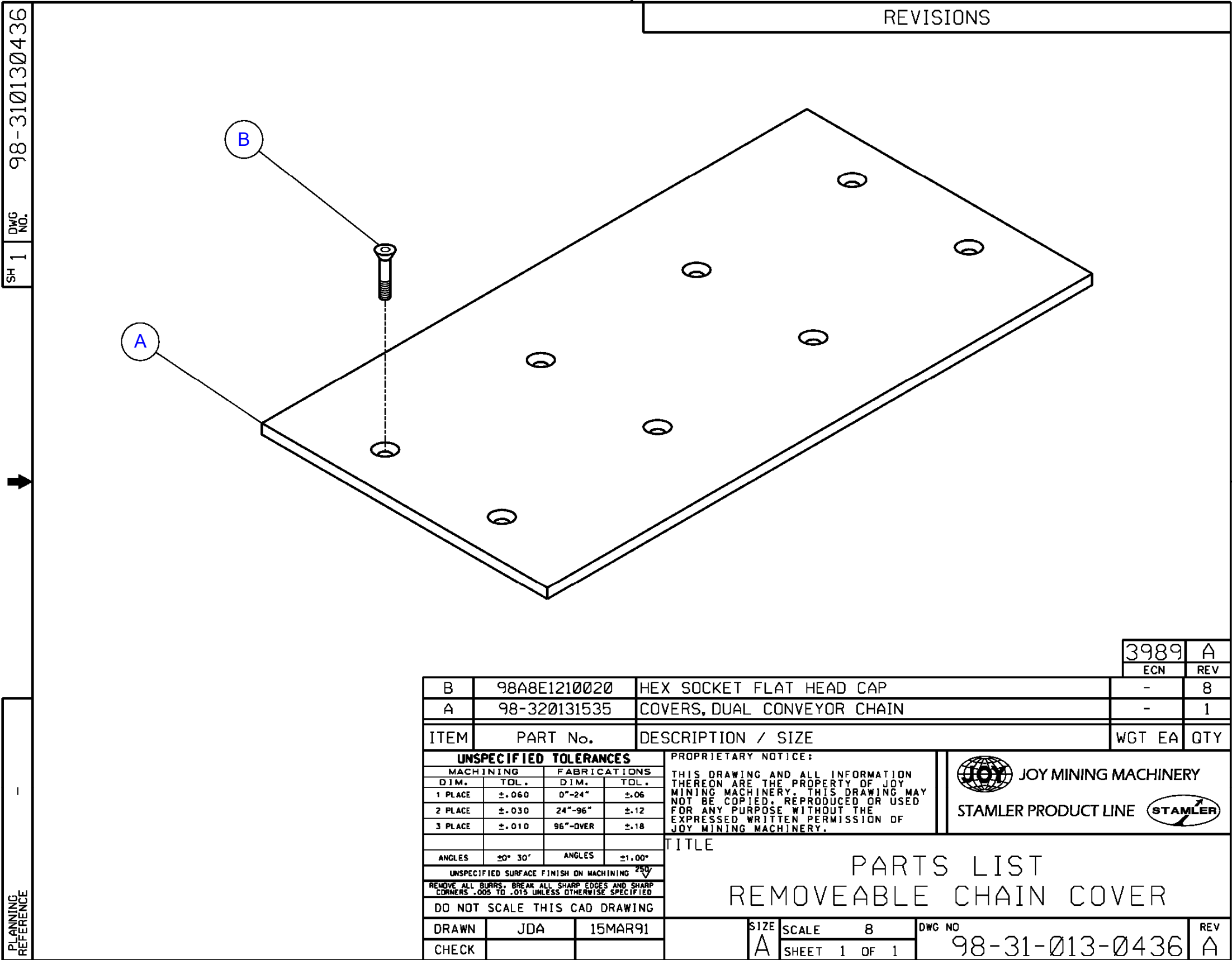
PLANNING
REFERENCE



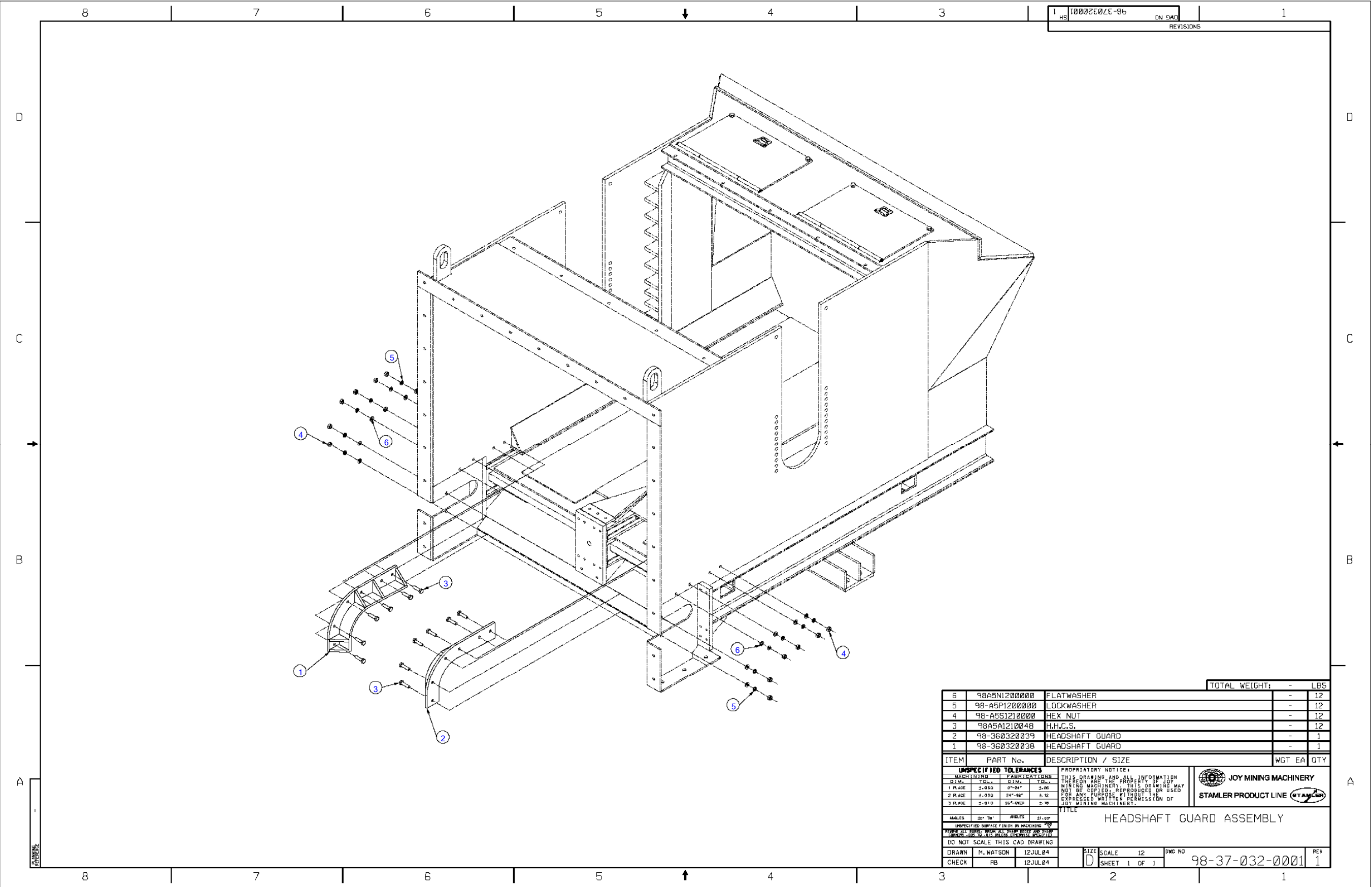
	Item	Part Number	Description	Qty	Units
	ASSY	98-330130260	ASSY,COVER-DISCH END,CENTER FRAME	1	EA
	1	98-360130337	• COVER,CNVR-DUAL	1	EA
	2	98-A8E1210024	• SCREW,HEX FLAT CAP *	3	EA
	3	98-A8C1210036	• SCREW,SOCKET CAP *	2	EA
	4	98-A5P1200000	• WASHER,LOCK *	2	EA







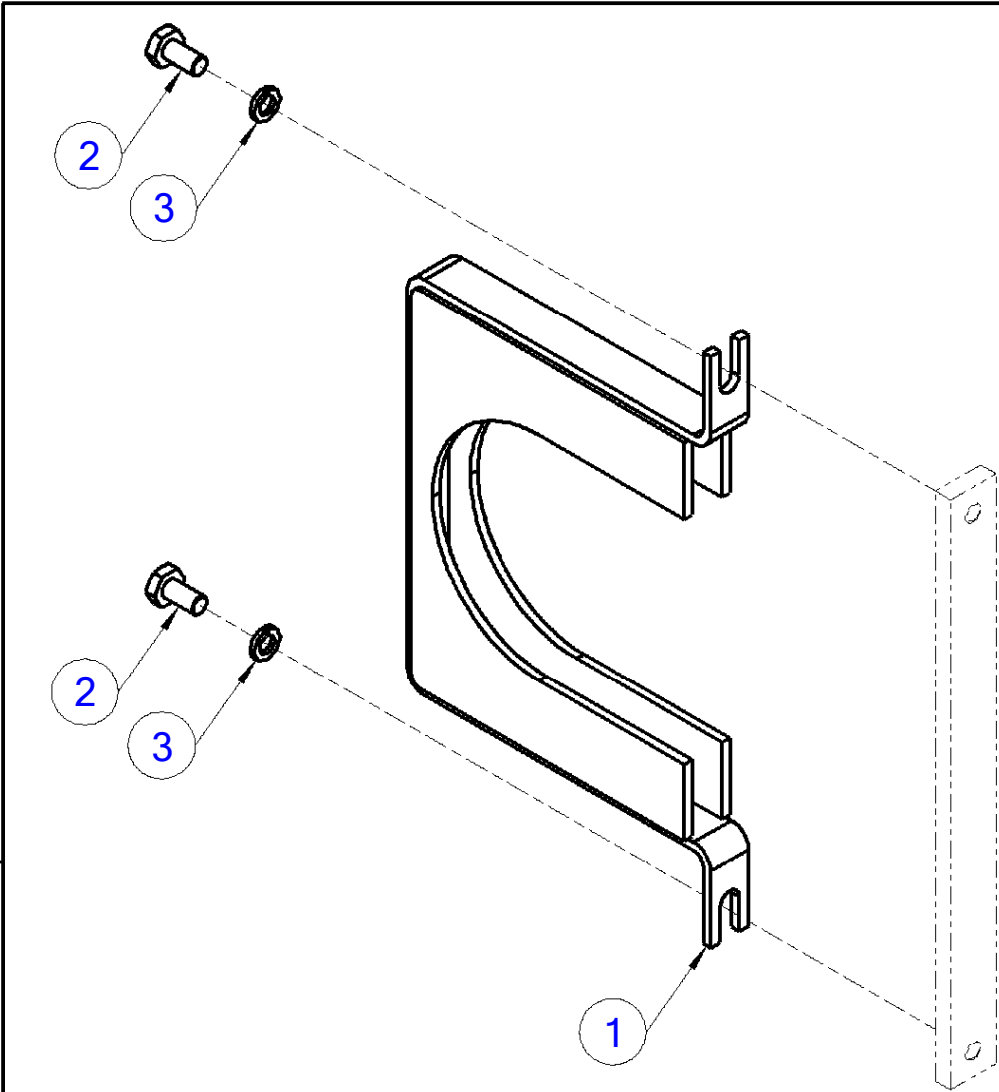
[illegible]



			TOTAL WEIGHT:		-	LBS
6	98A5N1200000	FLATWASHER	-	-	12	
5	98-A5P1200000	LOCKWASHER	-	-	12	
4	98-A5S1210000	HEX NUT	-	-	12	
3	98A5A1210048	H.H.C.S.	-	-	12	
2	98-360320039	HEADSHAFT GUARD	-	-	1	
1	98-360320038	HEADSHAFT GUARD	-	-	1	
ITEM	PART No.	DESCRIPTION / SIZE	WGT	EA	QTY	
UNSPECIFIED TOLERANCES		PROPRIETARY NOTICE:				
MACH (NINE) FABRICATIONS		THIS DRAWING AND ALL INFORMATION THEREON ARE THE PROPERTY OF JOY MINING MACHINERY. THIS DRAWING MAY NOT BE COPIED, REPRODUCED OR USED FOR ANY PURPOSE WITHOUT THE EXPRESS WRITTEN PERMISSION OF JOY MINING MACHINERY.	 JOY MINING MACHINERY STAMLER PRODUCT LINE 			
DIM. TOL. DIM. TOL.						
1 PLACE	±.000 0°-24° ±.00					
2 PLACE	±.010 24°-76° ±.12					
3 PLACE	±.010 96°-OVER ±.18					
ANGLES 25°-30°	ANGLES ±1.00°	TITLE				
HEADSHAFT GUARD ASSEMBLY						
UNSPECIFIED SURFACE FINISH OR WORKING FINISH						
DO NOT SCALE THIS CAD DRAWING						
DRAWN	M. WATSON	12JUL04	SIZE	SCALE	12	
CHECK	FB	12JUL04	D	SHEET	1 OF 1	
				DWG NO	98-37-032-0001	
				REV	1	



	Item	Part Number	Description	Qty	Units
	ASSY	98-370320001	ASSY, GUARD-HEADSHAFT CHAIN	1	EA
	1	98-360320038	• GUARD,HEADSHAFT CHAIN	1	EA
	2	98-360320039	• GUARD,HEADSHAFT CHAIN	1	EA
	3	98-A5A1210048	• SCREW,HEX CAP *	12	EA
	4	98-A5S1210000	• NUT,HEAVY HEX	12	EA
	5	98-A5P1200000	• WASHER,LOCK *	12	EA
	6	98-A5N1200000	• WASHER,FLAT *	12	EA



REVISIONS					
REV:	1	ECM:	1164541	DATE:	19FEB09
		BY:	MS		
1. AS FIRST DRAWN.					

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	100373699	GUARD,BEARING & HYDR MOTOR	1	EA	35.48	78.06
0002	00900042-0014	BOLT,HEX HD,1/2"-13 UNC,1" LG,GR 8,BL	2	EA	0.80	1.77
0003	98-A5P0800000	WASHER,LOCK *	2	EA	0.10	0.21
TOTAL WEIGHT:					36.38	80.04

UNSPECIFIED TOLERANCES					
METRIC		IMPERIAL			
MACHINING	±0.5mm	MACHINING		FABRICATIONS	
ANGULAR	±1/2°	DIM.	TOL.	DIM.	TOL.
FABRICATIONS		1 PLACE	±.060	0"-24"	±.06
0-300mm	± 1mm	2 PLACE	±.030	24"-96"	±.12
301mm-1000mm	± 2mm	3 PLACE	±.010	96"-OVER	±.18
1001mm-3000mm	± 3mm				
3001mm-OVER	± 5mm	ANGLES	±0° 30'	ANGLES	±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING 250					
REMOVE ALL BURRS. BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .015 UNLESS OTHERWISE SPECIFIED					
DO NOT SCALE THIS CAD DRAWING		DRAWN	MS	19FEB09	
		CHECK	KC	19FEB09	

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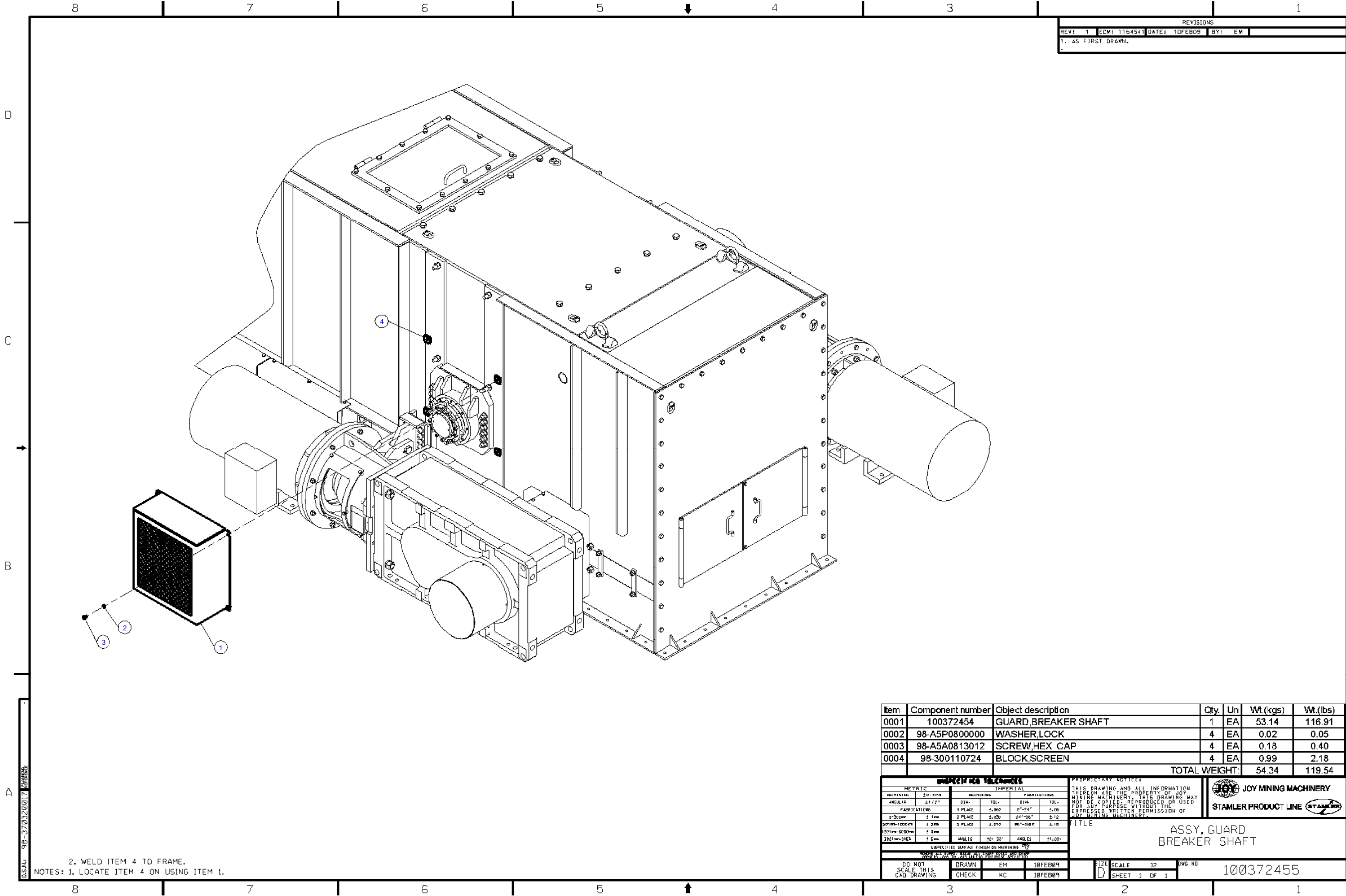
TITLE ASSY-GUARD-SHAFT					
SIZE A		SCALE 5	DWG NO 100373902		
		SHEET 1 OF 1			

D.S.N.: 98-310320106

PLANNING REFERENCE -



	Item	Part Number	Description	Qty	Units
	ASSY	100373902	ASSY, GUARD-SHAFT	1	EA
	1	100373699	• GUARD, BEARING & HYDR MOTOR	1	EA
	2	00900042-0014	• BOLT, HEX HD, 1/2"-13 UNC, 1" LG, GR 8, BL	2	EA
	3	98-A5P0800000	• WASHER, LOCK	2	EA



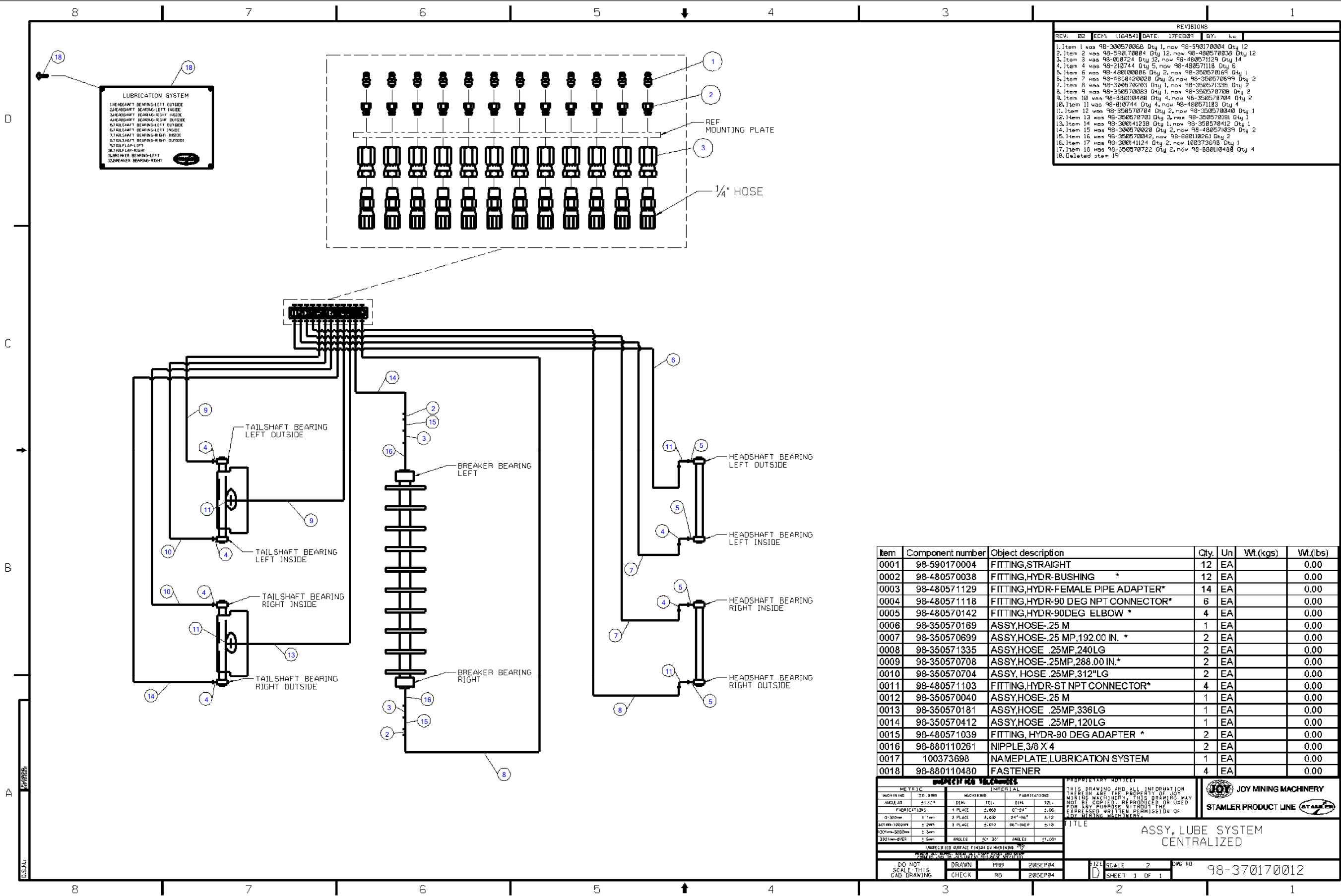
REVISIONS			
REV: 1	ECM: 1164541	DATE: 10FEB09	BY: EM
1. AS FIRST DRAWN.			

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	100372454	GUARD,BREAKER SHAFT	1	EA	53.14	116.91
0002	98-A5P0800000	WASHER,LOCK	4	EA	0.02	0.05
0003	98-A5A0813012	SCREW,HEX CAP	4	EA	0.18	0.40
0004	98-300110724	BLOCK,SCREEN	4	EA	0.99	2.18
TOTAL WEIGHT:					54.34	119.54

UNSPECIFIED SURFACE FINISH				PROPRIETARY NOTICE			
METRIC		INCHES		THIS DRAWING AND ALL INFORMATION THEREON ARE THE PROPERTY OF JOY MINING MACHINERY. THIS DRAWING MAY NOT BE COPIED, REPRODUCED OR USED FOR ANY PURPOSE WITHOUT THE EXPRESSED WRITTEN PERMISSION OF JOY MINING MACHINERY.			
MACHINING	±0.075	MACHINING	±0.003	JOY MINING MACHINERY STAMLER PRODUCT LINE			
ANGULAR	±1/2°	ANGULAR	±0.005				
FABRICATIONS		FABRICATIONS					
1 PLACE	±0.005	1 PLACE	±0.005				
2 PLACE	±0.005	2 PLACE	±0.005	TITLE ASSY, GUARD BREAKER SHAFT			
3 PLACE	±0.010	3 PLACE	±0.010				
4 PLACE	±0.015	4 PLACE	±0.015				
5 PLACE	±0.020	5 PLACE	±0.020				
6 PLACE	±0.025	6 PLACE	±0.025	DO NOT SCALE THIS CAB DRAWING			
7 PLACE	±0.030	7 PLACE	±0.030				
8 PLACE	±0.035	8 PLACE	±0.035				
9 PLACE	±0.040	9 PLACE	±0.040				
10 PLACE	±0.045	10 PLACE	±0.045	DRAWN EM 38FEB09			
11 PLACE	±0.050	11 PLACE	±0.050				
12 PLACE	±0.055	12 PLACE	±0.055				
13 PLACE	±0.060	13 PLACE	±0.060				
14 PLACE	±0.065	14 PLACE	±0.065	SIZE SCALE 32 DWG NO 100372455			
15 PLACE	±0.070	15 PLACE	±0.070				
16 PLACE	±0.075	16 PLACE	±0.075				
17 PLACE	±0.080	17 PLACE	±0.080				
18 PLACE	±0.085	18 PLACE	±0.085	SHEET 1 OF 1			
19 PLACE	±0.090	19 PLACE	±0.090				
20 PLACE	±0.095	20 PLACE	±0.095				
21 PLACE	±0.100	21 PLACE	±0.100				
22 PLACE	±0.105	22 PLACE	±0.105	CHECK KC 38FEB09			
23 PLACE	±0.110	23 PLACE	±0.110				
24 PLACE	±0.115	24 PLACE	±0.115				
25 PLACE	±0.120	25 PLACE	±0.120				
26 PLACE	±0.125	26 PLACE	±0.125	UNSPECIFIED SURFACE FINISH ON MACHINING			
27 PLACE	±0.130	27 PLACE	±0.130				
28 PLACE	±0.135	28 PLACE	±0.135				
29 PLACE	±0.140	29 PLACE	±0.140				
30 PLACE	±0.145	30 PLACE	±0.145	UNSPECIFIED SURFACE FINISH ON MACHINING			
31 PLACE	±0.150	31 PLACE	±0.150				
32 PLACE	±0.155	32 PLACE	±0.155				
33 PLACE	±0.160	33 PLACE	±0.160				
34 PLACE	±0.165	34 PLACE	±0.165	UNSPECIFIED SURFACE FINISH ON MACHINING			
35 PLACE	±0.170	35 PLACE	±0.170				
36 PLACE	±0.175	36 PLACE	±0.175				
37 PLACE	±0.180	37 PLACE	±0.180				
38 PLACE	±0.185	38 PLACE	±0.185	UNSPECIFIED SURFACE FINISH ON MACHINING			
39 PLACE	±0.190	39 PLACE	±0.190				
40 PLACE	±0.195	40 PLACE	±0.195				
41 PLACE	±0.200	41 PLACE	±0.200				
42 PLACE	±0.205	42 PLACE	±0.205	UNSPECIFIED SURFACE FINISH ON MACHINING			
43 PLACE	±0.210	43 PLACE	±0.210				
44 PLACE	±0.215	44 PLACE	±0.215				
45 PLACE	±0.220	45 PLACE	±0.220				
46 PLACE	±0.225	46 PLACE	±0.225	UNSPECIFIED SURFACE FINISH ON MACHINING			
47 PLACE	±0.230	47 PLACE	±0.230				
48 PLACE	±0.235	48 PLACE	±0.235				
49 PLACE	±0.240	49 PLACE	±0.240				
50 PLACE	±0.245	50 PLACE	±0.245	UNSPECIFIED SURFACE FINISH ON MACHINING			
51 PLACE	±0.250	51 PLACE	±0.250				
52 PLACE	±0.255	52 PLACE	±0.255				
53 PLACE	±0.260	53 PLACE	±0.260				
54 PLACE	±0.265	54 PLACE	±0.265	UNSPECIFIED SURFACE FINISH ON MACHINING			
55 PLACE	±0.270	55 PLACE	±0.270				
56 PLACE	±0.275	56 PLACE	±0.275				
57 PLACE	±0.280	57 PLACE	±0.280				
58 PLACE	±0.285	58 PLACE	±0.285	UNSPECIFIED SURFACE FINISH ON MACHINING			
59 PLACE	±0.290	59 PLACE	±0.290				
60 PLACE	±0.295	60 PLACE	±0.295				
61 PLACE	±0.300	61 PLACE	±0.300				
62 PLACE	±0.305	62 PLACE	±0.305	UNSPECIFIED SURFACE FINISH ON MACHINING			
63 PLACE	±0.310	63 PLACE	±0.310				
64 PLACE	±0.315	64 PLACE	±0.315				
65 PLACE	±0.320	65 PLACE	±0.320				
66 PLACE	±0.325	66 PLACE	±0.325	UNSPECIFIED SURFACE FINISH ON MACHINING			
67 PLACE	±0.330	67 PLACE	±0.330				
68 PLACE	±0.335	68 PLACE	±0.335				
69 PLACE	±0.340	69 PLACE	±0.340				
70 PLACE	±0.345	70 PLACE	±0.345	UNSPECIFIED SURFACE FINISH ON MACHINING			
71 PLACE	±0.350	71 PLACE	±0.350				
72 PLACE	±0.355	72 PLACE	±0.355				
73 PLACE	±0.360	73 PLACE	±0.360				
74 PLACE	±0.365	74 PLACE	±0.365	UNSPECIFIED SURFACE FINISH ON MACHINING			
75 PLACE	±0.370	75 PLACE	±0.370				
76 PLACE	±0.375	76 PLACE	±0.375				
77 PLACE	±0.380	77 PLACE	±0.380				
78 PLACE	±0.385	78 PLACE	±0.385	UNSPECIFIED SURFACE FINISH ON MACHINING			
79 PLACE	±0.390	79 PLACE	±0.390				
80 PLACE	±0.395	80 PLACE	±0.395				
81 PLACE	±0.400	81 PLACE	±0.400				
82 PLACE	±0.405	82 PLACE	±0.405	UNSPECIFIED SURFACE FINISH ON MACHINING			
83 PLACE	±0.410	83 PLACE	±0.410				
84 PLACE	±0.415	84 PLACE	±0.415				
85 PLACE	±0.420	85 PLACE	±0.420				
86 PLACE	±0.425	86 PLACE	±0.425	UNSPECIFIED SURFACE FINISH ON MACHINING			
87 PLACE	±0.430	87 PLACE	±0.430				
88 PLACE	±0.435	88 PLACE	±0.435				
89 PLACE	±0.440	89 PLACE	±0.440				
90 PLACE	±0.445	90 PLACE	±0.445	UNSPECIFIED SURFACE FINISH ON MACHINING			
91 PLACE	±0.450	91 PLACE	±0.450				
92 PLACE	±0.455	92 PLACE	±0.455				
93 PLACE	±0.460	93 PLACE	±0.460				
94 PLACE	±0.465	94 PLACE	±0.465	UNSPECIFIED SURFACE FINISH ON MACHINING			
95 PLACE	±0.470	95 PLACE	±0.470				
96 PLACE	±0.475	96 PLACE	±0.475				
97 PLACE	±0.480	97 PLACE	±0.480				
98 PLACE	±0.485	98 PLACE	±0.485	UNSPECIFIED SURFACE FINISH ON MACHINING			
99 PLACE	±0.490	99 PLACE	±0.490				
100 PLACE	±0.495	100 PLACE	±0.495				
101 PLACE	±0.500	101 PLACE	±0.500				
102 PLACE	±0.505	102 PLACE	±0.505	UNSPECIFIED SURFACE FINISH ON MACHINING			
103 PLACE	±0.510	103 PLACE	±0.510				
104 PLACE	±0.515	104 PLACE	±0.515				
105 PLACE	±0.520	105 PLACE	±0.520				
106 PLACE	±0.525	106 PLACE	±0.525	UNSPECIFIED SURFACE FINISH ON MACHINING			
107 PLACE	±0.530	107 PLACE	±0.530				
108 PLACE	±0.535	108 PLACE	±0.535				
109 PLACE	±0.540	109 PLACE	±0.540				
110 PLACE	±0.545	110 PLACE	±0.545	UNSPECIFIED SURFACE FINISH ON MACHINING			
111 PLACE	±0.550	111 PLACE	±0.550				
112 PLACE	±0.555	112 PLACE	±0.555				
113 PLACE	±0.560	113 PLACE	±0.560				
114 PLACE	±0.565	114 PLACE	±0.565	UNSPECIFIED SURFACE FINISH ON MACHINING			
115 PLACE	±0.570	115 PLACE	±0.570				
116 PLACE	±0.575	116 PLACE	±0.575				
117 PLACE	±0.580	117 PLACE	±0.580				
118 PLACE	±0.585	118 PLACE	±0.585	UNSPECIFIED SURFACE FINISH ON MACHINING			
119 PLACE	±0.590	119 PLACE	±0.590				
120 PLACE	±0.595	120 PLACE	±0.595				
121 PLACE	±0.600	121 PLACE	±0.600				
122 PLACE	±0.605	122 PLACE	±0.605	UNSPECIFIED SURFACE FINISH ON MACHINING			
123 PLACE	±0.610	123 PLACE	±0.610				
124 PLACE	±0.615	124 PLACE	±0.615				
125 PLACE	±0.620	125 PLACE	±0.620				
126 PLACE	±0.625	126 PLACE	±0.625	UNSPECIFIED SURFACE FINISH ON MACHINING			
127 PLACE	±0.630	127 PLACE	±0.630				
128 PLACE	±0.635	128 PLACE	±0.635				
129 PLACE	±0.640	129 PLACE	±0.640				
130 PLACE	±0.645	130 PLACE	±0.645	UNSPECIFIED SURFACE FINISH ON MACHINING			
131 PLACE	±0.650	131 PLACE	±0.650				
132 PLACE	±0.655	132 PLACE	±0.655				
133 PLACE	±0.660	133 PLACE	±0.660				
134 PLACE	±0.665	134 PLACE	±0.665	UNSPECIFIED SURFACE FINISH ON MACHINING			
135 PLACE	±0.670	135 PLACE	±0.670				
136 PLACE	±0.675	136 PLACE	±0.675	UNSPECIFIED SURFACE FINISH ON MACHINING			
137 PLACE	±0.680	137 PLACE	±0.680				

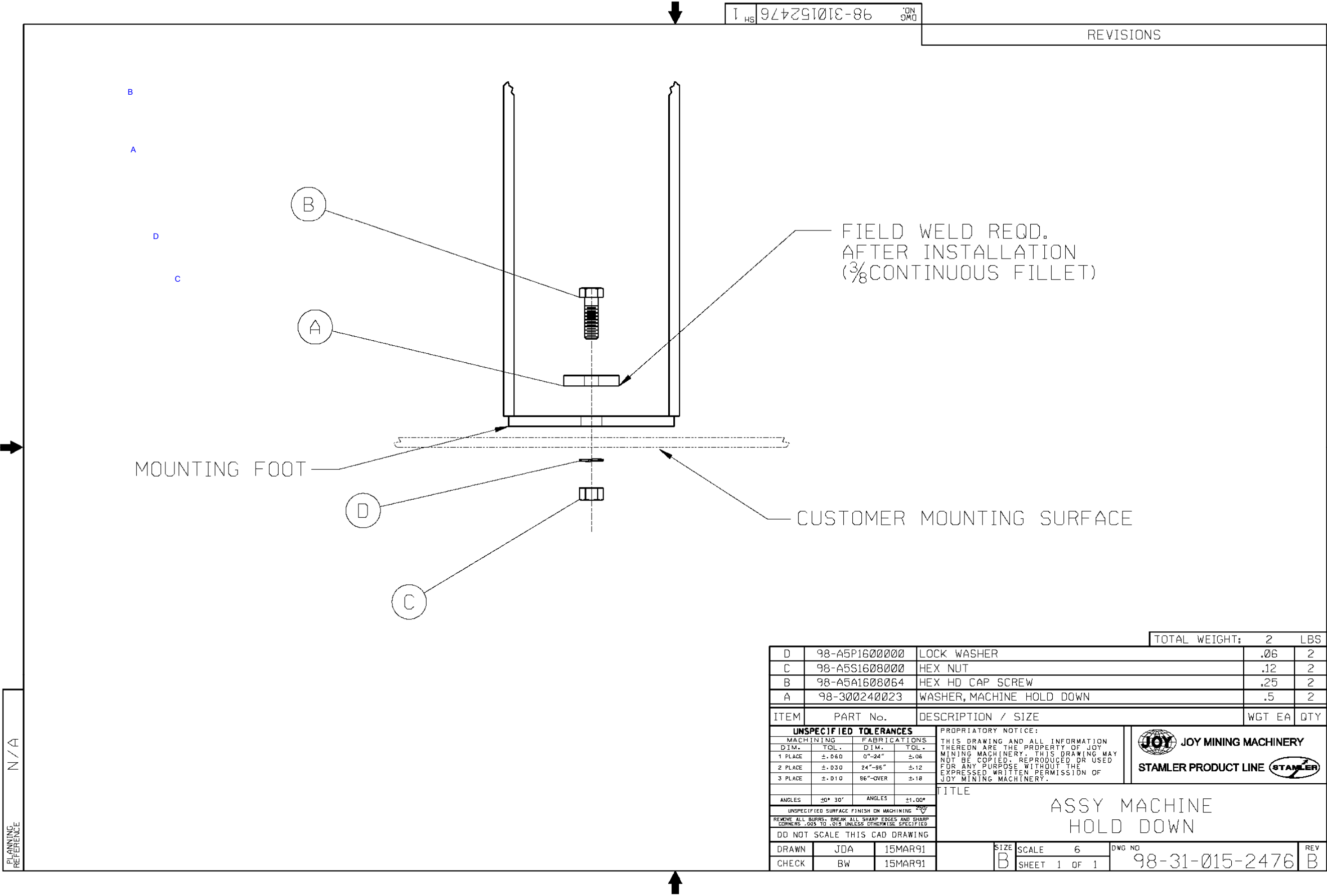


	Item	Part Number	Description	Qty	Units
	ASSY	100372455	ASSY, GUARD-BREAKER SHAFT	1	EA
	1	100372454	• GUARD, BREAKER SHAFT	1	EA
	2	98-A5P0800000	• WASHER, LOCK	4	EA
	3	98-A5A0813012	• SCREW, HEX CAP *	4	EA
	4	98-300110724	• BLOCK, SCREEN	4	EA





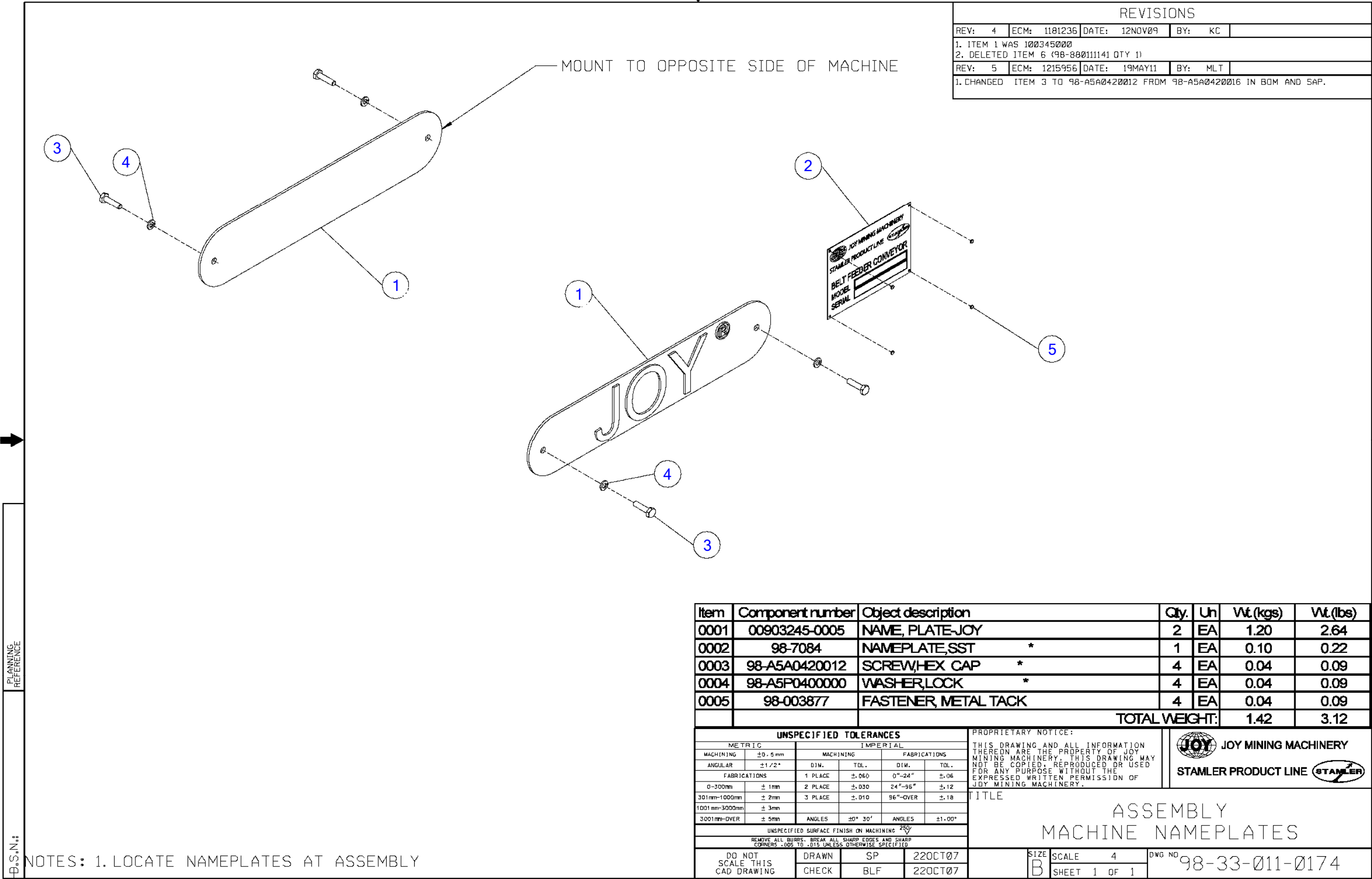
	Item	Part Number	Description	Qty	Units
	ASSY	98-370170012	ASSY,LUBE SYS-CENTRALIZED	1	EA
	1	98-590170004	• FITTING,STRAIGHT	12	EA
	2	98-480570038	• FITTING,HYDR-BUSHING *	12	EA
	3	98-480571129	• FITTING,HYDR-FEMALE PIPE ADAPTER*	14	EA
	4	98-480571118	• FITTING,HYDR-90 DEG NPT CONNECTOR*	6	EA
	5	98-480570142	• FITTING,HYDR-90DEG ELBOW *	4	EA
	6	98-350570169	• ASSY,HOSE-.25 M	1	EA
	7	98-350570699	• ASSY,HOSE-.25 MP,192.00 IN. *	2	EA
	8	98-350571335	• ASSY,HOSE .25 MNPT, 240" LG.	2	EA
	9	98-350570708	• ASSY,HOSE-.25MP,288.00 IN.*	2	EA
	10	98-350570704	• ASSY, HOSE .25MP,312"LG	2	EA
	11	98-480571103	• FITTING,HYDR-ST NPT CONNECTOR*	4	EA
	12	98-350570040	• ASSY,HOSE-.25 M	1	EA
	13	98-350570181	• ASSY,HOSE .25MP,336LG	1	EA
	14	98-350570412	• ASSY,HOSE .25MP,120LG	1	EA
	15	98-480571039	• FITTING, HYDR-90 DEG ADAPTER	2	EA
	16	98-880110261	• NIPPLE,3/8 X 4	2	EA
	17	100373698	• NAMEPLATE,LUBRICATION SYSTEM	1	EA
	18	98-880110480	• FASTENER	4	EA



D	98-A5P1600000	LOCK WASHER		.06	2
C	98-A5S1608000	HEX NUT		.12	2
B	98-A5A1608064	HEX HD CAP SCREW		.25	2
A	98-300240023	WASHER, MACHINE HOLD DOWN		.5	2
ITEM	PART No.	DESCRIPTION / SIZE			WGT EA QTY
UNSPECIFIED TOLERANCES		PROPRIETARY NOTICE:			 JOY MINING MACHINERY STAMLER PRODUCT LINE 
MACHINING		THIS DRAWING AND ALL INFORMATION THEREON ARE THE PROPERTY OF JOY MINING MACHINERY. THIS DRAWING MAY NOT BE COPIED, REPRODUCED OR USED FOR ANY PURPOSE WITHOUT THE EXPRESSED WRITTEN PERMISSION OF JOY MINING MACHINERY.			
DIM.	TOL.	DIM.	TOL.		
1 PLACE	±.060	0°-24°	±.06		
2 PLACE	±.030	24°-96°	±.12		
3 PLACE	±.010	96°-OVER	±.18		
ANGLES		±0°-30°		ANGLES	±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING		TITLE			
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .015 UNLESS OTHERWISE SPECIFIED		ASSY MACHINE HOLD DOWN			
DO NOT SCALE THIS CAD DRAWING					
DRAWN	JDA	15MAR91	SIZE	SCALE 6	DWG NO
CHECK	BW	15MAR91	B	SHEET 1 OF 1	98-31-015-2476
					REV B



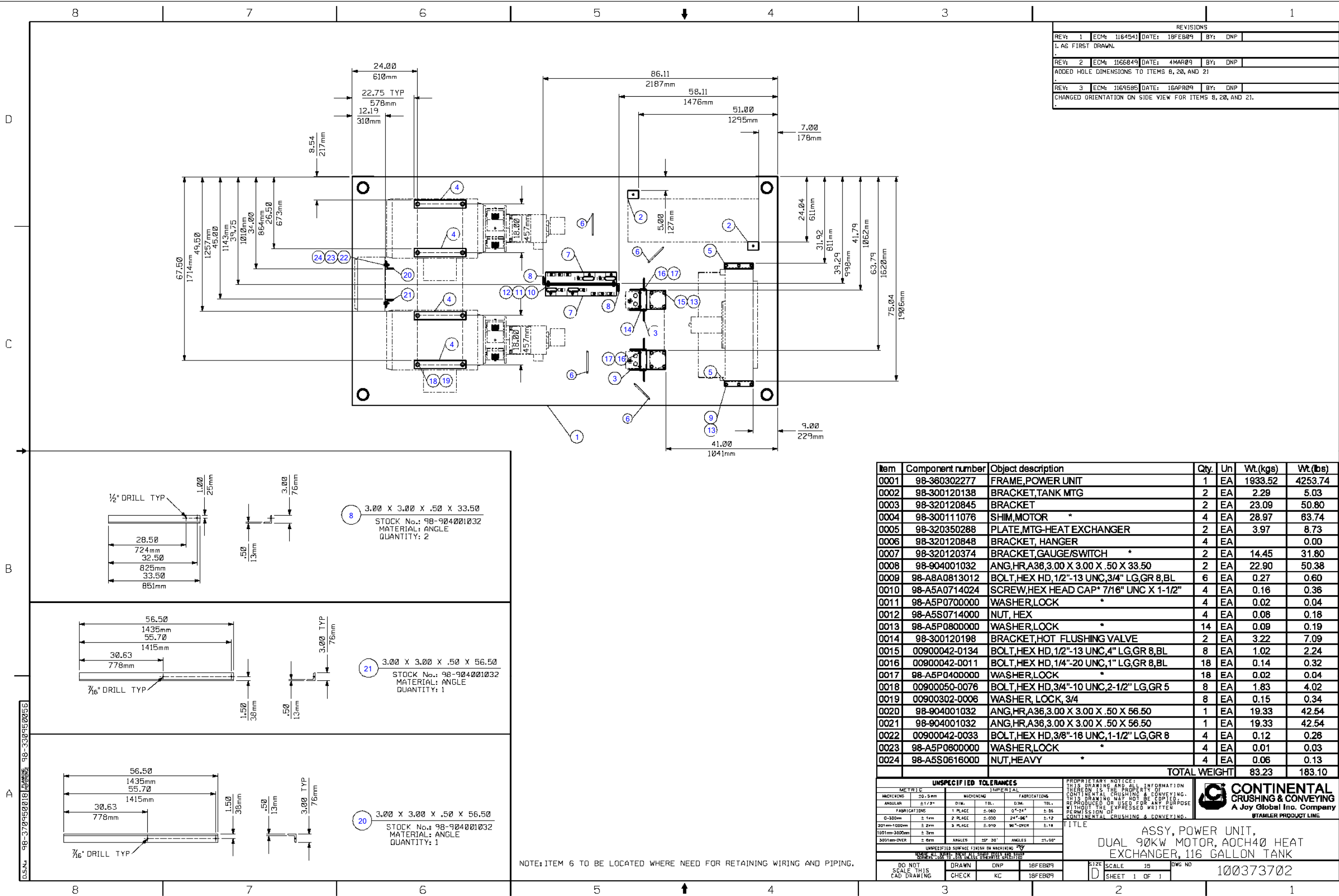
Reference:98-310152476



NOTES: 1. LOCATE NAMEPLATES AT ASSEMBLY

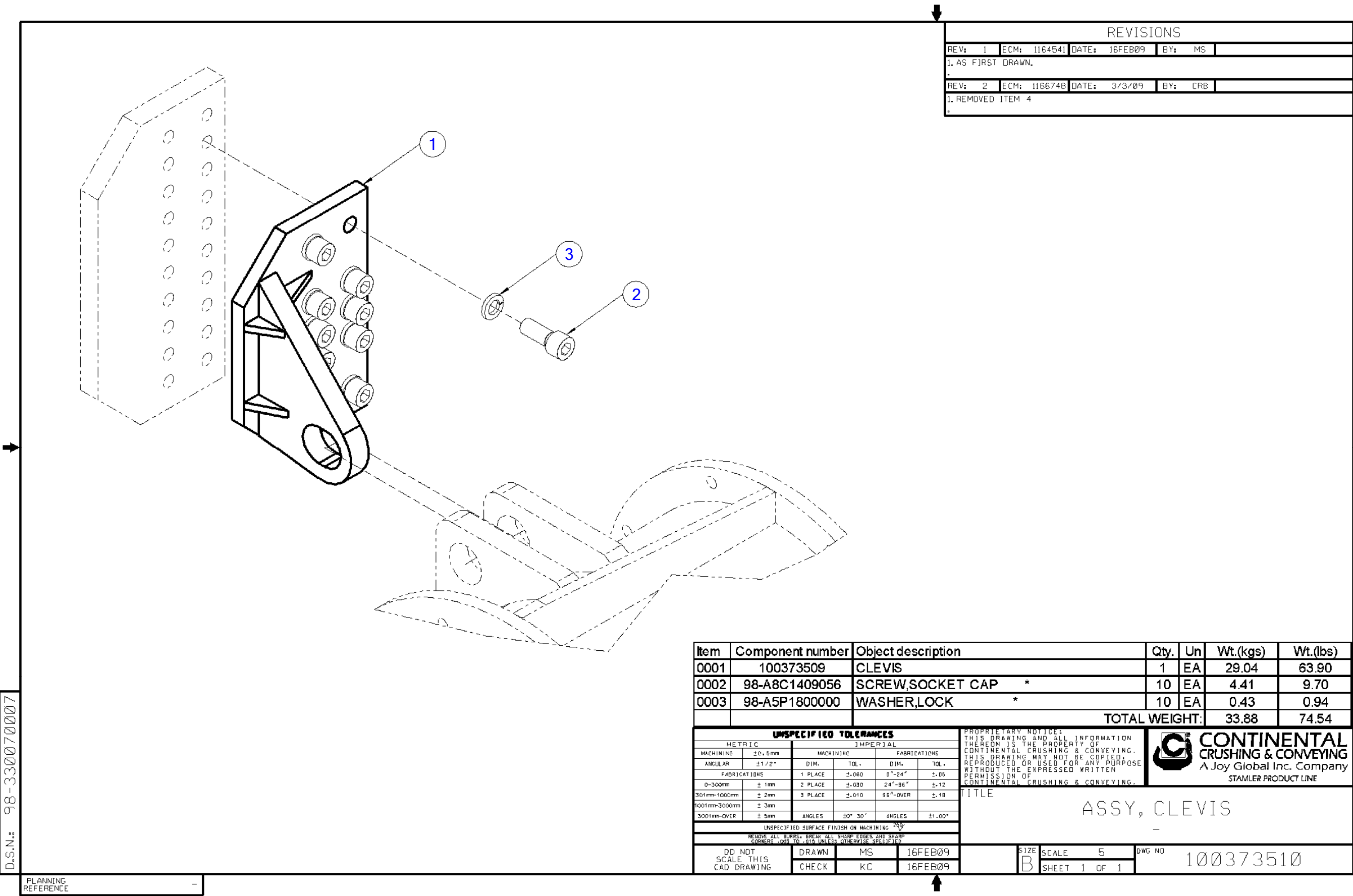


	Item	Part Number	Description	Qty	Units
	ASSY	98-330110174	ASSY,MACHINE NAMEPLATES	1	EA
	1	00903245-0005	• PLATE, NAME, JOY LOGO	2	EA
	2	98-7084	• NAMEPLATE,SST *	1	EA
	3	00900012-0001	• BOLT,HEX HD,1/4"-20 UNC,3/4" LG,GR 5,B	4	EA
	4	98-A5P0400000	• WASHER,LOCK *	4	EA
	5	98-003877	• FASTENER, METAL TACK	4	EA





	Item	Part Number	Description	Qty	Units
	ASSY	100373702	ASSY,POWER UNIT*	1	EA
	1	98-360302277	• FRAME,POWER UNIT	1	EA
	2	98-300120138	• BRACKET,TANK MTG	2	EA
	3	98-320120845	• BRACKET	2	EA
	4	98-300111076	• SHIM,MOTOR *	4	EA
	5	98-320350288	• PLATE,MTG-HEAT EXCHANGER	2	EA
	6	98-320120848	• BRACKET, HANGER	4	EA
	7	98-320120374	• BRACKET,GAUGE/SWITCH *	2	EA
	8	98-904001032	• ANG,HR,A36,3X3X.5X240	26	KG
	9	98-A8A0813012	• BOLT,HEX HD,1/2"-13 UNC,3/4" LG,GR 8,B	6	EA
	10	98-A5A0714024	• SCREW,HEX HEAD CAP* 7/16" UNC X 1-1/2"	4	EA
	11	98-A5P0700000	• WASHER,LOCK	4	EA
	12	98-A5S0714000	• NUT, HEX	4	EA
	13	98-A5P0800000	• WASHER,LOCK	14	EA
	14	98-300120198	• BRACKET,HOT FLUSHING VALVE	2	EA
	15	00900042-0134	• BOLT,HEX HD,1/2"-13 UNC,4" LG,GR 8,BL	8	EA
	16	00900042-0011	• BOLT,HEX HD,1/4"-20 UNC,1" LG,GR 8,BL	18	EA
	17	98-A5P0400000	• WASHER,LOCK *	18	EA
	18	00900050-0076	• BOLT,HEX HD,3/4"-10 UNC,2-1/2" LG,GR 5	8	EA
	19	00900302-0006	• WASHER,SPRING,3/4",BLACK	8	EA
	20	98-904001032	• ANG,HR,A36,3X3X.5X240	22	KG
	21	98-904001032	• ANG,HR,A36,3X3X.5X240	22	KG
	22	00900042-0033	• BOLT,HEX HD,3/8"-16 UNC,1-1/2" LG,GR 8	4	EA
	23	98-A5P0600000	• WASHER,LOCK *	4	EA
	24	98-A5S0616000	• NUT,HEAVY *	4	EA



REVISIONS				
REV:	1	ECM:	1164541	DATE: 16FEB09 BY: MS
1. AS FIRST DRAWN.				
REV:	2	ECM:	1166748	DATE: 3/3/09 BY: CRB
1. REMOVED ITEM 4				

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	100373509	CLEVIS	1	EA	29.04	63.90
0002	98-A8C1409056	SCREW,SOCKET CAP *	10	EA	4.41	9.70
0003	98-A5P1800000	WASHER,LOCK *	10	EA	0.43	0.94
TOTAL WEIGHT:					33.88	74.54

UNSPECIFIED TOLERANCES					
METRIC		IMPERIAL			
MACHINING	±0.5mm	MACHINING		FABRICATIONS	
ANGULAR	±1/2°	DIM.	TOL.	DIM.	TOL.
FABRICATIONS		1 PLACE	±.060	0°-24°	±.06
0-300mm	± 1mm	2 PLACE	±.030	24°-96°	±.12
301mm-1000mm	± 2mm	3 PLACE	±.010	96°-OVER	±.18
1001mm-3000mm	± 3mm				
3001mm-OVER	± 5mm	ANGLES	±0° 30'	ANGLES	±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING 250					
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS TO R.0125 UNLESS OTHERWISE SPECIFIED					
DD NOT SCALE THIS CAD DRAWING	DRAWN	MS	16FEB09	SIZE B	SCALE 5
	CHECK	KC	16FEB09		DWG NO 100373510

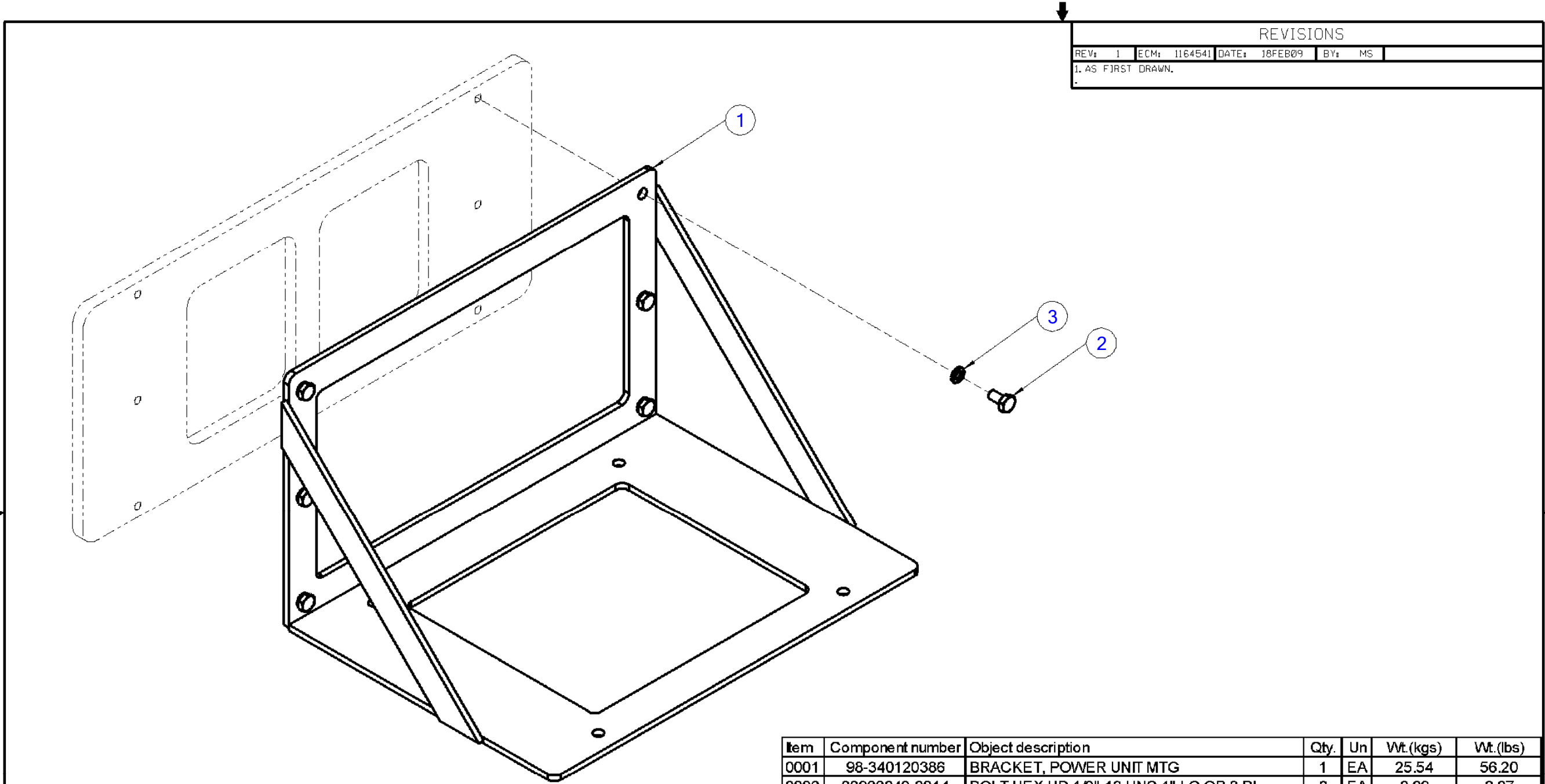
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CONTINENTAL CRUSHING & CONVEYING
A Joy Global Inc. Company
STAMLER PRODUCT LINE

TITLE ASSY, CLEVIS



	Item	Part Number	Description	Qty	Units
	ASSY	100373510	ASSY,CLEVIS	1	EA
	1	100373509	• CLEVIS	1	EA
	2	98-A8C1409056	• SCREW,SOCKET CAP *	10	EA
	3	98-A5P1800000	• WASHER,LOCK *	10	EA



REVISIONS					
REV#	1	ECM#	1164541	DATE#	18FEB09
		BY#	MS		
1. AS FIRST DRAWN.					

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	98-340120386	BRACKET, POWER UNIT MTG	1	EA	25.54	56.20
0002	00900042-0014	BOLT,HEX HD,1/2"-13 UNC,1" LG,GR 8,BL	6	EA	0.03	0.07
0003	98-A5P0800000	WASHER,LOCK *	6	EA	0.04	0.08
TOTAL WEIGHT:					25.61	56.35

UNSPECIFIED TOLERANCES					
METRIC		IMPERIAL			
MACHINING	±0.5mm	MACHINING		FABRICATIONS	
ANGULAR	±1/2°	DIM.	TOL.	DIM.	TOL.
FABRICATIONS		1 PLACE	±.060	0°-24°	±.06
0-300mm	± 1mm	2 PLACE	±.030	24°-86°	±.12
301mm-1000mm	± 2mm	3 PLACE	±.010	96°-OVER	±.18
1001mm-3000mm	± 3mm				
3001mm-OVER	± 5mm	ANGLES	±0° 30'	ANGLES	±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING ²⁵⁰					
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS TO .015 UNLESS OTHERWISE SPECIFIED					
DD NOT SCALE THIS CAD DRAWING		DRAWN	MS	18FEB09	
		CHECK	KC	18FEB09	

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CONTINENTAL CRUSHING & CONVEYING.



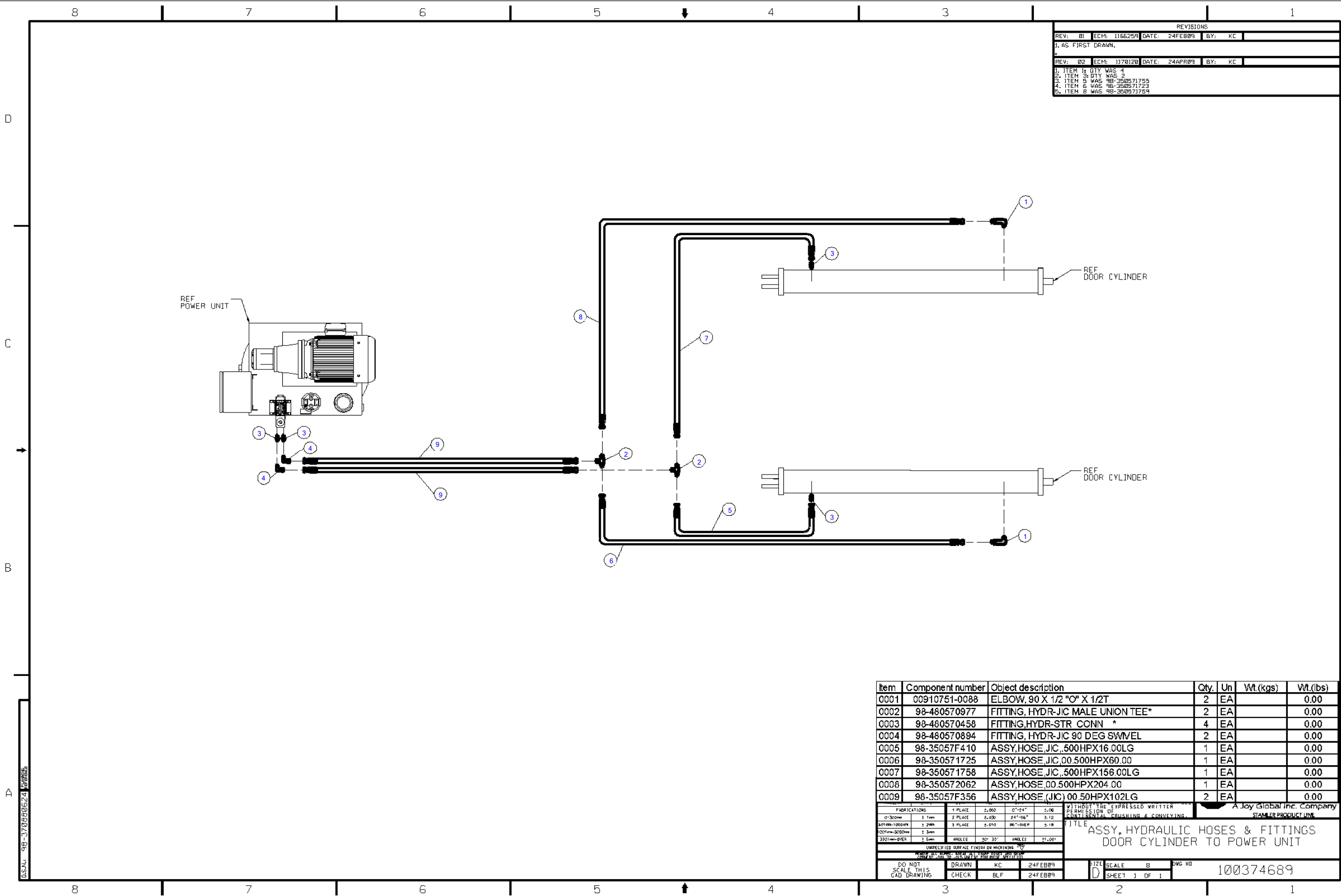
CONTINENTAL
CRUSHING & CONVEYING
A Joy Global Inc. Company
STAMLER PRODUCT LINE

TITLE		ASSEMBLY, POWER UNIT MTG BRACKET			
SIZE	B	SCALE	5	DWG NO	100373830
		SHEET	1 OF 1		

D.S.N.: 98-330120029

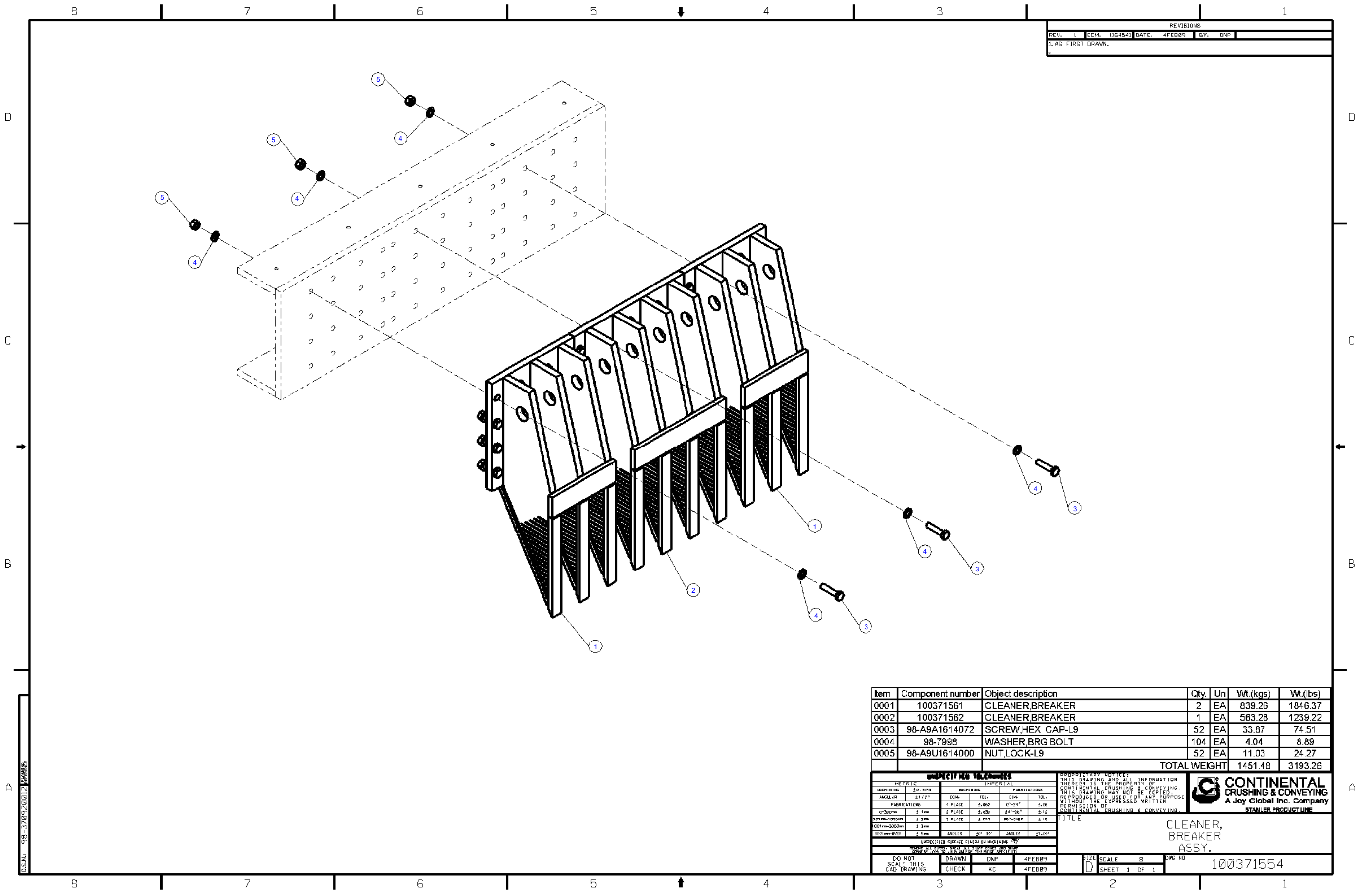
PLANNING
REFERENCE

[illegible]





	Item	Part Number	Description	Qty	Units
	ASSY	100374689	HYDRAULICS,HOSE/FTGS-DOOR CYL TO PWR UN	1	EA
	1	00910751-0088	• ADAPTOR,90,#08SAE,M,3/4"JIC,M	2	EA
	2	98-480570977	• FITTING, HYDR-JIC MALE UNION TEE*	2	EA
	3	98-480570458	• FITTING,HYDR-STR CONN *	4	EA
	4	98-480570894	• FITTING, HYDR-JIC 90 DEG SWIVEL	2	EA
	5	98-35057F410	• ASSY,HOSE,JIC,00.500HPX16.00	1	EA
	6	98-350571725	• ASSY,HOSE,JIC,.500HPX60.00LG	1	EA
	7	98-350571758	• ASSY,HOSE,JIC,.500HPX156.00LG	1	EA
	8	98-350572062	• ASSY,HOSE .50HP(3500PSI),204LG	1	EA
	9	98-35057F356	• ASSY,HOSE,(JIC) 00.50HPX102LG	2	EA



REVISIONS			
REV:	1	ECH: 1164541	DATE: 4FEB09 BY: DNP
1. AS FIRST DRAWN.			

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	100371561	CLEANER,BREAKER	2	EA	839.26	1846.37
0002	100371562	CLEANER,BREAKER	1	EA	563.28	1239.22
0003	98-A9A1614072	SCREW,HEX CAP-L9	52	EA	33.87	74.51
0004	98-7998	WASHER,BRG BOLT	104	EA	4.04	8.89
0005	98-A9U1614000	NUT,LOCK-L9	52	EA	11.03	24.27
TOTAL WEIGHT					1451.48	3193.26

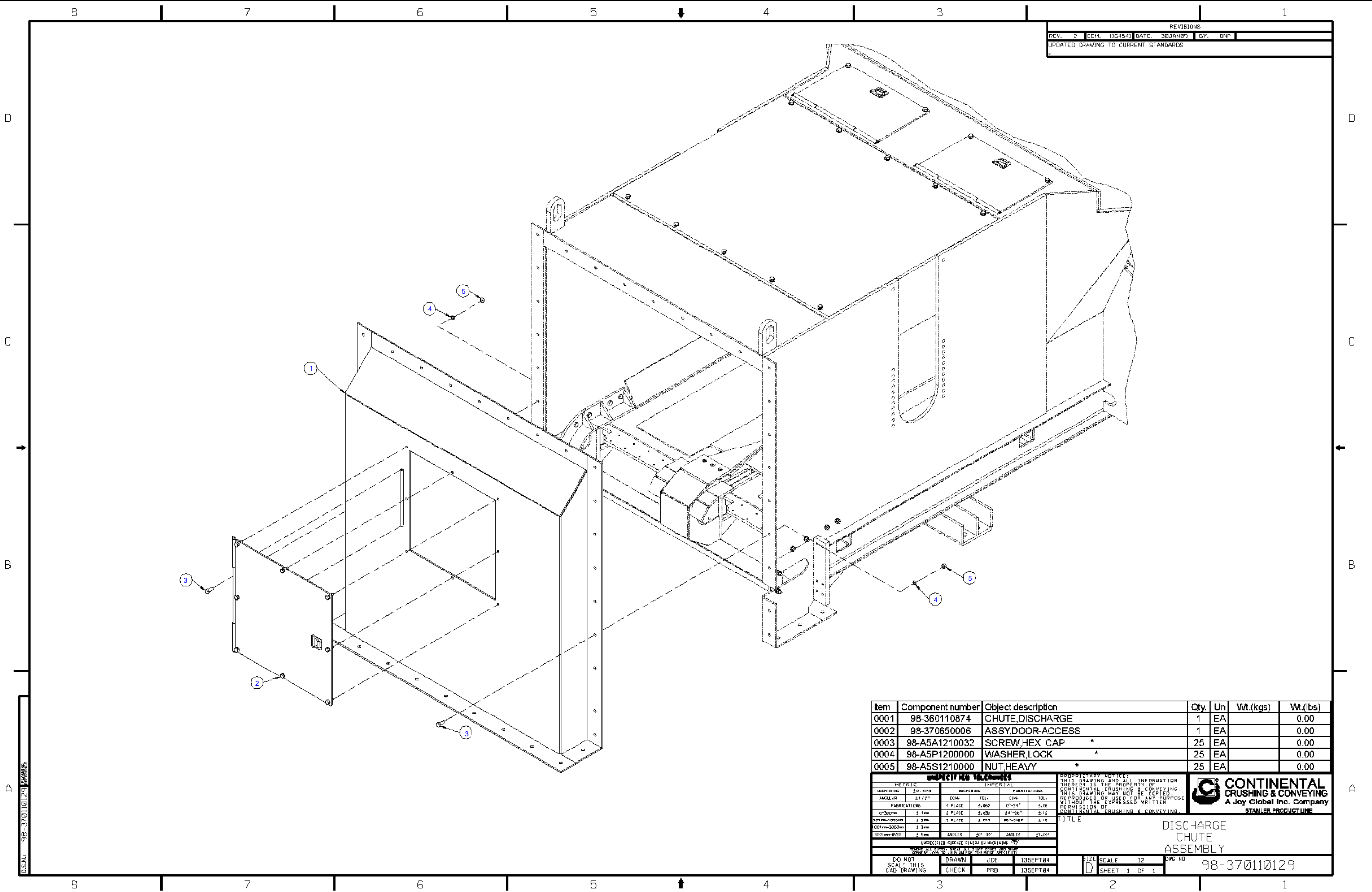
UNSPECIFIED SURFACE FINISH			
METRIC		INFERIAL	
MACHINING	±0.075	MACHINING	±0.003
ANGULAR	±1/2°	ANGULAR	±1/2°
FABRICATIONS		FABRICATIONS	
1 PLACE	±0.06	1 PLACE	±0.003
2 PLACE	±0.03	2 PLACE	±0.003
3 PLACE	±0.01	3 PLACE	±0.003
4 PLACE	±0.005	4 PLACE	±0.003
5 PLACE	±0.002	5 PLACE	±0.003
6 PLACE	±0.001	6 PLACE	±0.003
7 PLACE	±0.0005	7 PLACE	±0.003
8 PLACE	±0.0002	8 PLACE	±0.003
9 PLACE	±0.0001	9 PLACE	±0.003
10 PLACE	±0.00005	10 PLACE	±0.003
11 PLACE	±0.00002	11 PLACE	±0.003
12 PLACE	±0.00001	12 PLACE	±0.003
13 PLACE	±0.000005	13 PLACE	±0.003
14 PLACE	±0.000002	14 PLACE	±0.003
15 PLACE	±0.000001	15 PLACE	±0.003
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29 PLACE	±0.00000000002	29 PLACE	±0.003
30 PLACE	±0.00000000001	30 PLACE	±0.003
31 PLACE	±0.000000000005	31 PLACE	±0.003
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50 PLACE	±0.000000000000000002	50 PLACE	±0.003
51 PLACE	±0.000000000000000001	51 PLACE	±0.003
52 PLACE	±0.0000000000000000005	52 PLACE	±0.003
53 PLACE	±0.0000000000000000002	53 PLACE	±0.003
54 PLACE	±0.0000000000000000001	54 PLACE	±0.003
55 PLACE	±0.00000000000000000005	55 PLACE	±0.003
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58 PLACE	±0.000000000000000000005	58 PLACE	±0.003
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60 PLACE	±0.000000000000000000001	60 PLACE	±0.003
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62 PLACE	±0.0000000000000000000002	62 PLACE	±0.003
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65 PLACE	±0.00000000000000000000002	65 PLACE	±0.003
66 PLACE	±0.00000000000000000000001	66 PLACE	±0.003
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85 PLACE	±0.000000000000000000000000000005	85 PLACE	±0.003
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96 PLACE	±0.000000000000000000000000000000001	96 PLACE	±0.003
97 PLACE	±0.0000000000000000000000000000000005	97 PLACE	±0.003
98 PLACE	±0.0000000000000000000000000000000002	98 PLACE	±0.003
99 PLACE	±0.0000000000000000000000000000000001	99 PLACE	±0.003
100 PLACE	±0.00000000000000000000000000000000005	100 PLACE	±0.003

DO NOT SCALE THIS CAD DRAWING	DRAWN	DNP	4FEB09	SIZE	SCALE	B	DWG NO	100371554
CHECK	KC	4FEB09	SHEET	1	OF	1		

CONTINENTAL CRUSHING & CONVEYING		A Joy Global Inc. Company	
STAMLER PRODUCT LINE		TITLE	
CLEANER, BREAKER ASSY.		CLEANER, BREAKER ASSY.	



	Item	Part Number	Description	Qty	Units
	ASSY	100371554	ASSY,CLEANER-BREAKER	1	EA
	1	100371561	• CLEANER,BREAKER	2	EA
	2	100371562	• CLEANER,BREAKER	1	EA
	3	98-A9A1614072	• SCREW,HEX CAP-L9 *	52	EA
	4	98-7998	• WASHER,BRG BOLT	104	EA
	5	98-A9U1614000	• NUT,LOCK-L9	52	EA



REVISIONS			
REV:	2	ECH: 1164541	DATE: 30JAN09 BY: DNP
UPDATED DRAWING TO CURRENT STANDARDS			

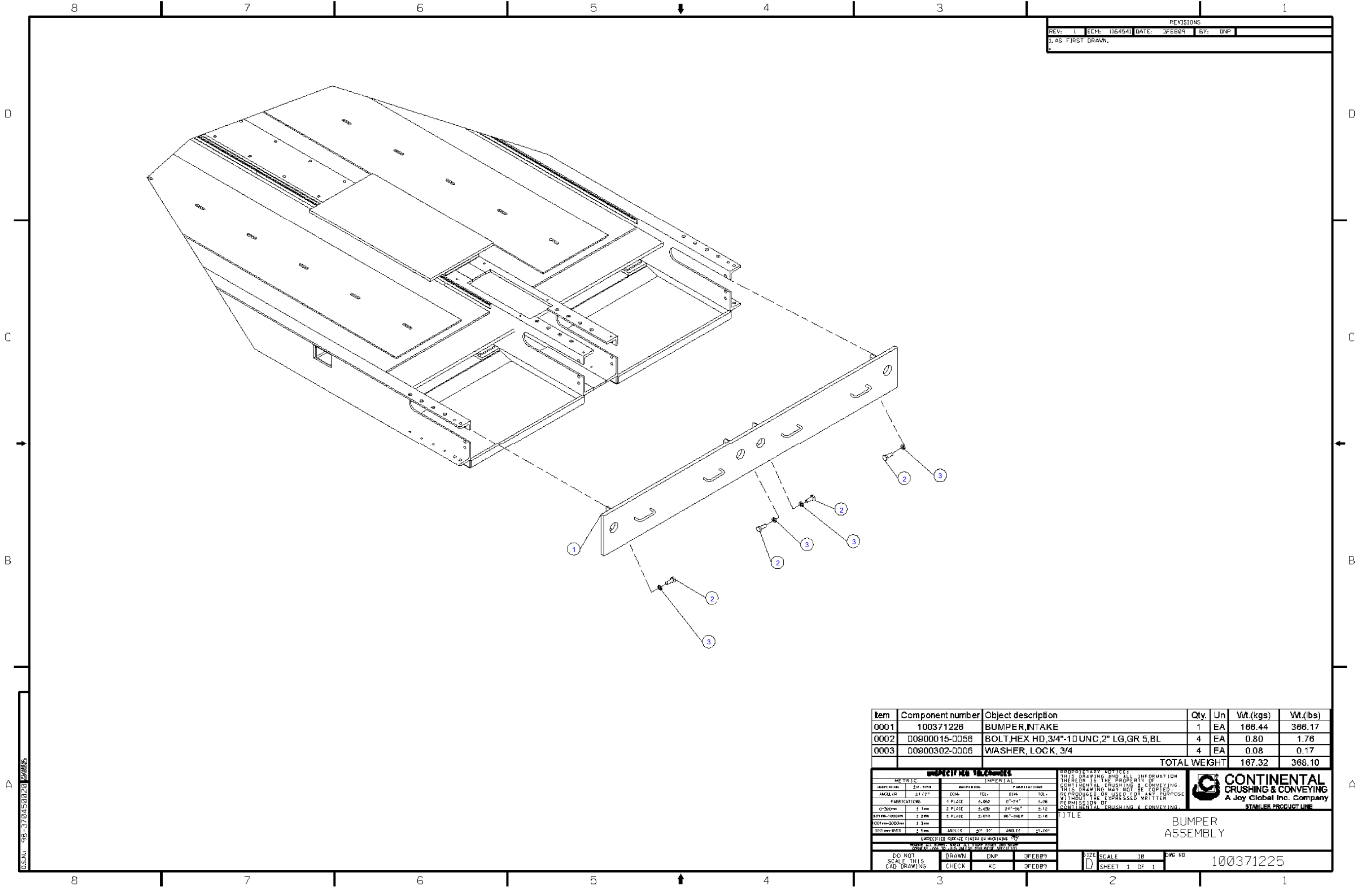
Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	98-360110874	CHUTE,DISCHARGE	1	EA		0.00
0002	98-370650006	ASSY,DOOR-ACCESS	1	EA		0.00
0003	98-A5A1210032	SCREW,HEX CAP *	25	EA		0.00
0004	98-A5P1200000	WASHER,LOCK *	25	EA		0.00
0005	98-A5S1210000	NUT,HEAVY *	25	EA		0.00

METRIC		INCHES	
MACHINING	±0.13mm	MACHINING	±0.005"
ANGULAR	±1/2°	ANGULAR	±1/2°
FABRICATIONS		FABRICATIONS	
1 PLACE	±0.00	1 PLACE	±0.00
2 PLACE	±0.00	2 PLACE	±0.00
3 PLACE	±0.00	3 PLACE	±0.00
4 PLACE	±0.00	4 PLACE	±0.00
5 PLACE	±0.00	5 PLACE	±0.00
6 PLACE	±0.00	6 PLACE	±0.00
7 PLACE	±0.00	7 PLACE	±0.00
8 PLACE	±0.00	8 PLACE	±0.00
9 PLACE	±0.00	9 PLACE	±0.00
10 PLACE	±0.00	10 PLACE	±0.00
11 PLACE	±0.00	11 PLACE	±0.00
12 PLACE	±0.00	12 PLACE	±0.00
13 PLACE	±0.00	13 PLACE	±0.00
14 PLACE	±0.00	14 PLACE	±0.00
15 PLACE	±0.00	15 PLACE	±0.00
16 PLACE	±0.00	16 PLACE	±0.00
17 PLACE	±0.00	17 PLACE	±0.00
18 PLACE	±0.00	18 PLACE	±0.00
19 PLACE	±0.00	19 PLACE	±0.00
20 PLACE	±0.00	20 PLACE	±0.00
21 PLACE	±0.00	21 PLACE	±0.00
22 PLACE	±0.00	22 PLACE	±0.00
23 PLACE	±0.00	23 PLACE	±0.00
24 PLACE	±0.00	24 PLACE	±0.00
25 PLACE	±0.00	25 PLACE	±0.00
26 PLACE	±0.00	26 PLACE	±0.00
27 PLACE	±0.00	27 PLACE	±0.00
28 PLACE	±0.00	28 PLACE	±0.00
29 PLACE	±0.00	29 PLACE	±0.00
30 PLACE	±0.00	30 PLACE	±0.00
31 PLACE	±0.00	31 PLACE	±0.00
32 PLACE	±0.00	32 PLACE	±0.00
33 PLACE	±0.00	33 PLACE	±0.00
34 PLACE	±0.00	34 PLACE	±0.00
35 PLACE	±0.00	35 PLACE	±0.00
36 PLACE	±0.00	36 PLACE	±0.00
37 PLACE	±0.00	37 PLACE	±0.00
38 PLACE	±0.00	38 PLACE	±0.00
39 PLACE	±0.00	39 PLACE	±0.00
40 PLACE	±0.00	40 PLACE	±0.00
41 PLACE	±0.00	41 PLACE	±0.00
42 PLACE	±0.00	42 PLACE	±0.00
43 PLACE	±0.00	43 PLACE	±0.00
44 PLACE	±0.00	44 PLACE	±0.00
45 PLACE	±0.00	45 PLACE	±0.00
46 PLACE	±0.00	46 PLACE	±0.00
47 PLACE	±0.00	47 PLACE	±0.00
48 PLACE	±0.00	48 PLACE	±0.00
49 PLACE	±0.00	49 PLACE	±0.00
50 PLACE	±0.00	50 PLACE	±0.00
51 PLACE	±0.00	51 PLACE	±0.00
52 PLACE	±0.00	52 PLACE	±0.00
53 PLACE	±0.00	53 PLACE	±0.00
54 PLACE	±0.00	54 PLACE	±0.00
55 PLACE	±0.00	55 PLACE	±0.00
56 PLACE	±0.00	56 PLACE	±0.00
57 PLACE	±0.00	57 PLACE	±0.00
58 PLACE	±0.00	58 PLACE	±0.00
59 PLACE	±0.00	59 PLACE	±0.00
60 PLACE	±0.00	60 PLACE	±0.00
61 PLACE	±0.00	61 PLACE	±0.00
62 PLACE	±0.00	62 PLACE	±0.00
63 PLACE	±0.00	63 PLACE	±0.00
64 PLACE	±0.00	64 PLACE	±0.00
65 PLACE	±0.00	65 PLACE	±0.00
66 PLACE	±0.00	66 PLACE	±0.00
67 PLACE	±0.00	67 PLACE	±0.00
68 PLACE	±0.00	68 PLACE	±0.00
69 PLACE	±0.00	69 PLACE	±0.00
70 PLACE	±0.00	70 PLACE	±0.00
71 PLACE	±0.00	71 PLACE	±0.00
72 PLACE	±0.00	72 PLACE	±0.00
73 PLACE	±0.00	73 PLACE	±0.00
74 PLACE	±0.00	74 PLACE	±0.00
75 PLACE	±0.00	75 PLACE	±0.00
76 PLACE	±0.00	76 PLACE	±0.00
77 PLACE	±0.00	77 PLACE	±0.00
78 PLACE	±0.00	78 PLACE	±0.00
79 PLACE	±0.00	79 PLACE	±0.00
80 PLACE	±0.00	80 PLACE	±0.00
81 PLACE	±0.00	81 PLACE	±0.00
82 PLACE	±0.00	82 PLACE	±0.00
83 PLACE	±0.00	83 PLACE	±0.00
84 PLACE	±0.00	84 PLACE	±0.00
85 PLACE	±0.00	85 PLACE	±0.00
86 PLACE	±0.00	86 PLACE	±0.00
87 PLACE	±0.00	87 PLACE	±0.00
88 PLACE	±0.00	88 PLACE	±0.00
89 PLACE	±0.00	89 PLACE	±0.00
90 PLACE	±0.00	90 PLACE	±0.00
91 PLACE	±0.00	91 PLACE	±0.00
92 PLACE	±0.00	92 PLACE	±0.00
93 PLACE	±0.00	93 PLACE	±0.00
94 PLACE	±0.00	94 PLACE	±0.00
95 PLACE	±0.00	95 PLACE	±0.00
96 PLACE	±0.00	96 PLACE	±0.00
97 PLACE	±0.00	97 PLACE	±0.00
98 PLACE	±0.00	98 PLACE	±0.00
99 PLACE	±0.00	99 PLACE	±0.00
100 PLACE	±0.00	100 PLACE	±0.00

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CONTINENTAL CRUSHING & CONVEYING A Joy Global Inc. Company	
STAMLER PRODUCT LINE	
TITLE	
DISCHARGE CHUTE ASSEMBLY	
DO NOT SCALE THIS CAD DRAWING	
DRAWN JDE 13SEPT04	
CHECK PRB 13SEPT04	
SIZE SCALE 32	
SHEET 1 OF 1	
DWG NO 98-370110129	



	Item	Part Number	Description	Qty	Units
	ASSY	98-370110129	ASSY,DISCHARGE CHUTE	1	EA
	1	98-360110874	• CHUTE,DISCHARGE	1	EA
	2	98-370650006	• ASSY,DOOR-ACCESS	1	EA
	3	98-A5A1210032	• SCREW,HEX CAP *	25	EA
	4	98-A5P1200000	• WASHER,LOCK *	25	EA
	5	98-A5S1210000	• NUT,HEAVY HEX	25	EA



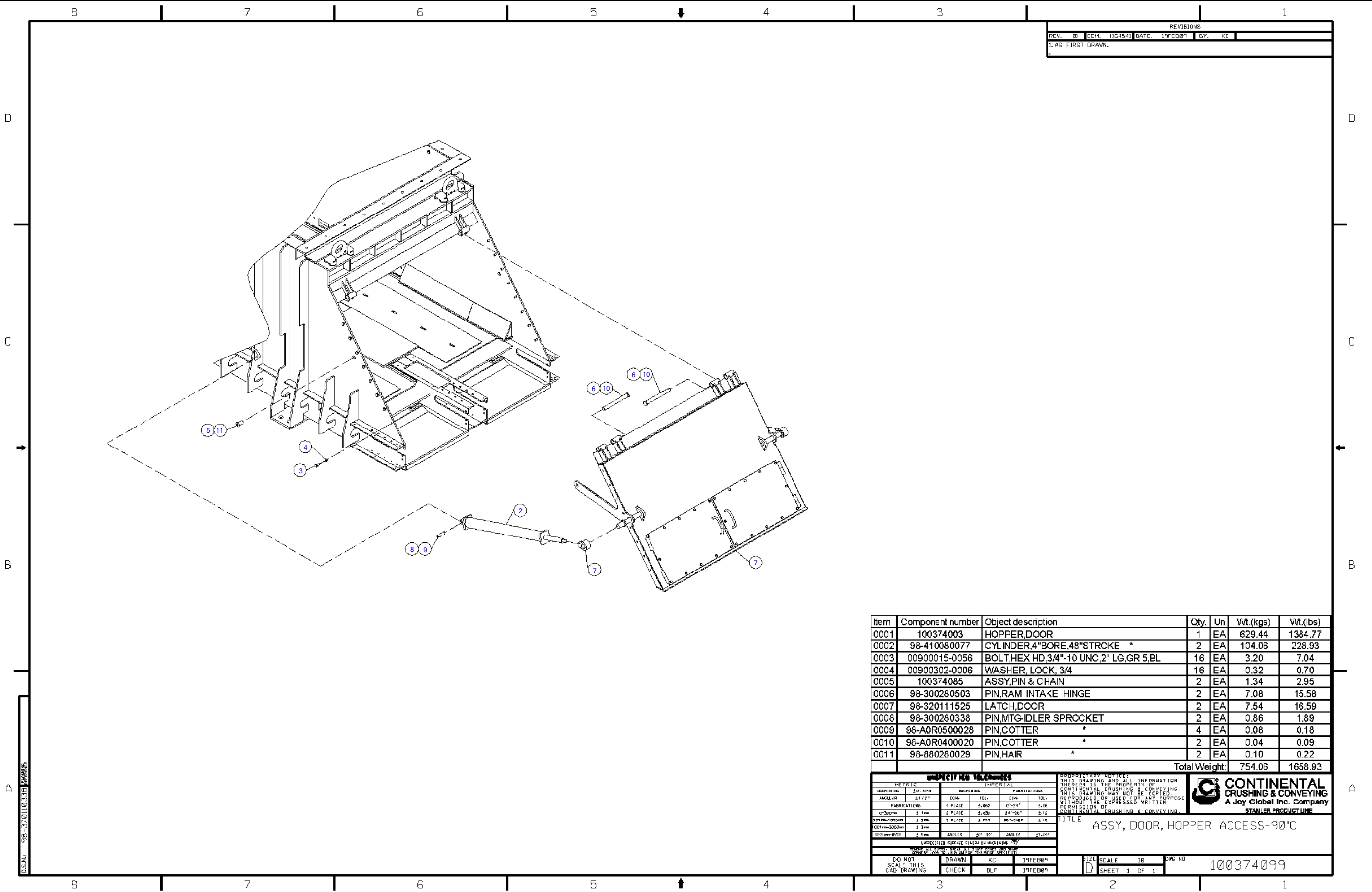
REVISIONS			
REV: 1	ECH: 1164541	DATE: 3FEB09	BY: DNP
1. AS FIRST DRAWN.			

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	100371226	BUMPER,INTAKE	1	EA	168.44	368.17
0002	00900015-0058	BOLT,HEX HD,3/4"-10 UNC,2" LG,GR 5,BL	4	EA	0.80	1.76
0003	00900302-0006	WASHER, LOCK, 3/4	4	EA	0.08	0.17
TOTAL WEIGHT					167.32	368.10

METRIC				IMPERIAL			
FINISHING	±0.075	FINISHING	±0.003	FINISHING	±0.003	FINISHING	±0.003
ANGULAR	±1/2°	ANGULAR	±1/2°	ANGULAR	±1/2°	ANGULAR	±1/2°
FABRICATIONS				FABRICATIONS			
1 PLACE		±0.00	0"-24"	1 PLACE		±0.00	0"-24"
2 PLACE		±0.00	24"-36"	2 PLACE		±0.00	24"-36"
3 PLACE		±0.00	36"-48"	3 PLACE		±0.00	36"-48"
4 PLACE		±0.00	48"-60"	4 PLACE		±0.00	48"-60"
5 PLACE		±0.00	60"-72"	5 PLACE		±0.00	60"-72"
6 PLACE		±0.00	72"-84"	6 PLACE		±0.00	72"-84"
7 PLACE		±0.00	84"-96"	7 PLACE		±0.00	84"-96"
8 PLACE		±0.00	96"-108"	8 PLACE		±0.00	96"-108"
9 PLACE		±0.00	108"-120"	9 PLACE		±0.00	108"-120"
10 PLACE		±0.00	120"-132"	10 PLACE		±0.00	120"-132"
11 PLACE		±0.00	132"-144"	11 PLACE		±0.00	132"-144"
12 PLACE		±0.00	144"-156"	12 PLACE		±0.00	144"-156"
13 PLACE		±0.00	156"-168"	13 PLACE		±0.00	156"-168"
14 PLACE		±0.00	168"-180"	14 PLACE		±0.00	168"-180"
15 PLACE		±0.00	180"-192"	15 PLACE		±0.00	180"-192"
16 PLACE		±0.00	192"-204"	16 PLACE		±0.00	192"-204"
17 PLACE		±0.00	204"-216"	17 PLACE		±0.00	204"-216"
18 PLACE		±0.00	216"-228"	18 PLACE		±0.00	216"-228"
19 PLACE		±0.00	228"-240"	19 PLACE		±0.00	228"-240"
20 PLACE		±0.00	240"-252"	20 PLACE		±0.00	240"-252"
21 PLACE		±0.00	252"-264"	21 PLACE		±0.00	252"-264"
22 PLACE		±0.00	264"-276"	22 PLACE		±0.00	264"-276"
23 PLACE		±0.00	276"-288"	23 PLACE		±0.00	276"-288"
24 PLACE		±0.00	288"-300"	24 PLACE		±0.00	288"-300"
25 PLACE		±0.00	300"-312"	25 PLACE		±0.00	300"-312"
26 PLACE		±0.00	312"-324"	26 PLACE		±0.00	312"-324"
27 PLACE		±0.00	324"-336"	27 PLACE		±0.00	324"-336"
28 PLACE		±0.00	336"-348"	28 PLACE		±0.00	336"-348"
29 PLACE		±0.00	348"-360"	29 PLACE		±0.00	348"-360"
30 PLACE		±0.00	360"-372"	30 PLACE		±0.00	360"-372"
31 PLACE		±0.00	372"-384"	31 PLACE		±0.00	372"-384"
32 PLACE		±0.00	384"-396"	32 PLACE		±0.00	384"-396"
33 PLACE		±0.00	396"-408"	33 PLACE		±0.00	396"-408"
34 PLACE		±0.00	408"-420"	34 PLACE		±0.00	408"-420"
35 PLACE		±0.00	420"-432"	35 PLACE		±0.00	420"-432"
36 PLACE		±0.00	432"-444"	36 PLACE		±0.00	432"-444"
37 PLACE		±0.00	444"-456"	37 PLACE		±0.00	444"-456"
38 PLACE		±0.00	456"-468"	38 PLACE		±0.00	456"-468"
39 PLACE		±0.00	468"-480"	39 PLACE		±0.00	468"-480"
40 PLACE		±0.00	480"-492"	40 PLACE		±0.00	480"-492"
41 PLACE		±0.00	492"-504"	41 PLACE		±0.00	492"-504"
42 PLACE		±0.00	504"-516"	42 PLACE		±0.00	504"-516"
43 PLACE		±0.00	516"-528"	43 PLACE		±0.00	516"-528"
44 PLACE		±0.00	528"-540"	44 PLACE		±0.00	528"-540"
45 PLACE		±0.00	540"-552"	45 PLACE		±0.00	540"-552"
46 PLACE		±0.00	552"-564"	46 PLACE		±0.00	552"-564"
47 PLACE		±0.00	564"-576"	47 PLACE		±0.00	564"-576"
48 PLACE		±0.00	576"-588"	48 PLACE		±0.00	576"-588"
49 PLACE		±0.00	588"-600"	49 PLACE		±0.00	588"-600"
50 PLACE		±0.00	600"-612"	50 PLACE		±0.00	600"-612"
51 PLACE		±0.00	612"-624"	51 PLACE		±0.00	612"-624"
52 PLACE		±0.00	624"-636"	52 PLACE		±0.00	624"-636"
53 PLACE		±0.00	636"-648"	53 PLACE		±0.00	636"-648"
54 PLACE		±0.00	648"-660"	54 PLACE		±0.00	648"-660"
55 PLACE		±0.00	660"-672"	55 PLACE		±0.00	660"-672"
56 PLACE		±0.00	672"-684"	56 PLACE		±0.00	672"-684"
57 PLACE		±0.00	684"-696"	57 PLACE		±0.00	684"-696"
58 PLACE		±0.00	696"-708"	58 PLACE		±0.00	696"-708"
59 PLACE		±0.00	708"-720"	59 PLACE		±0.00	708"-720"
60 PLACE		±0.00	720"-732"	60 PLACE		±0.00	720"-732"
61 PLACE		±0.00	732"-744"	61 PLACE		±0.00	732"-744"
62 PLACE		±0.00	744"-756"	62 PLACE		±0.00	744"-756"
63 PLACE		±0.00	756"-768"	63 PLACE		±0.00	756"-768"
64 PLACE		±0.00	768"-780"	64 PLACE		±0.00	768"-780"
65 PLACE		±0.00	780"-792"	65 PLACE		±0.00	780"-792"
66 PLACE		±0.00	792"-804"	66 PLACE		±0.00	792"-804"
67 PLACE		±0.00	804"-816"	67 PLACE		±0.00	804"-816"
68 PLACE		±0.00	816"-828"	68 PLACE		±0.00	816"-828"
69 PLACE		±0.00	828"-840"	69 PLACE		±0.00	828"-840"
70 PLACE		±0.00	840"-852"	70 PLACE		±0.00	840"-852"
71 PLACE		±0.00	852"-864"	71 PLACE		±0.00	852"-864"
72 PLACE		±0.00	864"-876"	72 PLACE		±0.00	864"-876"
73 PLACE		±0.00	876"-888"	73 PLACE		±0.00	876"-888"
74 PLACE		±0.00	888"-900"	74 PLACE		±0.00	888"-900"
75 PLACE		±0.00	900"-912"	75 PLACE		±0.00	900"-912"
76 PLACE		±0.00	912"-924"	76 PLACE		±0.00	912"-924"
77 PLACE		±0.00	924"-936"	77 PLACE		±0.00	924"-936"
78 PLACE		±0.00	936"-948"	78 PLACE		±0.00	936"-948"
79 PLACE		±0.00	948"-960"	79 PLACE		±0.00	948"-960"
80 PLACE		±0.00	960"-972"	80 PLACE		±0.00	960"-972"
81 PLACE		±0.00	972"-984"	81 PLACE		±0.00	972"-984"
82 PLACE		±0.00	984"-996"	82 PLACE		±0.00	984"-996"
83 PLACE		±0.00	996"-1008"	83 PLACE		±0.00	996"-1008"
84 PLACE		±0.00	1008"-1020"	84 PLACE		±0.00	1008"-1020"
85 PLACE		±0.00	1020"-1032"	85 PLACE		±0.00	1020"-1032"
86 PLACE		±0.00	1032"-1044"	86 PLACE		±0.00	1032"-1044"
87 PLACE		±0.00	1044"-1056"	87 PLACE		±0.00	1044"-1056"
88 PLACE		±0.00	1056"-1068"	88 PLACE		±0.00	1056"-1068"
89 PLACE		±0.00	1068"-1080"	89 PLACE		±0.00	1068"-1080"
90 PLACE		±0.00	1080"-1092"	90 PLACE		±0.00	1080"-1092"
91 PLACE		±0.00	1092"-1104"	91 PLACE		±0.00	1092"-1104"
92 PLACE		±0.00	1104"-1116"	92 PLACE		±0.00	1104"-1116"
93 PLACE		±0.00	1116"-1128"	93 PLACE		±0.00	1116"-1128"
94 PLACE		±0.00	1128"-1140"	94 PLACE		±0.00	1128"-1140"
95 PLACE		±0.00	1140"-1152"	95 PLACE		±0.00	1140"-1152"
96 PLACE		±0.00	1152"-1164"	96 PLACE		±0.00	1152"-1164"
97 PLACE		±0.00	1164"-1176"	97 PLACE		±0.00	1164"-1176"
98 PLACE		±0.00	1176"-1188"	98 PLACE		±0.00	1176"-1188"
99 PLACE		±0.00	1188"-1200"	99 PLACE		±0.00	1188"-1200"
100 PLACE		±0.00	1200"-1212"	100 PLACE		±0.00	1200"-1212"
101 PLACE		±0.00	1212"-1224"	101 PLACE		±0.00	1212"-1224"
102 PLACE		±0.00	1224"-1236"	102 PLACE		±0.00	1224"-1236"
103 PLACE		±0.00	1236"-1248"	103 PLACE		±0.00	1236"-1248"
104 PLACE		±0.00	1248"-1260"	104 PLACE		±0.00	1248"-1260"
105 PLACE		±0.00	1260"-1272"	105 PLACE		±0.00	1260"-1272"
106 PLACE		±0.00	1272"-1284"	106 PLACE		±0.00	1272"-1284"
107 PLACE		±0.00	1284"-1296"	107 PLACE		±0.00	1284"-1296"
108 PLACE		±0.00	1296"-1308"	108 PLACE		±0.00	1296"-1308"
109 PLACE		±0.00	1308"-1320"	109 PLACE		±0.00	1308"-1320"
110 PLACE		±0.00	1320"-1332"	110 PLACE		±0.00	1320"-1332"
111 PLACE		±0.00	1332"-1344"	111 PLACE		±0.00	1332"-1344"
112 PLACE		±0.00	1344"-1356"	112 PLACE		±0.00	1344"-1356"
113 PLACE		±0.00	1356"-1368"	113 PLACE		±0.00	1356"-1368"
114 PLACE		±0.00	1368"-1380"	114 PLACE		±0.00	1368"-1380"
115 PLACE		±0.00	1380"-1392"	115 PLACE		±0.00	1380"-1392"
116 PLACE		±0.00	1392"-1404"	116 PLACE		±0.00	1392"-1404"
117 PLACE		±0.00	1404"-1416"	117 PLACE		±0.00	1404"-1416"
118 PLACE		±0.00	1416"-1428"	118 PLACE		±0.00	1416"-1428"
119 PLACE		±0.00	1428"-1440"	119 PLACE		±0.00	1428"-1440"
120 PLACE		±0.00	1440"-1452"	120 PLACE		±0.00	1440"-1452"
121 PLACE		±0.00	1452"-1464"	121 PLACE		±0.00	1452"-1464"
122 PLACE		±0.00	1464"-1476"	122 PLACE		±0.00	1464"-1476"
123 PLACE		±0.00	1476"-1488"	123 PLACE		±0.00	1476"-1488"
124 PLACE		±0.00	1488"-1500"	124 PLACE		±0.00	1488"-1500"
125 PLACE		±0.00	1500"-1512"	125 PLACE		±0.00	1500"-1512"
126 PLACE		±0.00	1512"-1524"	126 PLACE		±0.00	1512"-1524"
127 PLACE		±0.00	1524"-1536"	127 PLACE		±0.00	1524"-1536"
128 PLACE		±0.00	1536"-1548"	128 PLACE		±0.00	1536"-1548"
129 PLACE		±0.00	1548"-1560"	129 PLACE		±0.00	1548"-1560"
130 PLACE		±0.00	1560"-1572"	130 PLACE		±0.00	1560"-1572"
131 PLACE		±0.00	1572"-1584"	131 PLACE		±0.00	1572"-1584"
132 PLACE		±0.00	1584"-1596"	132 PLACE		±0.00	1584"-1596"
133 PLACE		±0.00	1596"-1608"	133 PLACE		±0.00	1596"-1608"
134 PLACE		±0.00	1608"-1620"	134 PLACE		±0.00	1608"-1620"
135 PLACE		±0.00	1620"-1632"	135 PLACE		±0.00	1620"-1632"
136 PLACE		±0.00	1632"-1644"	136 PLACE		±0.00	1632"-1644"
137 PLACE		±0.00	1644"-1656"	137 PLACE		±0.00	1644"-1656"
138 PLACE		±0.00	1656"-1668"	138 PLACE		±0.00	1656"-1668"
139 PLACE		±0.00	1668"-1680"	139 PLACE		±0.00	1668"-1680"
140 PLACE		±0.00	1680"-1692"	140 PLACE		±0.00	1680"-1692"
141 PLACE		±0.00	1692"-1704"	141 PLACE		±0.00	1692"-1704"
142 PLACE		±0.00	1704"-1716"	142 PLACE		±0.00	1704"-1716"
143 PLACE		±0.00	1716"-1728"	143 PLACE		±0.00	1716"-1728"
144 PLACE		±0.00	1728"-1740"	144 PLACE		±0.00	1728"-1740"
145 PLACE		±0.00	1740"-1752"	145 PLACE		±0.00	1740"-1752"
146 PLACE		±0.00	1752"-1764"	146 PLACE		±0.00	1752"-1764"
147 PLACE		±0.00	1764"-1776"	147 PLACE		±0.00	1764"-1776"
148 PLACE		±0.00	1776"-1788"	148 PLACE		±0.00	1776"-1788"
149 PLACE		±0.00	1788"-1800"	149 PLACE		±0.00	1788"-1800"
150 PLACE		±0.00	1800"-1812"	150 PLACE		±0.00	1800"-1812"
151 PLACE		±0.00	1812"-1824"	151 PLACE		±0.00	1812"-1824"
152 PLACE		±0.00	1824"-1836"	152 PLACE		±0.00	1824"-1836"
153 PLACE		±0.00	1836"-1848"	153 PLACE		±0.00	1836"-1848"
154 PLACE		±0.00	1848"-1860"	154 PLACE		±0.00	1848"-1860"
155 PLACE		±0.00	1860"-1872"	155 PLACE		±0.00	1860"-1872"
156 PLACE		±0.00	1872"-1884"	156 PLACE		±0.00	1872"-1884"
157 PLACE		±0.00	1884"-1896"	157 PLACE		±0.00	1884"-1896"
158 PLACE		±0.00	1896"-1908"	158 PLACE		±0.00	1896"-1908"
159 PLACE		±0.00	1908"-1920"	159 PLACE		±0.00	1908"-1920"
160 PLACE		±0.00	1920"-1932"	160 PLACE		±0.00	1920"-1932"
161 PLACE		±0.00	1932"-1944"	161 PLACE		±0.00	1932"-1944"
162 PLACE		±0.00	1944"-1956"	162 PLACE		±0.00	1944"-1956"
163 PLACE		±0.00	1956"-1968"	163 PLACE		±0.00	1956"-1968"
164 PLACE		±0.00	1968"-198				



	Item	Part Number	Description	Qty	Units
	ASSY	100371225	ASSY,BUMPER	1	EA
	1	100371226	• BUMPER,INTAKE	1	EA
	2	00900015-0056	• BOLT,HEX HD,3/4"-10 UNC,2" LG,GR 5,BL	4	EA
	3	00900302-0006	• WASHER,SPRING,3/4",BLACK	4	EA



REVISIONS			
REV:	01	ECH: 1164541	DATE: 19FEB09 BY: KC
1. AS FIRST DRAWN.			

Item	Component number	Object description	Qty.	Un	Wt.(kgs)	Wt.(lbs)
0001	100374003	HOPPER,DOOR	1	EA	629.44	1384.77
0002	98-410080077	CYLINDER,4"BORE,48"STROKE *	2	EA	104.06	228.93
0003	00900015-0056	BOLT,HEX HD,3/4"-10 UNC,2" LG,GR 5,BL	16	EA	3.20	7.04
0004	00900302-0006	WASHER, LOCK, 3/4	16	EA	0.32	0.70
0005	100374085	ASSY,PIN & CHAIN	2	EA	1.34	2.95
0006	98-300280503	PIN,RAM INTAKE HINGE	2	EA	7.08	15.58
0007	98-320111525	LATCH,DOOR	2	EA	7.54	16.59
0008	98-300280338	PIN,MTG-IDLER SPROCKET	2	EA	0.86	1.89
0009	98-A0R0500028	PIN,COTTER *	4	EA	0.08	0.18
0010	98-A0R0400020	PIN,COTTER *	2	EA	0.04	0.09
0011	98-880280029	PIN,HAIR *	2	EA	0.10	0.22
Total Weight:					754.06	1658.93

METRIC				IMPERIAL			
FINISHING	2.0-3.0mm	MACHINING	0.127mm	FINISHING	0.005"	MACHINING	0.002"
ANGULAR	±1/2°	TOL.	±0.05	ANGULAR	±1/2°	TOL.	±0.05
FABRICATIONS				FABRICATIONS			
0-300mm	±1mm	1 PLACE	±0.05	0"-24"	±0.06	1 PLACE	±0.06
300mm-1000mm	±2mm	2 PLACE	±0.10	24"-96"	±0.12	2 PLACE	±0.12
1000mm-3000mm	±3mm	3 PLACE	±0.15	96"-384"	±0.18	3 PLACE	±0.18
3000mm-OVER	±5mm	4 PLACE	±0.20	384"-OVER	±0.25	4 PLACE	±0.25
UNSPECIFIED SURFACE FINISH ON MACHINING				UNSPECIFIED SURFACE FINISH ON MACHINING			
DO NOT SCALE THIS CAD DRAWING				DO NOT SCALE THIS CAD DRAWING			
DRAWN		KC		19FEB09		SIZE SCALE 3B	
CHECK		BLF		19FEB09		DWG NO 100374099	
SHEET 1		OF 1		D		SHEET 1	



	Item	Part Number	Description	Qty	Units
	ASSY	100374099	ASSY,DOOR-HOPPER ACCESS	1	EA
	1	100374003	• HOPPER,DOOR	1	EA
	2	98-410080077	• CYLINDER,4"BORE,48"STROKE *	2	EA
	3	00900015-0056	• BOLT,HEX HD,3/4"-10 UNC,2" LG,GR 5,BL	16	EA
	4	00900302-0006	• WASHER,SPRING,3/4",BLACK	16	EA
	5	100374085	• ASSY,PIN & CHAIN	2	EA
	6	98-300280503	• PIN,RAM INTAKE HINGE	2	EA
	7	98-320111525	• LATCH,DOOR	2	EA
	8	98-300280338	• PIN,MTG-IDLER SPROCKET	2	EA
	9	98-A0R0500028	• PIN,COTTER	4	EA
	10	98-A0R0400020	• PIN,COTTER	2	EA
	11	98-880280029	• PIN,HAIR *	2	EA

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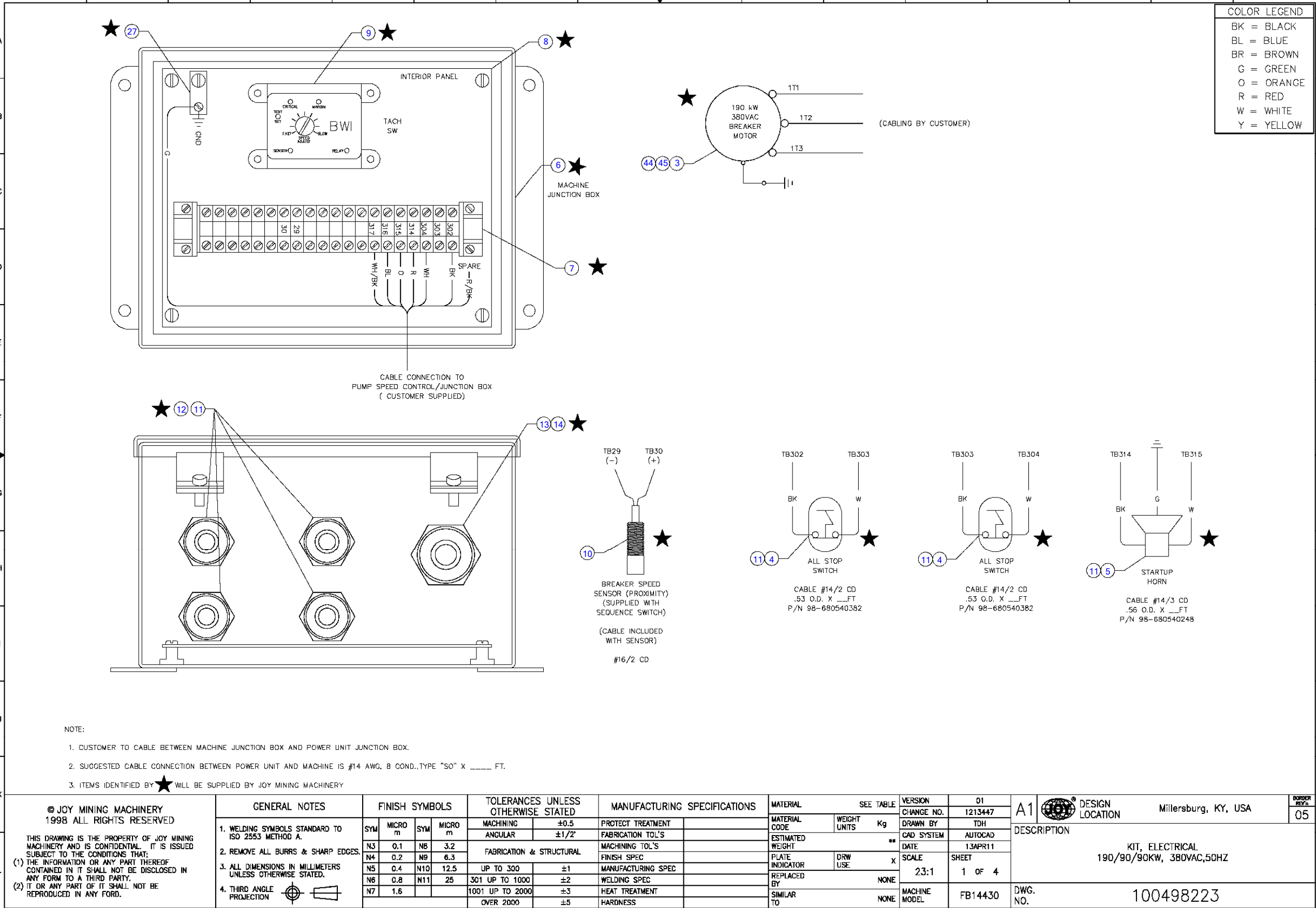


Description		Page
Section 3.0 Electrical Assemblies		
• Kit - Electrical (190/90/90 kw, 380vac, 50hz)		1-6-1
• • Kit - Electrical (Suggested Schematic)		1-6-2

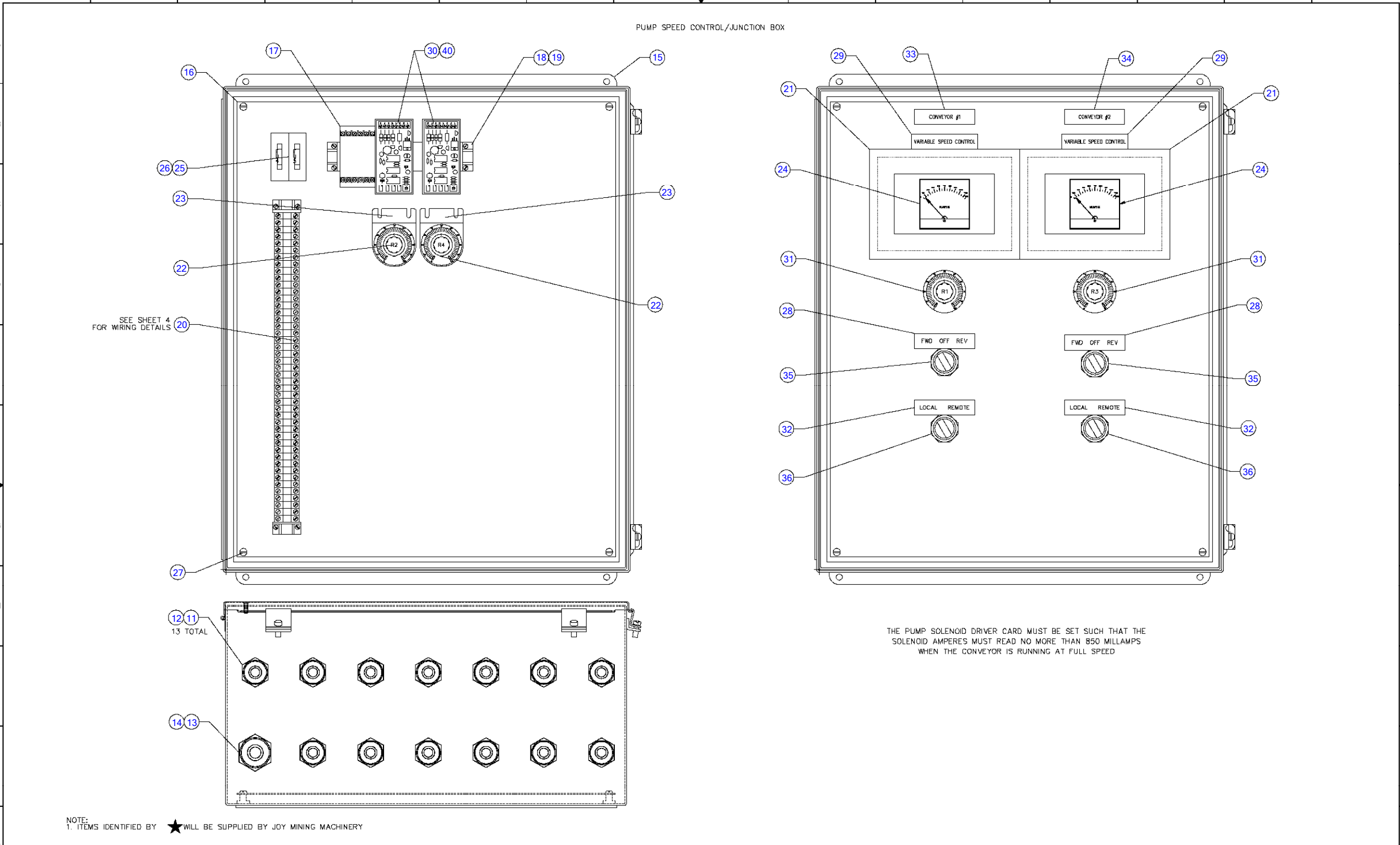
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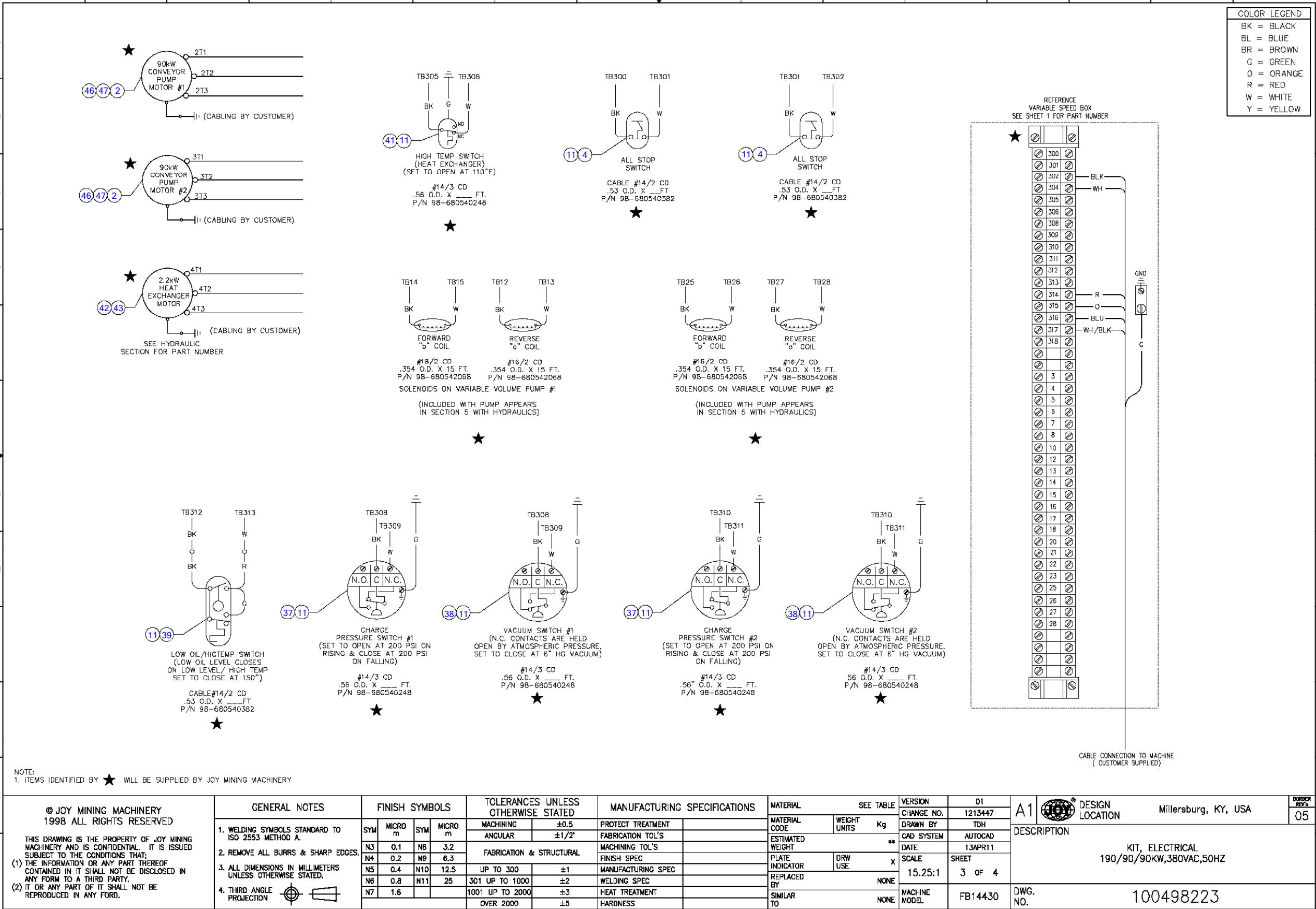
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												MATERIAL CODE	WEIGHT UNITS	Kg	CHANGE NO.	1213447			
				ESTIMATED WEIGHT		**	DRAWN BY	TDH											
				PLATE INDICATOR	DRW USE	X	CAD SYSTEM	AUTOCAD											
				REPLACED BY	NONE		DATE	13APR11											
												SIMILAR TO		NONE	MACHINE MODEL	FB14430	DWG. NO.		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				

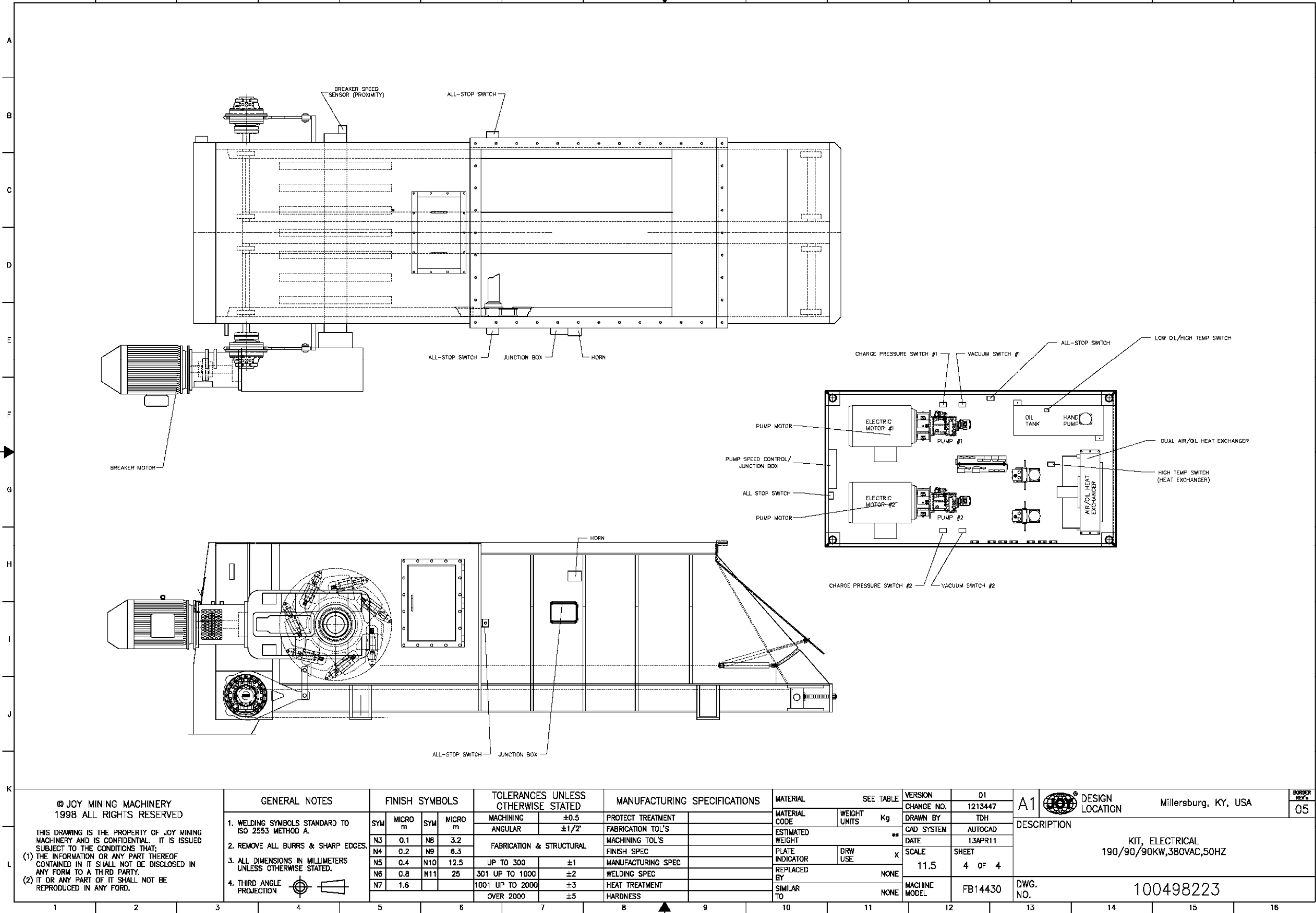


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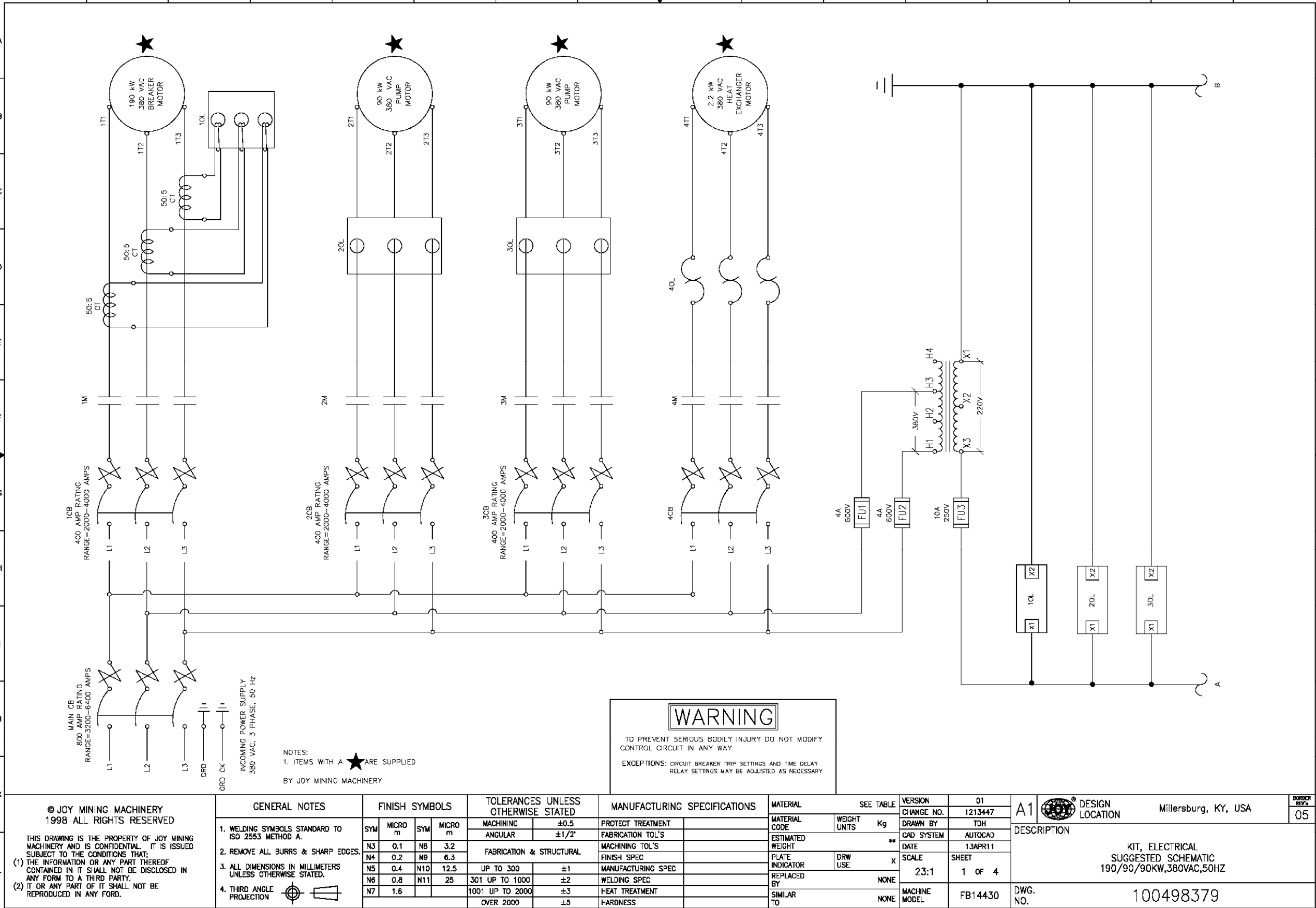


	Item	Part Number	Description	Qty	Units
	ASSY	100498223	KIT,ELECTRICAL,190/90/90/KW, 380VAC,50HZ	1	EA
	1	100498379	• KIT,ELECTRICAL,SUGGESTED SCHEMATIC	1	EA
	2	98-790550447	• MOTOR,90KW(125HP),380VAC,50HZ,3PH	2	EA
	3	98-790550556	• MOTOR,190KW(250HP),380VAC,50HZ,3PH	1	EA
	4	98-680542207	• PUSHBUTTON	4	EA
	5	98-680543312	• HORN	1	EA
	6	98-690490046	• ENCLOSURE	1	EA
	7	98-690540121	• TERMINAL BLOCK	1	EA
	8	98-680310019	• PANEL	1	EA
	9	100371779	• DETECTOR,MOTION-RAMP UP TIMER	1	EA
	10	98-680541314	• SENSOR, PROXIMITY	1	EA
	11	98-680544009	• CONNECTOR,CABLE	28	EA
	12	98-SL1	• NUT,SEAL	24	EA
	13	98-CGB396	• CONNECTOR	2	EA
	14	98-SL3	• NUT,SEAL 1"	2	EA
	15	98-690490119	• ENCLOSURE	1	EA
	16	98-680310013	• PANEL	1	EA
	17	98-680544033	• POWER SUPPLY	1	EA
	18	98-680541749	• MOUNTING,DIN RAIL	1	M
	19	98-680542029	• CLAMP,END *	2	EA
	20	98-690540357	• BLOCK, TERMINAL 45SM PT	1	EA
	21	98-690540114	• WINDOW	2	EA
	22	98-680541098	• POTENTIOMETER,W/KNOB (500 OHM)	2	EA
	23	98-300120248	• BRACKET	2	EA
	24	98-680540730	• AMMETER,0-1500 DC	2	EA
	25	00600014-0041	• FUSE	4	EA
	26	00600007-1083	• BLOCK, FUSE	1	EA
	27	98-SLU125	• GROUND LUG	2	EA
	28	98-300140921	• NAMEPLATE	2	EA
	29	98-300140923	• NAMEPLATE	2	EA
	30	98-680544011	• AMPLIFIER,SOLENOID DRIVE CARD	2	EA
	31	98-680542221	• POTENTIOMETER	2	EA
	32	98-300140714	• NAMEPLATE *	2	EA
	33	98-300142726	• NAMEPLATE *	1	EA
	34	98-300142727	• NAMEPLATE *	1	EA
	35	98-680543157	• SWITCH,COMPLETE SELECTOR	2	EA
	36	98-680543156	• SWITCH,SELECTOR COMPLETE	2	EA
	37	98-680542213	• SWITCH,PRESSURE	2	EA
	38	98-680542214	• SWITCH,VACUUM	2	EA
	39	98-680543193	• SWITCH,OIL LEVEL/TEMP	1	EA
	40	98-680544012	• HOLDER,SOLENOID CARD	2	EA
	41	98-680542683	• SWITCH,TEMPERATURE	1	EA
	42	98-CGB296	• CONNECTOR,CABLE	1	EA
	43	98-680540126	• NUT,LOCK	1	EA
	44	98-SHC1091	• CONNECTOR,CABLE	1	EA
	45	98-48057F114	• FITTING,HYDR-NPT BUSHING*	1	EA
	46	98-CGB6913	• CONNECTOR,CABLE	2	EA
	47	98-480570077	• FITTING,HYDR-BUSHING *	2	EA

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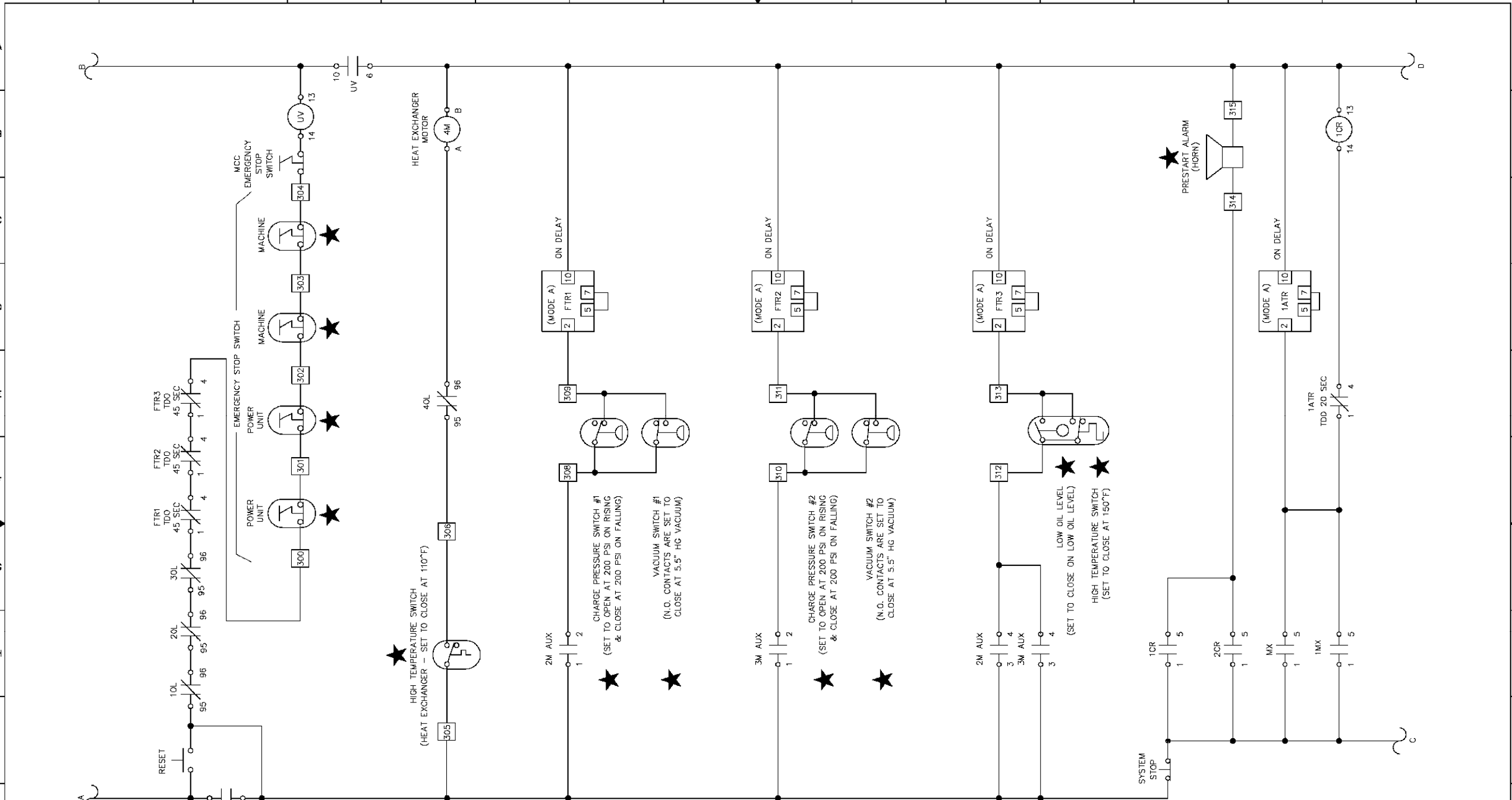


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NOTES:
1. ITEMS WITH A ★ ARE SUPPLIED
BY JOY MINING MACHINERY

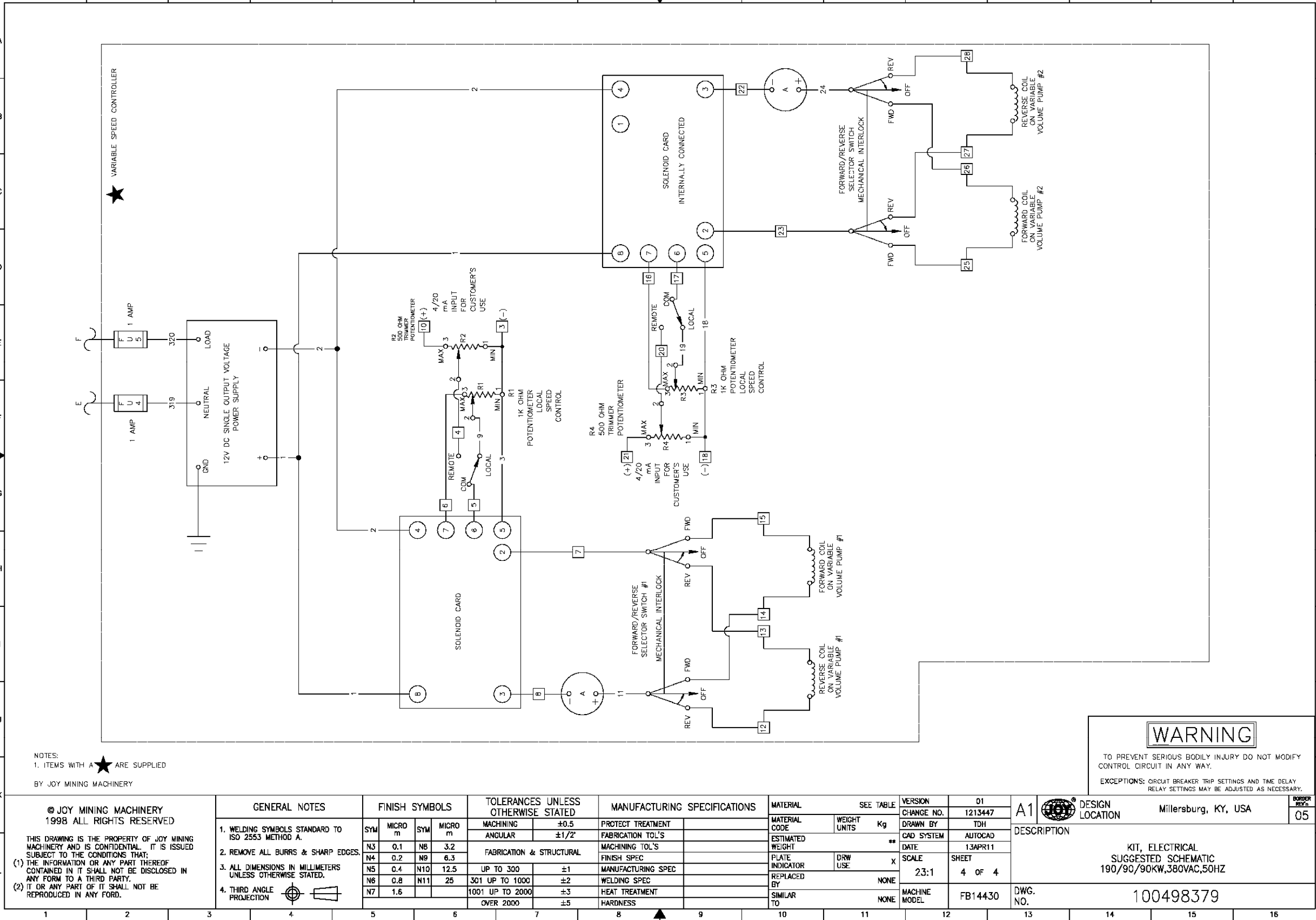
WARNING

TO PREVENT SERIOUS BODILY INJURY DO NOT MODIFY
CONTROL CIRCUIT IN ANY WAY.
EXCEPTIONS: CIRCUIT BREAKER TRIP SETTINGS AND TIME DELAY
RELAY SETTINGS MAY BE ADJUSTED AS NECESSARY.

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	1. WELDING SYMBOLS STANDARD TO ISO 2553 METHOD A.	SYM	MICRO m	SYM	MICRO m	MACHINING	±0.5	PROTECT TREATMENT	MATERIAL CODE	WEIGHT UNITS	Kg	CHANGE NO.			1213447	DESCRIPTION KIT, ELECTRICAL SUGGESTED SCHEMATIC 190/90/90KW,380VAC,50HZ DWG. NO. 100498379
						ANGULAR	±1/2°	FABRICATION TOL'S				DRAWN BY			TDH	
		N3	0.1	N8	3.2	FABRICATION & STRUCTURAL	MACHINING TOL'S	ESTIMATED WEIGHT	**	CAD SYSTEM	AUTOCAD					
										FINISH SPEC	DATE	13APR11				
		N4	0.2	N9	6.3	UP TO 300	±1	MANUFACTURING SPEC	PLATE INDICATOR	DRW USE	X	SCALE			SHEET	
		N6	0.8	N11	25	301 UP TO 1000	±2	WELDING SPEC	REPLACED BY	NONE	23:1	2 OF 4				
	N7	1.6			1001 UP TO 2000	±3	HEAT TREATMENT	SIMILAR TO	NONE	MACHINE MODEL	FB14430					
					OVER 2000	±5	HARDNESS									
	4. THIRD ANGLE PROJECTION															



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	Item	Part Number	Description	Qty	Units
	ASSY	100498379	KIT,ELECTRICAL - SUGGESTED SCHEMATIC	1	EA

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	Description	Page
	Section 4.0 Hydraulic Assemblies	
	• Kit - Hydraulic	1-7-1
	• • 180cc Series Pump	1-7-2
	• • Fitting Kit - Hardware	1-7-3
	• • Single Pump	1-7-4
	• • Fitting Kit - Flushing Valve	1-7-5
	• • Fitting Kit - CA Series	1-7-6
	• • Fitting Kit - Charge Filter	1-7-7
	• • Pall Filter Assembly	1-7-8
	• • Fitting Kit - Heat Exchanger	1-7-9
	• • Fitting Kit - Gauge & Switch	1-7-10
	• • Hydraulic Tank Assembly	1-7-11
	• • Hand Pump	1-7-12

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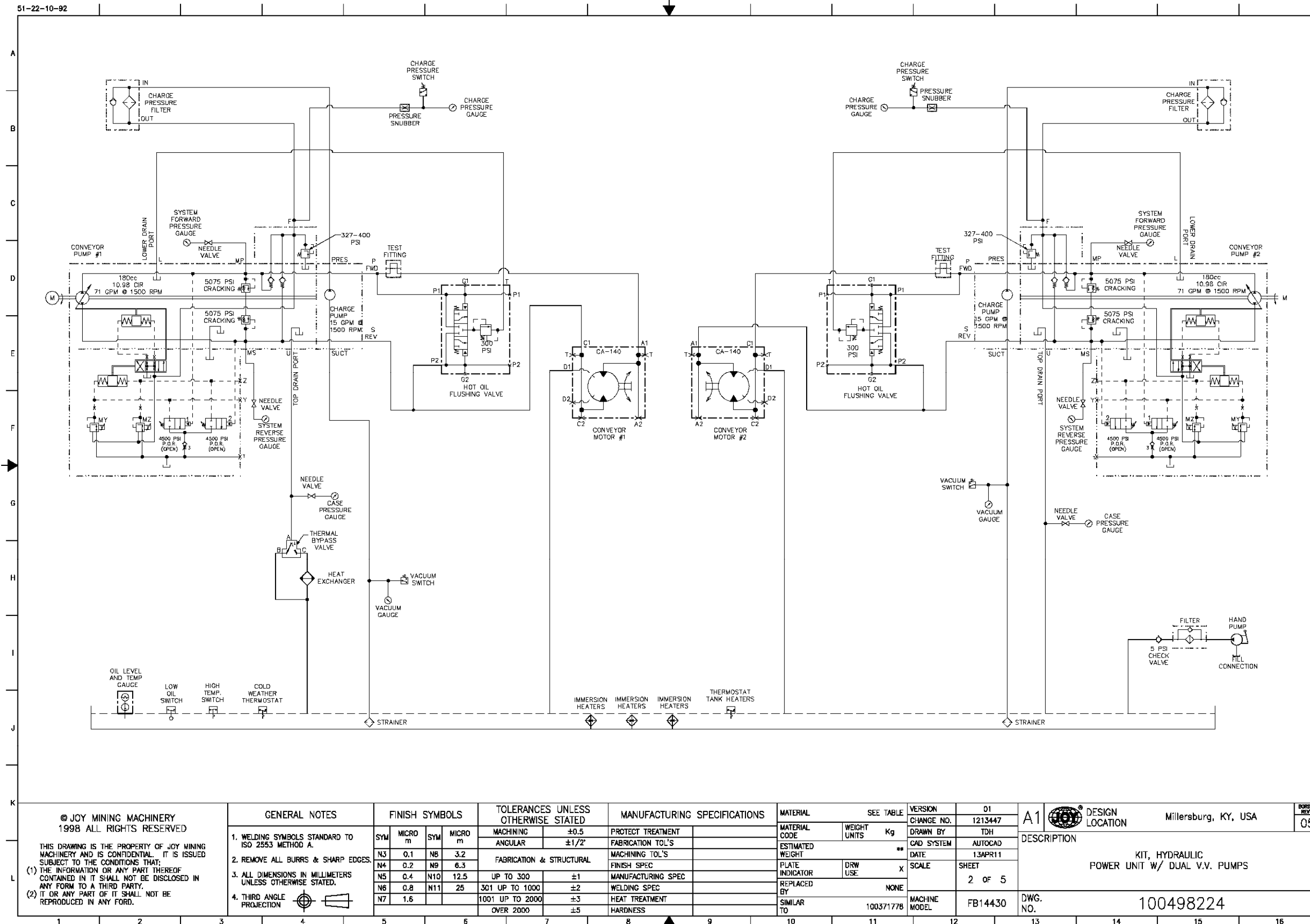
HOSE GUIDELINE

ITEM	USE	LENGTH	HOSE ASSY.	RAW HOSE	HOSE ENDS
A 2 "	A TANK TO GEAR PUMP #1 "SUCTION" PORT	—	—	98-480570161	N/A
	B TANK TO GEAR PUMP #2 "SUCTION" PORT	—	—	98-480570161	N/A
C 1 "	A VARIABLE VOLUME PUMP #1 "S" PORT TEE TO POWER UNIT EDGE	—	—	98-480570287	98-490570145*
	B POWER UNIT EDGE TO FEEDER EDGE (INTERCONNECTING HOSE)	240"	98-350571682	98-480570287	98-490570145*
	C FEEDER EDGE TO CONVEYOR MOTOR #1 "C1" PORT	—	—	98-480570287	98-490570145*
	D VARIABLE VOLUME PUMP #1 "P" PORT TEST FITTING TO POWER UNIT EDGE	—	—	98-480570287	98-490570145*
	E POWER UNIT EDGE TO FEEDER EDGE (INTERCONNECTING HOSE)	240"	98-350571682	98-480570287	98-490570145*
	F FEEDER EDGE TO CONVEYOR MOTOR #1 "A1" PORT	—	—	98-480570287	98-490570145*
	G VARIABLE VOLUME PUMP #2 "S" PORT TEE TO POWER UNIT EDGE	—	—	98-480570287	98-490570145*
	H POWER UNIT EDGE TO FEEDER EDGE (INTERCONNECTING HOSE)	240"	98-350571682	98-480570287	98-490570145*
	I FEEDER EDGE TO CONVEYOR MOTOR #2 "A1" PORT	—	—	98-480570287	98-490570145*
	J VARIABLE VOLUME PUMP #2 "P" PORT TEST FITTING TO POWER UNIT EDGE	—	—	98-480570287	98-490570145*
	K POWER UNIT EDGE TO FEEDER EDGE (INTERCONNECTING HOSE)	240"	98-350571682	98-480570287	98-490570145*
	L FEEDER EDGE TO CONVEYOR MOTOR #2 "C1" PORT	—	—	98-480570287	98-490570145*
	M VARIABLE VOLUME PUMP #1 "U" PORT (UPPER DRAIN) TO BYPASS VALVE "A" PORT	—	—	98-480570165	98-490570073
	N BYPASS VALVE "B" PORT TO HEAT EXCHANGER "OUT" PORT TEE	—	—	98-480570165	98-490570073
	P HEAT EXCHANGER "OUT" PORT TEE TO TANK TEE	—	—	98-480570165	98-490570073
	R VARIABLE VOLUME PUMP #2 "U" PORT (UPPER DRAIN) TO TANK TEE	—	—	98-480570165	98-490570073
D 1 "	A GEAR PUMP #1 "P" PORT TO CHARGE FILTER #1 "IN" PORT	—	—	98-480570198	98-490570090
	B CHARGE FILTER #1 "OUT" PORT TO VARIABLE VOLUME PUMP #1 "F" PORT	—	—	98-480570198	98-490570090
	C GEAR PUMP #2 "P" PORT TO CHARGE FILTER #2 "IN" PORT	—	—	98-480570198	98-490570090
	D CHARGE FILTER #2 "OUT" PORT TO VARIABLE VOLUME PUMP #2 "F" PORT	—	—	98-480570198	98-490570090
	E FLUSHING VALVE #1 "T" PORT TO CONVEYOR PUMP #1 "L" PORT (LOWER DRAIN)	—	—	98-480570198	98-490570090
	F FLUSHING VALVE #1 "P1" PORT TO VARIABLE VOLUME PUMP #1 "P" PORT TEE	—	—	98-480570286	98-490570136*
	G FLUSHING VALVE #1 "P2" PORT TO VARIABLE VOLUME #1 "S" PORT TEE	—	—	98-480570286	98-490570136*
	H FLUSHING VALVE #2 "T" PORT TO CONVEYOR PUMP #2 "L" PORT (LOWER DRAIN)	—	—	98-480570198	98-490570090
	J FLUSHING VALVE #2 "P1" PORT TO VARIABLE VOLUME PUMP #2 "P" PORT TEE	—	—	98-480570286	98-490570136*
	K FLUSHING VALVE #2 "P2" PORT TO VARIABLE VOLUME PUMP #2 "S" PORT TEE	—	—	98-480570286	98-490570136*
	L POWER UNIT EDGE TO FEEDER EDGE (INTERCONNECTING HOSE)	240"	98-350571204	98-480570198	98-490570090
	M POWER UNIT EDGE TO TANK	—	—	98-480570198	98-490570090
	N CONVEYOR MOTOR #1 DRAIN PORT TO FEEDER EDGE	—	—	98-480570198	98-490570090
	P POWER UNIT EDGE TO FEEDER EDGE (INTERCONNECTING HOSE)	240"	98-350571204	98-480570198	98-490570090
	R POWER UNIT EDGE TO TANK	—	—	98-480570198	98-490570090
	S CONVEYOR MOTOR #2 DRAIN PORT TO FEEDER EDGE	—	—	98-480570198	98-490570090
E 3/4 "	A CHECK VALVE ON TANK TO FILL FILTER "OUT" PORT	—	—	98-480570163	98-490570055
	B HAND PUMP "IN" PORT TO FILL CONNECTION	—	—	98-480570163	98-490570055
H 1/4 "	A VARIABLE VOLUME PUMP #1 "Mp" PORT TO FORWARD PRESSURE GAGE #1	—	—	98-480570190	98-490570046
	B VARIABLE VOLUME PUMP #1 "Ms" PORT TO REVERSE PRESSURE GAGE #1	—	—	98-480570190	98-490570046
	C VARIABLE VOLUME PUMP #1 "U" PORT TO CASE PRESSURE GAGE #1	—	—	98-480570190	98-490570046
	D VARIABLE VOLUME PUMP #1 "F" PORT TO CHARGE PRESSURE GAGE #1 TEE	—	—	98-480570190	98-490570046
	E CHARGE PRESSURE GAGE #1 TEE TO CHARGE PRESSURE SWITCH #1	—	—	98-480570190	98-490570046
	F GEAR PUMP #1 "S" PORT TO VACUUM GAGE #1 TEE	—	—	98-480570190	98-490570046
	G VACUUM GAUGE #1 TEE TO VACUUM SWITCH #1	—	—	98-480570190	98-490570046
	H VARIABLE VOLUME PUMP #2 "Mp" PORT TO FORWARD PRESSURE GAGE #2	—	—	98-480570190	98-490570046
	J VARIABLE VOLUME PUMP #2 "Ms" PORT TO REVERSE PRESSURE GAGE #2	—	—	98-480570190	98-490570046
	K VARIABLE VOLUME PUMP #2 "U" PORT TO CASE PRESSURE GAGE #2	—	—	98-480570190	98-490570046
	L VARIABLE VOLUME PUMP #2 "F" PORT TO CHARGE PRESSURE GAGE #2 TEE	—	—	98-480570190	98-490570046
	M CHARGE PRESSURE GAGE #2 TEE TO CHARGE PRESSURE SWITCH #2	—	—	98-480570190	98-490570046
	N GEAR PUMP #2 "S" PORT TO VACUUM GAUGE #2 TEE	—	—	98-480570190	98-490570046
	P VACUUM GAUGE #2 TEE TO VACUUM SWITCH #2	—	—	98-480570190	98-490570046

NOTES:

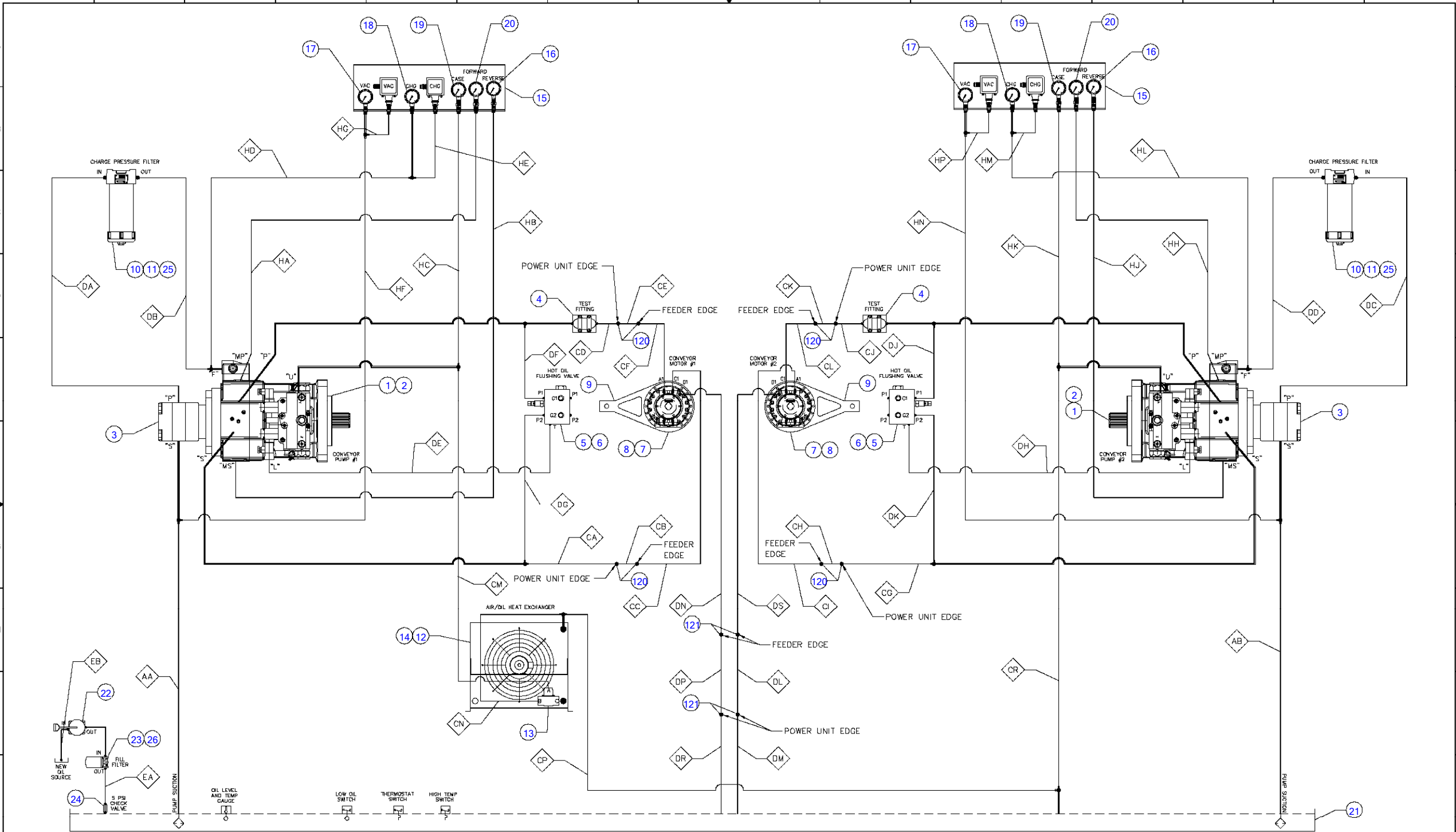
- REPLACEMENT CHARGE PRESSURE FILTER ELEMENTS - P/N 48011F111 (2 REQUIRED)
- REPLACEMENT FILL FILTER ELEMENT - P/N 490520049 (1 REQUIRED)
- REPLACEMENT MOUNTING BRACKET FOR GAGES AND SWITCHES - P/N 320120374
- HOT OIL FLUSHING VALVE P/N 490560095 - REPLACEMENT 30 GPM RELIEF CARTRIDGE P/N 480560355 AND REPLACEMENT SHUTTLE CARTRIDGE P/N 480560376
- DOCUMENT LENGTHS & RETURN TO ENGINEERING FOR PROPER DOCUMENTATION
- * DENOTES SEAL-LOK FITTINGS FOR CIRCUIT

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							DRAWN BY AUTOCAD	TOH		
							DATE 13APR11	SHEET 1 OF 5		
							SCALE 5.75:1	MACHINE MODEL FB14430		





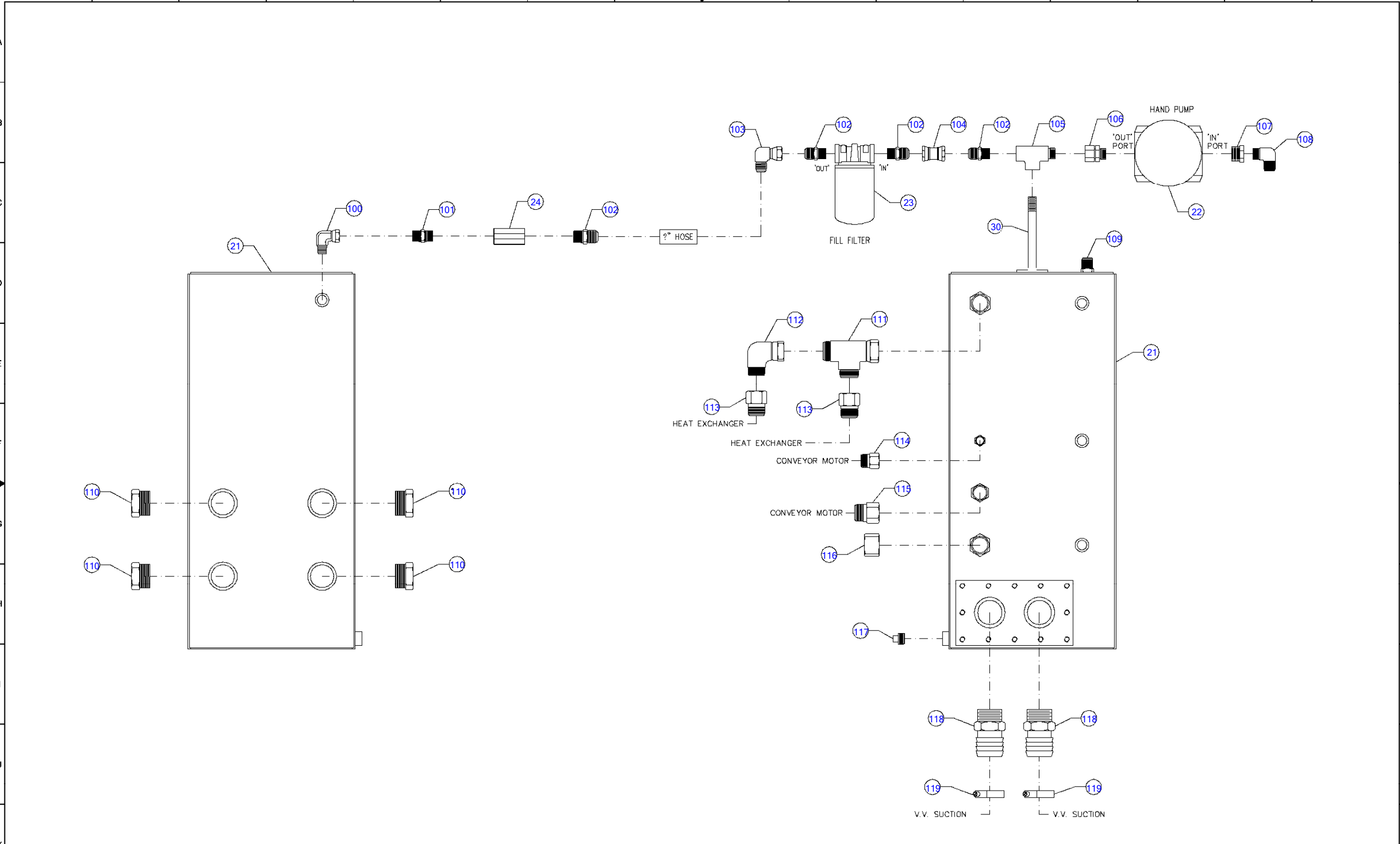
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	1. WELDING SYMBOLS STANDARD TO ISO 2553 METHOD A.		SYM	MICRO m	SYM	MICRO m	MACHINING	±0.5	PROTECT TREATMENT		MATERIAL CODE	WEIGHT UNITS	Kg		CHANGE NO.	1213447
	2. REMOVE ALL BURRS & SHARP EDGES.		N3	0.1	N8	3.2	ANGULAR	±1/2°	FABRICATION TOL'S		ESTIMATED WEIGHT		**		DRAWN BY	TDH
	3. ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE STATED.		N4	0.2	N9	6.3	FABRICATION & STRUCTURAL		MACHINING TOL'S		PLATE INDICATOR	DRW USE	X		CAD SYSTEM	AUTOCAD
	4. THIRD ANGLE PROJECTION 		N5	0.4	N10	12.5	UP TO 300	±1	MANUFACTURING SPEC		REPLACED BY	NONE			DATE	13APR11
			N6	0.8	N11	25	301 UP TO 1000	±2	WELDING SPEC						SCALE	SHEET
			N7	1.6			1001 UP TO 2000	±3	HEAT TREATMENT						6:1	3 OF 5
							OVER 2000	±5	HARDNESS						MACHINE MODEL	FB14430

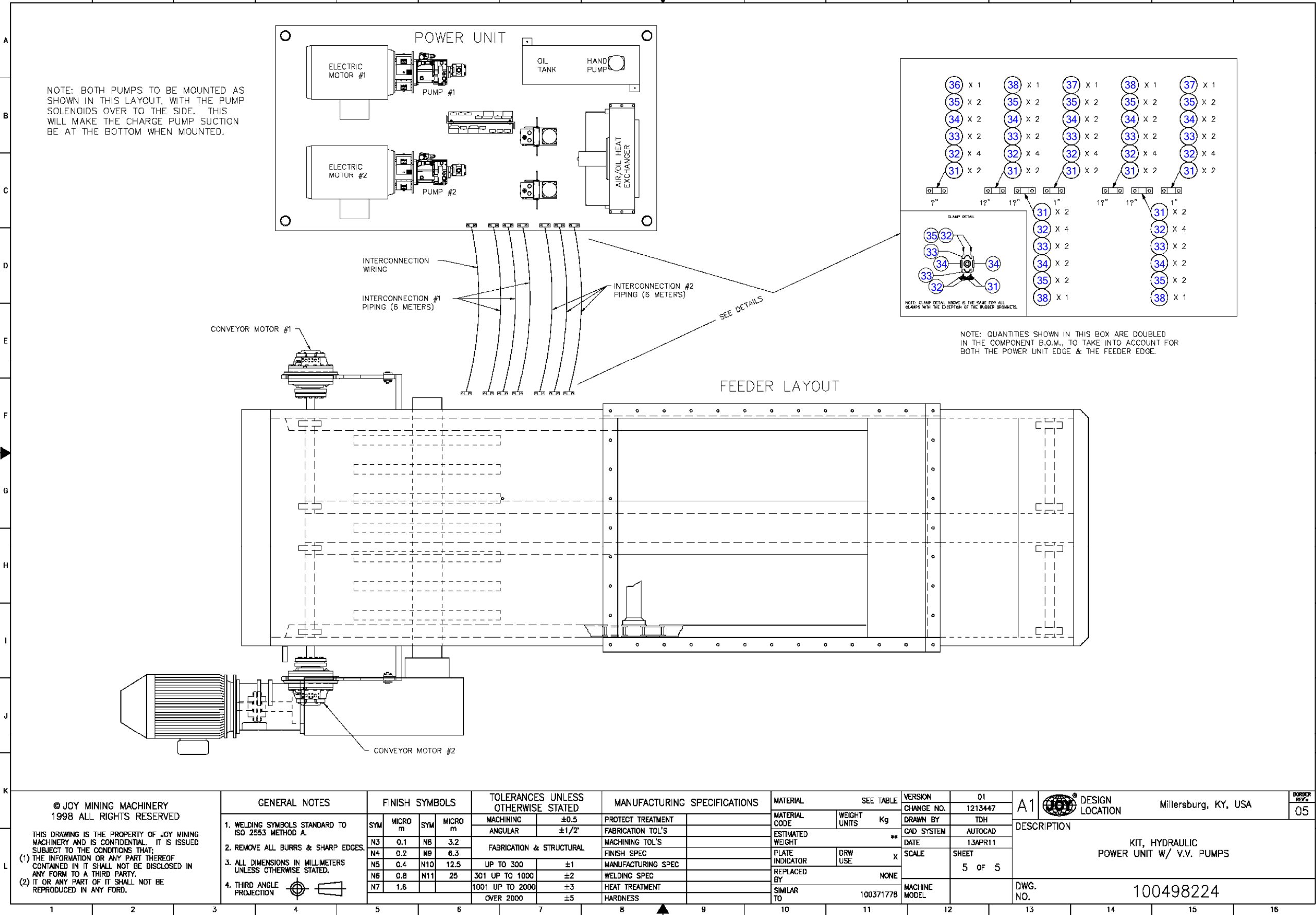


51-22-10-92



© JOY MINING MACHINERY 1998 ALL RIGHTS RESERVED			GENERAL NOTES		FINISH SYMBOLS		TOLERANCES UNLESS OTHERWISE STATED		MANUFACTURING SPECIFICATIONS		MATERIAL		SEE TABLE		VERSION		01		A1		DESIGN LOCATION		Millersburg, KY, USA		BORDER REVISION		05					
THIS DRAWING IS THE PROPERTY OF JOY MINING MACHINERY AND IS CONFIDENTIAL. IT IS ISSUED SUBJECT TO THE CONDITIONS THAT: (1) THE INFORMATION OR ANY PART THEREOF CONTAINED IN IT SHALL NOT BE DISCLOSED IN ANY FORM TO A THIRD PARTY. (2) IT OR ANY PART OF IT SHALL NOT BE REPRODUCED IN ANY FORM.			1. WELDING SYMBOLS STANDARD TO ISO 2553 METHOD A. 2. REMOVE ALL BURRS & SHARP EDGES. 3. ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE STATED. 4. THIRD ANGLE PROJECTION 		SYM		MICRO m		SYM		MICRO m		MACHINING		±0.5		PROTECT TREATMENT				MATERIAL CODE		WEIGHT UNITS		Kg		CHANGE NO.		1213447			
					N3		0.1		N8		3.2		ANGULAR		±1/2°		FABRICATION TOL'S				ESTIMATED WEIGHT				**		CAD SYSTEM		AUTOCAD			
					N4		0.2		N9		6.3		FABRICATION & STRUCTURAL				FINISH SPEC				PLATE INDICATOR		DRW USE		X		DATE		13APR11			
					N5		0.4		N10		12.5		UP TO 300		±1		MANUFACTURING SPEC				REPLACED BY		NONE				SCALE		SHEET			
					N6		0.8		N11		25		301 UP TO 1000		±2		WELDING SPEC				SIMILAR TO		100371778				4 OF 5		MACHINE MODEL		FB14340	
					N7		1.6						1001 UP TO 2000		±3		HEAT TREATMENT										DWG. NO.		100498224			
													OVER 2000		±5		HARDNESS															
1		2		3		4		5		6		7		8		9		10		11		12		13		14		15		16		

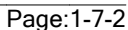
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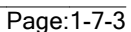
	Item	Part Number	Description	Qty	Units
	ASSY	100498224	KIT,HYD.	1	EA
	1	98-79026F049	• PUMP,180CC SERIES) *	2	EA
	2	98-390890337	• KIT,HARDWARE,HYD. FITTINGS(PUMP)	2	EA
	3	98-33026F004	• PUMP,SINGLE *	2	EA
	4	03069230-0234	• STATION,TEST (1-1/4 O-RING PORTS)	2	EA
	5	98-490560095	• VALVE,FLUSHING *	2	EA
	6	98-390890305	• KIT,FTG-FLUSHING VALVE #490560095 ENG/S	2	EA
	7	98-79023F005	• MOTOR,HYDRAULIC	2	EA
	8	98-390890306	• KIT,FTG-CA-SERIES HYDR MOT/ABVGRD APPL*	2	EA
	9	98-780670002	• ARM,HYDR MOTOR TORQUE	2	EA
	10	98-49052F004	• ASSY,PALL FILTER (CCC NAMEBRANDED)	2	EA
	11	98-390890304	• KIT,FTG-CHARGE FILTER #490520094 ENG/SN	2	EA
	12	98-890110195	• HEAT EXCHANGER,AIR/OIL	1	EA
	13	98-480560324	• VALVE,BYPASS *	1	EA
	14	100313436	• KIT,FITTING-HEAT EXCHANGER	1	EA
	15	98-39089F054	• KIT,FITTING-GAUGE & SWITCH ENG REQ.	2	EA
	16	98-300140832	• NAMEPLATE	2	EA
	17	98-300140791	• NAMEPLATE	2	EA
	18	98-300140792	• NAMEPLATE	2	EA
	19	98-300140790	• NAMEPLATE	2	EA
	20	98-300141978	• NAMEPLATE *	2	EA
	21	98-310180273	• ASSY,TANK-HYDR 116 GAL	1	EA
	22	98-730260036	• PUMP,HAND	1	EA
	23	98-490520011	• FILTER	1	EA
	24	98-480560343	• VALVE,CHECK	1	EA
	25	98-48011F111	• ELEMENT,PALL NAMEBRANDED	2	EA
	26	98-490520049	• ELEMENT,FILTER *	1	EA
	27	98-880110917	• FITTING,HYDR-SAE ORING KIT	1	EA
	28	98-880110918	• FITTING,HYDR-ORING KIT	1	EA
	29	98-880110919	• KIT,SEAL LOCK O RING	1	EA
	30	98-8656	• BRACKET,MTG-HAND PUMP	1	EA
	31	98-880110145	• ADAPTER,WELD	28	EA
	32	00900042-0013	• BOLT,HEX HD,3/8"-16 UNC,1" LG,GR 8,BL	54	EA
	33	98-880110140	• UNIT,CLAMPING *	28	EA
	34	98-880110142	• NUT,SOCKET STACKING *	28	EA
	35	98-A5P0600000	• WASHER,LOCK *	28	EA
	36	98-880110161	• BUSHING,SPLIT *	2	EA
	37	98-CSB3220	• SPLIT BUSHING	4	EA
	38	98-880110141	• BUSHING,SPLIT *	8	EA
	100	98-480571124	• FITTING, HYDR-90 DEG NPT CONNECTOR*	1	EA
	101	98-480570066	• FITTING,HYDR-NIPPLE	1	EA
	102	98-480571029	• FITTING,HYDR-ST ADAPTER *	4	EA
	103	98-480570964	• FITTING, HYDR-JIC SWIVEL	1	EA
	104	98-480571081	• FITTING,HYDR-ST FJIC UNION *	1	EA
	105	98-480570017	• FITTING,HYDR-STREET TEE *	1	EA
	106	98-480570536	• FITTING,HYDR-MALE STR ADAPT*	1	EA
	107	98-480570046	• FITTING,HYDR-BUSHING *	1	EA
	108	98-480571049	• FITTING,HYDR-MALE ELBOW *	1	EA
	109	98-480571003	• FITTING,HYDR-MALE JIC PLUG	1	EA
	110	98-480100009	• FITTING,HYDR-2"PIPE PLUG	4	EA
	111	98-480570890	• FITTING, HYDR-SWVL NUT RUN TEE	1	EA
	112	98-480570967	• FITTING, HYDR-JIC SWIVEL	1	EA
	113	98-480570910	• FITTING, HYDR-JIC REDUCER	2	EA
	114	98-480570903	• FLTTING, HYDR-JIC EXPANDER	1	EA
	115	98-480570908	• FITTING, HYDR-JIC REDUCER	1	EA
	116	98-480571013	• FITTING HYDR-FEMALE JIC CAP	1	EA
	117	98-480100003	• PLUG,3/4" PIPE	1	EA
	118	98-490570068	• FITTING,HYDR-HOSE END	2	EA
	119	98-490570066	• FITTING,HYDR-HOSE *	2	EA
	120	98-480570441	• FITTING,HYDR-UNION *	8	EA
	121	98-480570996	• FITTING,HYDR-JIC UNION *	4	EA





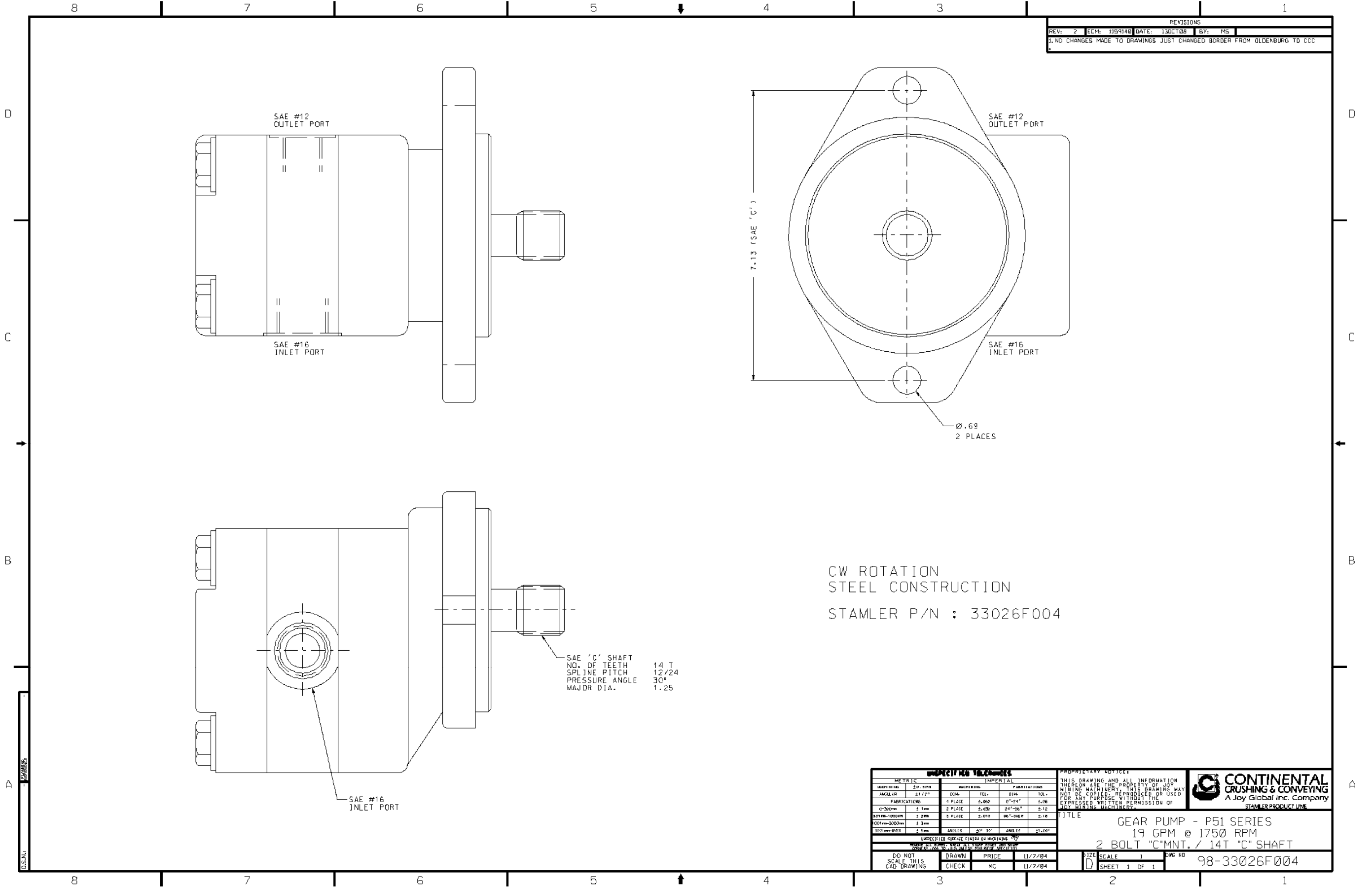
Joy Mining Machinery
FB14430 - BF-32F-60-76F - STATIONARY FEEDER BREAKER - NUSA TAMBANG PRATAMA
PUMP

Item	Part Number	Description	Qty	Units
ASSY	98-79026F049	PUMP	1	EA





	Item	Part Number	Description	Qty	Units
	ASSY	98-390890337	KIT,HARDWARE,HYD. FITTINGS(PUMP)	1	EA
	3	98-490570066	• FITTING,HYDR-HOSE *	1	EA
	4	00925441-1623	• BOLT,HEX HD,M16 X 2,45,GR 10.9	2	EA
	5	98-M4P1600000	• WASHER,LOCK *	2	EA
	6	98-48057F082	• KIT,SF 1.5CODE 62(METRIC)*	2	EA
	7	98-480570338	• FITTING,HYDR-FLNG/90DEG ELBOW*	2	EA
	8	98-480570472	• FITTING,HYDR-TUBE RDCR *	2	EA
	9	98-480570461	• FITTING,HYDR-SEAL UNION *	1	EA
	10	98-480570343	• FITTING,HYDR-90DEG *	1	EA
	11	98-480570426	• FITTING,HYDR-STR CONN	2	EA
	12	98-48057F086	• FITTING,METRIC JIC ADAPTER*	4	EA
	13	98-480570183	• FITTING,HYDR-SWVL RUN TEE*	2	EA
	14	98-480570276	• FITTING,HYDR-SWIVEL *	4	EA
	15	98-48057F049	• FITTING,HYDR-MALE PORT COUPLING*	4	EA
	16	98-48057F084	• FITTING,HYDR-METRIC JIC ST.ADAPT*	3	EA
	17	98-480571071	• FITTING,HYDR-SWIVEL ELBOW	2	EA
	18	98-480570905	• FITTING, HYDR-JIC REDUCER	1	EA
	19	98-300570049	• FITTING,HYDR-90DEG NUT ELBOW	1	EA
	20	98-480571039	• FITTING, HYDR-90 DEG ADAPTER	3	EA
	21	98-480570906	• FITTING, HYDR-JIC EXPANDER	3	EA
	22	98-480570909	• FITTING, HYDR-JIC EXPANDER	1	EA
	23	98-300570010	• FITTING,HYDR-SWVL 90DEG ELBOW	1	EA
	24	98-480570910	• FITTING, HYDR-JIC REDUCER	1	EA
	25	98-480570339	• FITTING,HYDR-SWVL RUN TEE*	2	EA
	26	98-480570340	• FITTING,HYDR-TUBE RDCR *	2	EA
	27	98-48057F092	• FITTING,SAE X MJIC ST*	1	EA
	28	98-480570789	• FITTING,HYDR-90DEG ELBOW*	1	EA
	29	98-480570550	• FITTING,HYDR-STR CONN	1	EA





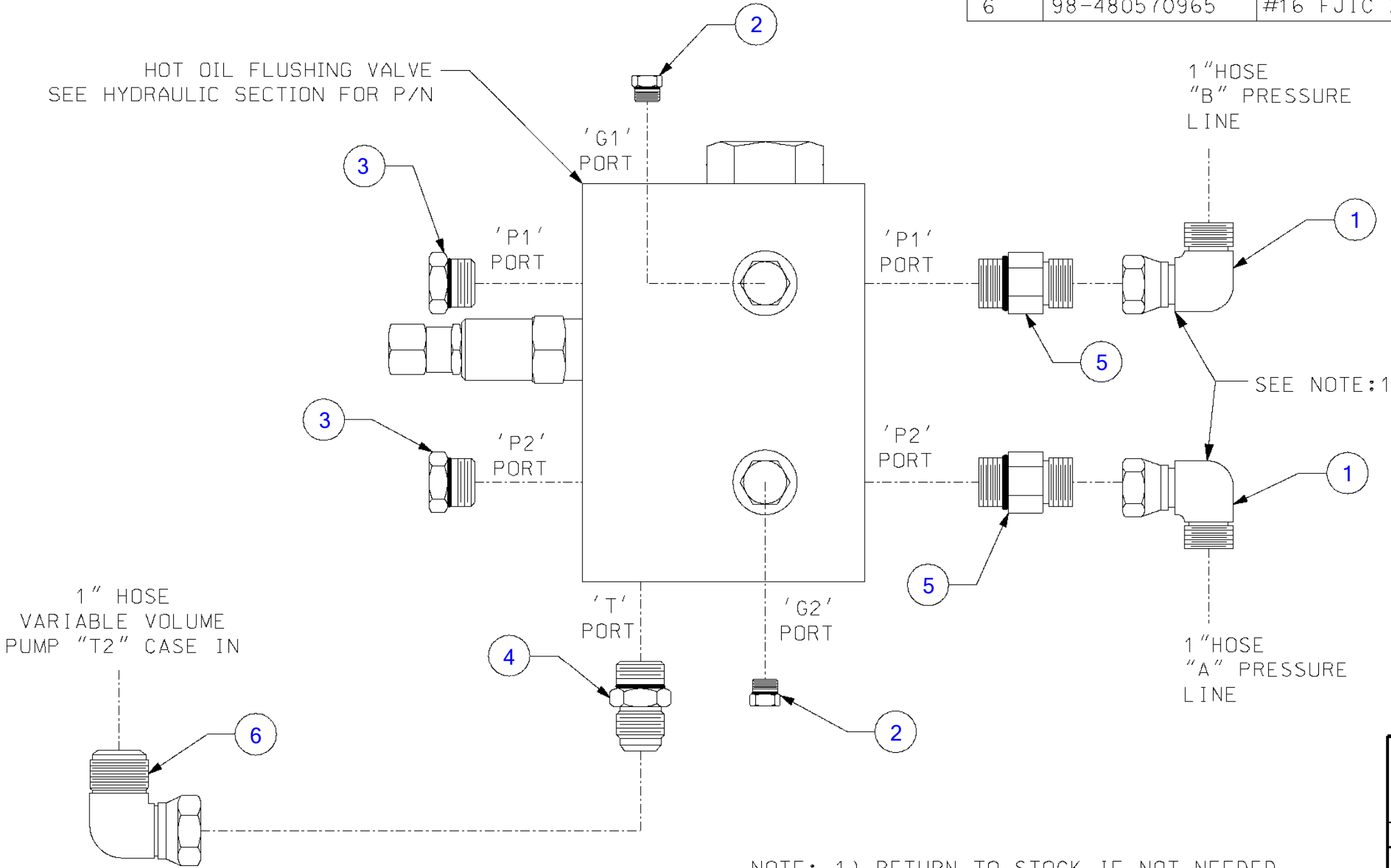
Joy Mining Machinery
FB14430 - BF-32F-60-76F - STATIONARY FEEDER BREAKER - NUSA TAMBANG PRATAMA
PUMP

Item	Part Number	Description	Qty	Units
ASSY	98-33026F004	PUMP	1	EA



REV: 02	ECM: 1169746	DATE: 21APR09	BY: TDH
1. ITEM#3 98-480571144 NOW NON-PRODUCTION/NON-COST REVELANT IN SAP.			

REF	PART NUMBER	DESCRIPTION	QTY
1	98-480570420	#16 SEAL LOK SWIVEL 90° ELBOW	2
2	98-480571141	SAE #6 PLUG	2
3	98-480571144	SAE #12 PLUG	2
4	98-480570550	SAE #12 X #16 MJIC ST. ADAPTER	1
5	98-480570337	SAE #12 X 1" MALE SEAL-LOK	2
6	98-480570965	#16 FJIC X #16 MJIC 90° ELBOW	1

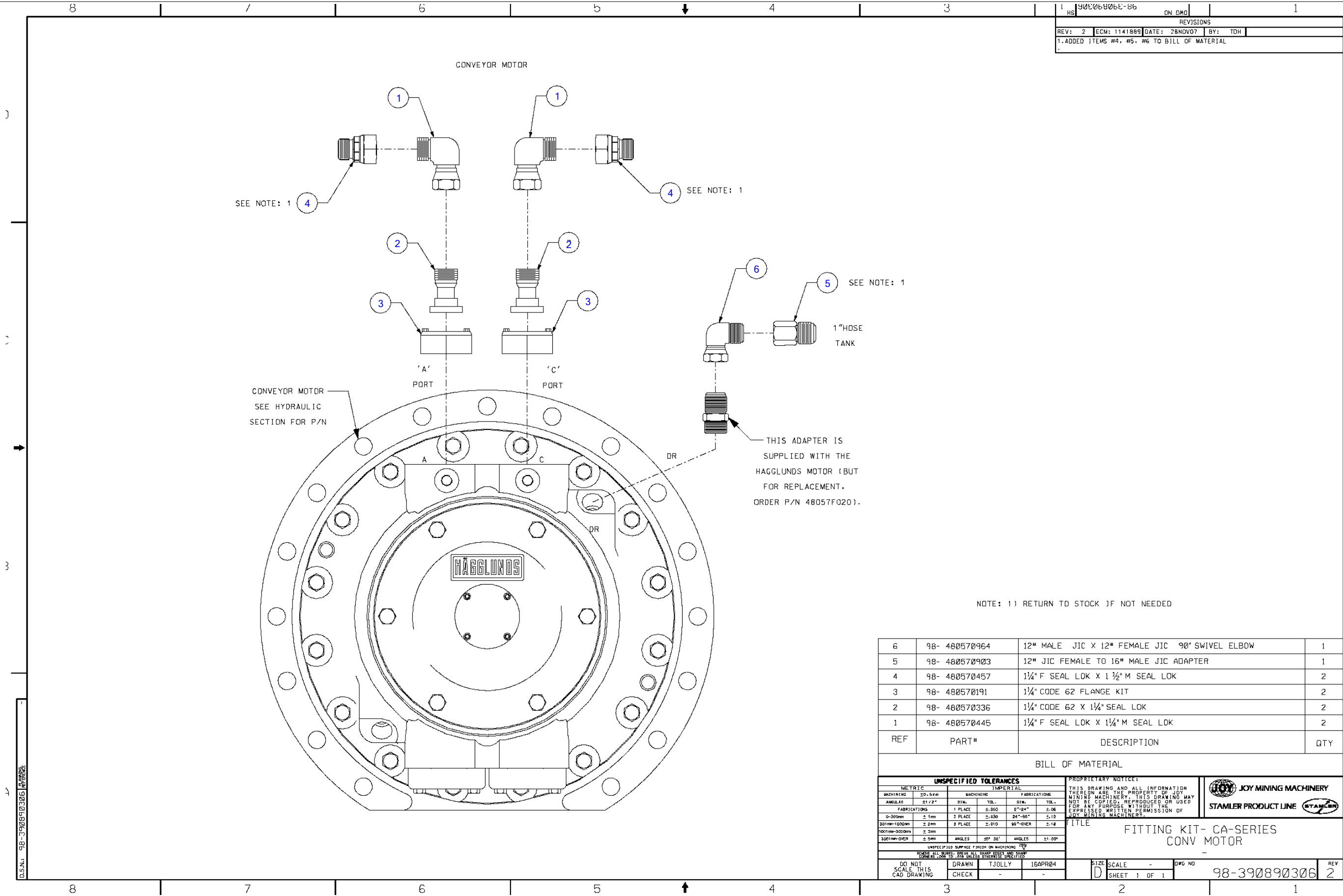


NOTE: 1) RETURN TO STOCK IF NOT NEEDED

ECN	2
REV	2
JOY JOY MINING MACHINERY	
STAMLER PRODUCT LINE STAMLER	
FITTING KIT-FLUSHING VALVE	
(REFERENCE FLUSHING VALVES	
98-490560095 & 98-490560242)	
DATE DRAWN	16APR04
DRAWN BY	TJOLLY
APPROVED BY	
REVIEWED BY	MB
SHEET 1 OF 1	
DWG NO. 98-39-089-0305	



	Item	Part Number	Description	Qty	Units
	ASSY	98-390890305	KIT,FTG-FLUSHING VALVE #490560095 ENG/S	1	EA
	1	98-480570420	• FITTING,HYDR-90DEG NUT ELBOW*	2	EA
	2	98-480571141	• FITTING,HYDR-SAE #6 HEX HD PLUG	2	EA
	3	98-480571144	• FITTING,HYDR-SAE #12 HEX HD PLUG	2	EA
	4	98-480570550	• FITTING,HYDR-STR CONN	1	EA
	5	98-480570337	• FITTING,HYDR-STR CONN	2	EA
	6	98-480570965	• FITTING, HYDR-JIC SWIVEL *	1	EA

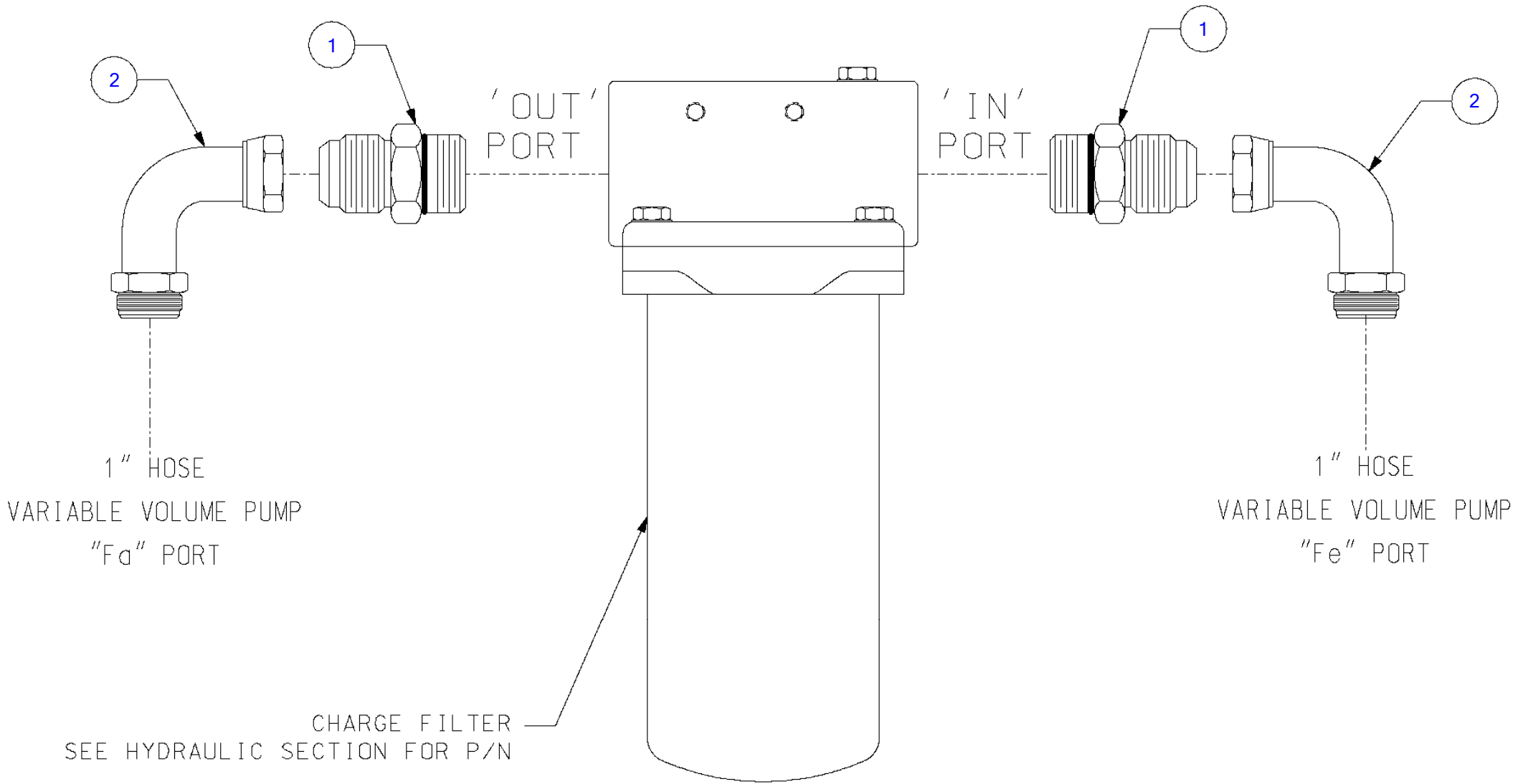




	Item	Part Number	Description	Qty	Units
	ASSY	98-390890306	KIT,FTG-CA-SERIES HYDR MOT/ABVGRD APPL*	1	EA
	1	98-480570445	• FITTING,HYDR-SWVL ELBOW *	2	EA
	2	98-480570336	• FITTING,HYDR-FLNG *	2	EA
	3	98-480570191	• FITTING,HYDR-FLANGE *	2	EA
	4	98-480570457	• FITTING,HYDR-TUBE RDCR *	2	EA
	5	98-480570903	• FLTTING, HYDR-JIC EXPANDER	1	EA
	6	98-480570964	• FITTING, HYDR-JIC SWIVEL	1	EA



REF	PART NUMBER	DESCRIPTION	QTY
1	48057F028	SAE #24 X SAE #16 MJIC ADAPTER	2
2	98-480571071	#16 FJIC X #16 MJIC 90° ELBOW	2



1	
ECN	REV
JOY MINING MACHINERY	
STAMLER PRODUCT LINE	
FITTING KIT-CHARGE FILTER	
DATE DRAWN	DRAWN BY
16APR04	TJOLLY
APPROVED BY	REVIEWED BY
	MB
SHEET 1 OF 1	DWG NO. 98-390890304



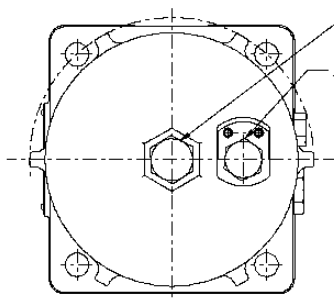
	Item	Part Number	Description	Qty	Units
	ASSY	98-390890304	KIT,FTG-CHARGE FILTER #490520094 ENG/SN	1	EA
	1	98-48057F028	• FITTING,HYDR-STR CONN	2	EA
	2	98-480571071	• FITTING,HYDR-SWIVEL ELBOW *	2	EA



1 HS 98-49-052-F004 1

POSSIBLE DIRT INDICATORS

- 1) ELECTRICAL (CSA / ATEX) - P/N 98-680543542
2) ELECTRICAL SWITCH - P/N 98-68054F013
3) VISUAL POP UP - P/N 98-48011F153



ALTERNATE (PLUGGED) DIRT INDICATOR PORT
(REPLACEMENT PLUG P/N 98-48011F152)

AIR BLEED PORT

SERVICE JOINT SEALS

REPL. BACK-UP RING
P/N 98-48011F170
REPL. O-RING
P/N 98-48011F169

COMPLETE SEAL KIT
P/N 98-39089F106

900 PSI WORKING PRESSURE
SAE #24 ORB PORTS
50 PSID BYPASS VALVE
PLUGGED INDICATOR PORT

PERFORMANCE RATINGS:

HOUSING PRESSURE RATINGS:
MAX. WORKING PRESSURE: 900 PSI (62 BAR)
RATED FATIGUE PRESSURE: 2400 PSI (165 BAR)
(10⁷ CYCLES PER NFPA T2.6.1 1974)

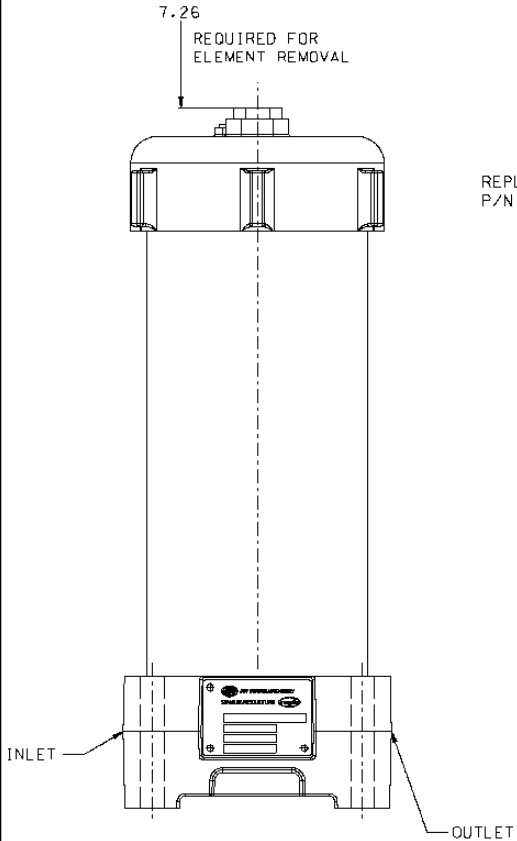
TEMPERATURE RANGE:
NITRILE SEALS: -45°F TO +225°F (-43°C TO +107°C)

FLUID COMPATABILITY:
PER ANSI/B93-23-1972 (ISO 2943)
SEALS: NITRILE: WATER-GLYCOL, EMULSIONS, WATER-OILS & PETRO FLUIDS.

MATERIALS OF CONSTRUCTION:
HOUSING:
HEAD: DUCTILE IRON (SGI CASTING)
TUBE: DRAWN STEEL
CAP: DUCTILE IRON
VALVE: STEEL

FINISH: CORROSION RESISTANT BLACK POLYURATHANE PAINT

ELEMENT:
ENDCAPS: STEEL W/ CORROSION RESISTANT PLATED FINISH
ORE TUBE: STEEL W/ CORROSION RESISTANT PLATED FINISH
MEDIA: INERT INORGANIC FIBERS IMPREGNATED

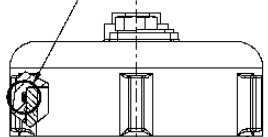


INLET

OUTLET

REPLACEMENT O-RING
P/N 98-480110126
REPLACEMENT O-RING
P/N 98-2014N9

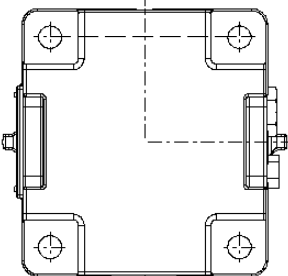
REPLACEMENT O-RING
P/N 98-480110126



SERVICE JOINT

PERMANENT JOINT

PRIMARY DIRT INDICATOR
PORT (PLUGGED) - (REPLACEMENT
PLUG P/N 98-48011F152)

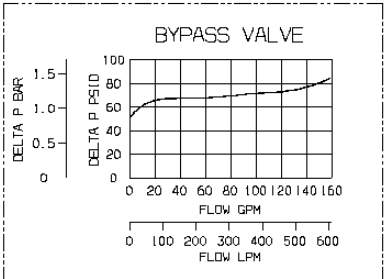
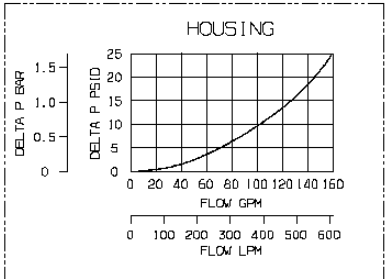


PERMANENT JOINT SEALS

REPL. BACK-UP RING
P/N 98-48011F171
REPL. O-RING
P/N 98-48011F042

PRESSURE DROP VS FLOW

(DATA BASED ON .9 SPECIFIC GRAVITY FLUID WITH A VISCOSITY OF 150 SUS (30 Cst.))



ELEMENT PRESURE DROP 0.07 PSI/GPM

UNSPECIFIED TOLERANCES			
MACHINING	FABRICATIONS		
DIM.	TOL.	DIM.	TOL.
1 PLACE	±.060	0"-24"	±.06
2 PLACE	±.030	24"-96"	±.12
3 PLACE	±.010	96"-OVER	±.18
ANGLES	±0° 30'	ANGLES	±1.00°
UNSPECIFIED SURFACE FINISH ON MACHINING .250			
REMOVE ALL BURRS, BREAK ALL SHARP EDGES AND SHARP CORNERS .005 TO .015 UNLESS OTHERWISE SPECIFIED			
DO NOT SCALE THIS CAD DRAWING			
DRAWN	PRICE	23MAR06	
CHECK	-	-	

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EXPRESSED WRITTEN PERMISSION OF
JOY MINING MACHINERY.



TITLE
PROPRIETARY PALL FILTER ASSY.
900 PSI WORKING PRESSURE

SIZE	SCALE	DWG NO	REV
B	4	98-49-052-F004	A
SHEET	1 OF 1		

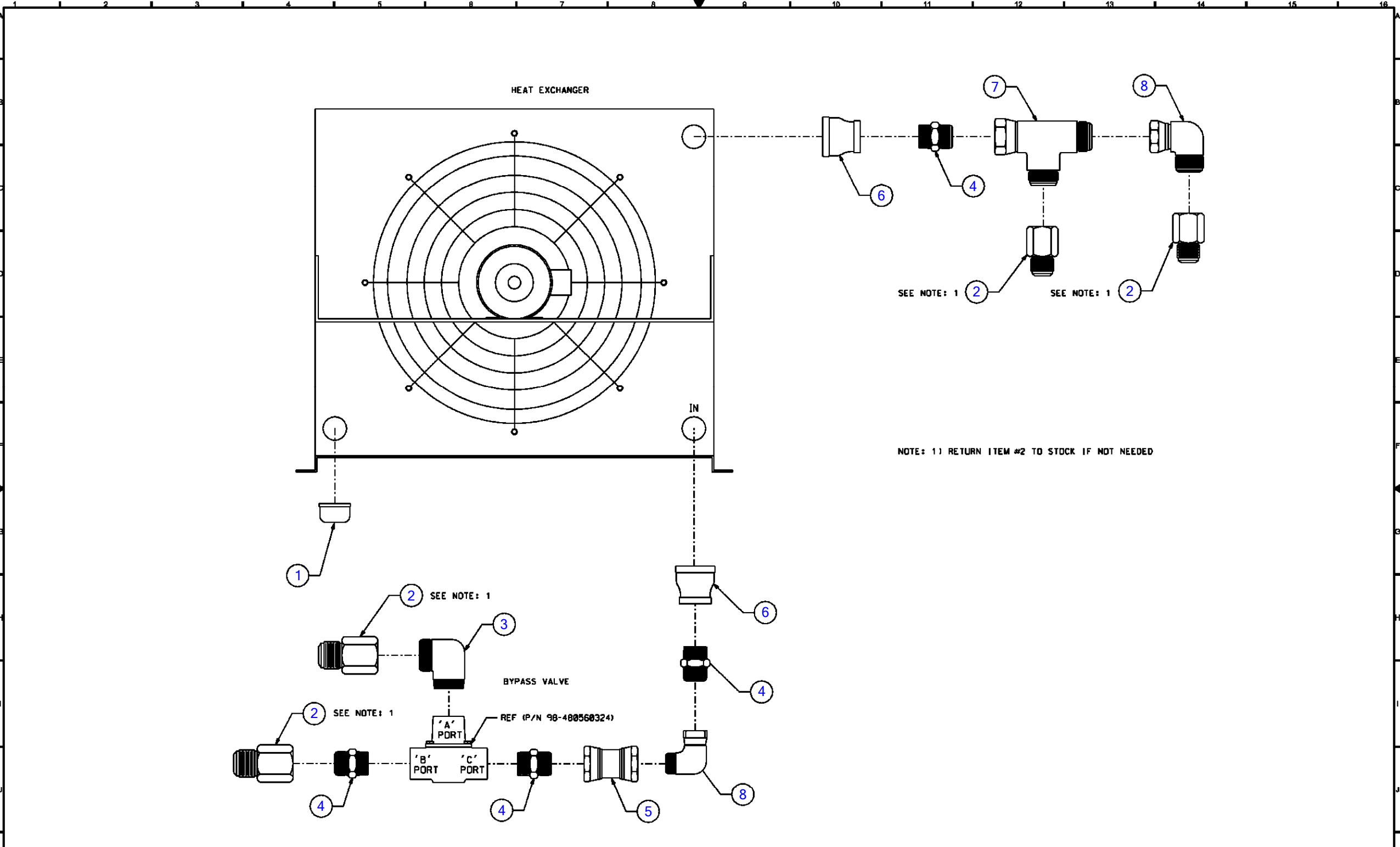
REPLACEMENT FILTER ELEMENT
P/N 98-48011F111



Item	Part Number	Description	Qty	Units
A	98-49052F004	ASSY,PALL NAMEBRANDED)*	1	EA



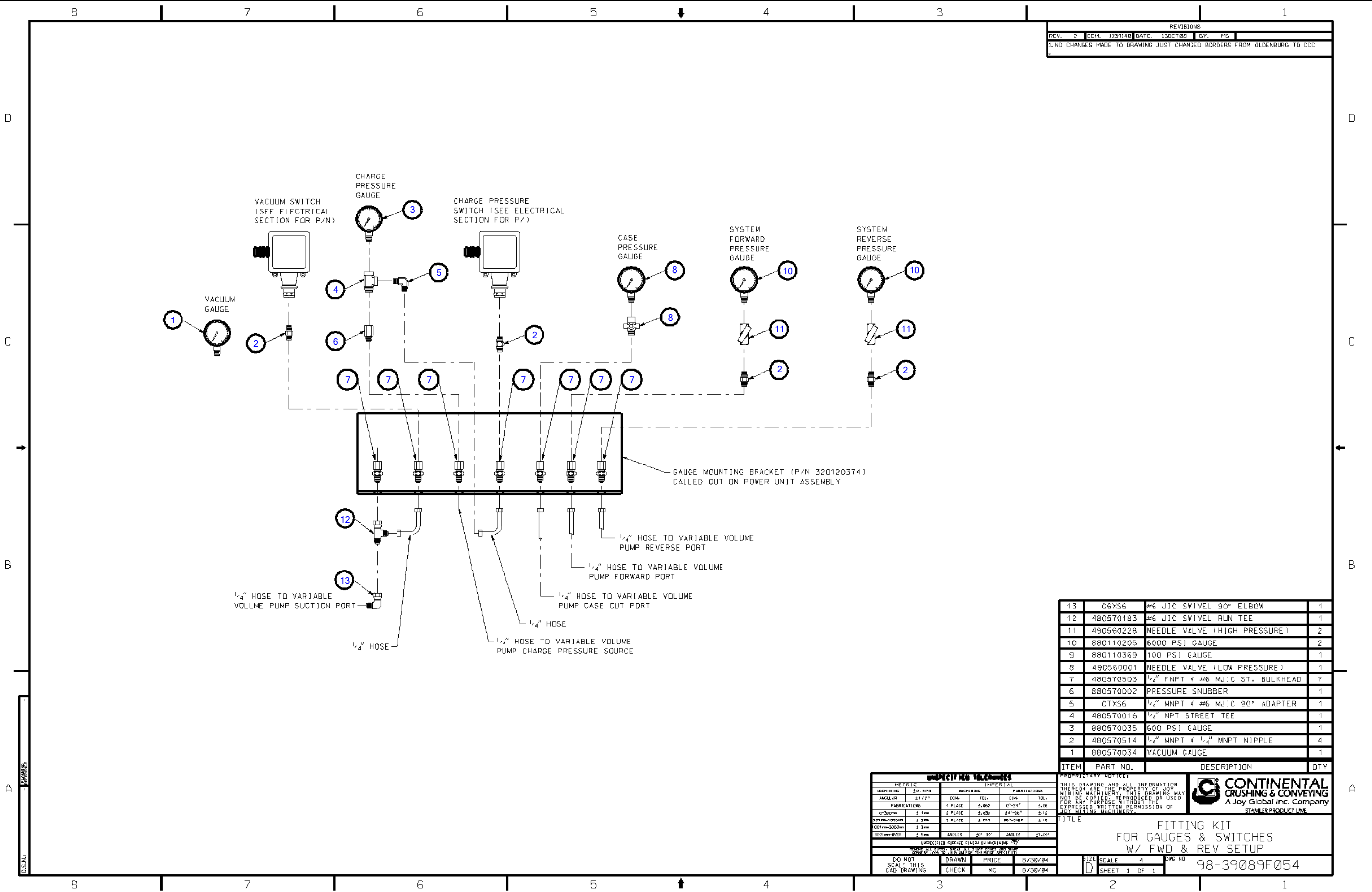
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						SYM	MICRO	SYM	MICRO																																																																																																				
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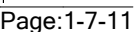


	Item	Part Number	Description	Qty	Units
	ASSY	100313436	KIT,FITTING-HEAT EXCHANGER	1	EA
	1	98-480570098	• FITTING,HYDR-PIPE *	1	EA
	2	98-480570910	• FITTING, HYDR-JIC REDUCER	4	EA
	3	98-480571053	• FITTING,HYDR-MALE ELBOW *	1	EA
	4	98-480571035	• FITTING,HYDR-ST ADAPTER *	4	EA
	5	98-480571084	• FITTING,HYDR-ST FJIC UNION *	1	EA
	6	98-480570095	• FITTING,HYDR-RDCR *	2	EA
	7	98-480570890	• FITTING, HYDR-SWVL NUT RUN TEE	1	EA
	8	98-480570967	• FITTING, HYDR-JIC SWIVEL *	2	EA



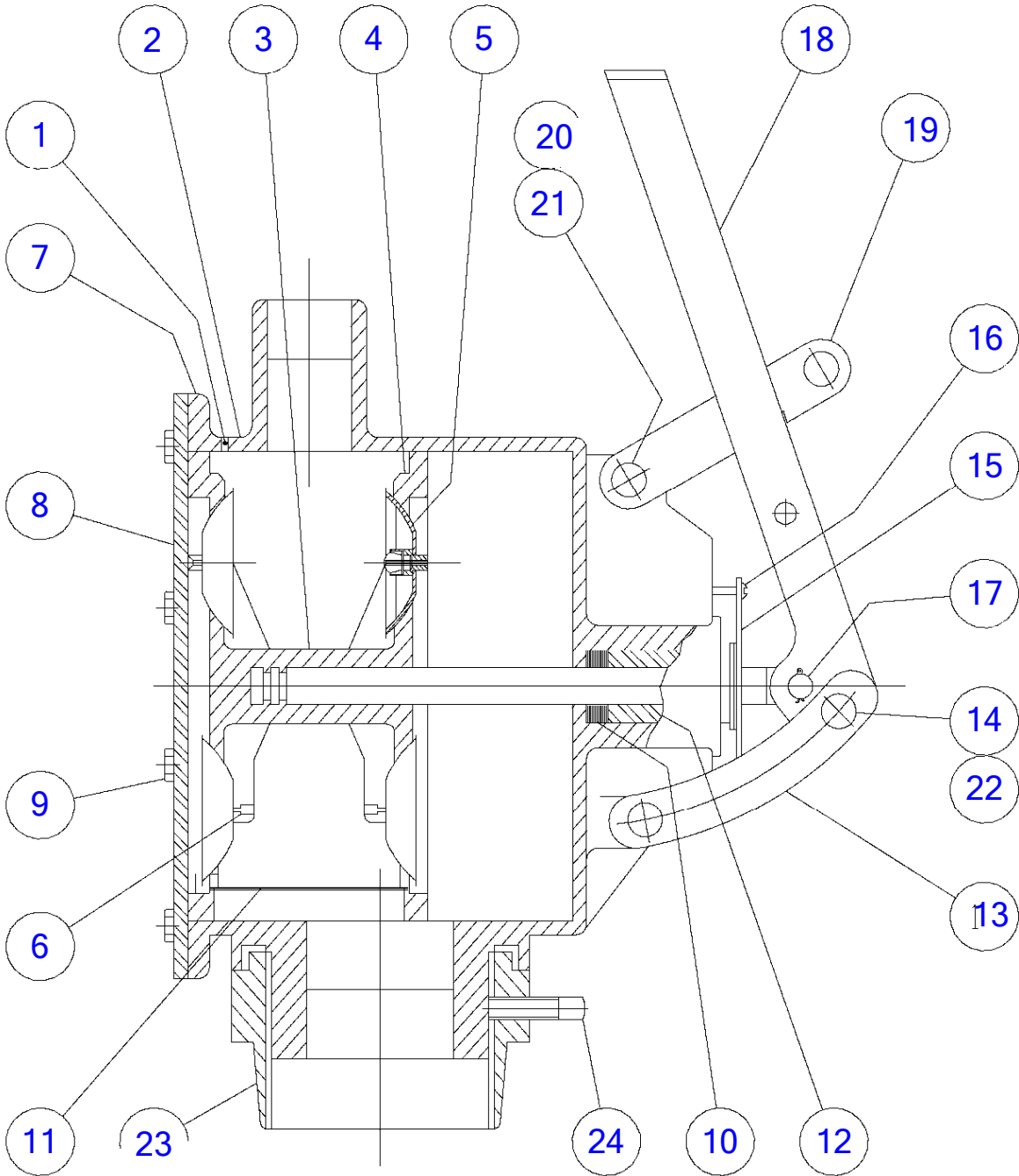


	Item	Part Number	Description	Qty	Units
	ASSY	98-39089F054	KIT,FITTING-GAUGE & SWITCH ENG REQ.	1	EA
	1	98-880570034	• GAUGE,VACUUM *	1	EA
	2	98-480570514	• FITTING,HYDR-MALE *	4	EA
	3	98-880570035	• GAUGE,PRESSURE *	1	EA
	4	98-480570016	• FITTING,HYDR-STR	1	EA
	5	98-480571039	• FITTING, HYDR-90 DEG ADAPTER	1	EA
	6	98-880570002	• SNUBBER,PRESSURE	1	EA
	7	98-480570503	• FITTING,HYDR-FEM CONN	7	EA
	8	98-490560001	• VALVE,NEEDLE	1	EA
	9	98-880110369	• GAUGE,PRESSURE *	1	EA
	10	98-880110205	• GAUGE,PRESSURE *	2	EA
	11	98-490560228	• VALVE,NEEDLE	2	EA
	12	98-480570183	• FITTING,HYDR-SWVL RUN TEE*	1	EA
	13	98-480570962	• FITTING, HYDR-JIC SWIVEL	1	EA





	Item	Part Number	Description	Qty	Units
	ASSY	98-310180273	ASSY,TANK-HYDR 116 GAL	1	EA
	A	98-370180460	• ASSY,TANK-HYDR GAL	1	EA
	B	98-880110053	• GAUGE,OIL LEVEL/TEMPERATURE	1	EA
	C	98-4190	• PLATE,MTG-HYDR SUCT STR	1	EA
	D	98-880110904	• PLUG,BREATHER	1	EA
	E	98-480570184	• FITTING,HYDR-90DEG ELBOW	1	EA
	F	98-320110467	• BAFFLE,HYDR TANK	1	EA
	G	98-A5A0813020	• SCREW,HEX CAP *	12	EA
	H	98-A5A0616012	• SCREW,HEX CAP *	28	EA
	J	98-A5P0800000	• WASHER,LOCK	14	EA
	K	98-A5P0600000	• WASHER,LOCK *	28	EA
	L	98-320110468	• BAFFLE,HYDR TANK	1	EA
	M	98-A5A0813024	• SCREW,HEX CAP *	2	EA
	P	98-8536	• ANGLE,HYDR MTG	2	EA
	Q	98-320570439	• FITTING,HYDR-TUBE CONN	1	EA
	R	98-480570141	• GLASS,SIGHT	1	EA
	S	98-300870001	• GASKET	1	EA
	T	98-880870001	• VELLUMOID *	3	M
	U	98-880110035	• NIPPLE,CLOSE-2" 2"LG *	1	EA
	V	98-490520009	• STRAINER *	1	EA



ITEM	PART No.	DESCRIPTION / SIZE	WGT EA	QTY
1	98-780110001	BODY, HAND PUMP	-	1
2	98-780110002	LINER, HAND PUMP BODY	-	1
3	98-780680001	ASSY, HAND PUMP - PISTON/SHAFT	-	1
4	98-780030001	RING, PISTON	-	2
5	98-780560001	ASSY, VALVE	-	4
6	98-780480002	HOLDER, SPRING	-	4
7	98-780870001	GASKET, HAND PUMP COVER	-	1
8	98-780130001	COVER, HAND PUMP	-	1
9	98-A5A0420008	SCREW, HEX HEAD CAP	-	8
10	98-780110003	PACKING	-	1
11	98-780110004	SCREEN, HAND PUMP	-	1
12	98-780430001	BEARING, HAND PUMP	-	1
13	98-780110005	LINK, HAND PUMP	-	2
14	98-780280001	PIN, LINK-CLEVIS	-	2
15	98-780140001	PLATE, HAND PUMP BEARING	-	1
16	98-A2FS324016	SCREW, SLOTTED ROUND HEAD MACHINE	-	2
17	98-780280002	PIN, HAND PUMP CLEVIS SHAFT	-	2
18	98-780590001	HANDLE, HAND PUMP	-	1
19	98-780110006	LINK, HAND PUMP LOCKING	-	1
20	98-780110007	RIVET, HAND PUMP	-	1
21	98-780110008	WASHER, TENSION	-	1
22	98-780280003	PIN, HAND PUMP COTTER	-	3
23	98-780110009	ADAPTER, HAND PUMP BUNA W/ SET SCREW	-	1
24	98-780110010	BREAKER, HAND PUMP VACUUM	-	1

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ECN	REV



JOY MINING MACHINERY

STAMLER PRODUCT LINE



PARTS LIST ILLUSTRATION -
HAND PUMP

DATE DRAWN	5DEC85	DRAWN BY	DH	APPROVED BY	
				REVIEWED BY	JM
SHEET 1 OF 1			DWG NO. 98-730260036		

REDRAWN ON CAD 24 JULY 86 BY K.SLEDD.



	Item	Part Number	Description	Qty	Units
	A	98-730260036	PUMP,HAND	1	EA
	1	98-780110001	BODY,HAND PUMP *	1	EA
	2	98-780110002	LINER,HAND PUMP BODY *	1	EA
	3	98-780680001	ASSY,PISTON/SHAFT-HAND PUMP*	1	EA
	4	98-780030001	RING,PISTON	2	EA
	5	98-780560001	ASSY, VALVE	4	EA
	6	98-780480002	HOLDER,SPRING*	4	EA
	7	98-780870001	GASKET,HAND COVER*	1	EA
	8	98-780130001	COVER,HAND PUMP*	1	EA
	9	98-A5A0420008	SCREW,HEX CAP *	8	EA
	10	98-780110003	PACKING,HAND *	1	EA
	11	98-780110004	SCREEN,HAND *	1	EA
	12	98-780430001	BEARING,HAND *	1	EA
	13	98-780110005	LINK,HAND *	2	EA
	14	98-780280001	PIN,LINK-CLEVIS	2	EA
	15	98-780140001	PLATE,HAND PUMP BEARING*	1	EA
	16	98-A2FS324016	SCREW,ROUND HEAD MACH-SLOTTED*	2	EA
	17	98-780280002	PIN,HAND CLEVIS SHAFT*	2	EA
	18	98-780590001	HANDLE,HAND PUMP*	1	EA
	19	98-780110006	LINK,HAND PUMP LOCKING *	1	EA
	20	98-780110007	RIVET,HAND PUMP *	1	EA
	21	98-780110008	WASHER,HAND TENSION *	1	EA
	22	98-780280003	PIN,HAND COTTER*	3	EA
	23	98-780110009	ADAPTER,HAND PUMP BUNA W/SET SCREW*	1	EA
	24	98-780110010	BREAKER,HAND PUMP VACUUM *	1	EA

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JOY MINING MACHINERY

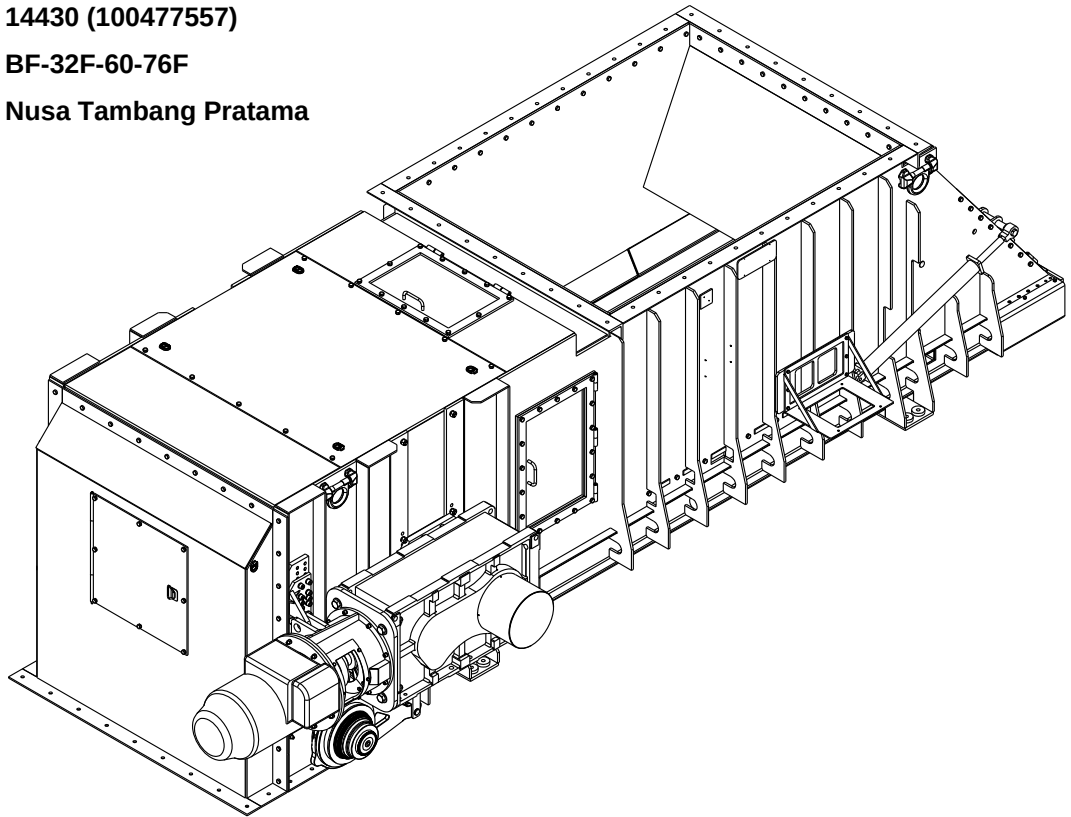
STAMLER PRODUCTS

Volume

1

STATIONARY BELT FEEDER BREAKER

Serial Number: 14430 (100477557)
Model Number: BF-32F-60-76F
Customer: Nusa Tambang Pratama



Service Manual

Revisions & Updates

Revision Level	Date	Description
A	17 June 2011	AS FIRST ISSUED



JOY MINING MACHINERY

STAMLER PRODUCTS

Belt Feeder Breaker Service Manual

JOY Mining Machinery
Stamler Products
100 Stamler Street
Millersburg, Kentucky 40348
Phone (859) 484-3431 ♦ Fax (859) 484-3102

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JOY MINING MACHINERY

STAMLER PRODUCTS

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JOY MINING MACHINERY
STAMLER PRODUCTS

Section

1

1. SAFETY

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General Safety Precautions

WARNING

TO PREVENT SERIOUS BODILY INJURY
OBSERVE THE FOLLOWING

1. Turn off circuit breaker & lockout before:

- A. Entering the hopper or going near the breaker.
- B. Removing or replacing a shear pin (if equipped).
- C. Touching conveyor chain, flights, couplings, drive chains, or other moving parts.
- D. Removing guards & covers. (Replace all guards & covers before restoring power.)

2. Do not:

- A. Handle energized trailing cable without approved protective gloves.
- B. Open any permissible enclosures, such as a motor controller, in an area where it is not safe to do so.
- C. Perform any maintenance that requires the machine to be running unless a second person is present to shutdown the machine in an emergency.
- D. Disconnect any hydraulic or grease fittings, which are under pressure.

3. Ensure that:

- A. All operators are familiar with how the machine operates and its safety provisions.
- B. Maintenance personnel periodically inspect that all safety devices are properly installed and functional.
- C. Only the proper sized shear pins are used (if equipped).
- D. Hopper, breaker and conveyor areas are clear and no one is in or near them, before starting the breaker or conveyor.
- E. Circuit breaker, which supplies power via the trailing cable to the machine, is open and tagged out before opening the motor controller enclosure.

Lockout Procedure

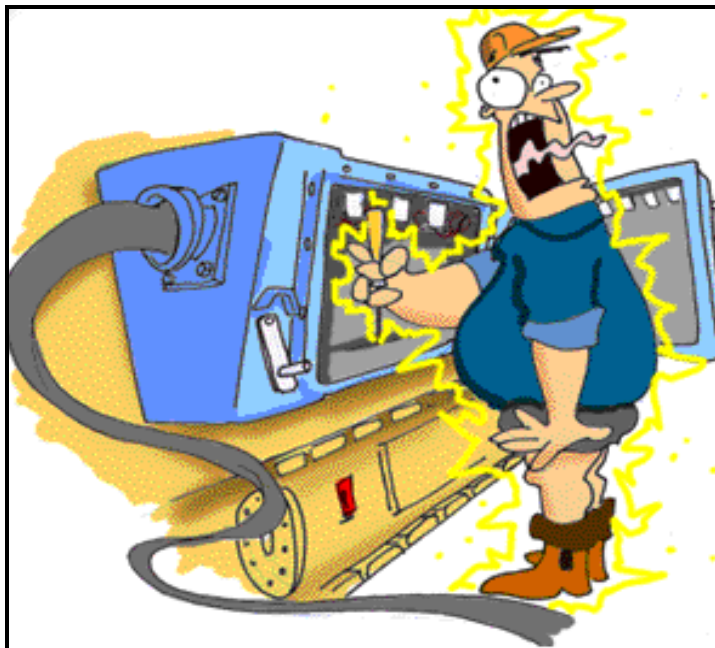
To insure that the power is off and cannot be inadvertently restored when personnel are repairing, adjusting, inspecting, or maintaining this equipment, the following lockout procedure is deemed vital to prevent injury or death to those involved.

The main circuit breaker, which is used to connect electrical power to the unit, is equipped with a padlock device. By turning this circuit breaker to the "off" position, attaching a padlock to the locking device, and locking the padlock, the machine is now isolated from all electrical power, except the line side of the circuit breaker. Also, if the person working on the machine has custody of the padlock key there is no way that power can be restored without their knowledge.

With the electrical power isolated in this way the machine will not start from any control station and all electrical circuits fed by this circuit breaker are no longer energized. Thus any maintenance tasks, which would otherwise be dangerous, can now be accomplished.

When instructions in the manual, or posted, call for "LOCKOUT THE CIRCUIT BREAKER" the following should be done:

1. Locate the circuit breaker, which when turned off removes all power to the electrical system. (This has not been supplied by Stamler, so it may not be located in the immediate area of the machine.)
2. Turn the circuit breaker off.
3. Attach a padlock to prevent the circuit breaker from being turned on.
4. Check that power has been removed from the system by testing with a voltmeter. Check the line side of the circuit breaker; voltage should be present and the meter should indicate this. Next check the load side of the circuit breaker on all three phases to ensure no voltage is present.
5. Maintain custody of the padlock key until the work is completed.
6. Visually inspect the interior and exterior areas of the machine to make sure that no one is in or near the machine before removing the padlock and restoring power to the machine.



Operational Safety Instructions

IT IS THE BASIC POLICY OF STAMLER TO MANUFACTURE PRODUCTS THAT ARE SAFE AND RELIABLE. HOWEVER, THERE IS ALWAYS AN ELEMENT OF RISK WHEN OPERATING AND MAINTAINING HEAVY-DUTY EQUIPMENT.

TO HELP MINIMIZE THE RISKS AND PROMOTE SAFETY AT ALL TIMES, THIS SECTION DETAILS A NUMBER OF SAFETY RULES WHICH SHOULD ALWAYS BE FOLLOWED AND OBEYED.

READ THIS ENTIRE SECTION AND REVIEW ALL OF THE SAFETY MESSAGES CAREFULLY. ENSURE THAT ALL PERSONNEL RESPONSIBLE FOR THE OPERATION AND MAINTENANCE OF THIS MACHINE HAVE READ AND UNDERSTAND THIS SECTION, AS WELL AS THE ENTIRE SERVICE MANUAL. IT IS IMPORTANT THAT THESE SAFETY RULES ARE APPLIED ON THE JOB.

IF ANY OF THE INFORMATION IN THIS SERVICE MANUAL CONTRADICTS ANY STATE OR FEDERAL MINING LAW OR REGULATION AS WELL AS ANY CORPORATE SAFETY RULES APPLIED BY THE PURCHASER, PLEASE CONTACT STAMLER IMMEDIATELY.

REMEMBER, ALL FEDERAL AND STATE MINING LAWS AND REGULATIONS MUST BE STRICTLY OBEYED.

Operating Safety Precautions

Always disconnect power and lockout the power supply before entering or performing maintenance on the equipment.

Personnel must be fully trained to operate this piece of equipment. Serious injury or death to the operator or individuals in the vicinity of the feeder breaker may result from the actions of an inexperienced, untrained operator.

Know where all of the machine's **ALL STOP** switches are located and how to properly operate them. This is very important in the case of an emergency.

To avoid serious injury or death from an over loaded machine, get to know the capabilities and limitations of the equipment and learn the most *efficient and safe operating techniques*.

To avoid serious injury or death from improper operation, comply with all instructions associated with the operation of this machine.

Inspect the machine daily for signs of damage, unusual wear, fluid leaks or faulty operation that may result in machine failure and ultimately, personal injury or death.

Never remove any safety covers or devices from the machine while operating. Carry no passengers. This equipment does not provide and is not approved for seating of an operator or passengers. Operation with unsecured passengers may result in serious injury or death to the individuals.

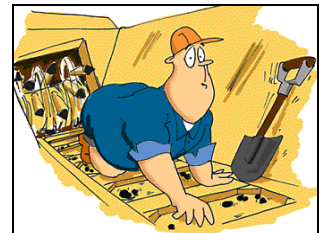
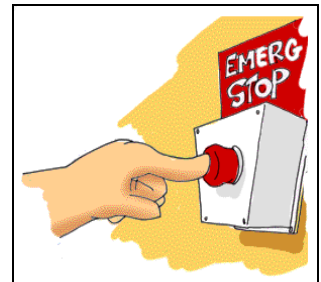
To avoid serious injury or death to self or others, stay alert. Operate the machine only when *physically fit and not under the influence of alcohol or drugs*.

Servicing Safety Precautions

To avoid serious injury or death, read and thoroughly understand maintenance procedures before performing them.

Never perform maintenance while the machine is moving or operating. Performing maintenance on a moving machine can result in crushing, entanglement in moving parts, electrocution, or serious burns to personnel.

Never perform maintenance on a hot machine. Performing maintenance on hot components can cause serious burns to skin. Allow machine to cool before performing maintenance.



Personal injury can result from unsafe maintenance practices. Before performing maintenance or repair work on any equipment, consult the manufacturer's service manual and follow the recommended procedures.

Keep your head, hands, legs, feet and loose clothing away from rotating and moving parts. Entanglement of body parts in rotating and moving parts can result.

Pay attention to warning labels on the machine.

When equipped with crawler drive units the machine is designed to turn a full 360°. Because of attachments and working location care should be taken to ensure that the area is clear and that any cables and hoses attached are able to travel with the machine. The equipment should never be turned more than is necessary, this could cause damage to the cables and hoses or put the operator, or others, at risk of injury.

Always use the proper tools for the job. Repair or replace any broken or damaged tools including lifting equipment immediately. Personal injury can result from the use of inadequate tools.

Pressurized hydraulic oil can penetrate skin and cause serious injury or death. Pressure can be maintained in hydraulic circuits long after the power source and pump have been shut down. Lower the hydraulic implements to the ground; provide support for items supported by the hydraulic cylinders and relieve all trapped pressures before performing any service work to the hydraulic components.

Pressurized hydraulic fluid escaping from a faulty component can penetrate the skin and cause serious injury or death. When troubleshooting a hydraulic system for leaks, use a piece of cardboard or other material as a detector rather than your hand. Do not change the pressure setting of any hydraulic valves. Changing pressure settings may result in pressures exceeding the ratings of the components.

Use caution when draining hot fluids from the machine. Allow machine to cool before draining fluids. Splashing hot fluid can cause serious burns.

When performing weld repairs, remove paint before welding or heating. Heated paint can create hazardous fumes. Perform weld repairs outside or in a well-ventilated area. Avoid breathing dust when sanding or grinding paint. An approved respirator should be worn when sanding or grinding. Remove stripper and solvent with soap and water and allow fumes to disperse for at least 15 minutes before welding or heating.

Fire Prevention

Clean the combustible debris from the machine after each operation. When working in a mining environment, it is impossible to prevent combustible debris from collecting in tight corners of the machine. This debris, in itself, may cause a fire; however, when mixed with oil or grease in a hot and confined place, the danger of fire is increased greatly.

Do not weld or heat near pressurized hydraulic lines. Damaging a line during heating or welding can result in flammable spray and serious burns. To reduce the chances for a fire to breakout, follow these preventive instructions:

1. Clean dust and combustible debris from the machine after each operation.
2. Inspect the machine daily for potential fire hazards and make any necessary repairs immediately.
3. Always ensure that excess grease and oil accumulation, including spillage, is cleaned up immediately.
4. Use only nonflammable solutions for cleaning the machine or machine components.
5. Store oil rags and other combustible materials in a safe and fireproof location.
6. Before performing repair work such as welding, the area surrounding the repair location should be cleaned, covered with a non-combustible material and a fire extinguisher positioned close by.

Fire Suppression System

When equipped with fire suppression the system uses nitrogen gas stored in sealed cylinders at extremely high pressure. Puncture of the seal will result in release of high-pressure gas that cannot be stopped. An uncontrolled release of this high-pressure gas can result in violent and rapid propulsion of the cylinder, capable of causing personal injury, death and/or equipment damage.



JOY MINING MACHINERY
STAMLER PRODUCTS

Section

2

2. OPERATION

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General Description – Belt Feeder Breaker

Stationary Belt Feeder Breaker, Dual Hydraulic Conveyors

The Stamler Stationary Belt Feeder Breaker is a conveyor system that incorporates a pick type breaker to reduce the size of the material being conveyed through the machine. The feeder breaker is a single piece of equipment that is installed at the customer's facility to receive material and size it before being conveyed to another area of the operation.

The Stamler Stationary Belt Feeder Breaker is designed and built specifically to suit the customer's operation. The machine consists of dual Drag Conveyors, a Rotary Pick Breaker, an Electrical System and a Hydraulic System. These systems are designed to provide dependable long lasting service with proper operation and preventative maintenance.

The feeder breaker design is such that the material is loaded into a hopper at the intake (receiving) end of the machine. The material is held in the intake hopper as it is being conveyed through the pick breaker, which sizes the material by the force of the sharp pointed picks striking down onto the material. The material, once reduced in size, is then moved through the machine and out of the discharge end to receiving equipment of the customer's choosing.

The feeder breaker must be **Operated** and **Maintained** as outlined in the Service Manual. To accomplish this all persons responsible for the operation and maintenance of this machine must read and understand the instructions and procedures described in the service manual. Should there be any items that are not understood or conflict with information in another section of the manual the manufacturer must be consulted. Also a machine "Startup Performance Sheet" must be filled out during commissioning and returned to Stamler.

Dual Drag Conveyors

- o Each conveyor consists of two strands of engineered chain with flights (cross members) attached at regular intervals, to move the material. The chain & flights are wrapped around and engage the sprockets attached to individual headshafts located at the discharge end.
- o A direct drive hydraulic motor coupled to each headshaft is used to drive the conveyors. Relief valves are provided in the hydraulic system to protect the conveyors in the event of a stall, etc.
- o A tailshaft is provided at the intake end of each conveyor as an idler to maintain the correct tension of the conveyor chains.
- o The conveyor deck is lined with abrasion resistant plate to resist wear.

Pick Breaker

- o The breaker is constructed of a solid steel shaft and breaker tube that has support rings welded to it. Pick holders are welded to the rings and replaceable picks are inserted into the holders. The breaker is suspended at each end by large roller bearings mounted in housings. The breaker bearing housings are bolted to either side of the machine frame. The breaker rotates with the material flow allowing the picks to strike the material as it flows towards the discharge end of the machine.
- o The breaker shaft is driven by a direct drive system consisting of an electric motor and speed reducer mounted to each end of the breaker shaft. Torque-limiting couplings are positioned in the breaker drive systems to provide relief protection in the event of a stall, etc.

Electric System

- o The electric system of the machine is designed and built to operate on “AC” electrical power, supplied to the machine by means of a “trailing cable” connected to the customer's power supply. The electrical system is used to control the functions of the machine:
 - Breaker Drive
 - Conveyor Drives
 - Hydraulics
 - Safety Devices (horns, lights, etc.)
 - Sensing Devices (breaker speed, oil level, oil temperature, etc.)
 - Heating/Cooling of the hydraulic oil
- o The functions of the electric system are controlled remotely by a 4-20mA signal or manually at the motor controller box, mounted on or near the machine or at a customer specified location. All stop pushbuttons are supplied on the motor controller and along the side of the machine, to shutdown the feeder breaker in the event of an emergency.

Hydraulic System

- o The hydraulic system of the feeder breaker is used to operate the following systems:
 - Conveyor Drives
- o Hydraulic pumps mounted to electric motors supply power for the hydraulic system. Hydraulic oil is stored in an oil reservoir (tank). The hydraulic system is designed with provisions for both heating and cooling of the hydraulic oil, by means of tank heaters and heat exchangers, depending on the working environment and the system operating temperature. The hydraulic system is supplied with filtration to clean the oil and help prevent contaminants from damaging the vital components of the system.
- o The hydraulic system is equipped with hydraulic pressure gauges to help provide safe, reliable operation of the feeder breaker. Hydraulic pressure gauges are installed so that operators may visually inspect the system pressures during operation. The gauges are used to monitor the following functions:
 - Pump Operating Pressures
 - Conveyor Drive System Pressures

Site Commissioning

Stationary Belt Feeder Breaker, Dual Hydraulic Conveyor

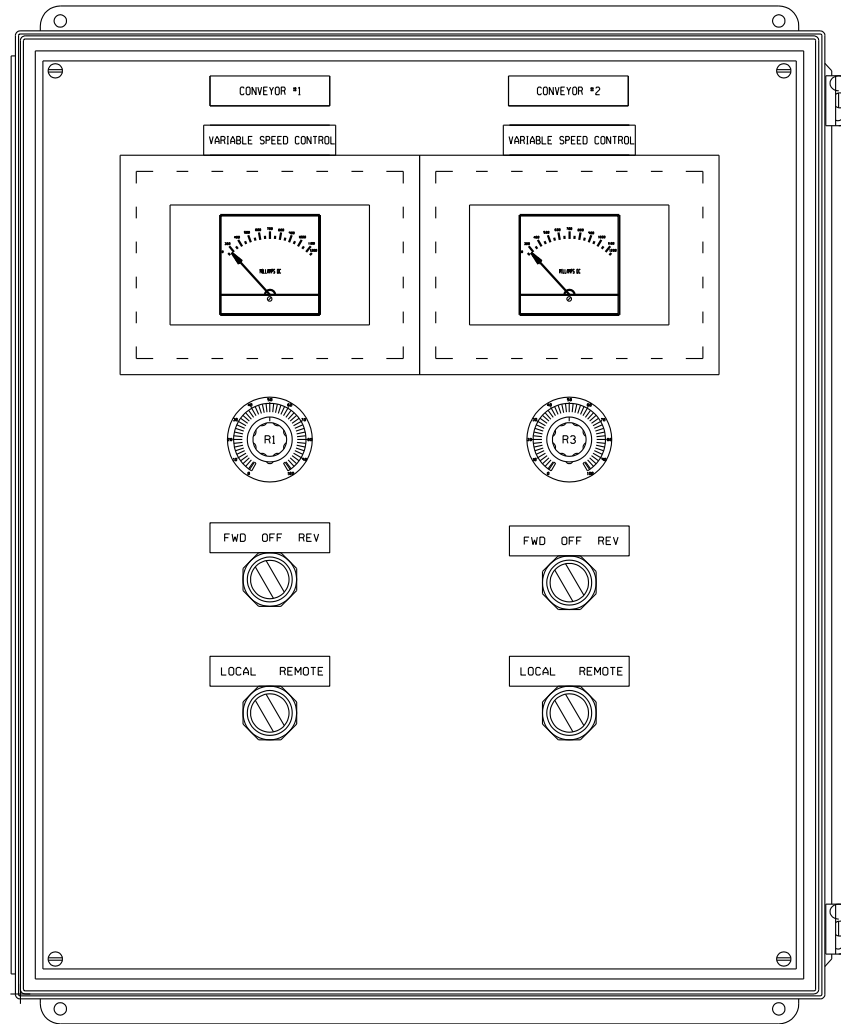
1. These steps are to be completed before the trailing cable is connected.
 - a. Ensure the machine is properly installed and securely fastened where it will be operating.
 - b. Ensure the each drag conveyor chain is properly tensioned.
 - c. Ensure the breaker and conveyor drives are tensioned properly.
 - d. Make sure all bolts between the motors, speed reducers, and their associated mounting plates are properly tightened.
 - e. Ensure the hydraulic oil reservoir (tank) is filled with the proper fluid.
 - f. Ensure all speed reducers are filled with the proper lubricant.
 - g. Ensure the cases of all hydraulic pumps are filled with hydraulic oil.
 - h. Lubricate all grease fittings on the machine.
 - i. Check the alignment of all couplings; lubricate any that are not an elastomer type.
 - j. Open all circuit breakers on the motor controller.
2. Connect the trailing cable and energize the high voltage section by closing the high voltage disconnect switch.
3. Close the necessary circuit breakers on the motor controller.
4. Start all of the electric motors and test run.
5. While running the hydraulic pump motor, check the hydraulic system.

Stopping the Machine:

1. Normal Stopping: Depress the system stop pushbutton.
2. Emergency Stopping: Depress any one of the all stop buttons located on the machine.

Operating Instructions – Belt Feeder Breaker

Machine Controls



Control Panel

The stationary feeder breaker includes a control panel mounted to the power unit or in another location of your choosing. The controls have been designed to allow operation from either from a linked control station or the local control box. Controls available locally include:

1. **REMOTE/LOCAL Mode Switch** – this two-position switch allows a local operator to switch between local control of the feeder breaker and control from a remote control station.
2. **FWD/OFF/REV Direction Switch** – in LOCAL mode, this three-position switch allows a local operator to control the direction of travel for both conveyors. The conveyors can be run in either the forward (towards the discharge end of the feeder breaker) or reverse (toward the intake end of the feeder breaker) directions, or the conveyors can be turned off.
3. **LOCAL SPEED CONTROL Pots** – these pot switches can be used to control each conveyor speed within its minimum and maximum design limits. Turning the dial to the right (in the clockwise direction) increases the corresponding conveyor's speed up to its maximum speed. Turning the dial to the left (in the counter-clockwise direction) decreases the conveyor's speed to its minimum speed.

Breaking and Conveying Material

WARNING

To avoid serious injury to the operator or other personnel do not start the machine until a check of the hopper, breaker, conveyor and area surrounding the feeder breaker has been completed to ensure all personnel are clear.

1. Ensure the belt speed switch is connected (if equipped) and that any receiving belt is actually operating.
2. Set the main circuit breaker for the machine to the ON position.
3. Set the mode selector switch on the control box to the REMOTE position and set the conveyor direction switch to FWD.
4. Start the machine using the controls provided.

NOTE

If the feeder breaker electrical system is equipped with a belt speed switch, it will prevent the conveyor from operating if the discharge belt conveyor is stopped.

Stopping

Depress the red pushbutton on any of the ALL STOP switches located on the machine. This will cause the electric motor to stop, ceasing all functions of the machine.

CAUTION

When using an ALL STOP switch to stop the machine, it must be remembered that this does not remove electrical power from the machine's electrical system. The operator must move the circuit breaker to the **OFF** position to remove power from the system.

Storage

If the machine is to be left for an extended period of time without operating, the circuit breaker should be set to the OFF position, or the power cable disconnected.

Instruction Sheet:

Number:	380210474
By:	SP
Date:	05/21/2001

All Stop Switches

The electrical system of the feeder breaker incorporates **All Stop** switches to shut down the machine in the event of an emergency. The all stop switches are large red pushbuttons and are easy to see. The all stop switches are easily accessible and are usually located on or near the machine in areas where personnel are required to work.

When any one of the all stop switches is engaged, power is removed from the control circuit. The power is removed by means of the “UV” relay opening (in the motor controller), which removes power to the remainder of the electrical circuit below the relay. When electrical power is removed from the circuit the electric motor(s) will shutdown causing the functions of the machine to stop operating.

The all stop switches used on the feeder breaker will be either a **detent** or **momentary** type. A **detent** switch will stay in the depressed position when engaged, until an operator returns the button to its original position by pulling the button out until it stops, resetting the switch. A **momentary** switch pushbutton will return to its original position once the operator releases the pushbutton, resetting the switch. Test the all stop switches supplied with the machine to determine which type has been used.

Restarting the machine

Check the area in and around the feeder breaker to make sure that the cause of the emergency shutdown has been corrected and all personnel are safe and clear of the machine.

1. Energize the electrical system.
2. Start the electric motor(s) and check that it is operating properly.

CAUTION

When the all stop is engaged the electric power is not removed from the feeder breaker, the circuit breaker is still closed (in the **ON** position) and energized. To remove the electric power from the machine the circuit breaker switch must be opened (in the **OFF** position), stopping the input of power. Electric power will still be available on the “line” side of the circuit breaker.

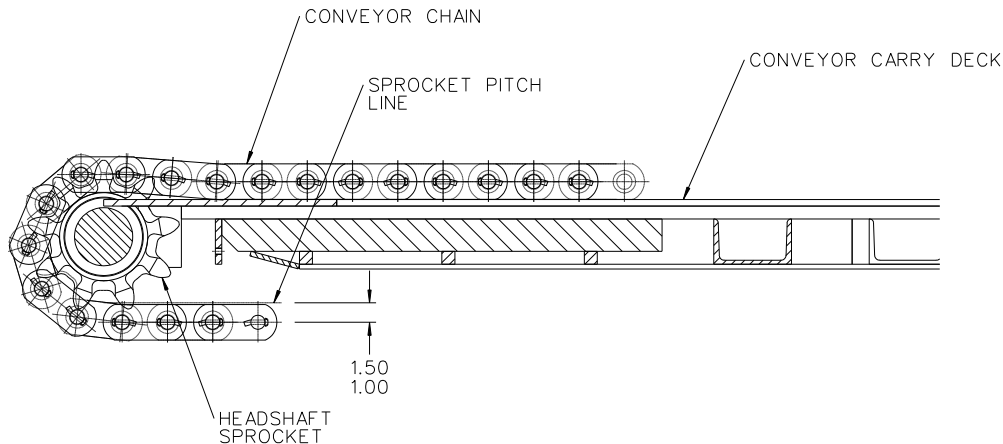
Instruction Sheet:

Number: 38021****
 By: SP
 Date: 04/25/2000

Conveyor Chain Tensioning

Proper conveyor chain tension is achieved by adjusting the position of the tailshaft with respect to the headshaft. The tailshaft is located at the intake end of the machine and acts as an idler. The position of the tailshaft can be changed by adjusting the Conveyor Take-up assembly. The take-up consists of threaded rods, which are connected to the tailshaft bearings. Once adjusted the position of the tailshaft is maintained by the use of large hex nuts and "jam" nuts on the ends of the threaded rods.

The conveyor chain is properly tensioned when a slight amount of slack exists in the conveyor chain and is visible behind the headshaft sprockets. If this is not the case the following instructions should be used to obtain the correct tension.



Conveyor Chain Tension

When changing the conveyor chain tension, the conveyor will need to be operated at times until the correct tension is achieved. The cavity immediately behind each tailshaft bearing must be free of obstructions to allow proper travel during tensioning. Ensure all personnel are made aware of this and all safety precautions are taken.

Chain Tension too Loose

1. Run the conveyor empty of material.
2. Shutdown the machine and lockout the electrical power.
3. Clear the area at the intake end of the machine and remove any covers necessary to access the tailshaft and take-up.
4. Using the wrenches provided loosen the "jam" hex nuts and move them back to give adequate room to adjust the tensioning hex nuts.
5. Using the wrench provided turn each (total of 4) adjustment hex nut an equal number of turns causing the tailshaft to move towards the bumper of the machine. Note it may be necessary to move one adjustment nut more than the other on the tailshaft to ensure that the tailshaft is aligned properly with the conveyor and headshaft.

CAUTION

To prevent serious injury or death, before proceeding, ensure that the hopper, breaker and conveyor areas of the machine are all clear and no one is in or near them.

6. Restore power to the machine.
7. Start the conveyor in the forward direction.
8. Make sure both sides of the conveyor chain are tensioned equally (this does not necessarily mean an equal number of turns per side). Correct conveyor chain tension is achieved when both sides of the chain have the correct amount of slack, visible at the headshaft, as noted in the illustration above.
9. Also ensure that the conveyor chain is not “kinked”, if so **over tension** the conveyor chain and then slack off until the correct tension is obtained.
10. Once the conveyor chain tension is correct tighten the “jam” hex nuts against the adjustment hex nuts to “lock” them in place so that they will not move during operation allowing the chain to become too slack.
11. Once the chain tension is correct replace all the guards/covers removed to access the take-up assembly.
12. Ensure that all steps have been completed, check that the area around and inside the machine is clear, restore power at the motor controller, and test run the conveyor in maintenance mode.

Chain Over Tensioned

1. Remove the access plate covering the shims and tailshaft lock-up bars.
2. Using the wrenches provided loosen the “jam” hex nuts and move them back to give adequate room to adjust the tensioning hex nuts.
3. Using the wrench provided turn each (total of 4) adjustment hex nut an equal number of turns causing the tailshaft to move away from the bumper of the machine. Note it may be necessary to move one adjustment nut more than the other on the tailshaft to ensure that the tailshaft is aligned properly with the conveyor and headshaft
4. Start the conveyor in the forward direction.
5. Make sure both sides of the tailshaft are tensioned equally; also make sure that the conveyor chain is not “kinked”, if so over tension the conveyor chain and then slack off, by opening the bleed valves, until the correct tension is obtained.
6. Stop the conveyor and lockout the electrical power.
7. Once the conveyor chain tension is correct tighten the “jam” hex nuts against the adjustment hex nuts to “lock” them in place so that they will not move during operation allowing the chain to become too slack.
8. Once the chain tension is correct replace all the guards/covers removed to access the take-up assembly.
9. Ensure that all steps have been completed, check that the area around and inside the machine is clear, restore power at the motor controller, and test run the conveyor in maintenance mode.

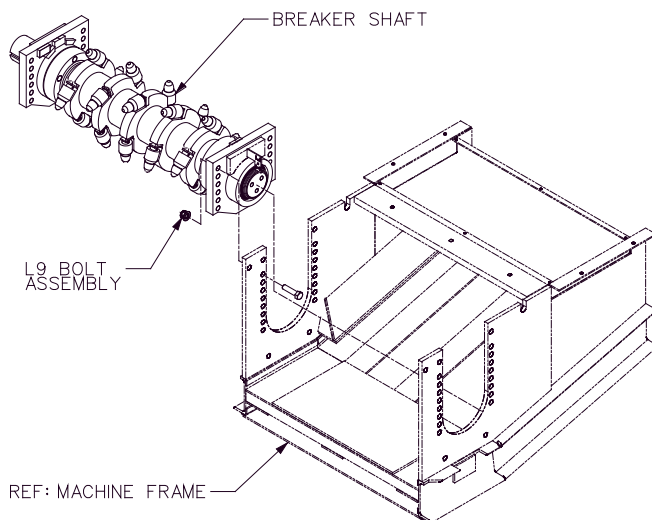
Instruction Sheet:

Number: 380210525
By: SP
Date: 08/06/2001

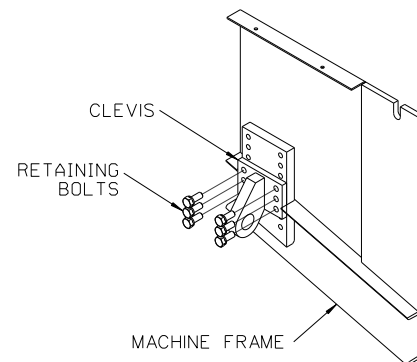
Breaker Height Adjustment

Before any adjustments are made to the operating height of the breaker shaft please consult the Engineering or Service departments at Stamler.

1. Before doing any work ensure there is an adequate supply of "L9" bolt assemblies (see parts book for quantities) to replace the existing fasteners about to be removed. The "L9" bolt assemblies can only be used one time due to the design of the locking device.
2. Shutdown the feeder breaker and lockout the electrical power, at the motor controller box.
3. Remove any guards, covers and components necessary to access the breaker shaft bearings.
4. Support the breaker shaft and drive assemblies. Attach lifting devices so that the weight of the breaker is shared equally to prevent the breaker shaft from tilting, lift the breaker shaft and drive assemblies until all the slack is removed.
5. With the breaker shaft and drives supported, remove the bolt assemblies and dispose of them so that they cannot be reused. Remove the bolts holding the breaker drive clevises to the sides of the machine frame but do not discard these bolts. Removing these bolts will allow the clevises to move with the breaker shaft and drive.
6. Slowly raise or lower the breaker shaft to the desired height, checking that the breaker does not interfere with any other part of the machine. Align the holes in the bearing housing with the mounting holes in the frame.
7. Install new "L9" bolt assemblies, apply the proper torque (see instructions), and reattach the clevises to the frame and any other mounting or supporting hardware.
8. Check that the breaker is properly secured and then release the tension on the lifting devices and remove.
9. Replace all guards and covers that were removed and check that all connections are secure.
10. Ensure that all steps have been completed, check that the area around and inside the machine is clear, restore power at the motor controller, and test run the breaker.



Breaker Height Adjustment



Breaker Drive Clevis

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Section

3

3. MAINTENANCE

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Instruction Sheet:

Number:	38021T003
By:	SP
Date:	05/21/2001

Headshaft Assembly Installation

The feeder breaker is equipped with two headshaft assemblies which are used to transfer power from the conveyor drives to the conveyor chain and flights. The headshaft assemblies have been installed and tested by Stamler and arrive ready for operation. If one of the headshaft assemblies or one of their components should become damaged and need to be repaired or replaced, remove/install using the following procedures:

Removing the Headshaft Assembly

1. Shutdown the machine and lockout the electrical system power.
2. Remove any guards and covers necessary to access the headshaft assembly.
3. Remove the hydraulic conveyor drive motor.
4. Release the tension and separate the conveyor chain and unwrap it from the headshaft sprockets.
5. Disconnect any lubrication lines attached to the headshaft bearings.
6. Remove the splined and smooth adapters from the headshaft. Wrap the adapters in rags or cardboard to prevent damage that may hinder the reassembly of the headshaft assembly.
7. Remove the keys from the headshaft.
8. Support the headshaft so that it does not fall when released, and remove the bolts securing the bearing mounting plates to the machine. Set the headshaft assembly in a suitable work area before disassembly begins.
9. Loosen the setscrews in the bearing set collars and remove the bearings from the headshaft.
10. Slide the spacers off the ends of the headshaft.
11. Remove the sprockets and keys from the headshaft.
12. Complete all repairs needed and reassemble the headshaft.

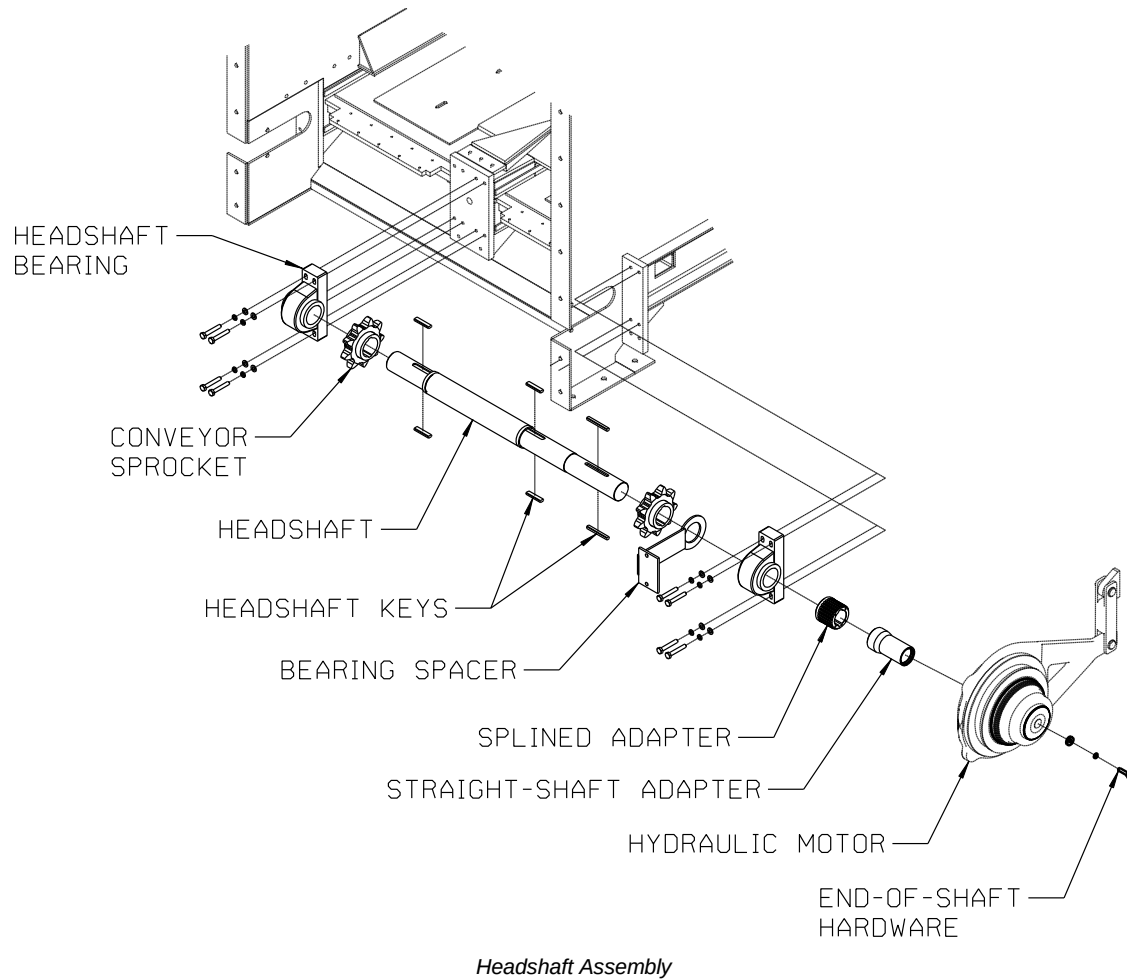
NOTE

While the headshaft is disassembled it is a good time to visually and mechanically inspect all of the components, not just the item(s) needing repaired, for abuse, excessive wear, cracks, mechanical malfunctions, etc. before reassembling and installing the headshaft.

Installing the Headshaft Assembly

1. Place the sprocket keys into the keyways on the headshaft. Slide the sprockets onto the headshaft, align with the keys, and continue pushing the sprockets until they stop against the spacer rings attached to the headshaft. Check the distance between the sprocket centers to ensure they are positioned correctly.
2. Slide the spacers onto the headshaft, against the sprockets.
3. Install the headshaft bearings. Slide the bearings onto the shaft and position approximately where they will mount onto the machine. Do not tighten the setscrews in the bearing set collars at this time.
4. Lift the headshaft into position, align the bearings with the mounting holes and insert the mounting fasteners and tighten.
5. Connect any lubrication lines that should be attached to the bearings.
6. Place the splined and straight adapter keys into the keyways on the headshaft. Slide the splined adapter onto the headshaft, align with the keys, and push it over the keys into position.
7. Install the hydraulic conveyor drive motor.
8. Wrap the conveyor chain around the headshaft sprockets and reconnect. Tension the conveyor chain.

9. Replace all of the guards and covers used around the headshaft assembly.
10. Set the conveyor drive to neutral (off).
11. Ensure that all steps have been completed, check that the area around and inside the machine is clear, restore power at the motor controller, and test run the conveyor.



Instruction Sheet:

Number: 380210581

By: SP

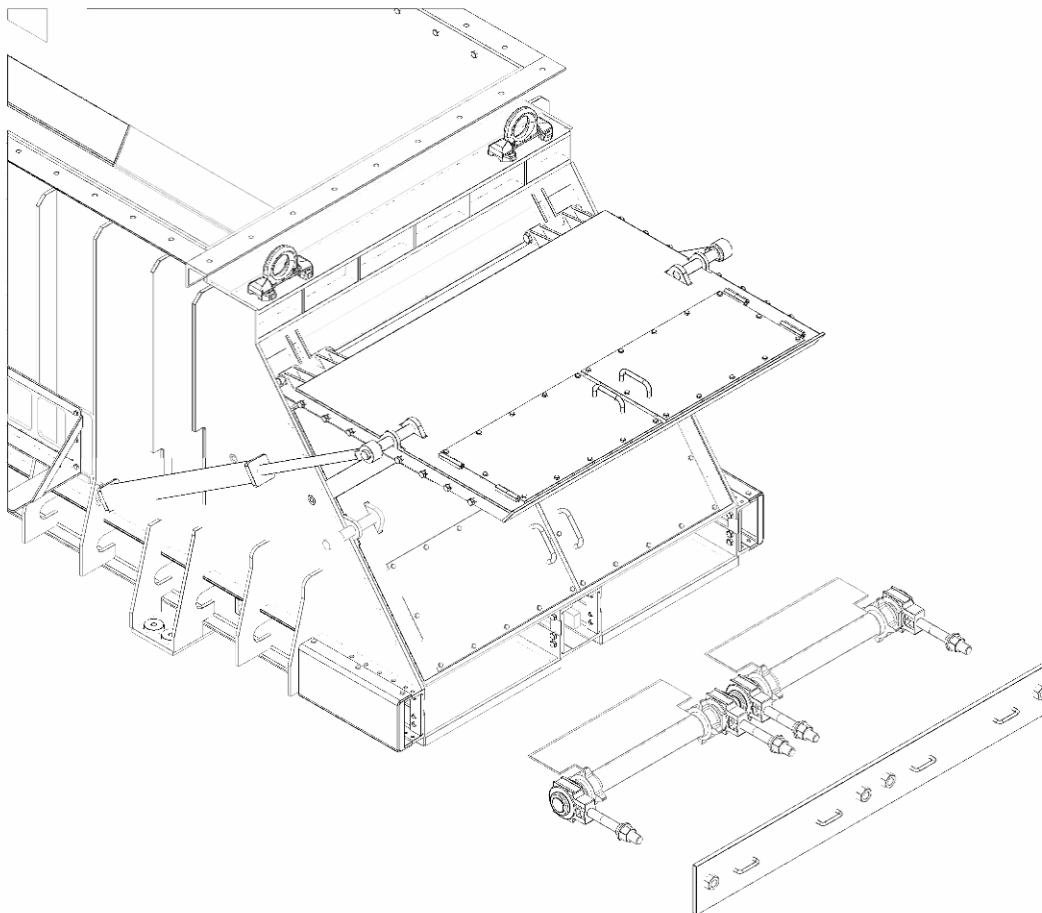
Date: 03/28/2003

Tailshaft Assembly Installation

The feeder breaker is equipped with tailshaft assemblies, which are used as idlers for the conveyor chains and flights. The tailshaft assemblies have been installed and tested by Stamler and arrive ready for operation. If the tailshaft assemblies or one of their components should become damaged and need to be repaired or replaced, remove/install using the following procedures:

Removing the Tailshaft Assembly

1. Shutdown the machine and lockout the electrical system power.
2. Remove the intake end covers that allow access to the conveyor chain and the tailshaft assembly.
3. Release the conveyor chain tension and separate the chain, unwrap from the tailshaft roller hubs. Ensure that the take-up bar is completely retracted from the tailshaft yokes.
4. Remove the lubrication lines connected to the tailshaft.
5. Attach properly sized lifting devices to the tailshaft. Pull the tailshaft until it is clear of the machine and lift. Set the tailshaft assembly in a suitable work area and begin disassembling the tailshaft.



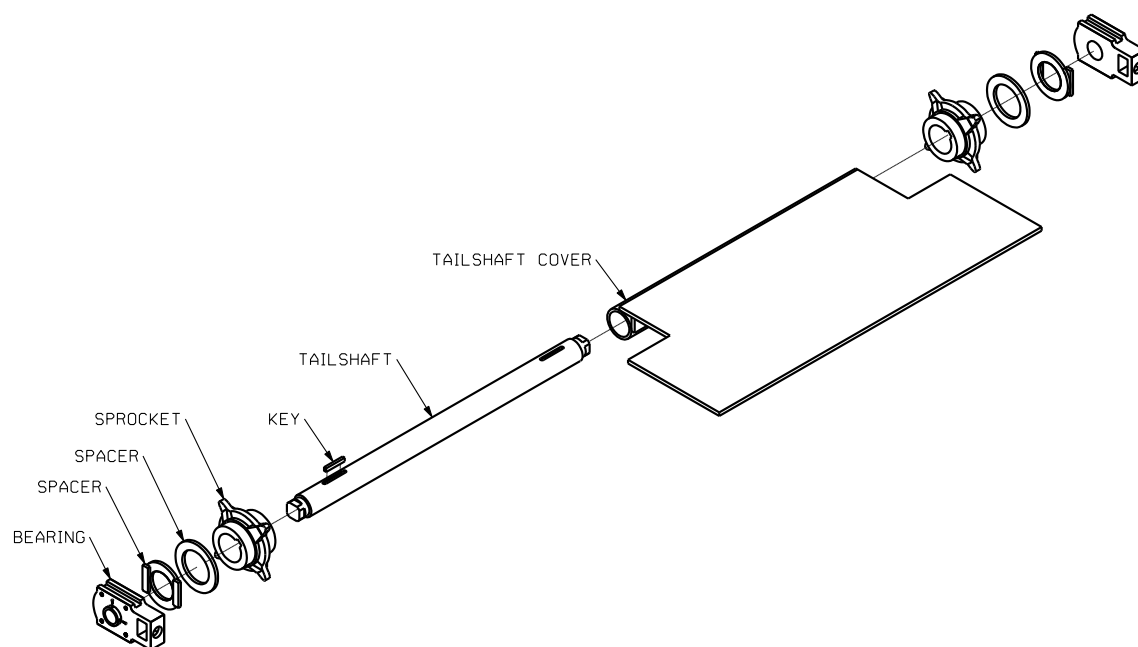
6. Loosen the setscrews in the bearing locking collars and remove the bearings from the tailshaft.
7. Remove the spacers.
8. Remove the sprockets and key from the tailshaft.
9. If necessary remove the tailshaft from the shaft cover.
10. Complete all repairs needed and reassemble the tailshaft.

NOTE

While the tailshaft is disassembled it is a good time to visually and mechanically inspect all of the components, not just the components needing repaired, for abuse, excessive wear, cracks, mechanical malfunctions, etc. before reassembling and installing the tailshaft.

Installing the Tailshaft Assembly

1. Assemble the tailshaft and cover if necessary.
2. Insert the key into the keyway slot on the tailshaft. Slide the sprockets over the ends of the shaft, align with the key, and push until the sprockets are the proper distance apart.
3. Slide the spacers onto the tailshaft, against the bearings. Ensure the rectangular tabs face away from the sprockets.
4. Slide the bearings onto the ends of the tailshaft and position approximately where they will mount into the machine. Do not tighten the setscrews in the bearing set collars at this time.
5. Lift the tailshaft assembly and lower back into the machine.
6. Align the grooves in the top and bottom of the bearing housing with the guides attached to the frame.
7. Compare the distances between the bearing centers and the sprocket centers to ensure the tailshaft is positioned correctly. When the tailshaft is in the proper position then tighten the setscrews in the bearing locking collars and lock the bearings to the shaft.
8. Push the shaft assembly towards the breaker as far as possible. Wrap the conveyor chain around the tailshaft sprockets and reconnect. Insert the lock-up bars into bearing housings and tension the conveyor chain.
9. Connect any lubrication lines that should be attached to the bearings.
10. Replace all of the guards and covers used around the tailshaft assembly.
11. Set the conveyor drive to neutral.
12. Ensure that all steps have been completed, check that the area around and inside the machine is clear, restore power at the motor controller, and test run the conveyor.



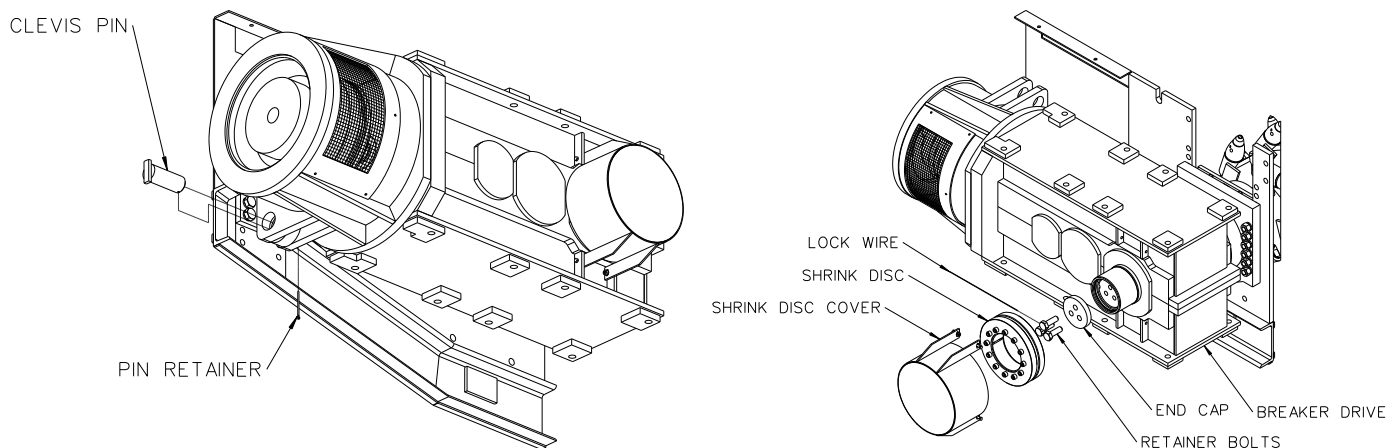
Tailshaft Assembly

Breaker Shaft Assembly Installation

The feeder breaker is equipped with a breaker shaft assembly, which is used to crush material as it is being conveyed through the machine. The breaker shaft assembly has been installed and tested by Stamler and arrives ready for operation. If the breaker shaft assembly or one of its components should become damaged and needs to be repaired or replaced, remove/install using the following:

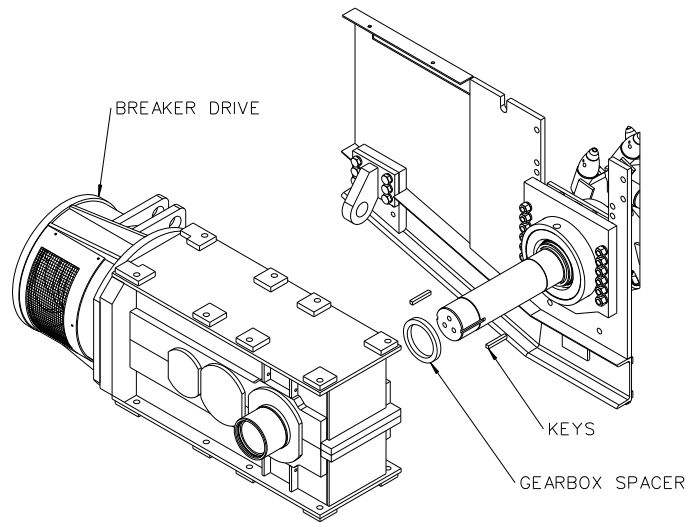
Removing the Breaker Drives

1. Before doing any work ensure there is an adequate supply of "L9" bolt assemblies (see parts book for quantities) to replace the existing fasteners about to be removed. The "L9" bolt assemblies can only be used one time due to the design of the locking device.
2. Shutdown the machine and lockout the electrical system power.
3. Remove any guards and covers necessary to access the breaker shaft assembly.
4. Begin removing the breaker drive components on each end of the breaker shaft. Attach adequately sized lifting devices to the breaker drive and lift until all slack is removed and the lifting device carries the weight of the breaker drive.
5. Disconnect all lubrication lines attached to any of the breaker shaft or breaker drive components.
6. Disconnect all cables and hoses connected to the drive components (motors, speed reducers, couplings, etc.).
7. Disconnect any mounting/supporting devices that connect the drives to the machine.
8. Remove the covers over the driven ends of the breaker shaft, exposing the gearbox shrink discs.
9. Loosen the shrink disc bolts until all the tension is removed from the disc and it slides easily off each end of the shaft. See the manufacturer's manual for more detailed instructions.
10. Remove the end cap or locking ring from the end of the shaft, and then slowly slide the gearbox off the end of the shaft.



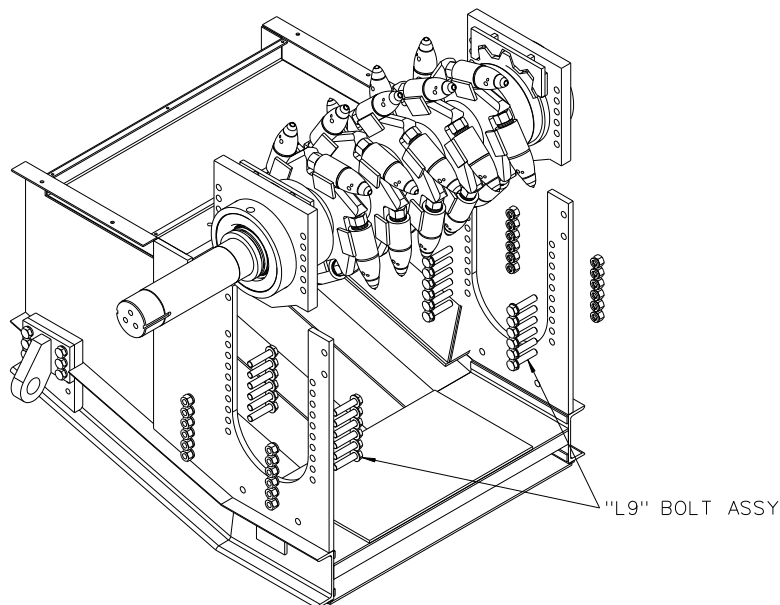
Removing the Breaker Drive

11. There is a spacer and two (2) keys on the end of the shaft. Make sure that the keys are accounted for when removing the breaker drive components, inspect the keys for wear/damage and replace as necessary.



Removing the Breaker Drive

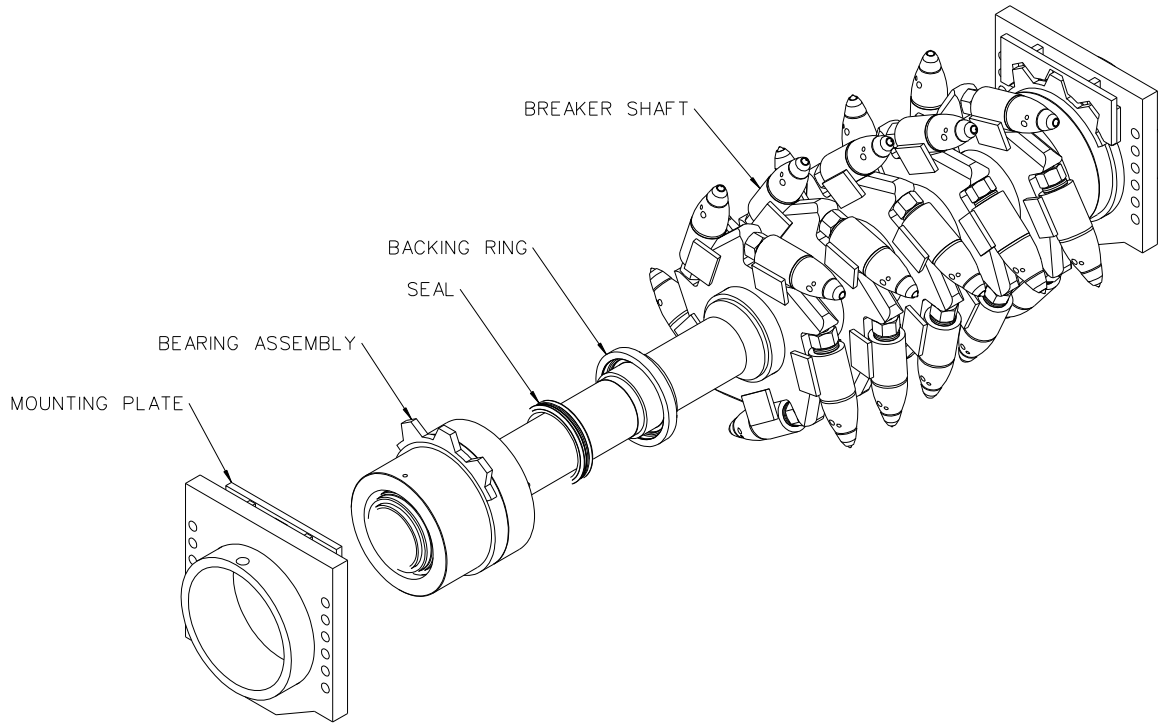
12. Support the breaker shaft. Attach lifting devices so that the weight of the breaker is shared equally to prevent the breaker shaft from tilting, lift the breaker shaft until all the slack is removed.
13. There may be a speed switch sensor attached to the breaker shaft mounting plate on one end of the shaft. The sensor may need to be removed before the breaker shaft can be removed from the machine.
14. Remove the "L9" bolt assemblies and dispose of them so that they cannot be reused.
15. Slowly lift the breaker shaft out of the machine until it is completely clear. Set the breaker shaft assembly in a suitable work area before disassembly begins.



Removing the Breaker Shaft

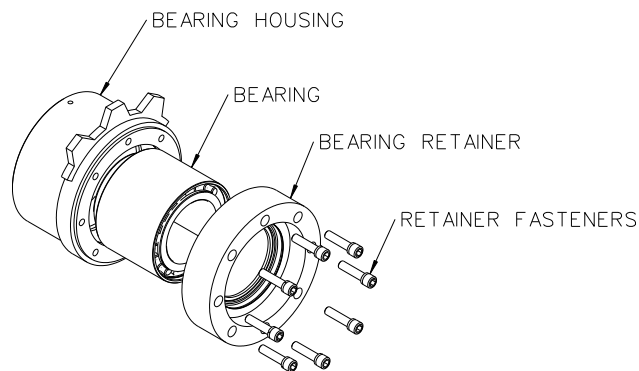
16. Disassemble the breaker shaft:
 - a. Slide the bearing mounting plate off of the bearing housing.
 - b. Slide the assembled bearing off the end of the breaker shaft.

- c. Remove the seal and the backing ring from the shaft. Inspect the seals for damage/wear and replace as necessary.
- d. Complete all repairs needed and reassemble the breaker shaft.



Breaker Shaft Disassembly

17. To disassemble the bearing, remove the bolts from the back of the bearing retainer plate. After removing the bolts the retaining plate will separate from the bearing housing exposing the bearing and allowing it to be removed.



Bearing Disassembly

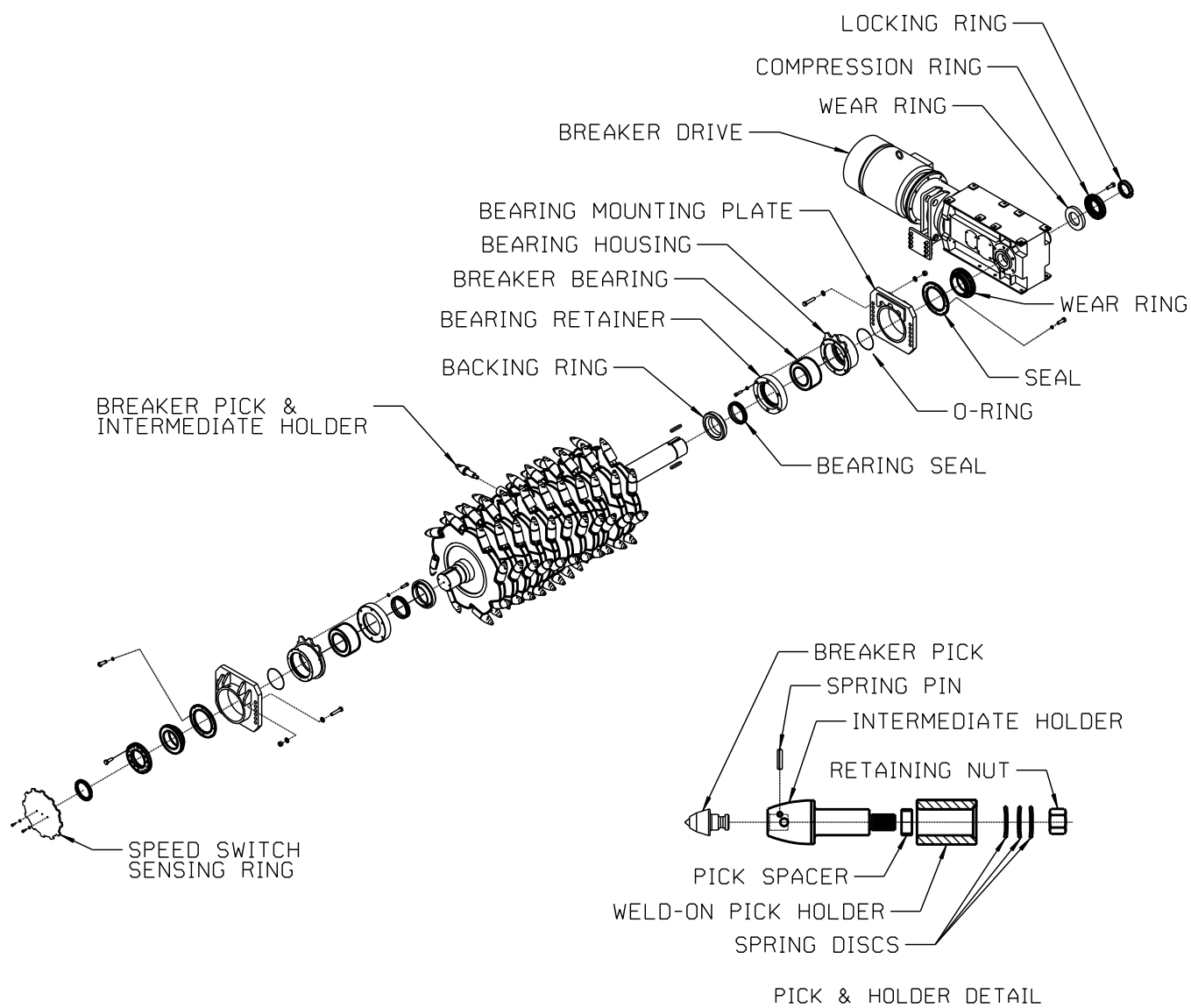
18. Complete all necessary repairs and re-assemble the breaker shaft.

NOTE

While the breaker shaft is disassembled it is a good time to visually and mechanically inspect all of the components of the breaker shaft and drive system, not just the ones needing repair, for abuse, excessive wear, cracks, mechanical malfunctions, etc. before reassembling and installing the breaker shaft and drive.

Installing the Breaker Shaft

1. Assemble the breaker shaft: (please refer to the previous illustrations)
 - a. Install the duo-cone seals into the backing ring and place them on the end of the shaft. Slide the backing ring towards the center of the shaft until it stops against the step of the shaft.
 - b. Pack the bearings with grease and insert them into the bearing housings.
 - c. Attach the bearing retainer caps to the bearing housings using the fasteners removed earlier.
 - d. Place the assembled bearing onto the end of the breaker shaft. Slide the bearings onto the shaft until it contacts the backing ring and seal. The seal should be inserted into the recess at the back of the retaining ring.
 - e. Place the bearing mounting plate over the bearing housing and engage the locking mechanism.
2. Attach lifting devices so that the weight of the breaker is shared equally to prevent the breaker shaft from tilting, lift the breaker shaft and place into the machine.
3. Align the bolt holes in the mounting plate and frame and insert four grade 5 bolts of the appropriate size into the bolt holes on each end of the breaker and tighten snug.
4. Check the alignment of the breaker and that it turns freely. Once everything is in line and checked insert the new "L9" bolt assemblies, replacing the eight temporary bolts, and tighten to the proper torque.
5. Install the breaker drive components:
 - a. Place the gearbox spacers onto the end of the shaft and insert the keys into the keyways on the end of the shaft.
 - b. Lift the breaker drive into place, align the keyways and the drive mounting/supporting devices and slide the gearbox onto the end of the breaker shaft.
 - c. Place the end-cap onto the end of the breaker shaft, attach using the fasteners removed earlier; this will push the gearbox into place and load the bearings. Place the shrink disc over the end of the gearbox "hollow shaft" and tighten per the instructions.
6. Align the holes in the clevis bracket, located on the gearbox "tunnel mount" adapter, insert the clevis pin and pin retainer clip. Replace all the guards and covers, before starting the machine.
7. Check to ensure that all supporting connections have been made, and that they are ready to accept the weight of the breaker drive components. Slowly release the tension on the lifting devices until all the weight of the drive components is resting on the supports. Remove the lifting devices.
8. Connect all cables, hoses and lubrication lines to the breaker shaft and breaker drive components that were removed for disassembly.
9. Ensure that all steps have been completed, check that the area around and inside the machine is clear, restore power at the motor controller, and test run the breaker.



Breaker Shaft Assembly

Breaker Pick Installation

The breaker is designed to use a breaker pick mounted in an intermediate pick holder mounted in a holder that is welded to the breaker shaft. The breaker pick is a normal wear item and needs to be replaced whenever it becomes worn (dull), remove/install using the following:

NOTE

Worn or dull breaker picks should be changed as soon as possible. Worn picks can cause a reduction in production, and cause sparking when striking hard material.

Removing a Breaker Pick

1. Shutdown the machine and lockout the electrical system power.
2. Remove any built-up material that is packed around the pick being removed.
3. Use a punch to remove the roll-pin that connects the breaker pick to the intermediate holder.
4. Remove the pick from the holder; it may be necessary to pry the pick out of the holder using a small crowbar or a screwdriver.
5. If the pick is difficult to remove, strike the pick with a hammer until it begins moving in the holder, use caution not to strike the pick hard enough to cause the pick to bend or wedge itself in the holder or damage the holder.

Installing a Breaker Pick

1. Check that the opening in the end of the holder is free of debris and able to accept a new pick.
2. Insert the new breaker pick into the holder; make sure that the pick is fully inserted into the holder or the holes of the pick and holder will not be aligned.
3. Insert the roll-pin into the holes in the pick and holder.

Inspect the Holder/Ring

Whenever it is necessary to replace a breaker pick, take the time to properly inspect the intermediate holder and weld-on holder for damage. If either holder is damaged or missing it will be necessary to replace it.

WARNING

Before performing any work to the feeder breaker, involving cutting/welding of metal, **review the state and federal mining laws** to ensure all safety regulations are being followed. Make all employees aware that the work is being performed and that all safety precautions are in place before beginning.

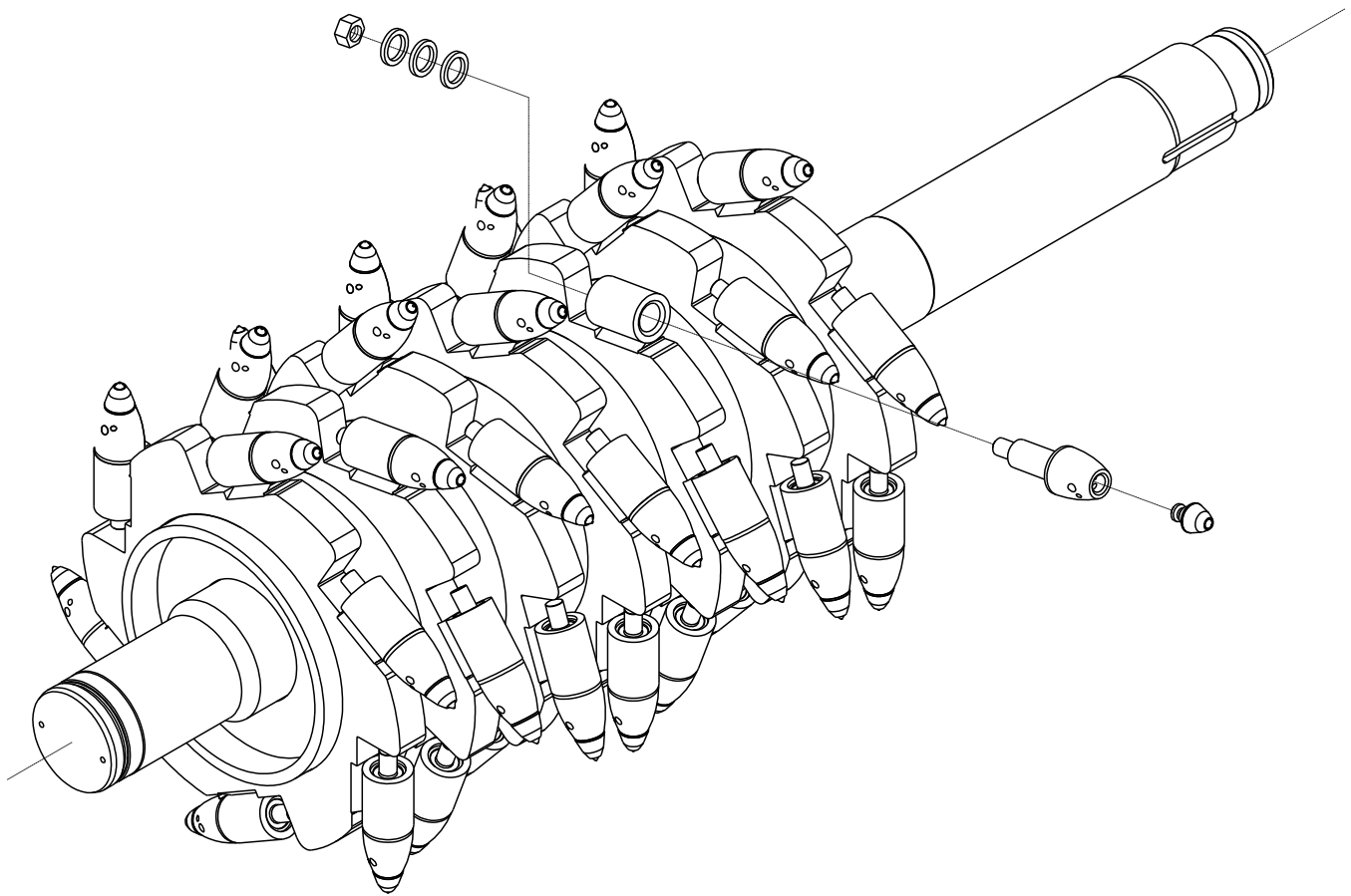
Should it become necessary to remove and/or replace an intermediate pick holder, use the following procedure:

Removing an Intermediate Pick Holder

1. Remove the breaker pick using the above procedure.
2. Using a wrench, loosen the locking nut that holds the intermediate holder in the weld-on holder.
3. Remove the intermediate holder from the weld-on holder, making sure not to lose the locking nut or the spring discs.

Installing an Intermediate Pick Holder

- a. Check that the weld-on holder is free of debris and make sure the threads of the locking nut are clear and undamaged.
- b. Insert the intermediate holder into the weld-on holder. Place the spring disks on the end of the intermediate holder and then tighten the locking nut.
- c. Install a breaker pick in the intermediate holder using the instructions above.



Pick Removal

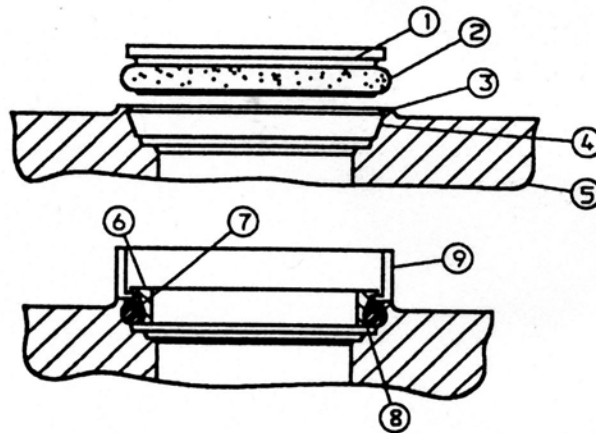
Duo-Cone Seal Installation

The breaker shaft assembly on the Stamler feeder breaker is designed to use duo-cone seals to protect the bearings, bushings, shafts, etc. from contaminants. The duo-cone seals are long lasting when install and maintained properly. To install new seals use to following:

Duo-Cone Seal Assembly Contents:

2 Metal Seal Rings
2 Rubber Toric Rings

Terminology:



- | | |
|--------------------------|-----------------------------|
| 1 - Seal Ring | 5 - Seal Ring Housing |
| 2 - Rubber Toric | 6 - Seal Ring Face |
| 3 - Housing Retainer Lip | 7 - Seal Ring Ramp |
| 4 - Housing Ramp | 8 - Seal Ring Retaining Lip |
| | 9 - Installation Tool |

Housing Preparation

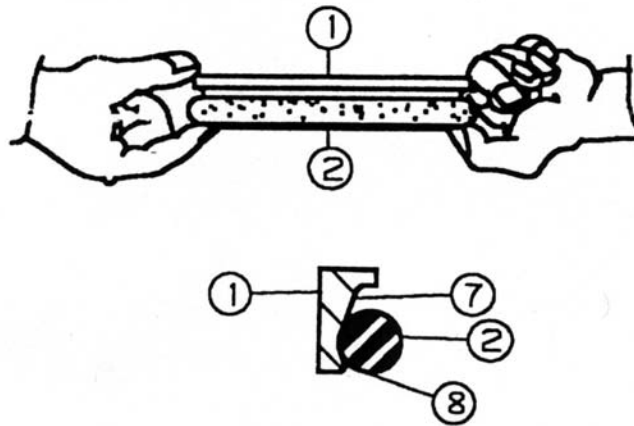
The parts of the housing that come in contact with the rubber toric rings (3 and 4) must be free of foreign materials (oil, grease, dirt, metal chips, dust, lint particles, etc.) before installing the seal. This should be done with a lint free wipe and non-petroleum base solvent. Dry with a clean wipe.

Seal Preparation

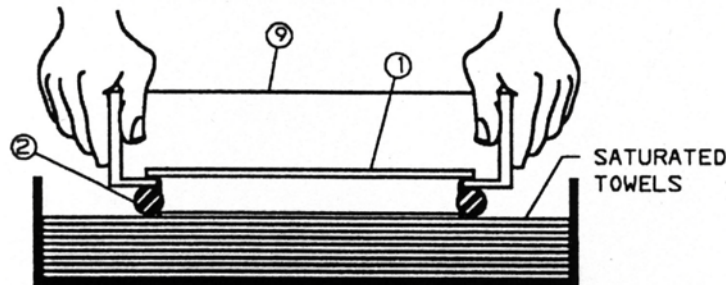
Remove any foreign material from the rubber torics (2) and the ramps (7) and lips (8) of both seal rings. This should also be done with a lint free wipe and a non-petroleum based solvent. Dry with a clean wipe.

Place the rubber torics on the metal seal ring at the bottom of the seal ring ramp and against the retaining lip (see illustration). Make sure the rubber toric is straight on the seal ring and not twisted. Be careful not to nick or cut the torics during this assembly, as this can cause leaks.

Seal Preparation



Put the installation tool onto the metal seal ring and rubber toric. Lightly dampen the lower half of the rubber toric with the appropriate assembly lubricant. Techniques to dampen the toric include wiping with a lint-free towel, lubricating using a clean foam brush, or dipping in a container lined with towels saturated in the assembly lubricant (as shown).



Approved Assembly Lubricants*

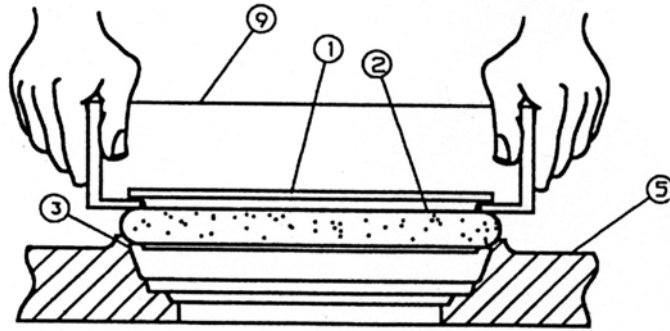
Isopropyl Alcohol**
Houghto-Grind 60 CT
Quacker® Solvo Clean 68-0

(*) Do not use Stansol or any other liquid that leaves an oil film or does not evaporate quickly.

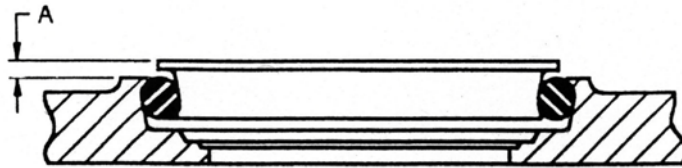
() All applicable safety and disposal guidelines for flammable liquids must be followed.**

Installation Process

With the lower half of the toric still wet, use the installation tool to position the seal ring and the rubber toric squarely against the housing (as shown). For smaller diameter seals, use sudden and even pressure to push the rubber toric under the retaining lip of the housing. For larger diameter seals, which will not press in with sudden and even pressure, it is acceptable to work the toric past the retaining lip by starting one side and tapping the opposite side of the installation tool with a rubber mallet until it is engaged past the retaining lip of the housing.



Check the assembled height (A) in at least four places, 90° apart, using either caliper, tool makers' ruler or any other calibrated measuring device. The difference in height around the ring must not be more than 1mm. If small adjustments are necessary, do not push or pull directly on the seal ring. Use the installation tool to push down and your fingers to pull up uniformly on the rubber toric and seal ring.



The rubber toric can twist if it is not wet all around during installation or if burrs or fins on the retaining lip of the housing. Twists, misalignment, and bulges of the toric will result in seal failure. If correct installation is not obvious, remove the seal from the housing and repeat the process.

The rubber torics must never slip on the ramps of either the seal rings or the housing. To prevent slippage, allow adequate evaporation time for the lubricants before proceeding with further assembly. Once correctly in place, the rubber toric must roll on the ramp only.

Wipe the face of each seal ring using a lint free wipe. No particles of any kind are permissible on the sealing surfaces. (Even a hair can hold the seal faces apart and cause a leak.) Apply a thin film of oil on the entire seal face of one or both seals using a clean finger or a lint free applicator. Oil must not contact surfaces other than the sealing faces.

Final Assembly

While completing the assembly of the unit, make sure that both housings are in correct alignment and are concentric. Slowly bring the two housings together. High impact can scratch or break the seal components.

If the rubber toric slips at any location, it will twist, causing the seal rings to cock. Cocked seals create uneven pressure on the seal faces, resulting in galling, scoring and leakage. Any wobbling action can cause dirt to enter by pumping mud past the torics.

Instruction Sheet:

Number:	380210478
By:	SP
Date:	05/21/2001

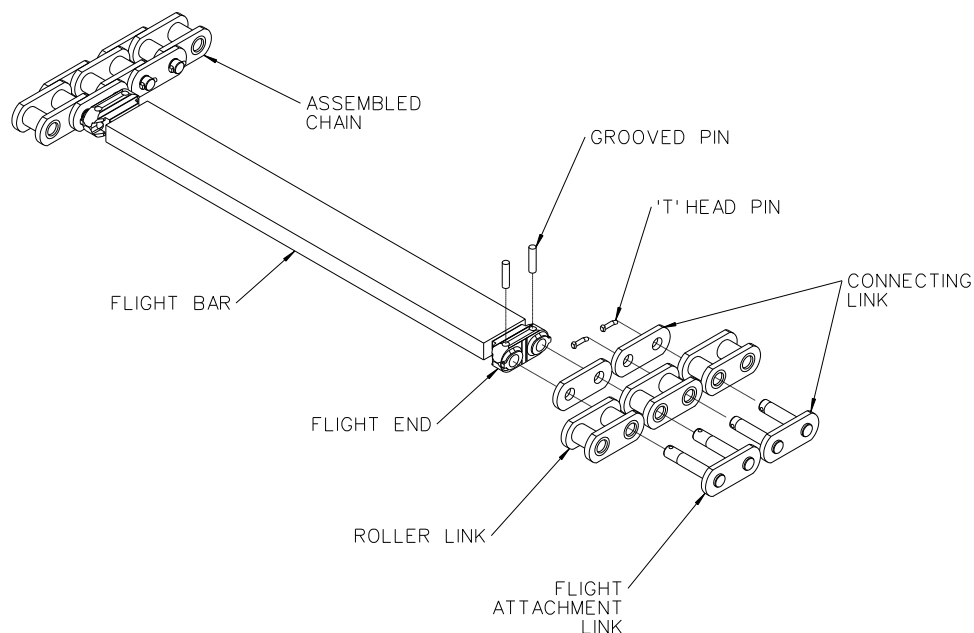
Conveyor Flight Replacement

Removing a Flight

1. Operate the conveyor until the flight being removed is in the intake hopper section of the machine and easily accessible.
2. Shutdown the machine and lockout the electrical system power.
3. Remove the chain cover, if equipped, over each end of the flight and set aside.
4. Release the tension on the conveyor chain. Ensure there is adequate slack in the chain to lift the flight.
5. Lift the conveyor chain and flight, one side at a time, and place supports under the flight bar and chains.
6. Remove the groove pins from each end of the flight.
7. Pull the flight connecting links away from the flights until they are completely extracted from the flight ends.
8. Remove the damaged flight.

Installing a Flight

1. Place a new flight in the space where the damaged one was removed. **Ensure the flight is installed correctly, if the flight is upside-down then the conveyor chain will not lay flat in the conveyor.**
2. Push the flight connecting links into the flight end until the vertical holes in the connecting link are inline with the vertical holes in the flight end.
3. Lift the conveyor chain and flight, remove the supports and lay the chain down on the conveyor deck.
4. Tension the conveyor chain.
5. Install the chain covers and tighten to the proper torque.
6. Ensure that all steps have been completed, check that the area around and inside the machine is clear, restore power at the motor controller, and test run the conveyor.



Conveyor Chain & Flights

Conveyor Chain Wear Limits – 3.50” Pitch

Most of the wear on the conveyor chain occurs between the inside of the bushing and the chain pin, this wear causes the chain to lengthen (elongation). The chain manufacturer has established that when elongation reaches 3% over the original length, the chain is worn out and should be replaced. At this point the surfaces of the hardened pins and bushing are close to being worn through; this causes the chain to no longer fit the sprocket properly.

Determining the Amount of Elongation

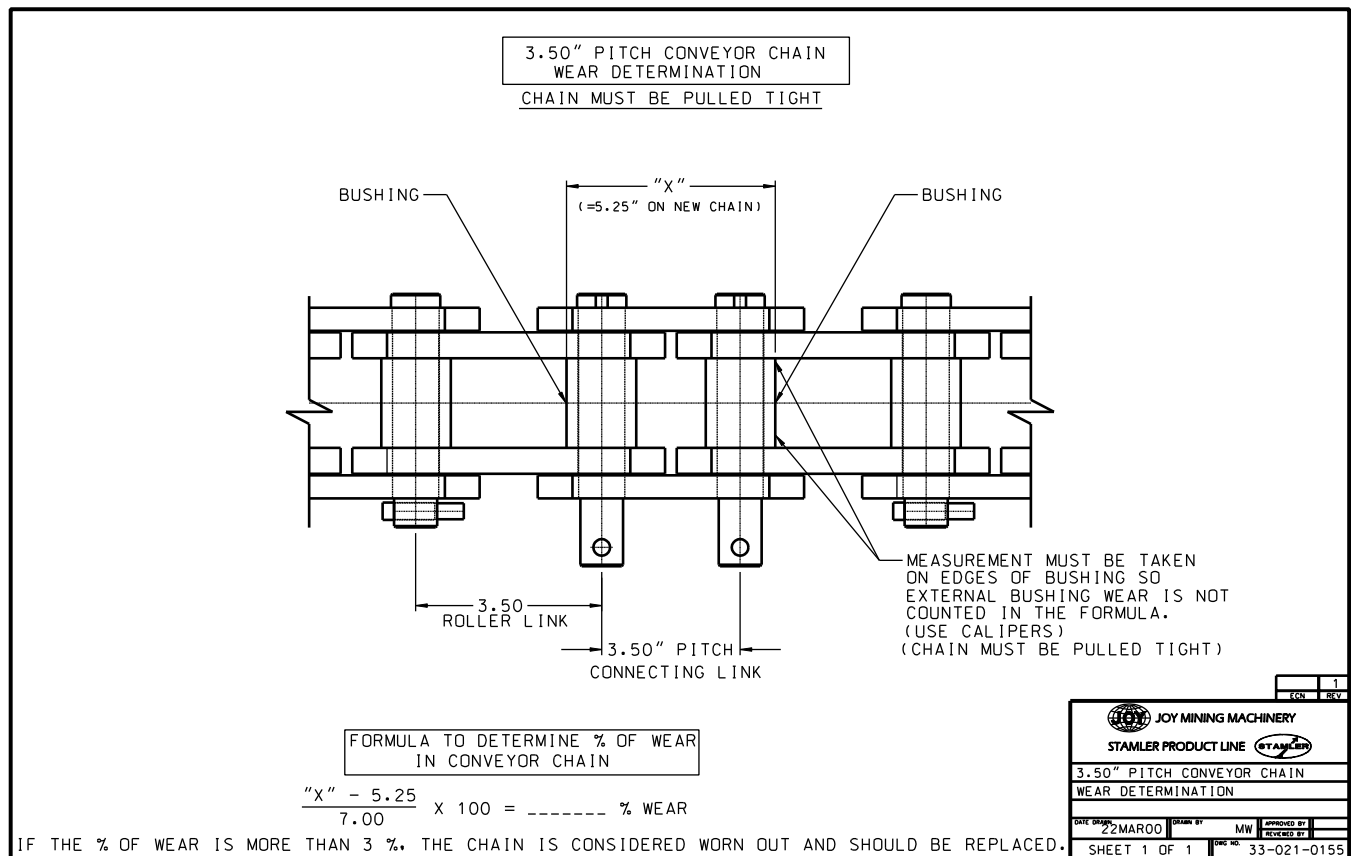
1. Tension the conveyor chain until it is completely tight (see instructions).
2. Measure the distance “X” (shown below) from one bushing to the next on the outside edges.
3. Use this formula to determine the percentage of elongation $[(X - 5.25) / 7.00] \times 100 = \text{Wear \%}$.

Example: $X = 5.44"$

$$5.44 - 5.25 = 0.19$$

$$(0.19 / 7.00 = 0.02714) \times 100 = 2.714\%$$

The chain is elongated by 2.714%, which means that the usable life of the chain is nearly complete.



Instruction Sheet:

Number: 380210138
 By: SP
 Date: 05/21/2001

Torque Settings “L9” Fasteners

Special “L9” grade fasteners are used to hold the breaker shaft in the machine frame and may also be used elsewhere on the machine. These fasteners are made of steel with a tensile strength of 180,000 PSI (by comparison grade 8 fasteners have a tensile strength of 150,000). The hex head nut is a locking nut of the deformed type, meaning the nuts and bolts can only be used one time (in the upper portion of the nut the last threads are compressed on two sides forming an elliptical shape). The bolts and washer can only be used with special hardened flat washers, if they are to work properly. When using “L9” fasteners the bolt must go through the locknut two full threads to have proper engagement and keep the nut from backing off. If this cannot be done the next longer bolt should be used.

NOTE

Do not use any type of lubricants on the fasteners. Do not weld the bolts, nuts or washers.

Identifying an “L9” Fastener

To determine if a fastener is “L9” grade locate the fastener in the parts list portion of the manual. L9 will be included as part of the description of the fastener. If the Stamler part number has a **9** as the second digit it is an “L9” grade fastener (example: **A9A0820024**).

Torque Values

Fastener Size	Torque by Bolt Head		Torque by Nut	
	N-m	Lbs/ft	N-m	Lbs/ft
$\frac{1}{2}$ - 20	129	95	142	105
$\frac{5}{8}$ - 11	231	170	251	185
$\frac{5}{8}$ - 18	258	190	278	205
$\frac{3}{4}$ - 16	447	330	481	355
$\frac{7}{8}$ - 14	705	520	793	585
1 - 14	949	700	1220	900
1 $\frac{1}{8}$ - 12	1559	1150	1797	1325
1 $\frac{1}{4}$ - 12	2169	1600	2373	1750



Instruction Sheet:

Number: 380210216
 By: SP
 Date: 05/21/2001

Torque Settings Gr. 5 & 8 Fasteners

The feeder breaker has been assembled using fasteners of both grade 5 and 8. When removing or replacing these fasteners the following torque values should be used to insure that the fasteners are properly secured. When using a bolt and nut together the bolt should protrude through the nut by a minimum of two (2) full threads to ensure proper engagement. If this cannot be accomplished the next longer bolt should be used.

Identifying the Fastener Grade

To determine the grade of a fastener locate it in the parts list portion of the manual. The grade (5 or 8) will be included as part of the description of the fastener. If the part number has a **5** as the second digit in the part number it is a grade **5** fastener. If the part number has an **8** as the second digit in the part number it is a grade **8** fastener (example: A5S1608032, A8A0813024).

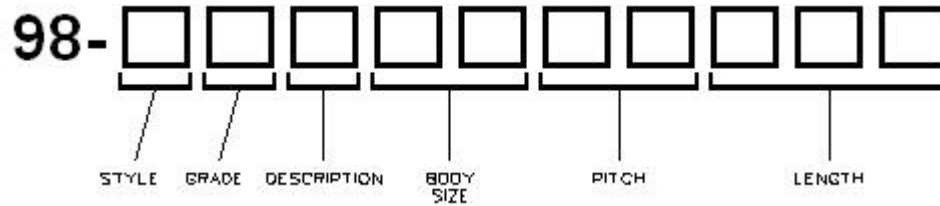
Torque Values

Fastener Size	Grade 5		Grade 8	
	N-m	Lbs/ft	N-m	Lbs/ft
$\frac{1}{4}$ - 20	8	6	12	9
$\frac{1}{4}$ - 28	10	7	14	10
$\frac{5}{16}$ - 18	18	13	24	18
$\frac{5}{16}$ - 24	19	14	27	20
$\frac{3}{8}$ - 16	31	23	48	35
$\frac{3}{8}$ - 24	34	25	48	35
$\frac{7}{16}$ - 20	48	35	75	55
$\frac{7}{16}$ - 20	54	40	81	60
$\frac{1}{2}$ - 13	75	55	109	80
$\frac{1}{2}$ - 20	88	65	122	90
$\frac{9}{16}$ - 12	109	80	149	110
$\frac{9}{16}$ - 18	122	90	239	130
$\frac{5}{8}$ - 11	149	110	231	170
$\frac{5}{8}$ - 18	239	130	244	180

Fastener Size	Grade 5		Grade 8	
	N-m	Lbs/ft	N-m	Lbs/ft
$\frac{3}{4}$ - 10	271	200	380	280
$\frac{3}{4}$ - 16	298	220	433	320
$\frac{7}{8}$ - 9	407	300	624	460
$\frac{7}{8}$ - 14	433	320	678	500
1 - 8	597	440	922	680
1 - 12	651	480	1003	740
1 $\frac{1}{8}$ - 7	814	600	1302	960
1 $\frac{1}{8}$ - 12	895	660	1464	1080
1 $\frac{1}{4}$ - 7	1139	840	1844	1360
1 $\frac{1}{4}$ - 12	1247	920	2033	1500
1 $\frac{3}{8}$ - 6	1491	1100	2413	1780
1 $\frac{3}{8}$ - 12	1708	1260	2766	2040
1 $\frac{1}{2}$ - 6	1980	1460	3200	2360
1 $\frac{1}{2}$ - 12	2224	1640	3607	2660

Fastener Code

The Stamler fastener code is a ten-digit part number, which describes the fastener. Below is a diagram showing the breakdown of the fastener code and what each part of the code identifies.



Style

A	American
M	Metric
G	Galvanized
S	Stainless Steel

Grade

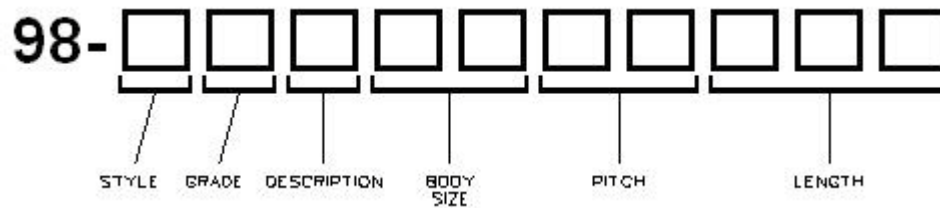
AMERICAN	
1	Special
2	SAE #2
3	SAE #3
5	SAE #5
8	SAE #8
9	L9

METRIC	
4	Metric 8.8
6	Metric 10.9
7	Metric 12.9

Description

A	Hex Head Capscrew
B	Hex Head Capscrew w/Drilled Head
C	Socket Head Capscrew
D	Socket Head Capscrew w/Drilled Head
E	Hex Socket Flat Head Capscrew
F	Round Head Machine Screw
G	Slotted Pan Head Machine Screw
H	Fillister Head Machine Screw
J	Eye Bolt
K	Hex Socket Setscrew w/Cup Point
L	Button Head Rivet
M	Flat Head Rivet
N	Flat Washer
P	Lock Washer – Regular
Q	Lock Washer – High Collar
R	Cotter Pin
S	Hex Head Nut - Heavy

T	Hex Head Nut – Regular
U	Hex Head Nut – Lock
V	Hex Head Nut – Jam
W	Split Roll Pin
X	Type “U” Drive Screw
Y	Threaded Rod
Z	Elevator Bolt
0	Shoulder Screw
1	Hex Socket Setscrew w/Oval Point
2	Hex Head Capscrew – All Thread
3	Slotted Hex Head Nut - Heavy
4	Hex Socket Button Head Capscrew
5	Carriage Bolt
6	A490 Type 1
7	Miscellaneous Fasteners
8	A325



Body Size

American:

For American (inch) fasteners denote the body size by the total number of 1/16's in the body size:

Example: 04 = 4/16" (1/4")
05 = 5/16"
06 = 6/16" (3/8")
07 = 7/16"
08 = 8/16" (1/2")

SAE	
S1	SAE #6
S2	SAE #8
S3	SAE #10
S4	SAE #12
S5	SAE #7

Metric

For metric fasteners use the body size in millimeters to denote the body size:

Example: 03 = M3
04 = M4
05 = M5
06 = M6
60 = M60

Pitch

American:

For thread pitches that are whole numbers use:

Example: 08 = 8 threads per inch
32 = 32 threads per inch

For thread pitches that have decimals use the metric code for the second digit:

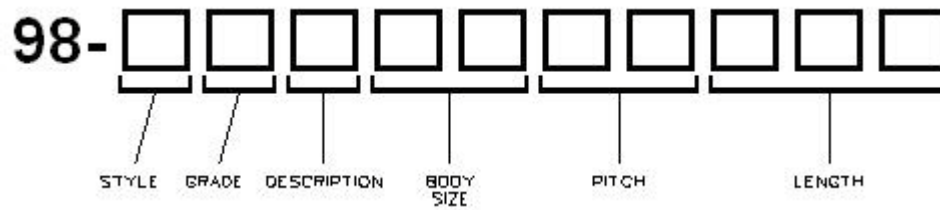
Example: 4L = 4 ½ threads per inch
2F = 2 ¼ threads per inch

Metric:

Use the following code to denote the thread pitch per millimeter for metric fasteners:

Example: BR = 1.75 threads per millimeter

First Digit	Second Digit	
A = 0	A = .00	L = .50
B = 1	B = .05	M = .55
C = 2	C = .10	N = .60
D = 3	D = .15	P = .65
E = 4	E = .20	Q = .70
F = 5	F = .25	R = .75
G = 6	G = .30	S = .80
H = 7	H = .35	T = .85
J = 8	J = .40	U = .90
K = 9	K = .45	V = .95



Length

American:

001 = 1/16"
002 = 1/8"
003 = 3/16"
004 = 1/4"
005 = 5/16"
006 = 3/8"
007 = 7/16"
008 = 1/2"
009 = 9/16"
010 = 5/8"
011 = 11/16"
012 = 3/4"
013 = 13/16"
014 = 7/8"
015 = 15/16"
16 = 1"

018 = 1 1/8"
020 = 1 1/4"
022 = 1 3/8"
024 = 1 1/2"
026 = 1 5/8"
028 = 1 3/4"
030 = 1 7/8"
032 = 2"
034 = 2 1/8"
036 = 2 1/4"
038 = 2 3/8"
040 = 2 1/2"
042 = 2 5/8"
044 = 2 3/4"
046 = 2 7/8"
048 = 3"

050 = 3 1/8"
052 = 3 1/4"
054 = 3 3/8"
056 = 3 1/2"
058 = 3 5/8"
060 = 3 3/4"
062 = 3 7/8"
064 = 4"
066 = 4 1/8"
068 = 4 1/4"
070 = 4 3/8"
072 = 4 1/2"
074 = 4 5/8"
076 = 4 3/4"
078 = 4 7/8"
080 = 5"

082 = 5 1/8"
084 = 5 1/4"
086 = 5 3/8"
088 = 5 1/2"
090 = 5 5/8"
092 = 5 3/4"
094 = 5 7/8"
096 = 6"
098 = 6 1/8"
100 = 6 1/4"
102 = 6 3/8"
104 = 6 1/2"
106 = 6 5/8"
108 = 6 3/4"
110 = 6 7/8"
112 = 7"

Metric:

005 = 5mm
006 = 6mm
007 = (7mm)
008 = 8mm
009 = (9mm)
010 = 10mm
011 = (11mm)
012 = 12mm
014 = 14mm
016 = 16mm
018 = (18mm)

020 = 20mm
022 = 22mm
025 = 25mm
028 = (28mm)
030 = 30mm
032 = (32mm)
035 = 35mm
038 = (38mm)
040 = 40mm
045 = 45mm
050 = 50mm

055 = 55mm
060 = 60mm
065 = 65mm
070 = 70mm
075 = 75mm
080 = 80mm
085 = 85mm
090 = 90mm
095 = 95mm
100 = 100mm
105 = 105mm

110 = 110mm
115 = 115mm
120 = 120mm
125 = (125mm)
130 = 130mm
140 = 140mm
150 = 150mm
160 = 160mm
170 = 170mm
180 = 180mm
190 = 190mm

Lengths in (parentheses) are non-preferred.

Note:

Lengths are per British Standard ISO Metric Bolt & Screw Length (Machinery's Handbook).



JOY MINING MACHINERY
STAMLER PRODUCTS

Section

4

4. ROUTINE MAINTENANCE & SPARES

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<u>Instruction Sheet:</u>	
Number:	380210054
By:	N/A
Date:	N/A

Preventative Maintenance

The following pages are a schedule or index of the recommended preventative or scheduled maintenance. Column one indicates the periodicity or frequency that the maintenance is performed. In some cases a definite frequency cannot be assigned in these cases it is assigned the category **Situation** and this situation is explained in column two. Column two is a brief description of the maintenance action. Refer to the proper section of this manual for further information regarding any of the maintenance actions listed.

Instruction Sheet:

Number: 380210004

By: N/A

Date: 10/17/1991

Preventative Maintenance – Breaker (Direct Drive)

Column 1								Column 2
Shift (8 hrs)	Daily (24 hrs)	Weekly (150 hrs)	Monthly (600 hrs)	Quarterly (1800 hrs)	Semiannually (3600 hrs)	Annually (7200 hrs)	Situation	<u>Breaker</u> Maintenance Action
X								Inspect picks and pick holder, repair as necessary. *
X								Inspect bearing cases for abnormal heat and rough operation. Check that grease hoses are connected to the swivels.
			X					Remove lock wire and check tightness of breaker shaft end bolts.
		X						Check that all bolts associated with the electric motor and speed reducer are tight.
							X	Check that all "L9" bolts used to hold the breaker shaft to the machine frame are tight.
		X						Lubricate the duo-cone seal type breaker bearings.

*Note: For maximum efficiency dull picks should be changed out, also dull picks can cause occasional sparking.

Instruction Sheet:

Number: 380210007

By: N/A

Date: N/A

Preventative Maintenance – Electrical (AC)

Column 1								Column 2
Shift (8 hrs)	Daily (24 hrs)	Weekly (150 hrs)	Monthly (600 hrs)	Quarterly (1800 hrs)	Semiannually (3600 hrs)	Annually (7200 hrs)	Situation	Electrical (AC) Maintenance Action
				X				Using a clamp on ammeter, verify that phase currents are balanced to all motors.
					X			Measure and record insulation resistance values of all motors.
						X		Check continuity of ground conductors.
				X				Visually inspect all electrical cabling and connectors for frayed or damaged areas or loose connections.
				X				Grease motor bearings (or 2000 hrs, whichever occurs first).
				X				Check that motor foundation bolts are tight.
X								Test operation of All Stop switch once per shift.
		X						Blow dust, dirt, debris, etc. out of motor cooling fins.
						X		Check vacuum contactors (if equipped).
X								Each Shift test operation of tape switch panic bar.
						X		Inspect the interior of the controller; ensure all connections are clean and tight.

Instruction Sheet:

Number: 380210009
 By: N/A
 Date: 03/28/1991

Preventative Maintenance - Mechanical

Column 1								Column 2
Shift (8 hrs)	Daily (24 hrs)	Weekly (150 hrs)	Monthly (600 hrs)	Quarterly (1800 hrs)	Semiannually (3600 hrs)	Annually (7200 hrs)	Situation	<u>Mechanical</u> Maintenance Action
<u>Conveyor:</u>								
X								Check conveyor chain for correct tension.
X								Inspect for bent or missing; flights, side plates, or pins – repair as necessary
<u>Headshaft:</u>								
X								Inspect bearing cases for abnormal heat and rough operation.
			X					Check tightness of headshaft end bolts (if applicable)
			X					If Timken bearings, check tightness of headshaft end bolts (if applicable)
<u>Tailshaft:</u>								
X								Inspect bearing cases for abnormal heat and rough operation.
			X					Check tightness of headshaft end bolts (if applicable)
			X					If Timken bearings, check tightness of headshaft end bolts (if applicable)
				X				Check sprockets for wear.
<u>Helical Gear Speed Reducer:</u>								
							X	After the first month of operation – drain oil; flush out gearbox with a light flushing oil and refill (see note 1).
				X				Take a sample of oil from the gear reducer (or 2500 hrs, whichever occurs first) (see note 2).
					X			Change oil in gear reducer.
					X			Clean filter element in breather cap of gear reducer.
				X				Grease “taconite” grease seals, if reducer is equipped.

Column 1								Column 2
Shift (8 hrs)	Daily (24 hrs)	Weekly (150 hrs)	Monthly (600 hrs)	Quarterly (1800 hrs)	Semiannually (3600 hrs)	Annually (7200 hrs)	Situation	<u>Mechanical</u> Maintenance Action
<u>Flexible Element Couplings:</u>								
				X				Check alignment, clamp ring fastener, torque and condition of flexible element on rubber element type coupling.
				X				Lubricate metal element coupling

Notes:

1. Presence of iron fines in the initial or break in oil is normal and is the reason for changing the oil at this time.
2. If oil sample indicates that the oil is contaminated or contains iron fines, oil should be changed. Machines in some applications may require the oil to be routinely changed more frequently than every 6 months or 2500 hrs. Frequency of the oil changes should be dictated by oil sampling results.
3. When using Mobil synthetic oil (in any gearbox) it can be used indefinitely as long as the laboratory oil analysis is below the following maximum levels:

Water = 0.5%

Iron = 40-50 ppm

Dirt = 15-20 ppm

Instruction Sheet:

Number: 380210010

By: N/A

Date: 10/19/1981

Preventative Maintenance - Lubrication

Column 1								Column 2
Shift (8 hrs)	Daily (24 hrs)	Weekly (150 hrs)	Monthly (600 hrs)	Quarterly (1800 hrs)	Semiannually (3600 hrs)	Annually (7200 hrs)	Situation	<u>Lubrication</u> Maintenance Action
<u>Centralized Lubrication:</u>								
X								Lubricate all centralized fittings except the one to the take-up cylinders.
		X						Inspect the system for damaged hoses or lines.
							X	Grease the fitting to the conveyor take-up cylinders.
<u>Feeder Breaker:</u>								
		X						Lubricate breaker bearings.
X								Lubricate headshaft bearings.
X								Lubricate tailshaft bearings.
		X						Lubricate tailshaft flaps.

Instruction Sheet:

Number: 380210011
 By: N/A
 Date: 10/17/1983

Preventative Maintenance – Miscellaneous

Column 1								Column 2
Shift (8 hrs)	Daily (24 hrs)	Weekly (150 hrs)	Monthly (600 hrs)	Quarterly (1800 hrs)	Semiannually (3600 hrs)	Annually (7200 hrs)	Situation	Miscellaneous Maintenance Action
<u>Water-Oil type Heat Exchanges</u>								
				X				Check zincs – replace if necessary.
						X		Clean oil sides with a degreasing agent.
						X		Chemically clean water sides.
<u>Air-Oil type Heat Exchangers</u>								
						X		Clean exterior cooling vanes and fan blades.
						X		Clean oil sides with a degreasing agent.
							X	Lubricate fan motor bearings – as per nameplate.
<u>Dust Suppression Water Strainer</u>								
			X					Flush out element.
<u>Fire Suppression Water Strainer</u>								
		X						Flush out element.
						X		Remove and inspect element.
<u>Dry Chemical Fire Extinguisher System</u>								
		X						Inspect system and weigh cartridge.
						X		Inspect system – check condition of dry chemical.
<u>Water Spray</u>								
					X			Inspect overall condition of system.
						X		Test operate system.

Instruction Sheet:

Number: 380210021
 By: JRB
 Date: 09/26/1988

Preventative Maintenance - Hydraulics

Column 1								Column 2
Shift (8 hrs)	Daily (24 hrs)	Weekly (150 hrs)	Monthly (600 hrs)	Quarterly (1800 hrs)	Semiannually (3600 hrs)	Annually (7200 hrs)	Situation	Hydraulics Maintenance Action
	X							Inspect all hoses and fittings for leaks.
	X							Inspect oil level in reservoir; add oil if necessary.
							X	Replace filter elements on pressure filters, every two weeks.
						X		Check all relief valve settings.
						X		Replace fill line filter elements.
						X		Inspect Chromalox heating elements for coatings and corrosion; clean if necessary.
Hydraulic Motor:								
							X	After the first 100 hrs of operation drain and flush the crankcase and refill
							X	Every 100 hrs check mounting bolts for tightness
							X	Every 400 hrs check the front cover screws for tightness.
							X	Every 10,000 hrs overhaul the hydraulic motor.
Hydraulic Pump:								
				X				Check mounting bolts for tightness.
Hydraulic Fluid:								
							X	Every 800 hrs take a sample of mineral or phosphate ester fluid and have it analyzed to determine if the fluid should be changed.
							X	Every 1600 hrs change the hydraulic fluid and the tank breather filter cap.
Air-Oil Heat Exchanger:								
						X		Clean exterior cooling vanes and fan blades.
						X		Clean oil sides with a degreasing agent.
							X	Lubricate the fan motor bearings per the motor nameplate.

Instruction Sheet:

Number: 380210097
 By: N/A
 Date: 02/11/1982

Recommended Spares - Breaker

The following table shows the components that the Stamler Engineering Department feel are necessary spares to have on-hand at the machine's operating location. The quantities listed for each component are derived based on the total number of machines that are ordered. For the correct part number of each component please refer to the parts manual portion of the manual.

Item	Quantity Recommended for Total Number of Units						
	1	2	3	4	5	6-10	11-20
<u>Breaker Complete:</u>							
Breaker Shaft Assembly	--	--	--	--	1	1	1
Bearing	2	2	2	2	4	6	6
Bit w/Retainer	30	30	30	30	10	200	200
Bit Holder	6	6	6	6	12	12	12
Bit Holder Ring	--	--	--	--	1	1	1
Grease Swivel	2	2	2	2	6	6	6
Shear Hub (see note 1)	1	1	1	1	2	2	2
Shear Sprocket (see note 1)	1	1	1	1	2	2	2
Shear Pin (see note 1)	30	30	30	30	100	200	200
Shear Pin Bushing (Hub) (see note 1)	3	3	3	3	3	6	6
Shear Pin Bushing (Sprocket) (see note 1)	5	5	5	5	5	10	10
Shear Hub Retainer (see note 1)	1	1	1	1	2	2	2
Bearing Housing	1	1	1	1	2	2	2
Bearing Housing Cap	1	1	1	1	2	2	2
<u>Breaker Drive:</u>							
Speed Reducer	1	1	1	1	1	2	2
Coupling (motor / speed reducer)	1	1	1	1	2	2	2
Coupling Element	2	2	2	2	4	6	6
Electric Motor	--	--	--	--	--	--	1
Drive Sprocket (see note 1)	1	1	1	1	2	2	2
Flywheel (see note 1)	1	1	1	1	1	1	2
Shrink Disc (see note 1)	1	1	1	1	1	2	2
Breaker Drive Chain (see note 1)	1	1	1	1	1	1	1
Roller Link	6	6	6	6	6	6	6
Connecting Link	6	6	6	6	6	6	6
Outboard Bearing (see note 1)	1	1	1	1	1	2	2
Idler Sprocket (see note 1)	1	1	1	1	1	2	2
Idler Bearing (see note 1)	1	1	1	1	1	2	2

Note 1: This item not supplied on all models of machines manufactured by Stamler.

Instruction Sheet:

Number: 380210096
 By: N/A
 Date: 2/03/1982

Recommended Spares – Conveyor (Hydraulically Driven)

The following table shows the components that the Stamler Engineering Department feel are necessary spares to have on-hand at the machine's operating location. The quantities listed for each component are derived based on the total number of machines that are ordered. For the correct part number of each component please refer to the parts manual portion of the manual.

Item	Quantity Recommended for Total Number of Units						
	1	2	3	4	5	6-10	11-20
<u>Head Shaft Assembly:</u>							
Head Shaft	1	1	1	1	2	2	2
Bearing	2	2	2	2	4	6	6
Drive Sprocket (see note 1)	2	2	2	2	4	6	6
Conveyor Sprocket	1	1	1	1	2	2	2
Coupling (motor / shaft)	1	1	1	1	2	2	2
Coupling Chain (see note 2)	1	1	1	1	2	2	2
<u>Tail Shaft Assembly:</u>							
Tail Shaft	1	1	1	1	2	2	2
Bearing	2	2	2	2	4	6	6
Conveyor Sprocket/Roller Bearing	2	2	2	2	4	6	6
Take-up Cylinder	--	--	--	--	1	1	1
Take-up Packing	2	2	2	2	4	6	6
<u>Conveyor Chain / Flight Assembly:</u>							
Conveyor Chain Complete	--	--	--	--	2	4	4
Connecting Link Regular	6	6	6	6	12	12	12
Connecting Link Flight Attachment	6	6	6	6	12	12	12
Roller Link	6	6	6	6	12	12	12
Flight	6	6	6	6	12	12	12
Flight End	30	30	30	30	50	100	100
<u>Hydraulic Conveyor Drive:</u>							
Hydraulic Motor	--	--	--	--	1	2	2
Sprocket (see note 1)	1	1	1	1	2	2	2
Drive Chain Complete (see note 1)	1	1	1	1	1	1	1
Roller Link (see note 1)	6	6	6	6	6	6	6
Connecting Link (see note 1)	6	6	6	6	6	6	6
<u>Hydraulic Conveyor Drive Pump:</u>							
Pump	1	1	1	1	2	2	2
Coupling (pump / motor)	1	1	1	1	2	2	2

Notes: 1. Used when the hydraulic motor drives the headshaft through a chain and sprocket arrangement.

2. Used when the headshaft is driven directly by a hydraulic motor, using a chain coupling between.

Instruction Sheet:

Number: 380210104

By: N/A

Date: 3/08/1982

Recommended Spares – Hydraulics

The following table shows the components that the Stamler Engineering Department feel are necessary spares to have on-hand at the machine's operating location. The quantities listed for each component are derived based on the total number of machines that are ordered. For the correct part number of each component please refer to the parts manual portion of the manual.

Item	Quantity Recommended for Total Number of Units						
	1	2	3	4	5	6-10	11-20
<u>Hydraulic System (Hydrostatically Driven Conveyor):</u>							
Suction Filter	2	2	2	2	2	4	4
Element	12	12	12	12	25	50	50
Pressure Filter	2	2	2	2	2	4	4
Element	12	12	12	12	25	50	50
B & R Block	1	1	1	1	1	2	2
Electric Controller	1	1	1	1	1	2	2
Heat Exchanger	1	1	1	1	1	2	2
Variable Displacement Pump	--	--	--	--	--	1	1
Charge Pump	1	1	1	1	1	2	2
Hose Set Complete	1	1	1	1	1	2	2

Instruction Sheet:

Number: 380210105
 By: N/A
 Date: 3/08/1982

Recommended Spares – Electrics

The following table shows the components that the Stamler Engineering Department feel are necessary spares to have on-hand at the machine's operating location. The quantities listed for each component are derived based on the total number of machines that are ordered. For the correct part number of each component please refer to the parts manual portion of the manual.

Item	Quantity Recommended for Total Number of Units						
	1	2	3	4	5	6-10	11-20
<u>Electric System (Hydrostatically Driven Conveyor):</u>							
Pressure Switch	1	1	1	1	1	2	2
Vacuum Switch	1	1	1	1	1	2	2
Heat Exchanger Motor	1	1	1	1	1	2	2
Proximity Switch	1	1	1	2	2	2	2
Motion Detector	1	1	1	2	2	2	2
High Temperature Switch	1	1	1	1	1	2	2
Low Oil Level Switch	1	1	1	1	1	2	2
All Stop Switch	1	1	1	1	1	2	2
Belt Switch	1	1	1	2	2	2	2
Servo Power Supply	--	--	--	--	--	1	1
Off Delay Timer	1	1	1	1	1	2	2
On Delay Timer	1	1	1	1	1	2	2
Starter	--	--	--	--	--	1	1
Coil	1	1	1	1	1	2	2
O.L. Heater	3	3	3	3	3	6	6
Selector Switch	1	1	1	1	1	2	2
UV Relay	--	--	--	--	--	1	1
Coil	1	1	1	1	1	2	2
Circuit Breaker	1	1	1	1	1	2	2

Lubrication Specifications

Stamler Specification:	Supplier:				
		Texaco	Mobil	Shell	BP Products
H1	-	Rando HD100	Vacuoline 525	Tellus 100	BP Energol HLP 100
H2	-	Hydrafluid 82 (Fire Resistant)	Pyroguard D	Irus F	BP Energol SFB 13
H3	-	Rando HD68	DTE 26	Tellus 68	BP Energol HLP 68
G1	-	Multigear 80W/90	Mobilube HD 80W/90	Spirax 80W/90	BP Hypogear GL5-EP SAE 90
G2	JOY S460J	Honor Cylinder Oil 68	600W Super Cylinder Oil	Valata J-680	BP Energol AC-C 460
G3	JOY SEP320J	Regal Oil R & Q 150	DTE Oil Extra Heavy	Turbo Oil 150	BP Energol CS 150
G4	JOY SEP320J	Regal Oil R & Q 220	DTE Oil BB	Turbo Oil 22	BP Energol CS 220
B1	-	Multifak EP-0	Mobilux EP-0	Alvania EP RO	BP Energrease PTO or BP Energrease FG-00-EP
B2	-	Multifak EP-0	Mobilux EP-0	Alvania EP RO	BP Energrease PTO or BP Energrease FG-00-EP
B3	-	N/A	Mobil Grease BRB No. 4	N/A	N/A
G5	-	50% Meropa 68 & 50% Meropa 150	Mobilgear 627	Omala 100	BP Energol GR-XP ISO 100
G6	-	Meropa 150	Mobilgear 629	Omala 150	BP Energol GR-XP ISO 150
G7	JOY SEP320J	Meropa 220	Mobilgear SHC 220	Omala 220	BP Energol GR-XP ISO 220
G8	JOY SEP320J	Meropa 68	Mobilgear 626	Omala 68	BP Energol GR-XP ISO 68

Recommended Lubricants

Ref No.	Component to be Lubricated	Specifications	Stamler Lube Spec.	
			H1 >32° F	H3 <32° F
1	Hydraulic System (mineral oil)	Non-detergent premium grade oil containing anti-wear, anti-rust, anti-oxidation and anti-foaming additives with a viscosity of 100 cSt @ 40° F (ISO 100) (see note 3)	See note 3	
2	Hydraulic System (fire resistant)	USBM code 30	H2	
3	Worm Gear Speed Reducer (Cone Drive)	AGMA 8 compound	G2 (see note 2)	
4	Helical Gear Speed Reducers (Falk & Flender)	15° to 60° F - AMB-AGMA 4 (see note 1)	G3 (see note 1)	
		50° to 125° F - AMB-AGMA 5 (see note 1)	G4 (see note 1)	
5	Helical Gear Speed Reducers (Link Belt)	-30° to 40° F - AMB-AGMA 2EP (see note 1)	G8 (see note 1)	
		32° to 100° F - AMB-AGMA 5EP (see note 1)	G7 (see note 1)	
6	Bearings (breaker shaft, head shaft, tail shaft) Take-up Cylinders	NLGI EP - 0	B1	
7	Falk Couplings	NLGI - 2	B2	
8	Gear Couplings (pump to motor)	NLGI EP - 0	B1	
9	Worm Thrust Bearing (On "Worm Over" Cone Drive speed reducer)	NLGI-4 (soda base Exxon Andok "C")	B3	
10	Drive Chains	AGMA 3 EP	G5	
11	"Taconite" Seals (on speed reducers)	NLGI - 2	B2	

Notes:

- JOY SEP320J synthetic oil is approved as a superior lubricant in all Flender, Falk, and Link Belt speed reducers (helical gears) when the ambient temperature is between -30° to 125° F.
- JOY S460J synthetic oil is approved as a superior lubricant in all worm gear speed reducers.
- Hydraulic System - When the ambient temperature is 32° F or lower, ISO 68 grade oil (Stamler Lube Spec "H3") should be used.
- Hydraulic System – synthetic oil is a superior hydraulic fluid and is highly recommended for all weather conditions. In extreme cold weather this is a **must use** fluid.

Approved Synthetic Hydraulic Fluids are:

- Exxon Teresstil SHP synthetic ISO grade 68
- Shell Tellus AW synthetic HD Oil ISO grade 68
- Mobil SHC 526 synthetic ISO grade 68

JOY Gear Lubricants - Product Data

Joy gear lubes are a family of high quality synthetic lubricants based on polyalphaolefin (PAO) base stocks, some containing extreme pressure (EP) additive technology. These lubricants provide outstanding thermal stability and high load carrying capacity in underground mining environments.

Advantages

- Maximum protection of gears and bearings against wear & shock loading
- Good thermal & oxidation stability
- Corrosion protection for all internal components
- High viscosity index for consistent performance over a wide range of operating temperatures
- Tested compatibility with seal materials used inside all Joy gear cases

GEARCASES	SEP320J	S150J	S460J	SEP680J
LONGWALL SHEARERS				
Ranging Arms	X			
Haulage Drives	X			
Lumpbreakers	X			
CONTINUOUS MINERS				
Cutter Case	X			
Gathering Head Cases	X			
Gathering Head Pots	X			
Traction Cases	X			
Traction Planetaries	X			
Rear Conveyor Drive Cases	X			
SHUTTLE CARS				
Traction Primary Cases	X			
Wheel Units	X			
Conveyor Cases (worm gears)			X	
FLEXIBLE CONVEYOR TRAINS (FCT)				
Belt Drives	X			
Traction Primary Cases	X			
Traction Secondary Cases	X			
Lumpbreakers	X			
Cooling Pump				
CHAIN HAULAGE SYSTEMS				
Conveyor Drives	X			
Breaker Drives	X			
Traction Cases	X			
BATTERY HAULERS				
Traction Cases		X		
FEEDER BREAKERS				
Conveyor & Breaker Cases (with helical gears only)	X			
Conveyor & Breaker Cases (with worm gears)			X	
Traction Planetaries	X			
ARMORED FACE CONVEYORS (AFC)				
EP & BP Cases	X			
Tail Transfer Drive				X

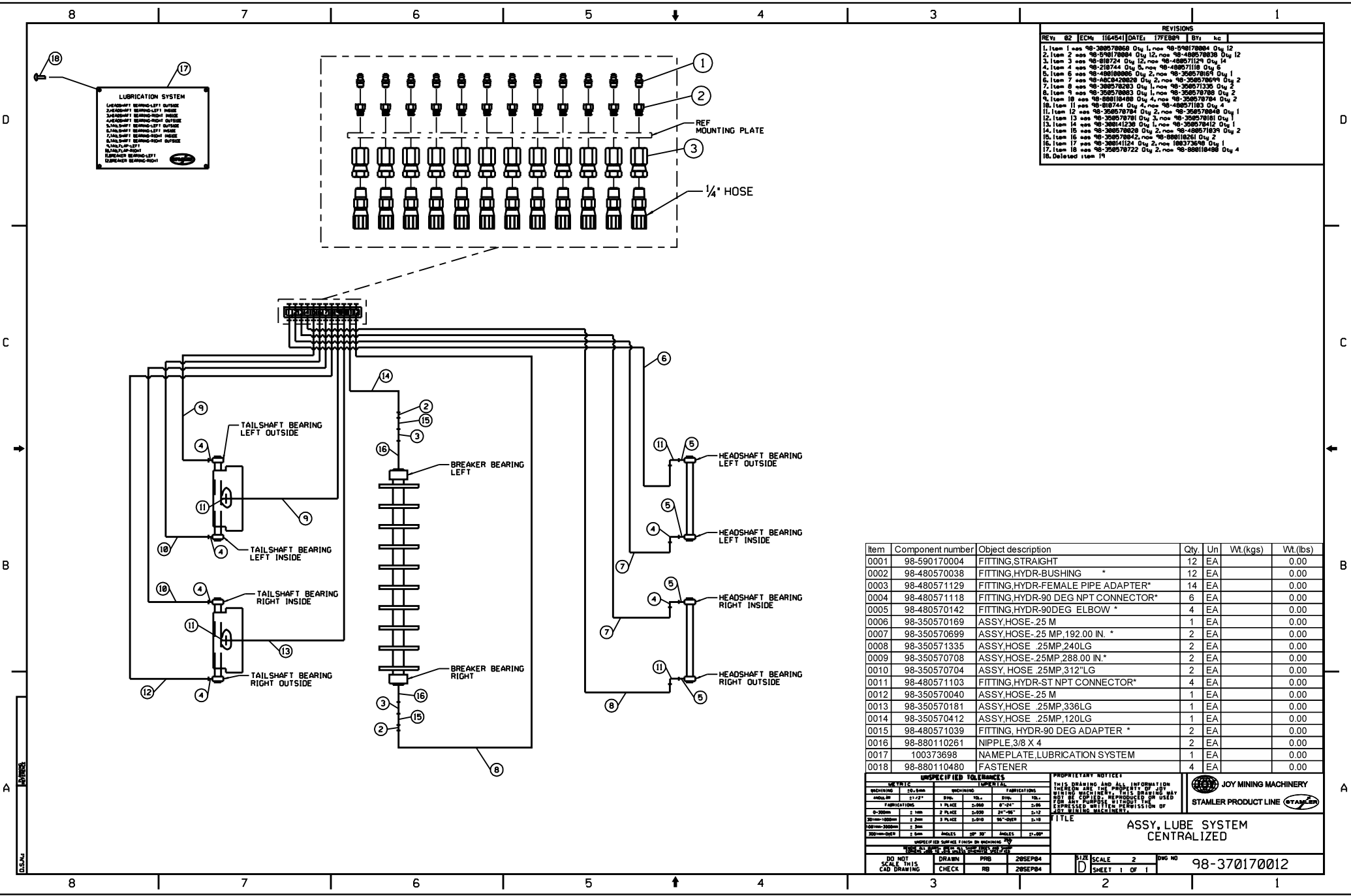
JOY Gear Lubricants - Product Data

Property	Test Method	S150J	SEP320J	S460J	SEP680J
ISO grade	----	150	320	460	680
EP Additives	----	no	yes	no	yes
Viscosity, 40°C, cSt	ASTM D-445	152	324	470	666
Viscosity, 100°C, cSt	ASTM D-445	20	35	47	61
Viscosity Index	ASTM D-2270	152	153	158	160
Flash Point, °C (°F)	ASTM D-92	249 (480)	265 (509)	248 (478)	274 (525)
Pour Point, °C (°F)	ASTM D-97	-43 (-45)	-45 (-49)	-40 (-40)	-45 (-49)
4-Ball Weld Load, kg	ASTM D-2783	210	250	210	240
FZG Scuffing, Pass (A/8.8/90)	ASTM D-5182 (DIN 51354)	11	14	11	13
Foam Tendency, ml Seq I, II, & III	ASTM D-892	0/0/0	0/0/0	0/0/0	0/10/0
Rust Protection	ASTM D-665	Pass	pass	pass	pass
Copper Corrosion, max	ASTM D-130 (3 hr 100°C)	1a	1b	1a	1b
Oxidation Stability Viscosity Increase	ASTM D2893 (312 hr @121°C)	2.3%	2.7%	2.1%	2.5%
Density, lb/gal @ 60°F kg/l @ 15.6°C		7.353 0.881	7.545 0.904	7.503 0.899	7.203 0.863



For additional technical information:
Contact Joy Mining Machinery at gearlube@joy.com or (814) 432-1878

Date Created: November 2009



REVISIONS				
REV.	BY	DATE	DESCRIPTION	BY
1	1	17FEB99	1. Item 1 was 98-350570004 Qty 1, now 98-590170004 Qty 12	1
2	2		2. Item 2 was 98-590170004 Qty 12, now 98-480570038 Qty 12	2
3	3		3. Item 3 was 98-010724 Qty 12, now 98-480571129 Qty 14	3
4	4		4. Item 4 was 98-210744 Qty 5, now 98-480571110 Qty 6	4
5	5		5. Item 5 was 98-480100005 Qty 2, now 98-350570057 Qty 1	5
6	6		6. Item 6 was 98-480100005 Qty 2, now 98-350570057 Qty 2	6
7	7		7. Item 7 was 98-350570283 Qty 1, now 98-350571335 Qty 2	7
8	8		8. Item 8 was 98-350570003 Qty 1, now 98-350570700 Qty 2	8
9	9		9. Item 9 was 98-00010488 Qty 4, now 98-350570784 Qty 2	9
10	10		10. Item 10 was 98-010744 Qty 4, now 98-480571163 Qty 4	10
11	11		11. Item 11 was 98-350570784 Qty 2, now 98-350570845 Qty 1	11
12	12		12. Item 12 was 98-350570784 Qty 3, now 98-350570810 Qty 1	12
13	13		13. Item 13 was 98-350571230 Qty 1, now 98-350570812 Qty 1	13
14	14		14. Item 14 was 98-350570828 Qty 2, now 98-480571039 Qty 2	14
15	15		15. Item 15 was 98-350570842, now 98-880110261 Qty 2	15
16	16		16. Item 16 was 98-350571124 Qty 2, now 100373698 Qty 1	16
17	17		17. Item 17 was 98-350570722 Qty 2, now 98-880110480 Qty 4	17
18	18		18. Deleted item 19	18

Item	Component number	Object description	Qty	Un	Wt (kgs)	Wt (lbs)
0001	98-590170004	FITTING, STRAIGHT	12	EA	0.00	0.00
0002	98-480570038	FITTING, HYDR. BUSHING *	12	EA	0.00	0.00
0003	98-480571129	FITTING, HYDR-FEMALE PIPE ADAPTER*	14	EA	0.00	0.00
0004	98-480571118	FITTING, HYDR-90 DEG NPT CONNECTOR*	6	EA	0.00	0.00
0005	98-480570142	FITTING, HYDR-90 DEG ELBOW *	4	EA	0.00	0.00
0006	98-350570169	ASSY, HOSE-25 M	1	EA	0.00	0.00
0007	98-350570699	ASSY, HOSE-25 MP, 192.00 IN. *	2	EA	0.00	0.00
0008	98-350571335	ASSY, HOSE 25MP, 240LG	2	EA	0.00	0.00
0009	98-350570708	ASSY, HOSE-25MP, 288.00 IN. *	2	EA	0.00	0.00
0010	98-350570704	ASSY, HOSE 25MP, 312" LG	2	EA	0.00	0.00
0011	98-480571103	FITTING, HYDR-ST NPT CONNECTOR*	4	EA	0.00	0.00
0012	98-350570040	ASSY, HOSE-25 M	1	EA	0.00	0.00
0013	98-350570181	ASSY, HOSE 25MP, 336LG	1	EA	0.00	0.00
0014	98-350570412	ASSY, HOSE 25MP, 120LG	1	EA	0.00	0.00
0015	98-480571039	FITTING, HYDR-90 DEG ADAPTER *	2	EA	0.00	0.00
0016	98-880110261	NIPPLE, 3/8 X 4	2	EA	0.00	0.00
0017	100373698	NAMEPLATE, LUBRICATION SYSTEM	1	EA	0.00	0.00
0018	98-880110480	FASTENER	4	EA	0.00	0.00

UNSPECIFIED TOLERANCES

UNITS	INCHES	MILLIMETERS
INCHES	1/16, 1/8, 1/4, 3/8, 1/2, 5/8, 3/4, 7/8, 1, 1 1/4, 1 1/2, 1 3/4, 2, 2 1/4, 2 1/2, 2 3/4, 3, 3 1/4, 3 1/2, 3 3/4, 4, 4 1/4, 4 1/2, 4 3/4, 5, 5 1/4, 5 1/2, 5 3/4, 6, 6 1/4, 6 1/2, 6 3/4, 7, 7 1/4, 7 1/2, 7 3/4, 8, 8 1/4, 8 1/2, 8 3/4, 9, 9 1/4, 9 1/2, 9 3/4, 10	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100

PROPRIETARY NOTICES

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JOY MINING MACHINERY

STAMPER PRODUCT LINE

ASSY. LUBE SYSTEM

CENTRALIZED

ITEM NO

98-370170012

DO NOT SCALE THIS CAD DRAWING	DRAWN	PRG	20SEP84	SIZE	SCALE	2	DWG NO	98-370170012
CHECK	RS	20SEP84		SHEET	1	OF	1	

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JOY MINING MACHINERY
STAMLER PRODUCTS

Section

5

5. TECHNICAL DATA

[Customer Contact Directory](#)

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[Machine Data Sheet](#)

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[General Arrangement Drawing: 98-370064734](#)

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Customer Contact Directory

Ordering of Repair Parts

Prime Contact: Donna Broderick – (859) 484-3125

Alternate Contact: Billy Joe Hawkins – (859) 484-3129

Performance/Service Problems

Prime Contact: Pravin Patel – (859) 484-3159

Alternate Contact: Bill Harney – (859) 484-3152

New Machine Sales

Prime Contact: James Lukins – (859) 484-3126

Alternate Contact: Billy Joe Hawkins – (859) 484-3129

NOTE:

All the persons listed can be reached at the telephone numbers below.

**Joy Mining Machinery
Stamler Product Line
100 Stamler Street
Millersburg, KY 40348
Phone (859) 484-3431 ♦ Fax (859) 484-3102**

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DATA SHEET



JOY MINING MACHINERY
A Joy Global Inc. Company
STAMLER PRODUCT LINE

HYDRAULIC CONVEYOR DRIVE - SURFACE

PROJECT LEADER: CHRIS BUCHANAN DATE COMPLETED: 20APR09
SERIAL NUMBER: FB14430
MINE: SAU – NUSA TAMBANG PRATAMA
MODEL: BF-32F-60-76F
G/A DRAWING #: 98-370064734
MACHINE LOCATION - COUNTRY: INDONESIA
STATE:

CONVEYOR HYDRAULIC MOTOR

PART NUMBER: 98-79023F005 RELIEF VLV SETTING: 4500 PSI
HEAD SHAFT RPM: 30.7 CONVEYOR FPM: 80.5
SPROCKET RATIO: N/A

BREAKER ELECTRIC MOTOR

PART NUMBER: 98-790550556 VOLTS: 380 VAC
HORSE POWER: 190 KW FRAME: 449TSD
SHAFT DIAMETER: 2.375" KEY: .625" SQ
RPM: 1500 ENCLOSURE: TEFC
MTR CABLE:

ADDITIONAL DATA:

BRKR SPD REDUCER: 98-190110393 BRKR SPKT RATIO: N/A
BRKR REDUCER RATIO: 41.45 : 1 BRKR SHEAR PIN DIA: N/A
BREAKER DIA: 50.00" BRKR BRG DIA: 8.00"
COUPLING: UJT328 SHEAR PIN RADIUS: N/A

DISTANCE, PICK POINT TO DECK PLATE: 4.36"

MOUNTING: FOOT GROUND CLEARANCE: N/A
O/A MACHINE WIDTH: 159.50"
HOPPER WIDTH: 114.00" (@BASE) HOPPER SLOPE: 90°
MACHINE HEIGHT: 95.00" FROM BASE OF: FOOT
CONVEYOR WIDTH: 90" FLIGHT SIZE: 1.75" X 4.00"
CONVEYOR CHAIN: 3.5 P FLIGHT SPACING: EVERY 6TH PITCH
H/S BRG DIA: 4.44" T/S BRG DIA: 3.44"
LUBRICATION: CENTRAL SIDEBOARDS: N/A
FIRE EXT SYSTEM: N/A DUST SUPPRESSION: N/A

POWER UNIT

HYD PUMP - ELECTRIC MOTOR

PART NUMBER: 98-790550447 VOLTS: 380 VAC
HORSE POWER: 90 KW FRAME: 444/5 TSC
SHAFT DIAMETER: 2.375" KEY: .625" SQ
RPM: 1500 ENCLOSURE: TEFC
MTR CABLE:

DATA SHEET



JOY MINING MACHINERY
A Joy Global Inc. Company
STAMLER PRODUCT LINE

HYDRAULIC CONVEYOR DRIVE - SURFACE

POWER UNIT (CONTINUED)

STARTER: BY OTHERS

HYDRAULIC PUMP#:	(2) 98-79026F049 / (2) 98-33026F004	MODEL @ GPM:	84 GPM / 16 GPM
HYD CONTROL VLV#:	N/A	BELT SWITCH:	N/A
HEAT X/C TYPE:	AIR/OIL	HEAT X/C P/N:	98-890110058
TANK HEATERS:	N/A	COUPLING:	98-330530048 (SIZE 2.50)

TYPE OF OIL FOR HYD SYSTEM: STAMLER LUBE SPEC H1 (OR APPROVED EQUIVALENT, SEE 98-380210288)

TYPE OF OIL FOR SPEED REDUCER: SEP320J STAMLER LUBE SPEC G3 (OR APPROVED EQUIVALENT, SEE 98-380210288)

SPECIAL LUBE REQUIREMENTS N/A

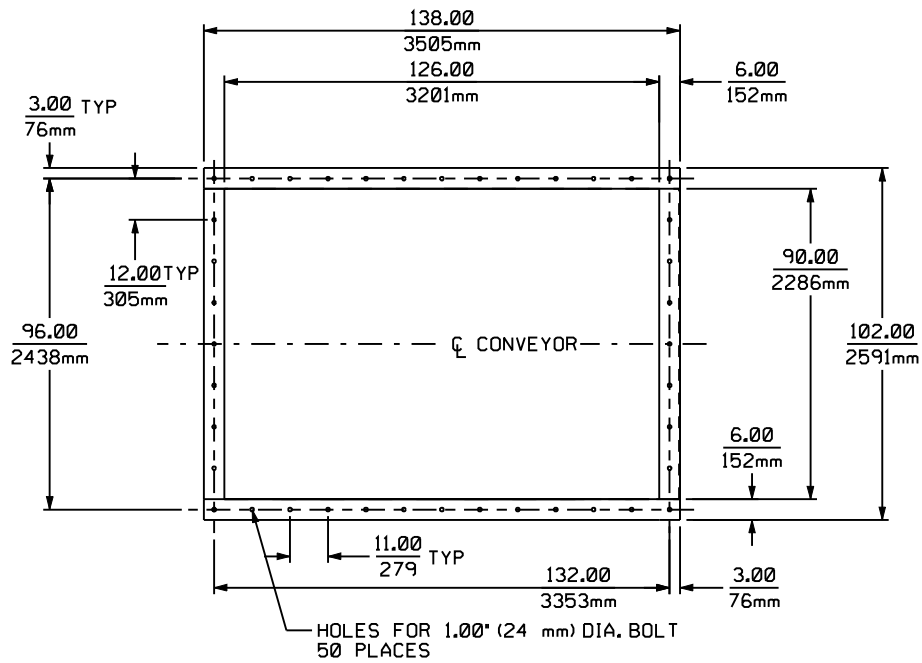
FRAME CONSTRUCTION

LENGTH:	156.00 (@BASE)	WIDTH:	84.00 (@BASE)
HEIGHT:	83.68	SUMP:	N/A
ROOF:	YES	MOUNTING:	FOOT
ENCLOSED:	N/A		

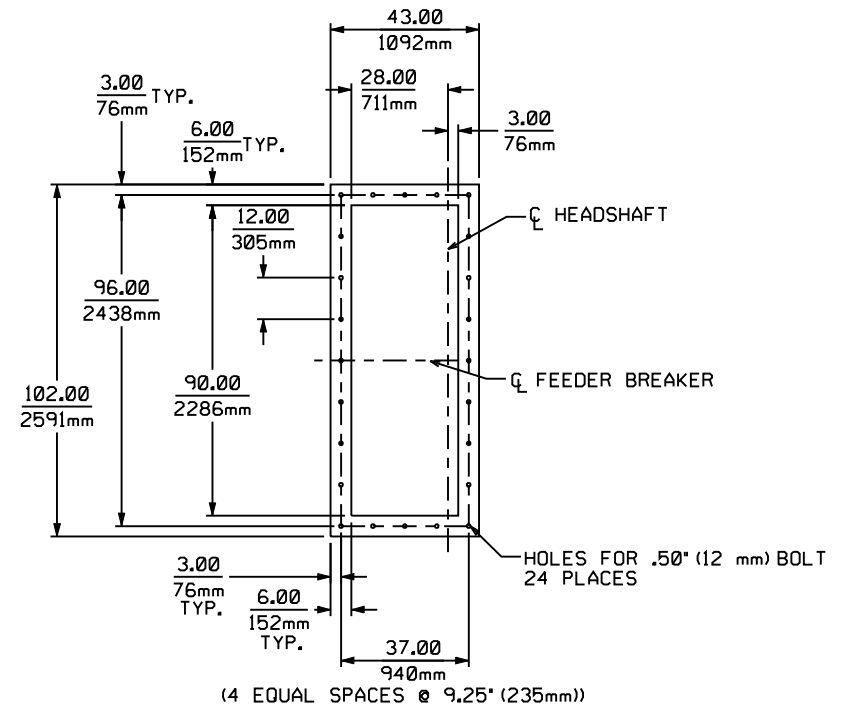
NOTES:

- 6 METER LONG CONNECTING HOSES BETWEEN POWER UNIT AND FRAME
- SCREW TAKE-UP ON CONVEYORS
- BREAKER CLEANER
- HYDRAULIC OPERATED REAR DOOR ON HOPPER
- SEPARATE POWER UNIT FOR HYDRAULIC HOPPER DOOR **(TO BE REMOVED FOR SHIPPING)**
- ADJUSTABLE INTAKE HOPPER FLANGE

2	PC224900/CE-86	ON	DRG	1
REVISIONS				
REV:	8	ECN:	1213187	DATE: 26-APR-11 BY: CRB
UPDATED VIEWS AND LOCATIONS OF LIFTING EYES				



HOPPER FLANGE
DETAIL



VIEW B-B

98-370064734	REV	8	SCALE	1:1	SHEET	2 OF 3	DATE	26-APR-11	BY	CRB	ECN	1213187	REV	8
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